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MIL-HDBK-17-2C
Volume 2 of 3
25 FEBRUARY 1994

MILITARY HANDBOOK

POLYMER MATRIX COMPOSITES

VOLUME II. MATERIAL PROPERTIES



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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20025

1. This Military Handbook is approved for use by all Departments and Agencies of the Department of Defense.

2. Every effort has been made to reflect the latest information on polymeric composites. The Handbook is continually reviewed and revised to insure its completeness and currentness. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Department of the Army, U.S. Army Research Laboratory, Materials Directorate, Attn: AMSRL-MA-TS, MIL-HDBK-17 Coordinator, Watertown, MA 02172-0001, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. Military Handbook 17 (Part 1) provides guidelines and material properties for polymer (organic) matrix composite materials. This handbook encompasses, but is not limited to, polymeric composites intended for aircraft and aerospace vehicles and military combat vehicle applications. Metal matrix composites (MMC) , ceramic matrix composites (CMC), and carbon/carbon composites (C/C) will be covered in separate military handbooks as developments occur.

4. This standardization handbook has been developed and is being maintained as a joint effort of the Department of Defense and the Federal Aviation Administration.

5. The information contained in this handbook was obtained from materials producers, industry, reports on Government sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity.

6. All information and data contained in this handbook have been coordinated with industry and the US Army, Navy, Air Force, NASA, and Federal Aviation Administration prior to publication.

7. Copies of this document and revisions thereto may be obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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SUMMARY OF CHANGES IN REVISION MIL-HDBK-17-2C

Chapter 1

Figure 1.5.1 on mechanical property notation was added.

In Section 1.6, a definition for traveller was added. The definitions for polymerization, addition polymerization, chain-growth polymerization, and step-growth polymerization were revised.

Chapters 4 and 10

The format for the presentation of data in the handbook has been revised. This revision affected all data presented in Chapters 4 and 10.

The density, fiber areal weight, and ply thickness information were removed from the header block on the summary page. A second page of summary information was added to include nominal values, measured ranges, and test methods for this information. Also added on the new second page of summary information, is a summary of thermophysical properties. Additional pages for summary tables for laminate and other properties will be appended as necessary.

A block of information on dates was added to the first summary page. The dates of component, prepreg, and composite manufacture may be relevant to the property values. The test methods which are acceptable for the handbook are those approved as the time of testing. The data documentation requirements applicable to a given data set are those approved when the data are submitted. The date of analysis reflects the approved analysis methods.

All data sets in this revision were re-analyzed in January 1993. This analysis incorporated the revision to the ANOVA method released in MIL-HDBK-17-1C, Change Notice 1, dated 4 November 1992.

The revision of the data tables included the change in notation included in MIL-HDBK-17-1C, Change Notice 1. Subscripts 1, 2, and 3 are used to indicate lamina properties for warp, fill, and z-directions for fabrics, as well as fiber, transverse, and z-directions in tape.

The revision to the data table format largely affected the header. The boldface index block has been expanded. The lay-up code was moved from the table header to the index block. The property, direction, temperatures, and moisture contents were added to the index block. In addition, a table number was added to the index block.

The test method was moved from the body of the data table to the header. The method of modulus calculation was added to the header.

When the ANOVA method was used to calculate a B-basis value, the estimated population standard deviation and the tolerance factor were listed as constants, C_1 and C_2.

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Chapter 1. GENERAL INFORMATION

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1.1 INTRODUCTION. The standardization of a statistically-based mechanical property data base, procedures used, and overall material guidelines for characterization of composite material systems is recognized as being beneficial to both manufacturers and governmental agencies. It is also recognized that a complete characterization of the capabilities of any engineering material system is primarily dependent on the inherent material physical and chemical composition which precede, and are independent of, specific applications. Therefore, at the material system characterization level, the data and guidelines contained in this handbook are applicable to military and commercial products and provide the technical basis for establishing statistically valid design values acceptable to certificating or procuring agencies.

This standardization handbook has been developed and is maintained as a joint effort of the Department of Defense and the Federal Aviation Administration. It is oriented toward the standardization of methods used to develop and analyze mechanical property data on current and emerging composite materials.

1.2 PURPOSE, SCOPE, AND ORGANIZATION OF VOLUME 2. The purpose of this handbook is to provide a standard source of statistically-based mechanical property data for current and emerging composite materials. Strength and strain-to-failure properties will be reported either in terms of B-values or S-values. A B-basis value is the mechanical property value above which at least 90 percent of the population is expected to fall, with a confidence of 95 percent. An S-basis value is the mechanical property value which is usually the specified minimum value of the appropriate government specification for the material. Physical, chemical, and mechanical values of the composite constituents - the fibers, matrix material, and prepreg - will be reported where applicable. Later chapters will include data summaries for the various composite systems. Individual chapters focus on particular reinforcement fibers.

Statistically-based strength properties will be defined for each composite material system over a range of potential usage conditions. The intent will be to provide data at the upper and lower limits of the potential environmental conditions for a particular material, so that application issues do not govern the mechanical property characterizations. If data are also available at intermediate environmental conditions, they will also be used to more exactly define the relationship between the material properties and the effect of the environment on those properties. The statistically-based strength data which are available are tabulated in Volume 2. These data are useful as a starting point for establishing structural design allowables when stress and strength analysis capabilities permit lamina level margin of safety checks. Depending on the application, some structural design allowables will have to be determined empirically at the laminate and composite level, since MIL-HDBK-17 does not provide these data. Additional information and properties will be added as they become available and are demonstrated to meet the guideline criteria.

All data included herein are based on test specimens only. Test specimen dimensions conform with those specified for the particular test method which is used. Standard test methods are recommended where possible (ASTM standards are the primary source). The designer and all users must be responsible for any translation of the data contained herein to other coupon dimensions, temperature, humidity, and other environmental conditions not covered in this document. Problems such as scale-up effects and the influence of the test method selected on properties are also not addressed in this document. The manner in which S-basis values are used is also up to the discretion of the designer. In general, decisions concerning which properties to use for a specific application or design are the responsibility of the designer and are outside the scope of this handbook.

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1.3 MATERIALS SYSTEMS CODES. The materials systems codes which are used in the handbook consist of a fiber system code and a matrix material code separated by a virgule (/). The codes for the fiber and matrix materials appear in Tables 1.3(a) and (b).

TABLE 1.3(a) *Fiber system codes.*

AlO	Alumina
Ar	Aramid
B	Boron
C	Carbon
DGl	D-Glass
EGl	E-Glass
Gl	Glass
Gr	Graphite
Li	Lithium
PAN	Polyacrylonitrile
PBT	Polybenzothiazole
Q	Quartz
Si	Silicon
SiC	Silicon carbide
SGl	S-Glass
Ti	Titanium
W	Tungsten

TABLE 1.3(b) *Matrix material codes.*

BMI	Bismaleimide
EP	Epoxy
PAI	Polyamide-imide
PEEK	Polyetheretherketone
PEI	Polyetherimide
PES	Polyethersulfone
PI	Polyimide
PPS	Polyphenylene sulfide
PSU	Polysulfone
SI	Silicone
TP	Thermoplastic
TPES	Thermoplastic polyester
	Fluorocarbon
	Phenolic
	Polybenzimidazole

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1.4 LAMINATE ORIENTATION CODES. The purpose of a laminate orientation code is to provide a simple, easily understood method of describing the lay-up of a laminate. The laminate orientation code is based largely on the code used in the Advanced Composites Design Guide (Reference 1.4(a)). The following information and the examples in Figure 1.4 describe the laminate orientation code that will be used in MIL-HDBK-17.

1. The orientation of each lamina with respect to the X-axis is indicated by the angle between the fiber direction and the X-axis. Positive angles are measured counter-clockwise from the X-axis when looking toward the lay-up surface (right-hand rule).

2. When indicating the lay-up of a weave, the angle is measured between the warp direction and the X-axis.

3. Orientations of successive laminae with different absolute values are separated by a virgule (/).

4. Two or more adjacent laminae with the same orientation are indicated by adding a subscript, to the angle of the first such lamina, equal to the number of repetitions of laminae with that orientation.

5. Laminae are listed in order from the first laid up to the last. Brackets are used to indicate the beginning and the end of the code.

6. A subscript of 'S' is used if the first half of the lay-up is indicated and the second half is symmetric with the first. When a symmetric lay-up with an odd number of laminae is shown, the layer which is not repeated is indicated by overlining the angle of that lamina.

7. A repeated set of laminae are enclosed in parentheses and the number of repetitions of the set indicated by a subscript.

8. The convention used for indicating materials is no subscript for a tape ply and a subscript "f" for a weave.

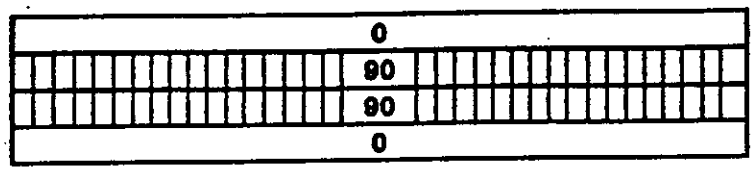
9. The laminate code for a hybrid has the different materials contained in the laminate indicated by subscripts on the laminae.

10. Since the majority of computer programs do not permit the use of subscripts and superscripts, the following modifications are recommended based on ASTM Committee E-49 guidelines (Reference 1.4(b)).

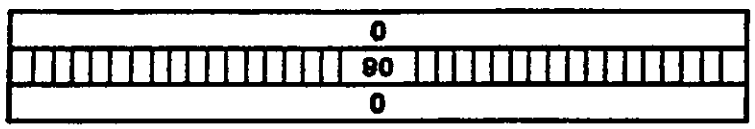
    a. Subscript information will be preceded by a colon (:), e.g., [90/0:2/45]:S.

    b. A bar over a ply (designating a non-repeated ply in a symmetric laminate) should be indicated by a backslash (\) after the ply, e.g., [0/45/90\]:S.

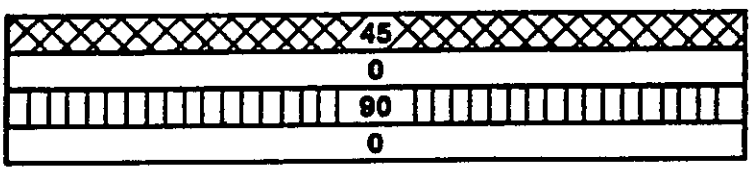
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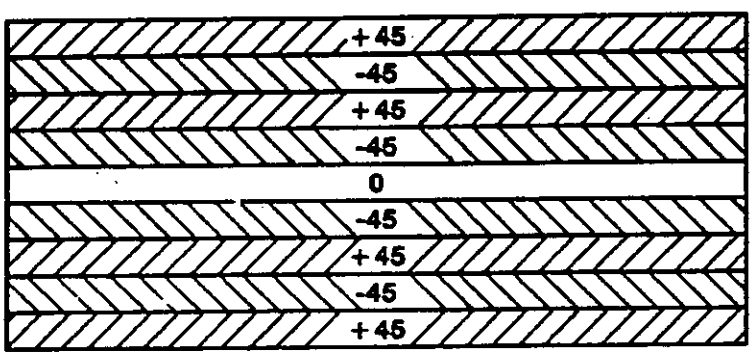
$[0/90]_s$



$[0/\overline{90}]_s$



$[45_f/0/90/0]$



$[(\pm 45)_2/\overline{0}]_s$

Figure 1.4 *Example laminate orientation codes.*

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1.5 SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS. This section defines the symbols and abbreviations which are used within MIL-HDBK-17 and describes the system of units which is maintained. Common usage is maintained where possible. References 1.5(a), 1.5(b), and 1.5(c) served as primary sources for this information.

1.5.1 Symbols and abbreviations. The symbols and abbreviations used in this document are defined in this section with the exception of statistical symbols. These latter symbols are defined in Chapter 8. The lamina/laminate coordinate axes used for all properties and a summary of the mechanical property notation are shown in Figure 1.5.1.

- The symbols f and m, when used as either subscripts or superscripts, always denote fiber and matrix, respectively.
- The type of stress (e.g., cy - compression yield) is always used in the superscript position.
- Direction indicators (e.g., x, y, z, 1, 2, 3, etc.) are always used in the subscript position.
- Ordinal indicators of laminae sequence (e.g., 1, 2, 3, etc.) are used in the superscript position and must be parenthesized to distinguish them from mathematical exponents.
- Other indicators may be used in either subscript or superscript position, as appropriate for clarity.
- Compound symbols (i.e., basic symbol plus indicators) which deviate from these rules are shown in their specific form in the following list.

The following general symbols and abbreviations are considered standard for use in MIL-HDBK-17. Where exceptions are made, they are noted in the text and tables.

A - (1) area (m^2,in^2)
- (2) ratio of alternating stress to mean stress
- (3) A-basis for mechanical property values

ASTM - American Society for Testing and Materials

a - (1) length dimension (mm,in)
- (2) acceleration (m/sec^2,ft/sec^2)
- (3) amplitude
- (4) crack or flaw dimension (mm,in)

B - (1) B-basis for mechanical property values
- (2) biaxial ratio

Btu - British thermal unit(s)

b - width dimension (mm,in), e.g. the width of a bearing or compression panel normal to load, or breadth of beam cross-section

C - (1) specific heat (kJ/kg °C,BTU/lb °F)
- (2) Celsius

CF - centrifugal force (N,lbf)

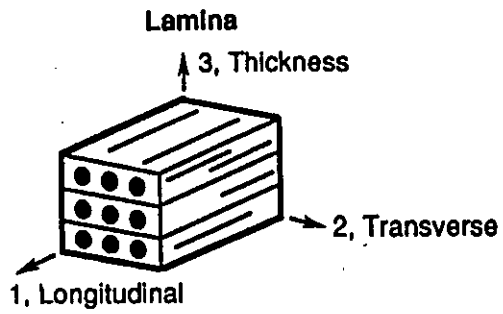
CG - (1) center of mass, "center of gravity"
- (2) area or volume centroid

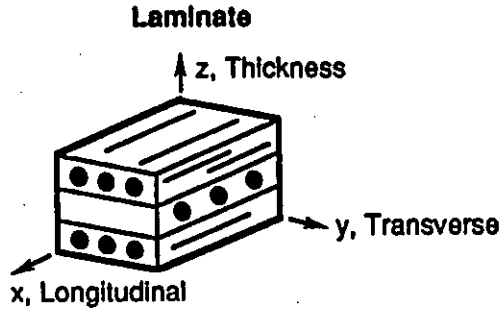
$\mathbb{C}_L$ - centerline

c - column buckling end-fixity coefficient

$\bar{c}$ - honeycomb sandwich core depth (mm,in)

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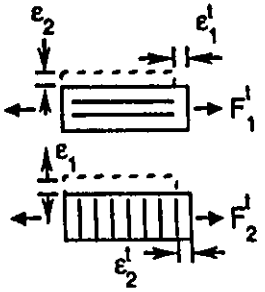
Notation = H_i^{jk}

Where,

H = { σ, τ; Applied Normal, Shear Stress
F; Allowable Stress
ε, γ; Extensional, Shear Strain
E, G; Young's, Shear Modulus
ν; Poisson's Ratio }

Note: ν_{12}^l = Major Poisson's Ratio = $-\frac{\varepsilon_2^l}{\varepsilon_1^l}$

ν_{21}^l = Minor Poisson's Ratio = $-\frac{\varepsilon_1^l}{\varepsilon_2^l}$



i = { 1; Longitudinal
2; Transverse
3; Thickness
12, 13, 32; Shear, Poisson's } Lamina

{ x; Longitudinal
y; Transverse
z; Thickness
xy, xz, zy; Shear, Poisson's } Laminate

j = { c; Compression
t; Tension
s; Shear }

k = { y; Yield
u; Ultimate, Not Used for Stiffness }

Examples, F_2^{tu} = Lamina Ultimate Transverse Tensile Allowable Stress

E_z^c = Laminate Compressive Young's Modulus, Thickness Direction

FIGURE 1.5.1 *Mechanical property notation.*

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cpm - cycles per minute
D - (1) diameter (mm,in)
- (2) hole or fastener diameter (mm,in)
- (3) plate stiffness (N-m,lbf-in)
d - mathematical operator denoting differential
E - modulus of elasticity in tension, average ratio of stress to strain for stress below proportional limit (GPa,Msi)
E' - storage modulus (GPa,Msi)
E" - loss modulus (GPa,Msi)
E_c - modulus of elasticity in compression, average ratio of stress to strain for stress below proportional limit (GPa,Msi)
E_c' - modulus of elasticity of honeycomb core normal to sandwich plane (GPa,Msi)
E^{sec} - secant modulus (GPa,Msi)
E^{tan} - tangent modulus (GPa,Msi)
e - minimum distance from a hole center to the edge of the sheet (mm,in)
e/D - ratio of edge distance to hole diameter (bearing strength)
F - (1) stress (MPa,ksi)
- (2) Fahrenheit
F^b - bending stress (MPa,ksi)
F^{ccr} - crushing or crippling stress (upper limit of column stress for failure) (MPa,ksi)
F^{su} - ultimate stress in pure shear (this value represents the average shear stress over the cross-section) (MPa,ksi)
f - (1) internal (or calculated) stress (MPa,ksi)
- (2) stress applied to the gross flawed section (MPa,ksi)
- (3) creep stress (MPa,ksi)
f^c - internal (or calculated) compressive stress (MPa,ksi)
f_c - (1) maximum stress at fracture (MPa,ksi)
- (2) gross stress limit (for screening elastic fracture data (MPa,ksi)
ft - foot, feet
G - modulus of rigidity (shear modulus) (GPa,Msi)
GPa - gigapascal(s)
g - (1) gram(s)
- (2) acceleration due to gravity (m/s^2,ft/s^2)
H/C - honeycomb (sandwich)
h - height dimension (mm,in) e.g. the height of a beam cross-section
hr - hour(s)
I - area moment of inertia (mm^4,in^4)
i - slope (due to bending) of neutral plane in a beam, in radians
in. - inch(es)
J - (1) torsion constant (= I_p for round tubes) (m^4,in^4)
- (2) Joule
K - (1) Kelvin
- (2) stress intensity factor (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
- (3) coefficient of thermal conductivity (W/m °C, BTU/ft^2/hr/in/°F)
- (4) correction factor
- (5) dielectric constant
K_{app} - apparent plane strain fracture toughness or residual strength (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_c - critical plane strain fracture toughness, a measure of fracture toughness at point of crack growth instability (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

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K_{Ic} - plane strain fracture toughness (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
K_N - empirically calculated fatigue notch factor
K_s - plate or cylinder shear buckling coefficient
K_t - (1) theoretical elastic stress concentration factor
- (2) t_w/c ratio in H/C sandwich
K_x,K_y - plate or cylinder compression buckling coefficient
k - strain at unit stress (m/m,in/in)
L - cylinder, beam, or column length (mm,in)
L' - effective column length (mm,in)
lb - pound
M - applied moment or couple (N-m,in-lbf)
Mg - megagram(s)
MPa - megapascal(s)
MS - military standard
M.S. - margin of safety
MW - molecular weight
MWD - molecular weight distribution
m - (1) mass (kg,lb)
- (2) number of half wave lengths
- (3) metre
- (4) slope
N - (1) number of fatigue cycles to failure
- (2) number of laminae in a laminate
- (3) distributed in-plane forces on a panel (lbf/in)
- (4) Newton
- (5) normalized
NA - neutral axis
NAS - National Aerospace Standard
n - (1) number of times in a set
- (2) number of half or total wavelengths
- (3) number of fatigue cycles endured
P - (1) applied load (N,lbf)
- (2) exposure parameter
- (3) probability
P^u - test ultimate load, (N,lb per fastener)
P^y - test yield load, (N,lb per fastener)
p - normal pressure (Pa,psi)
psi - pounds per square inch
Q - area static moment of a cross-section (mm^3,in^3)
q - shear flow (N/m,lbf/in)
R - (1) algebraic ratio of minimum load to maximum load in cyclic loading
- (2) reduced ratio
RA - reduction of area
R.H. - relative humidity
RMS - root-mean-square
RT - room temperature
r - (1) radius (mm,in)
- (2) root radius (mm,in)
- (3) reduced ratio (regression analysis)
S - (1) shear force (N,lbf)
- (2) nominal stress in fatigue (MPa,ksi)

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- (3) S-basis for mechanical property values

S_a - stress amplitude in fatigue (MPa,ksi)

S_e - fatigue limit (MPa,ksi)

S_m - mean stress in fatigue (MPa,ksi)

S_{max} - highest algebraic value of stress in the stress cycle (MPa,ksi)

S_{min} - lowest algebraic value of stress in the stress cycle (MPa,ksi)

S_R - algebraic difference between the minimum and maximum stresses in one cycle (MPa,ksi)

S.F. - safety factor

s - (1) arc length (mm,in)

- (2) H/C sandwich cell size (mm,in)

T - (1) temperature (°C,°F)

- (2) applied torsional moment (N-m,in-lbf)

T_d - thermal decomposition temperature (°C,°F)

T_F - exposure temperature (°C,°F)

T_g - glass transition temperature (°C,°F)

T_m - melting temperature (°C,°F)

t - (1) thickness (mm,in)

- (2) exposure time (s)

- (3) elapsed time (s)

V - (1) volume (mm^3,in^3)

- (2) shear force (N,lbf)

W - (1) weight (N,lbf)

- (2) width (mm,in)

- (3) Watt

x - distance along a coordinate axis

Y - nondimensional factor relating component geometry and flaw size

y - (1) deflection (due to bending) of elastic curve of a beam (mm,in)

- (2) distance from neutral axis to given point

- (3) distance along a coordinate axis

Z - section modulus, I/y (mm^3,in^3)

α - coefficient of thermal expansion (m/m/°C,in/in/°F)

γ - shear strain (m/m,in/in)

Δ - difference (used as prefix to quantitative symbols)

δ - elongation or deflection (mm,in)

ϵ - strain (m/m,in/in)

ϵ - elastic strain (m/m,in/in)

ϵ - plastic strain (m/m,in/in)

η - plasticity reduction factor

$[\eta]$ - intrinsic viscosity

η^* - dynamic complex viscosity

ν - Poisson's ratio

ρ - (1) density (kg/m3,lb/in3)

- (2) radius of gyration (mm,in)

ρ_c - H/C sandwich core density (kg/m^3,lb/in^3)

Σ - total, summation

σ - standard deviation

σ_{ij}, τ_{ij} - stress in j direction on surface whose outer normal is in i direction (i, j = 1, 2, 3 or x, y, z) (MPa,ksi)

T - applied shear stress (MPa,ksi)

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ω - angular velocity (radians/s)
∞ - infinity

1.5.1.1 *Constituent properties.* The following symbols apply specifically to the constituent properties of a typical composite material.

E^f - Young's modulus of filament material (MPa,ksi)
E^m - Young's modulus of matrix material (MPa,ksi)
E_x^g - Young's modulus of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (MPa,ksi)
E_y^g - Young's modulus of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (MPa,ksi)
G^f - shear modulus of filament material (MPa,ksi)
G^m - shear modulus of matrix (MPa,ksi)
G_{xy}^g - shear modulus of impregnated glass scrim cloth (MPa,ksi)
G_{cx}' - shear modulus of sandwich core along X-axis (MPa,ksi)
G_{cy}' - shear modulus of sandwich core along Y axis (MPa,ksi)
ℓ - filament length (mm,in)
α^f - coefficient of thermal expansion for filament material (m/m/°C,in/in/°F)
α^m - coefficient of thermal expansion for matrix material (m/m/°C,in/in/°F)
α_x^g - coefficient of thermal expansion of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (m/m/°C,in/in/°F)
α_y^g - coefficient of thermal expansion of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (m/m/°C,in/in/°F)
ν^f - Poisson's ratio of filament material
ν^m - Poisson's ratio of matrix material
ν_{xy}^g - glass scrim cloth Poisson's ratio relating to contraction in the transverse (or fill) direction as a result of extension in the longitudinal (or warp) direction
ν_{yx}^g - glass scrim cloth Poisson's ratio relating to contraction in the longitudinal (or warp) direction as a result of extension in the transverse (or fill) direction
σ - applied axial stress at a point, as used in micromechanics analysis (MPa,ksi)
τ - applied shear stress at a point, as used in micromechanics analysis (MPa,ksi)

1.5.1.2 *Laminae and laminates.* The following symbols, abbreviations, and notations apply to composite laminae and laminates. At the present time the focus in MIL-HDBK-17 is on laminae properties. However, commonly used nomenclature for both laminae and laminates are included here to avoid potential confusion.

A_{ij} (i,j = 1,2,6) - extensional rigidities (N/m,lbf/in)
B_{ij} (i,j = 1,2,6) - coupling matrix (N,lbf)
C_{ij} (i,j = 1,2,6) - elements of stiffness matrix (Pa,psi)
D_x, D_y - flexural rigidities (N-m,lbf-in)
D_{xy} - twisting rigidity (N-m,lbf-in)
D_{ij} (i,j = 1,2,6) - flexural rigidities (N-m,lbf-in)
E_1 - Young's modulus of lamina parallel to filament or warp direction (GPa,Msi)

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E_2 - Young's modulus of lamina transverse to filament or warp direction (GPa,Msi)
E_x - Young's modulus of laminate along x reference axis (GPa,Msi)
E_y - Young's modulus of laminate along y reference axis (GPa,Msi)
G_{12} - shear modulus of lamina in 12 plane (GPa,Msi)
G_{xy} - shear modulus of laminate in xy reference plane (GPa,Msi)
h_i - thickness of ith ply or lamina (mm,in)
M_x, M_y, M_{xy} - bending and twisting moment components (N-m/m, in-lbf/in in plate and shell analysis)
n_f - number of filaments per unit length per lamina
Q_x, Q_y - shear force parallel to z axis of sections of a plate perpendicular to x and y axes, respectively (N/m,lbf/in)
Q_{ij} (i,j = 1,2,6) - reduced stiffness matrix (Pa,psi)
u_x, u_y, u_z - components of the displacement vector (mm,in)
u_x^o, u_y^o, u_z^o - components of the displacement vector at the laminate's midsurface (mm,in)
V_v - void content (% by volume)
V_f - filament content or fiber volume (% by volume)
V_g - glass scrim cloth content (% by volume)
V_m - matrix content (% by volume)
V_x, V_y - edge or support shear force (N/m,lbf/in)
W_f - filament content (% by weight)
W_g - glass scrim cloth content (% by weight)
W_m - matrix content (% by weight)
W_s - weight of laminate per unit surface area (N/m^2,lbf/in^2)
α_1 - lamina coefficient of thermal expansion along 1 axis (m/m/°C,in/in/°F)
α_2 - lamina coefficient of thermal expansion along 2 axis (m/m/°C,in/in/°F)
α_x - laminate coefficient of thermal expansion along general reference x axis (m/m/°C,in/in/°F)
α_y - laminate coefficient of thermal expansion along general reference y axis (m/m/°C,in/in/°F)
α_{xy} - laminate shear distortion coefficient of thermal expansion (m/m/°C,in/in/°F)
θ - angular orientation of a lamina in a laminate, i.e., angle between 1 and x axes (°)
λ_{xy} - product of ν_{xy} and ν_{yx}
ν_{12} - Poisson's ratio relating contraction in the 2 direction as a result of extension in the 1 direction.
ν_{21} - Poisson's ratio relating contraction in the 1 direction as a result of extension in the 2 direction.
ν_{xy} - Poisson's ratio relating contraction in the y direction as a result of extension in the x direction
ν_{yx} - Poisson's ratio relating contraction in the x direction as a result of extension in the y direction
ρ_c - density of a single lamina (kg/m^3,lb/in^3)
$\overline{\rho}_c$ - density of a laminate (kg/m^3,lb/in^3)
ϕ - (1) general angular coordinate, (°)
- (2) angle between x and load axes in off-axis loading (°)

1.5.1.3 *Subscripts.* The following subscript notations are considered standard in MIL-HDBK-17.

1, 2, 3 - laminae natural orthogonal coordinates (1 is filament or warp direction)

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A - axial
a - (1) adhesive
- (2) alternating
app - apparent
c - composite system, specific filament/matrix composition. Composite as a whole, contrasted to individual constituents. Also, sandwich core when used in conjunction with prime (')
- (4) critical
cf - centrifugal force
e - fatigue or endurance
eff - effective
eq - equivalent
f - filament
g - glass scrim cloth
H - hoop
i - i^{th} position in a sequence
L - lateral
m - (1) matrix
- (2) mean
max - maximum
min - minimum
n - (1) n^{th} (last) position in a sequence
- (2) normal
p - polar
s - symmetric
st - stiffener
T - transverse
t - value of parameter at time t
x, y, z - general coordinate system
Σ - total, or summation
o - initial or reference datum
( ) - format for indicating specific, temperature associated with term in parentheses. RT - room temperature (21°C,70°F); all other temperatures in °F unless specified.

1.5.1.4 *Superscripts.* The following superscript notations are considered standard in MIL-HDBK-17.

b - bending
br - bearing
c - (1) compression
- (2) creep
cc - compression crippling
cr - compression buckling
e - elastic
f - filament
g - glass scrim cloth
is - interlaminar shear
(i) - i^{th} ply or lamina
lim - limit, used to indicate limit loading
m - matrix
p - plastic

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pl	- proportional limit
rup	- rupture
s	- shear
scr	- shear buckling
sec	- secant (modulus)
T	- temperature or thermal
t	- tension
tan	- tangent (modulus)
u	- ultimate
y	- yield
'	- secondary (modulus), or denotes properties of H/C core when used with subscript c

1.5.2 System of units. To comply with Department of Defense Instructive 5000.2, Part 6, Section M, "Use of the Metric System," dated February 23, 1991, the data in MIL-HDBK-17 are generally presented in both the International System of Units (SI units) and the U. S. Customary (English) system of units. ASTM E-380, Standard for Metric Practice, provides guidance for the application for SI units which are intended as a basis for worldwide standardization of measurement units (Reference 1.5.2(a)). Further guidelines on the use of the SI system of units and conversion factors are contained in the following publications (References 1.5.2(b) - (e)):

(1) DARCOM P 706-470, *Engineering Design Handbook: Metric Conversion Guide*, July 1976.

(2) NBS Special Publication 330, "The International System of Units (SI)," National Bureau of Standards, 1986 edition.

(3) NBS Letter Circular LC 1035, "Units and Systems of Weights and Measures, Their Origin, Development, and Present Status," National Bureau of Standards, November 1985.

(4) NASA Special Publication 7012, "The International System of Units Physical Constants and Conversion Factors", 1964.

English to SI conversion factors pertinent to MIL-HDBK-17 data are contained in Table 1.5.2.

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TABLE 1.5.2 *English to SI conversion factors.*

To convert from	to	Multiply by
Btu (thermochemical)/in^2-s	watt/meter^2 (W/m^2)	1.634 246 E+06
Btu-in/(s-ft^2-°F)	W/(m K)	5.192 204 E+02
degree Fahrenheit	degree Celsius (°C)	T = (T - 32)/1.8
degree Fahrenheit	kelvin (K)	T = (T + 459.67)/1.8
foot	meter (m)	3.048 000 E−01
ft^2	m^2	9.290 304 E−02
foot/second	meter/second (m/s)	3.048 000 E−01
ft/s^2	m/s^2	3.048 000 E−01
inch	meter (m)	2.540 000 E−02
in.^{2}	meter^2 (m^2)	6.451 600 E−04
in.^{3}	m^3	1.638 706 E−05
kilogram-force (kgf)	newton (N)	9.806 650 E+00
kgf/m^2	pascal (Pa)	9.806 650 E+00
kip (1000 lbf)	newton (N)	4.448 222 E+03
ksi (kip/in^2)	MPa	6.894 757 E+00
lbf-in	N-m	1.129 848 E−01
lbf-ft	N-m	1.355 818 E+00
lbf/in^2 (psi)	pascal (Pa)	6.894 757 E+03
lb/in^2	gm/m^2	7.030 696 E+05
lb/in^3	kg/m^3	2.767 990 E+04
Msi (10^6 psi)	GPa	6.894 757 E+00
pound-force (lbf)	newton (N)	4.488 222 E+00
pound-mass (lb avoirdupois)	kilogram (kg)	4.535 924 E−01
torr	pascal (Pa)	1.333 22 E+02

*The letter "E" following the conversion factor stands for exponent and the two digits after the letter "E" indicate the power of 10 by which the number is to be multiplied.

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1.6 DEFINITIONS. The following definitions are used within MIL-HDBK-17. This glossary of terms is not totally comprehensive but it does represent nearly all commonly used terms. Where exceptions are made, they are noted in the text and tables. For ease of identification the definitions have been organized alphabetically.

A-Basis (or A-Value) -- The mechanical property value above which 99 percent of the population of values is expected to fall, with a confidence of 95 percent. Note - A-values are not currently presented in MIL-HDBK-17. *This is a 95% lower confidence limit on the first percentile.*

A-Stage -- An early stage in the reaction of thermosetting resins in which the material is still soluble in certain liquids and may be liquid or capable of becoming liquid upon heating. (Sometimes referred to as resol.)

Absorption -- A process in which one material (the absorbent) takes in or absorbs another (the absorbate).

Accelerator -- A material which, when mixed with a catalyzed resin, will speed up the chemical *reaction between the catalyst and the resin.*

Accuracy -- The degree of conformity of a measured or calculated value to some recognized standard or specified value. Accuracy involves the systematic error of an operation.

Addition Polymerization -- Polymerization by a repeated addition process in which monomers are linked together to form a polymer without splitting off of water or other simple molecules.

Adhesion -- The state in which two surfaces are held together at an interface by forces or interlocking action or both.

Adhesive -- A substance capable of holding two materials together by surface attachment. In the handbook, the term is used specifically to designate structural adhesives, those which produce attachments capable of transmitting significant structural loads.

ADK -- Notation used for the k-sample Anderson-Darling statistic, which is used to test the hypothesis that k batches have the same distribution.

Aliquot -- A small, representative portion of a larger sample.

Aging -- The effect, on materials, of exposure to an environment for a period of time; the process of exposing materials to an environment for an interval of time.

Ambient -- The surrounding environmental conditions such as pressure or temperature.

Anelasticity -- A characteristic exhibited by certain materials in which strain is a function of both stress and time, such that, while no permanent deformations are involved, a finite time is required to establish equilibrium between stress and strain in both the loading and unloading directions.

Angleply -- Same as **Crossply**.

Anisotropic -- Not isotropic; having mechanical and/or physical properties which vary with direction relative to natural reference axes inherent in the material.

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Aramid -- A manufactured fiber in which the fiber-forming substance consisting of a long-chain synthetic aromatic polyamide in which at least 85% of the amide (-CONH-) linkages are attached directly to two aromatic rings.

Areal Weight of Fiber -- The weight of fiber per unit area of prepreg. This is often expressed as grams per square meter. See Table 1.6.2 for conversion factors.

Artificial Weathering -- Exposure to laboratory conditions which may be cyclic, involving changes in temperature, relative humidity, radiant energy and any other elements found in the atmosphere in various geographical areas.

Aspect Ratio -- In an essentially two-dimensional rectangular structure (e.g., a panel), the ratio of the long dimension to the short dimension. However, in compression loading, it is sometimes considered to be the ratio of the load direction dimension to the transverse dimension. Also, in fiber micro-mechanics, it is referred to as the ratio of length to diameter.

Autoclave -- A closed vessel for producing an environment of fluid pressure, with or without heat, to an enclosed object which is undergoing a chemical reaction or other operation.

Autoclave Molding -- A process similar to the pressure bag technique. The lay-up is covered by a pressure bag, and the entire assembly is placed in an autoclave capable of providing heat and pressure for curing the part. The pressure bag is normally vented to the outside.

B-Basis (or B-Value) -- The mechanical property value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent, specifically the same as A-Basis except on the tenth percentile. (See Volume 1, Section 8.5.1.1)

B-Stage -- An intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells when in contact with certain liquids but does not entirely fuse or dissolve. Materials are usually precured to this stage to facilitate handling and processing prior to final cure. (Sometimes referred to as resitol.)

Bag Molding -- A method of molding or laminating which involves the application of fluid pressure to a flexible material which transmits the pressure to the material being molded or bonded. Fluid pressure usually is applied by means of air, steam, water or vacuum.

Balanced Laminate -- A composite laminate in which all laminae at angles other than 0 degrees and 90 degrees occur only in ± pairs (not necessarily adjacent).

Batch (or Lot) -- For fibers and resins, a quantity of material formed during the same process and having identical characteristics throughout. For prepregs, laminae, and laminates, material made from one batch of fiber and one batch of resin.

Bearing Area -- The product of the pin diameter and the specimen thickness.

Bearing Load -- A compressive load on an interface.

Bearing Yield Strength -- The bearing stress at which a material exhibits a specified limiting deviation from the proportionality of bearing stress to bearing strain.

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Bend Test -- A test of ductility by bending or folding, usually with steadily applied forces. In some instances the test may involve blows to a specimen having a cross section that is essentially uniform over a length several times as great as the largest dimension of the cross section.

Binder -- A bonding resin used to hold strands together in a mat or preform during manufacture of a molded object.

Binomial Random Variable -- The number of successes in independent trials where the probability of success is the same for each trial.

Birefringence -- The difference between the two principal refractive indices (of a fiber) or the ratio between the retardation and thickness of a material at a given point.

Bleeder Cloth -- A nonstructural layer of material used in the manufacture of composite parts to allow the escape of excess gas and resin during cure. The bleeder cloth is removed after the curing process and is not part of the final composite.

Bond -- The adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

Braiding -- A textile process where two or more strands, yarns or tapes are intertwined in the bias direction to form an integrated structure.

Broadgoods -- A term loosely applied to prepreg material greater than about 12 inches in width, usually furnished by suppliers in continuous rolls. The term is currently used to designate both collimated uniaxial tape and woven fabric prepregs.

Buckling (Composite) -- A mode of structural response characterized by an out-of-plane material deflection due to compressive action on the structural element involved. In advanced composites, buckling may take the form not only of conventional general instability and local instability but also a micro-instability of individual fibers.

Bundle -- A general term for a collection of essentially parallel filaments or fibers.

C-Stage -- The final stage of the curing reaction of a thermosetting resin in which the material has become practically infusable and insoluble. (Normally considered fully cured and sometimes referred to as resite.)

Carbon Fibers -- Fibers produced by the pyrolysis of organic precursor fibers such as rayon, polyacrylonitrile (PAN), and pitch in an inert atmosphere. The term is often used interchangeably with "graphite"; however, carbon fibers and graphite fibers differ in the temperature at which the fibers are made and heat-treated, and the amount of carbon produced. Carbon fibers typically are carbonized at about 2400°F (1300°C) and assay at 93 to 95% carbon, while graphite fibers are graphitized at 3450 to 5450°F (1900 to 3000°C) and assay at more than 99% elemental carbon.

Caul Plates -- Smooth metal plates, free of surface defects, the same size and shape as a composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and to provide a smooth surface on the finished laminate.

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Censoring -- Data is right (left) censored at M, if, whenever an observation is less than or equal to M (greater than or equal to M), the actual value of the observation is recorded. If the observation exceeds (is less than) M, the observation is recorded as M.

Chain-Growth Polymerization -- One of the two principal polymerization mechanisms. In chain-growth polymerization, the reactive groups are continuously regenerated during the growth process. Once started, the polymer molecule grows rapidly by a chain of reactions emanating from a particular reactive initiator which may be a free radical, cation or anion.

Chromatogram -- A plot of detector response against peak volume of solution (eluate) emerging from the system for each of the constituents which have been separated.

Circuit -- One complete traverse of the fiber feed mechanism of a winding machine; one complete traverse of a winding band from one arbitrary point along the winding path to another point on a plane through the starting point and perpendicular to the axis.

Cocuring -- The act of curing a composite laminate and simultaneously bonding it to some other prepared surface during the same cure cycle (see **Secondary Bonding**).

Coefficient of Linear Thermal Expansion -- The change in length per unit length resulting from a one-degree rise in temperature.

Coefficient of Variation -- The ratio of the population (or sample) standard deviation to the population (or sample) mean.

Collimated -- Rendered parallel.

Compatible -- The ability of different resin systems to be processed in contact with each other without degradation of end product properties. (See **Compatible**, Volume 1, Section 8.5.1.1)

Composite Class -- As used in the handbook, a major subdivision of composite construction in which the class is defined by the fiber system and the matrix class, e.g., organic-matrix filamentary laminate.

Composite Material -- Composites are considered to be combinations of materials differing in composition or form on a macroscale. The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other although they act in concert. Normally, the components can be physically identified and exhibit an interface between one another.

Compound -- An intimate mixture of polymer or polymers with all the materials necessary for the finished product.

Condensation Polymerization -- This is a special type of step-growth polymerization characterized by the formation of water or other simple molecules during the stepwise addition of reactive groups.

Confidence Coefficient -- See **Confidence Interval**.

Confidence Interval -- A confidence interval is defined by a statement of one of the following forms:

$$(1)\ P\{a<\theta\} \leq 1-\alpha$$
$$(2)\ P\{\theta<b\} \leq 1-\alpha$$

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(3) $P\{a<\theta<b\} \leq 1-\alpha$

where $1-\alpha$ is called the confidence coefficient. A statement of type (1) or (2) is called a one-sided confidence interval and a statement of type (3) is called a two-sided confidence interval. In (1) a is a lower confidence limit and in (2) b is an upper confidence limit. With probability at least $1-\alpha$, the confidence interval will contain the parameter θ.

Constituent -- In general, an element of a larger grouping. In advanced composites, the principal constituents are the fibers and the matrix.

Continuous Filament -- A yarn or strand in which the individual filaments are substantially the same length as the strand.

Coupling Agent -- Any chemical substance designed to react with both the reinforcement and matrix phases of a composite material to form or promote a stronger bond at the interface. Coupling agents are applied to the reinforcement phase from an aqueous or organic solution or from a gas phase, or added to the matrix as an integral blend.

Crazing -- Apparent fine cracks at or under the surface of an organic matrix.

Creep -- The time dependent part of strain resulting from an applied stress.

Creep, Rate Of -- The slope of the creep-time curve at a given time.

Critical Value(s) -- When testing a one-sided statistical hypothesis, a critical value is the value such that, if the test statistic is greater than (less than) the critical value, the hypothesis is rejected. When testing a two-sided statistical hypothesis, two critical values are determined. If the test statistic is either less than the smaller critical value or greater than the larger critical value, then the hypothesis is rejected. In both cases, the critical value chosen depends on the desired risk (often 0.05) of rejecting the hypothesis when it is true.

Crossply -- Any filamentary laminate which is not uniaxial. Same as Angleply. In some references, the term crossply is used to designate only those laminates in which the laminae are at right angles to one another, while the term angleply is used for all others. In the handbook, the two terms are used synonymously. The reservation of a separate terminology for only one of several basic orientations is unwarranted because a laminate orientation code is used.

Cumulative Distribution Function -- See Volume 1, Section 8.5.1.1.

Cure -- To change the properties of a thermosetting resin irreversibly by chemical reaction, i.e., condensation, ring closure, or addition. Cure may be accomplished by addition of curing (cross-linking) agents, with or without catalyst, and with or without heat. Cure may occur also by addition, such as occurs with anhydride cures for epoxy resin systems.

Cure Cycle -- The schedule of time periods at specified conditions to which a reacting thermosetting material is subjected in order to reach a specified property level.

Cure Stress -- A residual internal stress produced during the curing cycle of composite structures. Normally, these stresses originate when different components of a lay-up have different thermal coefficients of expansion.

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Debond -- A deliberate separation of a bonded joint or interface, usually for repair or rework purposes. (See **Disbond, Unbond**).

Deformation -- The change in shape of a specimen caused by the application of a load or force.
Degradation -- A deleterious change in chemical structure, physical properties or appearance.

Delamination -- The separation of the layers of material in a laminate. This may be local or may cover a large area of the laminate. It may occur at any time in the cure or subsequent life of the laminate and may arise from a wide variety of causes.

Denier -- A direct numbering system for expressing linear density, equal to the mass in grams per 9000 meters of yarn, filament, fiber, or other textile strand.

Density -- The mass per unit volume.

Desorption -- A process in which an absorbed or adsorbed material is released from another material. Desorption is the reverse of absorption, adsorption, or both.

Deviation -- Variation from a specified dimension or requirement, usually defining the upper and lower limits.

Dielectric Constant -- The ratio of the capacity of a condenser having a dielectric constant between the plates to that of the same condenser when the dielectric is replaced by a vacuum; a measure of the electrical charge stored per unit volume at unit potential.

Dielectric Strength -- The average potential per unit thickness at which failure of the dielectric material occurs.

Disbond -- An area within a bonded interface between two adherends in which an adhesion failure or separation has occurred. It may occur at any time during the life of the structure and may arise from a wide variety of causes. Also, colloquially, an area of separation between two laminae in the finished laminate (in this case the term "delamination" is normally preferred.) (See **Debond, Unbond, Delamination.**)

Distribution -- A formula which gives the probability that a value will fall within prescribed limits. (See **Normal, Weibull, and Lognormal Distributions**, also Volume 1, Section 8.5.1.1).

Dry Fiber Area -- Area of fiber not totally encapsulated by resin.

Ductility -- The ability of a material to deform plastically before fracturing.

Elasticity -- The property of a material which allows it to recover its original size and shape immediately after removal of the force causing deformation.

Elongation -- The increase in gage length or extension of a specimen during a tension test, usually expressed as a percentage of the original gage length.

Eluate -- The liquid emerging from a column (in liquid chromatography).

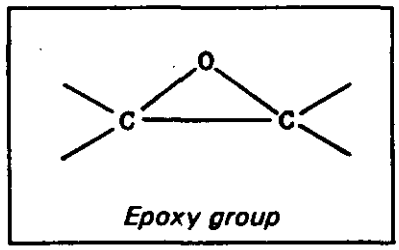
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Eluent -- The mobile phase used to sweep or elute the sample (solute) components into, through, and out of the column.

End -- A single fiber, strand, roving or yarn being or already incorporated into a product. An end may be an individual warp yarn or cord in a woven fabric. In referring to aramid and glass fibers, an end is usually an untwisted bundle of continuous filaments.

Epoxy Equivalent Weight -- The number of grams of resin which contain one chemical equivalent of the epoxy group.

Epoxy Resin -- Resins which may be of widely different structures but are characterized by the presence of the epoxy group. (The epoxy or epoxide group is usually present as a glycidyl ether, glycidyl amine, or as part of an aliphatic ring system. The aromatic type epoxy resins are normally used in composites.)



Epoxy group

Extensometer -- A device for measuring linear strain.

F-Distribution -- See Volume 1, Section 8.5.1.1.

Fabric, Nonwoven -- A textile structure produced by bonding or interlocking of fibers, or both, accomplished by mechanical, chemical, thermal, or solvent means, and combinations thereof.

Fabric, Woven -- A generic material construction consisting of interlaced yarns or fibers, usually a planar structure. Specifically, as used in this handbook, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina. In a fabric lamina, the warp direction is considered the longitudinal direction, analogous to the filament direction in a filamentary lamina.

Fiber -- A general term used to refer to filamentary materials. Often, fiber is used synonymously with filament. It is a general term for a filament of finite length.

Fiber Content -- The amount of fiber present in a composite. This is usually expressed as a percentage volume fraction or weight fraction of the composite.

Fiber Count -- The number of fibers per unit width of ply present in a specified section of a composite.

Fiber Direction -- The orientation or alignment of the longitudinal axis of the fiber with respect to a stated reference axis.

Fiber System -- The type and arrangement of fibrous material which comprises the fiber constituent of an advanced composite. Examples of fiber systems are collimated filaments or filament yarns, woven fabric, randomly oriented short-fiber ribbons, random fiber mats, whiskers, etc.

Filament -- The smallest unit of a fibrous material. The basic units formed during spinning and which are gathered into strands of fiber, (for use in composites). Filaments usually are of extreme length and of very small diameter. Filaments normally are not used individually. Some textile filaments can function as a yarn when they are of sufficient strength and flexibility.

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Filamentary Composites -- A major form of advanced composites in which the fiber constituent consists of continuous filaments. Specifically, a filamentary composite is a laminate comprised of a number of laminae, each of which consists of a nonwoven, parallel, uniaxial, planar array of filaments (or filament yarns) embedded in the selected matrix material. Individual laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Filament Winding -- A reinforced-plastics process that employs a series of continuous, resin-impregnated fibers applied to a mandrel in a predetermined geometrical relationship under controlled tension.

Filament Wound -- Pertaining to an object created by the filament winding method of fabrication.

Fill -- Yarn oriented at right angles to the warp in a woven fabric.

Filler -- A relatively inert substance added to a material to alter its physical, mechanical, thermal, electrical, and other properties or to lower cost. Sometimes the term is used specifically to mean particulate additives.

Finish (or Size System) -- A material, with which filaments are treated, which contains a coupling agent to improve the bond between the filament surface and the resin matrix in a composite material. In addition, finishes often contain ingredients which provide lubricity to the filament surface, preventing abrasive damage during handling, and a binder which promotes strand integrity and facilitates packing of the filaments.

Fixed Effect -- A common systematic (nonrandom) shift in a group of measurements which is due to a controlled level change of a treatment or condition. (See Volume 1, Section 8.5.1.1.)

Flash -- Excess material which forms at the parting line of a mold or die, or which is extruded from a closed mold.

Fracture Ductility -- The true plastic strain at fracture.

Gage Length -- the original length of that portion of the specimen over which strain or change of length is determined.

Gel -- The initial jelly-like solid phase that develops during formation of a resin from a liquid. Also, a semi-solid system consisting of a network of solid aggregates in which liquid is held.

Gel Coat -- A quick-setting resin used in molding processes to provide an improved surface for the composite; it is the first resin applied to the mold after the mold-release agent.

Gel Point -- The stage at which a liquid begins to exhibit pseudo-elastic properties. (This can be seen from the inflection point on a viscosity-time plot.)

Gel Time -- The period of time from a pre-determined starting point to the onset of gelation (gel point) as defined by a specific test method.

Glass -- An inorganic product of fusion which has cooled to a rigid condition without crystallizing. In the handbook, all reference to glass will be to the fibrous form as used in filaments, woven fabric, yarns, mats, chopped fibers, etc.

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Glass Cloth -- Conventionally-woven glass fiber material (see Scrim).

Glass Fibers -- A fiber spun from an inorganic product of fusion which has cooled to a rigid condition without crystallizing.

Glass Transition -- The reversible change in an amorphous polymer or in amorphous regions of a partially crystalline polymer from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one.

Glass Transition Temperature -- The approximate midpoint of the temperature range over which the glass transition takes place.

Graphite Fibers -- See Carbon Fibers.

Greige -- Fabric that has received no finish.

Hand Lay-up -- A process in which components are applied either to a mold or a working surface, and the successive plies are built up and worked by hand.

Hardness -- Resistance to deformation; usually measured by indention. Types of standard tests include Brinell, Rockwell, Knoop, and Vickers.

Heat Cleaned -- Glass or other fibers which have been exposed to elevated temperatures to remove preliminary sizings or binders which are not compatible with the resin system to be applied.

Heterogeneous -- Descriptive term for a material consisting of dissimilar constituents separately identifiable; a medium consisting of regions of unlike properties separated by internal boundaries. (Note that all nonhomogeneous materials are not necessarily heterogeneous).

Homogeneous -- Descriptive term for a material of uniform composition throughout; a medium which has no internal physical boundaries; a material whose properties are constant at every point, i.e., constant with respect to spatial coordinates (but not necessarily with respect to directional coordinates).

Horizontal Shear -- Sometimes used to indicate interlaminar shear. This is not an approved term for use in this handbook.

Humidity, Relative -- The ratio of the pressure of water vapor present to the pressure of saturated water vapor at the same temperature.

Hybrid -- A composite laminate comprised of laminae of two or more composite material systems. Or, a combination of two or more different fibers such as carbon and glass or carbon and aramid into a structure (tapes, fabrics and other forms may be combined).

Hygroscopic -- Capable of absorbing and retaining atmospheric moisture.

Hysteresis -- The energy absorbed in a complete cycle of loading and unloading.

Inclusion -- A physical and mechanical discontinuity occurring within a material or part, usually consisting of solid, encapsulated foreign material. Inclusions are often capable of transmitting some structural stresses and energy fields, but in a noticeably different manner from the parent material.

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Integral Composite Structure -- Composite structure in which several structural elements, which would conventionally be assembled by bonding or with mechanical fasteners after separate fabrication, are instead laid up and cured as a single, complex, continuous structure; e.g., spars, ribs, and one stiffened cover of a wing box fabricated as a single integral part. The term is sometimes applied more loosely to any composite structure not assembled by mechanical fasteners.

Interface -- The boundary between the individual, physically distinguishable constituents of a composite.

Interlaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., shear stress) referenced as existing or occurring between two or more adjacent laminae.

Interlaminar Shear -- Shearing force tending to produce a relative displacement between two laminae in a laminate along the plane of their interface.

Intermediate Bearing Stress -- The bearing stress at the point on the bearing load-deformation curve where the tangent is equal to the bearing stress divided by a designated percentage (usually 4%) of the original hole diameter.

Intralaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., temperature gradient) existing entirely within a single lamina without reference to any adjacent laminae.

Isotropic -- Having uniform properties in all directions. The measured properties of an isotropic material are independent of the axis of testing.

Knuckle Area -- The area of transition between sections of different geometry in a filament wound part.

k-Sample Data -- A collection of data consisting of values observed when sampling from k batches.

Lamina -- A single ply or layer in a laminate made up of a series of layers.

Laminae -- Plural of lamina.

Laminate -- A product made by bonding together two or more layers or laminae of material or materials.

Laminate Orientation -- The configuration of a crossplied composite laminate with regard to the angles of crossplying, the number of laminae at each angle, and the exact sequence of the lamina lay-up.

Lattice Pattern -- A pattern of filament winding with a fixed arrangement of open voids.

Lay-up -- A process of fabrication involving the assembly of successive layers of resin-impregnated material.

Lognormal Distribution -- A probability distribution for which the probability that an observation selected at random from this population falls between a and b ($0 < a < b < B$) is given by the area

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under the normal distribution between ℓog a and ℓog b. The common (base 10) or the natural (base e) logarithm may be used. (See Volume 1, Section 8.5.1.1.)

Lower Confidence Bound -- See **Confidence Interval.**

Macro -- In relation to composites, denotes the gross properties of a composite as a structural element but does not consider the individual properties or identity of the constituents.

Macrostrain -- The mean strain over any finite gage length of measurement which is large in comparison to the material's interatomic distance.

Mandrel -- A form fixture or male mold used for the base in the production of a part by lay-up or filament winding.

Mat -- A fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder.

Material Acceptance -- The testing of incoming material to ensure that it meets requirements.

Material Qualification -- The procedures used to accept a material by a company or organization for production use.

Material System -- A specific composite material made from specifically identified constituents in specific geometric proportions and arrangements and possessed of numerically defined properties.

Material System Class -- As used in this handbook, a group consisting of material systems categorized by the same generic constituent materials, but without defining the constituents uniquely; e.g., the carbon/epoxy class.

Material Variability -- A source of variability due to variations in the spatial, consistency, and processing aspects of the material. (See Volume 1, Section 8.5.1.1.)

Matrix -- The essentially homogeneous material in which the fiber system of a composite is embedded.

Mean -- See **Sample Mean** and **Population Mean.**

Mechanical Properties -- The properties of a material that are associated with elastic and inelastic reaction when force is applied, or the properties involving the relationship between stress and strain.

Median -- See **Sample Median** and **Population Median.**

Micro -- In relation to composites, denotes the properties of the constituents, i.e., matrix and reinforcement and interface only, as well as their effects on the composite properties.

Microstrain -- The strain over a gage length comparable to the material's interatomic distance.

Modulus, Chord -- The slope of the chord drawn between any two specified points on the stress-strain curve.

Modulus, Initial -- The slope of the initial straight portion of a stress-strain or load-elongation curve.

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Modulus, Secant -- The slope of the secant drawn from the origin to any specified point on the stress-strain curve.

Modulus, Tangent -- The ratio of change in stress to change in strain derived from the tangent to any point on a stress-strain curve.

Modulus, Young's -- The ratio of change in stress to change in strain below the elastic limit of a material. (Applicable to tension and compression).

Modulus of Rigidity (also Shear Modulus or Torsional Modulus) -- The ratio of stress to strain below the proportional limit for shear or torsional stress.

Modulus of Rupture, in Bending -- The maximum tensile or compressive stress (whichever causes failure) value in the extreme fiber of a beam loaded to failure in bending. The value is computed from the flexure equation:

$$F^b = \frac{Mc}{I} \qquad 1.6(a)$$

where M = maximum bending moment computed from the maximum load and the original moment arm,
c = initial distance from the neutral axis to the extreme fiber where failure occurs,
I = the initial moment of inertia of the cross section about its neutral axis.

Modulus of Rupture, in Torsion -- The maximum shear stress in the extreme fiber of a member of circular cross section loaded to failure in torsion calculated from the equation:

$$F^s = \frac{Tr}{J} \qquad 1.6(b)$$

where T = maximum twisting moment,
r = original outer radius,
J = polar moment of inertia of the original cross section.

Moisture Content -- The amount of moisture in a material determined under prescribed condition and expressed as a percentage of the mass of the moist specimen, i.e., the mass of the dry substance plus the moisture present.

Moisture Equilibrium -- The condition reached by a sample when it no longer takes up moisture from, or gives up moisture to, the surrounding environment.

Mold Release Agent -- A lubricant applied to mold surfaces to facilitate release of the molded article.

Molded Edge -- An edge which is not physically altered after molding for use in final form and particularly one which does not have fiber ends along its length.

Molding -- The forming of a polymer or composite into a solid mass of prescribed shape and size by the application of pressure and heat.

Monolayer -- The basic laminate unit from which crossplied or other laminates are constructed.

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Monomer -- A compound consisting of molecules each of which can provide one or more constitutional units.

NDE -- Nondestructive evaluation. Broadly considered synonymous with NDI.

NDI -- Nondestructive Inspection. A process or procedure for determining the quality or characteristics of a material, part, or assembly without permanently altering the subject or its properties.

NDT -- Nondestructive Testing. Broadly considered synonymous with NDI.

Necking -- A localized reduction in cross-sectional area which may occur in a material under tensile stress.

Negatively Skewed -- A distribution is said to be negatively skewed if the distribution is not symmetric and the longest tail is on the left.

Nominal Specimen Thickness -- The nominal ply thickness multiplied by the number of plies.

Nominal Value -- A value assigned for the purpose of a convenient designation. A nominal value exists in name only.

Normal Distribution -- A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \qquad 1.6(c)$$

between a and b. (See Volume 1, Section 8.5.1.1.)

Normalized Stress -- Stress calculated by multiplying the raw stress value by the ratio of measured fiber volume to the nominal fiber volume. This ratio is often approximated by the ratio of the measured specimen thickness to the nominal specimen thickness. Stresses for fiber-dominated failure modes are often normalized.

Observed Significance Level (OSL) -- The probability of observing a more extreme value of the test statistic when the null hypotheses is true.

Oligomer -- A polymer consisting of only a few monomer units such as a dimer, trimer, etc. or their mixtures.

One-Sided Tolerance Limit Factor -- See **Tolerance Limit Factor**.

Orthotropic -- Having three mutually perpendicular planes of elastic symmetry.

Oven Dry -- The condition of a material that has been heated under prescribed conditions of temperature and humidity until there is no further significant change in its mass.

PAN Fibers -- Reinforcement fiber derived from the controlled pyrolysis of poly(acrylonitrile) fiber.

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Parallel Laminate -- A laminate of woven fabric in which the plies are aligned in the same position as originally aligned in the fabric roll.

Peel Ply -- A layer of resin free material used to protect a laminate for later secondary bonding.

pH -- A measure of acidity or alkalinity of a solution, with neutrality represented by a value of 7, with increasing acidity corresponding to progressively smaller values, and increasing alkalinity corresponding to progressively higher values.

Pick Count -- The number of filling yarns per inch of woven fabric.

Pitch Fibers -- Reinforcement fiber derived from petroleum or coal tar pitch.

Plastic -- A material that contains one or more organic polymers of large molecular weight, is solid in its finished state, and, at some state in its manufacture or processing into finished articles, can be shaped by flow.

Plasticizer -- A material of lower molecular weight added to a polymer to separate the molecular chains. This results in a depression of the glass transition temperature, reduced stiffness and brittleness, and improved processability. (Note, many polymeric materials do not need a plasticizer.)

Poisson's Ratio -- The absolute value of the ratio of transverse strain to the corresponding axial strain resulting from uniformly distributed axial stress below the proportional limit of the material.

Polymer -- An organic material composed of molecules characterized by the repetition of one or more types of monomeric units.

Polymerization -- A chemical reaction in which the molecules of monomers are linked together to form polymers via two principal reaction mechanisms. Addition polymerizations proceed by chain growth and most condensation polymerizations through step growth.

Population -- The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for carbon/epoxy system A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population. (See Volume 1, Section 8.5.1.1.)

Population Mean -- The average of all potential measurements in a given population weighted by their relative frequencies in the population. (See Volume 1, Section 8.5.1.1.)

Population Median -- That value in the population such that the probability of exceeding it is 0.5 and the probability of being less than it is 0.5. (See Volume 1, Section 8.5.1.1.)

Population Variance -- A measure of dispersion in the population.

Porosity -- A condition of trapped pockets of air, gas, or vacuum within a solid material, usually expressed as a percentage of the total nonsolid volume to the total volume (solid plus nonsolid) of a unit quantity of material.

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Positively Skewed -- A distribution is said to be positively skewed if the distribution is not symmetric and the longest tail is on the right.

Postcure -- Additional elevated temperature cure, usually without pressure, to improve final properties or complete the cure or both.

Pot Life -- The period of time during which a reacting thermosetting composition remains suitable for its intended processing after mixing with a reaction initiating agent.

Precision -- The degree of agreement within a set of observations or test results obtained. Precision involves repeatability and reproducibility.

Precursor (for Carbon or Graphite Fiber) -- Either the PAN or pitch fibers from which carbon and graphite fibers are derived.

Preply -- A composite material lamina in the raw material stage ready to be fabricated into a finished laminate. The lamina is usually combined with other raw laminae prior to fabrication. A preply includes all of the fiber system placed in position relative to all or part of the required matrix material that together will comprise the finished lamina. An organic matrix preply is called a prepreg.

Prepreg -- Ready to mold or cure material in sheet form which may be fiber, cloth, or mat impregnated with resin and stored for use. The resin is partially cured to a B-stage and supplied to the fabricator for lay-up and cure.

Pressure -- The force or load per unit area.

Probability Density Function -- See Volume 1, Section 8.5.1.1.

Proportional Limit -- The maximum stress that a material is capable of sustaining without any deviation from the proportionality of stress to strain (also known as Hooke's law).

Quasi-Isotropic Laminate -- A laminate approximating isotropy by orientation of plies in several or more directions.

Random Effect -- A common shift in a group of measurements due to a random level change of a usually uncontrollable factor. (See Volume 1, Section 8.5.1.1.)

Random Error -- That part of data variation due to level changes in uncontrolled factors which affect each observation separately and independently. (See Volume 1, Section 8.5.1.1.)

Reduction of Area -- The difference between the original cross sectional area of a tension test specimen and the area of its smallest cross section, usually expressed as a percentage of the original area.

Refractive Index - The ratio of the velocity of light (of specified wavelength) in air to its velocity in the substance under examination. Also defined as the sine of the angle of incidence divided by the sine of the angle of refraction as light passes from air into the substance.

Reinforced Plastic -- A plastic with relatively high stiffness or very high strength fibers embedded in the composition. This improves some mechanical properties over that of the base resin.

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Release Agent -- See **Mold Release Agent**.

Resilience -- A property of a material which is able to do work against restraining forces during return from a deformed condition.

Resin -- A solid or pseudo-solid organic material usually of high molecular weight which exhibits a tendency to flow when subjected to stress, usually has a softening or melting range, and fractures conchoidally.

Resin Content -- The amount of matrix present in a composite either by percent weight or percent volume.

Resin Starved Area -- Area of composite part where the resin has a non-continuous smooth coverage of the fiber.

Resin System -- A mixture of resin, with ingredients such as catalyst, initiator, diluents, etc. required for the intended processing and final product.

Roving -- A number of strands, tows, or ends collected into a parallel bundle with little or no twist.

S-Basis (or S-Value) -- The mechanical property value which is usually the specified minimum value of the appropriate government specification or SAE Aerospace Material Specification for this material.

Sample -- A small portion of a material or product intended to be representative of the whole. Statistically, a sample is the collection of measurements taken from a specified population. (See Volume 1, Section 8.5.1.1.)

Sample Mean -- The arithmetic average of the measurements in a sample. The sample mean is an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Median -- Order the observation from smallest to largest. Then the sample median is the value of the middle observation if the sample size is odd; the average of the two central observations if n is even. If the population is symmetric about its mean, the sample median is also an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Standard Deviation -- The square root of the sample variance. (See Volume 1, Section 8.5.1.1.)

Sample Variance -- The sum of the squared deviations from the sample mean, divided by n-1. (See Volume 1, Section 8.5.1.1.)

Sandwich Construction -- A structural panel concept consisting in its simplest form of two relatively thin, parallel sheets of structural material bonded to, and separated by, a relatively thick, light-weight core.

Saturation -- An equilibrium condition in which the net rate of absorption under prescribed conditions falls essentially to zero.

Scrim (also called **Glass Cloth, Carrier**) -- A low cost fabric woven into an open mesh construction, used in the processing of tape or other B-stage material to facilitate handling.

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Secondary Bonding -- The joining together, by the process of adhesive bonding, of two or more already-cured composite parts, during which the only chemical or thermal reaction occurring is the curing of the adhesive itself.

Selvedge -- The woven edge portion of a fabric parallel to the warp.

Set -- The strain remaining after complete release of the force producing the deformation.

Shear Fracture (for crystalline type materials) -- A mode of fracture resulting from translation along slip planes which are preferentially oriented in the direction of the shearing stress.

Shelf Life -- The length of time a material, substance, product, or reagent can be stored under specified environmental conditions and continue to meet all applicable specification requirements and/or remain suitable for its intended function.

Significant -- Statistically, the value of a test statistic is significant if the probability of a value at least as extreme is less than or equal to a predetermined number called the significance level of the test.

Significant Digit -- Any digit that is necessary to define a value or quantity.

Size System -- See **Finish**.

Skewness -- See **Positively Skewed, Negatively Skewed**.

Slenderness Ratio -- The unsupported effective length of a uniform column divided by the least radius of gyration of the cross-sectional area.

Sliver -- A continuous strand of loosely assembled fiber that is approximately uniform in cross-sectional area and has no twist.

Solute -- The dissolved material.

Specific Gravity -- The ratio of the weight of any volume of a substance to the weight of an equal volume of another substance taken as standard at a constant or stated temperature. Solids and liquids are usually compared with water at 4°C (39°F).

Specific Heat -- The quantity of heat required to raise the temperature of a unit mass of a substance one degree under specified conditions.

Specimen -- A piece or portion of a sample or other material taken to be tested. Specimens normally are prepared to conform with the applicable test method.

Standard Deviation -- See **Sample Standard Deviation**.

Staple -- Either naturally occurring fibers or lengths cut from filaments.

Step-Growth Polymerization -- One of the two principal polymerization mechanisms. In sep-growth polymerization, the reaction grows by combination of monomer, oligomer, or polymer molecules through the consumption of reactive groups. Since average molecular weight increases with monomer consumption, high molecular weight polymers are formed only at high degrees of conversion.

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Strain -- the per unit change, due to force, in the size or shape of a body referred to its original size or shape. Strain is a nondimensional quantity, but it is frequently expressed in inches per inch, meters per meter, or percent.

Strand -- Normally an untwisted bundle or assembly of continuous filaments used as a unit, including slivers, tow, ends, yarn, etc. Sometimes a single fiber or filament is called a strand.

Strength -- the maximum stress which a material is capable of sustaining.

Stress -- The intensity at a point in a body of the forces or components of forces that act on a given plane through the point. Stress is expressed in force per unit area (pounds-force per square inch, megapascals, etc.).

Stress Relaxation -- The time dependent decrease in stress in a solid under given constraint conditions.

Stress-Strain Curve (Diagram) -- A graphical representation showing the relationship between the change in dimension of the specimen in the direction of the externally applied stress and the magnitude of the applied stress. Values of stress usually are plotted as ordinates (vertically) and strain values as abscissa (horizontally).

Surfacing Mat -- A thin mat of fine fibers used primarily to produce a smooth surface on an organic matrix composite.

Symmetrical Laminate -- A composite laminate in which the sequence of plies below the laminate midplane is a mirror image of the stacking sequence above the midplane.

Tack -- Stickiness of the prepreg.

Tape -- Prepreg fabricated in widths up to 12 inches wide for carbon and 3 inches for boron. Cross stitched carbon tapes up to 60 inches wide are available commercially in some cases.

Tenacity -- The tensile stress expressed as force per unit linear density of the unstrained specimen i.e., grams-force per denier or grams-force per tex.

Tex -- A unit for expressing linear density equal to the mass or weight in grams of 1000 meters of filament, fiber, yarn or other textile strand.

Thermal Conductivity -- Ability of a material to conduct heat. The physical constant for quantity of heat that passes through unit cube of a substance in unit time when the difference in temperature of two faces is one degree.

Thermoplastic -- A plastic that repeatedly can be softened by heating and hardened by cooling through a temperature range characteristic of the plastic, and when in the softened stage, can be shaped by flow into articles by molding or extrusion.

Thermoset -- A plastic that is substantially infusible and insoluble after having been cured by heat or other means.

Tolerance -- The total amount by which a quantity is allowed to vary.

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Tolerance Limit -- A lower (upper) confidence limit on a specified percentile of a distribution. For example, the B-basis value is a 95% lower confidence limit on the tenth percentile of a distribution.

Tolerance Limit Factor -- The factor which is multiplied by the estimate of variability in computing the tolerance limit.

Toughness -- A measure of a material's ability to absorb work, or the actual work per unit volume or unit mass of material that is required to rupture it. Toughness is proportional to the area under the load-elongation curve from the origin to the breaking point.

Tow -- An untwisted bundle of continuous filaments. Commonly used in referring to man-made fibers, particularly carbon and graphite fibers, in the composites industry.

Transformation -- A transformation of data values is a change in the units of measurement accomplished by applying a mathematical function to all data values. For example, if the data is given by x, then $y = x + 1$, x, $1/x$, $\ell og\ x$, and cos x are transformations.

Transition, First Order -- A change of state associated with crystallization or melting in a polymer.

Transversely Isotropic -- Descriptive term for a material exhibiting a special case of orthotropy in which properties are identical in two orthotropic dimensions, but not the third; having identical properties in both transverse directions but not the longitudinal direction.

Traveller -- A small piece of the same product (panel, tube, etc.) as the test specimen, used for example to measure moisture content as a result of conditioning.

Twist -- The number of turns about its axis per unit of length in a yarn or other textile strand. It may be expressed as turns per inch (tpi).

Twist, Direction of -- The direction of twist in yarns and other textile strands is indicated by the capital letters S and Z. Yarn has S twist if, when held in a vertical position, the visible spirals or helices around its central axis are in the direction of slope of the central portion of the letter S, and Z twist is in the other direction.

Typical Basis -- A typical property value is a sample mean. Note that the typical value is defined as the simple arithmetic mean which has a statistical connotation of 50% reliability with a 50% confidence.

Unbond -- An area within a bonded interface between two adherends in which the intended bonding action failed to take place. Also used to denote specific areas deliberately prevented from bonding in order to simulate a defective bond, such as in the generation of quality standards specimens. (See **Disbond, Debond**).

Unidirectional Laminate -- A laminate with nonwoven reinforcements and all layers laid up in the same direction.

Upper Confidence Limit -- See **Confidence Interval**.

Vacuum Bag Molding -- A process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges.

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Variance -- See Sample Variance.

Viscosity -- The property of resistance to flow exhibited within the body of a material.

Void -- A physical and mechanical discontinuity occurring within a material or part which may be two-dimensional (e.g., disbonds, delaminations) or three-dimensional (e.g., vacuum-, air-, or gas-filled pockets). Porosity is an aggregation of micro-voids. Voids are essentially incapable of transmitting structural stresses or nonradiative energy fields. (See Inclusion.)

Warp -- The longitudinally oriented yarn in a woven fabric (see Fill); a group of yarns in long lengths and approximately parallel.

Weibull Distribution (Two - Parameter) -- A probability distribution for which the probability that a randomly selected observation from this population lies between *a* and *b* ($0 < a < b < \infty$) is given by Equation 1.6(d) where α is called the scale parameter and β is called the shape parameter. (See Volume 1, Section 8.5.1.1.)

$$\exp\left[-\left(\frac{a}{\alpha}\right)^{\beta}\right] - \exp\left[-\left(\frac{b}{\alpha}\right)^{\beta}\right] \qquad 1.6(d)$$

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while the reinforcement is put in place.

Wet Strength -- The strength of an organic matrix composite when the matrix resin is saturated with absorbed moisture. (See **Saturation**).

Wet Winding -- A method of filament winding in which the fiber reinforcement is coated with the resin system as a liquid just prior to wrapping on a mandrel.

Whisker -- A short single crystal fiber or filament. Whisker diameters range from 1 to 25 microns, with aspect ratios between 100 and 15,000.

Work Life -- The period during which a compound, after mixing with a catalyst, solvent, or other compounding ingredient, remains suitable for its intended use.

Woven Fabric Composite -- A major form of advanced composites in which the fiber constituent consists of woven fabric. A woven fabric composite normally is a laminate comprised of a number of laminae, each of which consists of one layer of fabric embedded in the selected matrix material. Individual fabric laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Yarn -- A generic term for strands or bundles of continuous filaments or fibers, usually twisted and suitable for making textile fabric.

Yarn, Plied -- Yarns made by collecting two or more single yarns together. Normally, the yarns are twisted together though sometimes they are collected without twist.

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Yield Point -- The first stress in a material at which an increase in strain occurs without an increase in stress. (The stress is less than the maximum attainable.) It should be noted that only materials that exhibit the unique phenomenon of yielding have a "yield point".

Yield Strength -- The stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain. (The deviation is expressed in terms of strain such as 0.2 percent for the Offset Method or 0.5 percent for the Total Extension Under Load Method.)

X-Axis -- In composite laminates, an axis in the plane of the laminate which is used as the 0 degree reference for designating the angle of a lamina.

X-Y Plane -- In composite laminates, the reference plane parallel to the plane of the laminate.

Y-Axis -- In composite laminates, the axis in the plane of the laminate which is perpendicular to the x-axis.

Z-Axis -- In composite laminates, the reference axis normal to the plane of the laminate.

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4.1 INTRODUCTION.

4.2 CARBON - EPOXY COMPOSITES.

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4.2.1 T-500 12k/976.*

MATERIAL:	T-500 12k/976 unidirectional tape	C/Ep T-500/976 Summary
PREPREG:	*Fiberite hy-E 3076P unidirectional tape*	
FIBER:	Union Carbide Thornel T-500 12k RESIN: Fiberite 976	
T_g(dry):	361°F T_g(wet): T_g METHOD:	
PROCESSING:	240°F, 1 hour, 85 psi; 350°F, 2 hours, 100 psi	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture	12/83	Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW	ETW
F_1^{tu}	(ksi)	297	I	214	I		
E_1^t	(Msi)	21.9	I	19.1	I		
F_2^{tu}	(ksi)	10.2	I	10.2	I		
E_2^t	(Msi)	1.33	I	1.47	I		
F_1^{cu}	(ksi)						
E_1^c	(Msi)						
F_2^{cu}	(ksi)						
E_2^c	(Msi)						
F_{12}^{su}	(ksi)						
G_{12}^s	(Msi)						
F_{13}^{su}	(ksi)						
G_{13}^s	(Msi)						
F_{23}^{su}	(ksi)						
G_{23}^s	(Msi)						

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.79		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.59	1.57 - 1.61	
Fiber Areal Weight	(g/m^2)	142	142 - 146	
Ply Thickness	(in)	0.0053	0.0050 - 0.0057	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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Table 4.2.1()
C/Ep 142-UT
T-500/976
Tension, 1-axis
$[0]_8$
75/D, -65/D, 200/D
Interim Data

MATERIAL: T-500 12k/976 unidirectional tape

RESIN CONTENT: 28 - 34 wt% COMP. DENSITY: 1.57 - 1.61 g/cc
FIBER VOLUME: 59 - 64 % VOID CONTENT: 0.3 - 1.7 %
PLY THICKNESS: 0.0050 - 0.0057 in.

TEST METHOD: D3039-76

MODULUS CALCULATION: Chord, 20 - 40 % of ultimate load

NORMALIZED BY: Specimen thickness to 60% fiber volume

		75		-65		250	
Temperature(°F)		75		-65		250	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		13		13		13	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	295	298	213	213	273	276
	Minimum	257	270	163	196	236	258
	Maximum	329	328	243	235	302	310
	C.V. (%)	6.41	5.74	9.78	5.02	7.39	6.05
	B-value	(2)		(2)		(2)	
	Distribution	ANOVA		Weibull		Weibull	
	C_1	20.5		221		282	
	C_2	4.64		13.1		15.7	
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
ϵ_1^{tu} ($\mu\epsilon$)	Mean		13000		10700		11800
	Minimum		11700		9300		10800
	Maximum		13900		12000		12900
	C.V. (%)		4.98		5.98		5.32
	B-value		(2)		(2)		(2)
	Distribution		ANOVA		Weibull		Weibull
	C_1		706		11000		12100
	C_2		4.75		18.8		21.6
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
E_1^t (Msi)	Mean	21.9	22.0	19.0	19.1	22.2	22.4
	Minimum	20.9	20.5	15.9	17.7	18.6	21.0
	Maximum	24.7	24.0	21.5	21.5	25.1	23.8
	C.V. (%)	4.42	4.15	8.11	5.76	6.91	4.17
	No. Specimens	15		15		15	
	No. Batches	3		3		3	

(1) As-fabricated.
(2) B-values are not presented for interim data.

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Table 4.2.1(b)
C/Ep 142-UT
T-500/976
Tension, 2-axis
$[0]_8$
75/D, -65/D, 200/D
Interim Data

MATERIAL: T-500 12k/976 unidirectional tape

RESIN CONTENT: 28 - 34 wt%
FIBER VOLUME: 59 - 64 %
PLY THICKNESS: 0.0050 - 0.0057 in.

COMP. DENSITY: 1.57 - 1.61 lb/in^3
VOID CONTENT: 0.3 - 1.7 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord, 20 - 40 % of ultimate load

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	250			
Moisture Content(%)		dry	dry	dry			
Equilibrium at T, RH		(1)	(1)	(1)			
Source Code		13	13	13			
F_2^{tu} (ksi)	Mean	10.2	10.3	7.90			
	Minimum	9.40	9.40	7.00			
	Maximum	11.3	12.1	8.80			
	C.V. (%)	5.59	6.61	5.35			
	B-value	(2)	(2)	(2)			
	Distribution	ANOVA	Lognormal	Weibull			
	C_1	0.594	2.33	8.09			
	C_2	3.48	0.0636	19.7			
	No. Specimens	15	15	15			
	No. Batches	3	3	3			
ϵ_2^{tu} ($\mu\epsilon$)	Mean	7750	7110	6930			
	Minimum	5800	6200	5900			
	Maximum	8900	8600	8000			
	C.V. (%)	10.3	8.28	8.32			
	B-value	(2)	(2)	(2)			
	Distribution	Weibull	Weibull	Weibull			
	C_1	8080	7390	7180			
	C_2	12.4	11.5	13.7			
	No. Specimens	15	15	15			
	No. Batches	3	3	3			
E_2^t (Msi)	Mean	1.3	1.5	1.2			
	Minimum	1.3	1.4	1.1			
	Maximum	1.7	1.6	1.3			
	C.V. (%)	7.8	4.8	7.0			
	No. Specimens	15	15	15			
	No. Batches	3	3	3			

(1) As fabricated.
(2) B-values are not presented for interim data.

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4.2.2 HITEX 33 6k/E7K8.*

MATERIAL:	HITEX 33 6k/E7K8 unidirectional tape	C/Ep HITEX 33/E7K8 Summary
PREPREG:	U.S. Polymeric HITEX 33 6k/E7K8 unidirectional tape, grade 145	
FIBER:	Hitco HITEX 33 6k, no twist RESIN: U.S. Polymeric E7K8	
T_g(dry):	T_g(wet): T_g METHOD:	
PROCESSING:	Autoclave cure: 300 - 310°F, 120 - 130 min., 55 psi	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88
Date of prepreg manufacture		Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	313	S	296	S	318	S	308	S
E_1^t	(Msi)	18.2	S	18.5	S	18.5	S	18.7	S
F_2^{tu}	(ksi)	6.90	S						
E_2^t	(Msi)	1.25	S						
F_1^{cu}	(ksi)	209	S	230.	S	198	S	136	S
E_1^c	(Msi)	17.1	S	17.9	S	18.0	S	17.6	S
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)	15.0	S			16.3	S	11.7	S
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.80		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.59	1.56 - 1.61	
Fiber Areal Weight	(g/m^2)	145		
Ply Thickness	(in)	0.0057	0.0053 - 0.0058	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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Table 4.2.2(a)
C/Ep 145-UT
HITEX 33/E7K8
Tension, 1-axis
$[0]_{10}$
75/D, -65/D, 75/1.5%
Screening Data

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 wt% COMP. DENSITY: 1.58 g/cc
FIBER VOLUME: 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75		-65		75	
Moisture Content(%)		dry		dry		1.5	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	313	304	296	288	318	310
	Minimum	292	283	267	259	280	272
	Maximum	339	330	327	319	345	335
	C.V. (%)	4.80	4.84	9.19	9.20	7.63	7.65
F_1^{tu}	B-value	(3)		(3)		(3)	
	Distribution	Weibull		Normal		Normal	
(ksi)	C_1	320		296		319	
	C_2	22.2		27.2		24.3	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean		15900		16100		
	Minimum		15200		15500		
	Maximum		17100		17000		
	C.V. (%)		4.81		3.61		
e_1^{tu}	B-value		(3)		(3)		
	Distribution		Normal		Normal		
($\mu\epsilon$)	C_1		15900		16200		
	C_2		765		582		
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean	18.2	17.7	18.5	18.0	18.5	18.0
	Minimum	17.5	17.0	18.1	17.7	18.3	17.8
	Maximum	19.0	18.5	18.6	18.1	18.7	18.2
E_1^t	C.V. (%)	2.58	2.60	1.06	1.07	0.79	0.79
(Msi)	No. Specimens	18		5		5	
	No. Batches	1		1		1	
	Mean		0.310				0.310
	No. Specimens	5				5	
ν_{12}^t	No. Batches	1				1	

(1) As-fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 wt% COMP. DENSITY: 1.58 g/cc
FIBER VOLUME: 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Fiber volume fraction to 60%

Table 4.2.2(b)
C/Ep 145-UT
HITEX 33/E7K8
Tension, 1-axis
$[0]_{10}$
180/1.5%
Screening Data

Temperature(°F)		180					
Moisture Content(%)		1.5					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
F_1^{tu} (ksi)	Mean	308	300				
	Minimum	296	288				
	Maximum	318	309				
	C.V. (%)	2.65	2.65				
	B-value	(2)					
	Distribution	Normal					
	C_1	308					
	C_2	8.17					
	No. Specimens	5					
	No. Batches	1					
ϵ_1^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^t (Msi)	Mean	18.7	18.2				
	Minimum	17.8	17.3				
	Maximum	19.5	19.0				
	C.V. (%)	3.64	3.65				
	No. Specimens	5					
	No. Batches	1					
ν_{12}^t	Mean		0.300				
	No. Specimens	5					
	No. Batches	1					

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are not presented for screening data.

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Table 4.2.2(c)
C/Ep 145-UT
HITEX 33/E7K8
Tension, 2-axis
$[0]_{20}$
75/D
Screening Data

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 wt%
FIBER VOLUME: 58 %
PLY THICKNESS: 0.0058 in.

COMP. DENSITY: 1.58 g/cc
VOID CONTENT: 0.39 %

TEST METHOD:
ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		20					
	Mean	6.90					
	Minimum	5.58					
	Maximum	8.07					
	C.V. (%)	11.2					
F_2^{tu}	B-value	(2)					
	Distribution	Weibull					
(ksi)	C_1	7.23					
	C_2	10.9					
	No. Specimens	20					
	No. Batches	1					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_2^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	1.25					
	Minimum	1.23					
	Maximum	1.27					
E_2^t	C.V. (%)	0.977					
(Msi)	No. Specimens	20					
	No. Batches	1					

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.2(d)
C/Ep 145-UT
HITEX 33/E7K8
Compression, 1-axis
$[0]_{10}$
75/D, -65/D, 75/1.5%
Screening Data

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 - 35 wt% COMP. DENSITY: 1.57 - 1.58 g/cc
FIBER VOLUME: 57 - 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75		-65		75	
Moisture Content(%)		dry		dry		1.5	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	209	204	230	224	198	193
	Minimum	168	164	209	204	178	174
	Maximum	234	228	254	248	217	211
	C.V. (%)	9.41	9.41	7.98	8.04	8.13	8.03
F_1^{cu}	B-value	(3)		(3)		(3)	
	Distribution	Weibull		Normal		Normal	
(ksi)	C_1	218		230		198	
	C_2	13.7		18.3		16.1	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean	12600		13600			
	Minimum	12000		13600			
	Maximum	13400		13700			
	C.V. (%)	2.92		0.48			
ϵ_1^{cu}	B-value	(3)		(3)			
	Distribution	Weibull		Normal			
($\mu\epsilon$)	C_1	12800		13600			
	C_2	35.7		65.7			
	No. Specimens	20		5			
	No. Batches	1		1			
	Mean	17.1	16.2	17.9	16.9	18.0	15.0
	Minimum	16.1	15.2	17.5	16.5	17.5	16.6
	Maximum	17.8	16.8	18.1	17.1	18.8	17.8
E_1^c	C.V. (%)	2.89	2.94	1.23	1.35	3.04	5.59
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	

(1) As-fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.2(e)
C/Ep 145-UT
HITEX 33/E7K8
Compression, 1-axis
$[0]_{10}$
180/1.5%
Screening Data

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 34 - 35 wt% COMP. DENSITY: 1.57 - 1.58 g/cc
FIBER VOLUME: 57 - 58 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180					
Moisture Content(%)		1.5					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	136	132				
	Minimum	111	108				
	Maximum	161	157				
	C.V. (%)	13.4	13.6				
F_1^{cu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	136					
	C_2	18.3					
	No. Specimens		5				
	No. Batches		1				
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	17.6	16.6				
	Minimum	17.0	16.1				
	Maximum	18.0	17.0				
E_1^c	C.V. (%)	2.47	2.47				
(Msi)	No. Specimens		5				
	No. Batches		1				

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are not presented for screening data.

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Table 4.2.2(f)
C/Ep 145-UT
HITEX 33/E7K8
Shear, 12-axis
$[(\pm 45)_2/45]_s$
75/D, 180/D, 75/1.5%, 180/1.5%
Screening Data

MATERIAL: HITEX 33 6k/E7K8 unidirectional tape

RESIN CONTENT: 29 - 30 wt%
FIBER VOLUME: 62 - 64 %
PLY THICKNESS: 0.0053 in.

COMP. DENSITY: 1.59 - 1.61 g/cc
VOID CONTENT: 0.05 - 0.91 %

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	180	75	180		
Moisture Content(%)		dry	dry	1.5	1.5		
Equilibrium at T, RH		(1)	(1)	(2)	(2)		
Source Code		20	20	20	20		
F_{12}^{su} (ksi)	Mean	15.0	13.2	16.3	11.7		
	Minimum	13.5	13.1	15.8	11.5		
	Maximum	15.8	13.3	16.7	11.9		
	C.V. (%)	3.52	0.655	2.20	1.27		
	B-value	(3)	(3)	(3)	(3)		
	Distribution	Weibull	Normal	Normal	Normal		
	C_1	15.2	13.2	16.3	11.7		
	C_2	34.8	0.0865	0.357	0.148		
	No. Specimens	20	5	5	5		
	No. Batches	1	1	1	1		
γ_{12}^{su} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
G_{12}^{s} (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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4.2.3 AS4 12k/E7K8.*

MATERIAL:	AS4 12k/E7K8 unidirectional tape		C/Ep AS4/E7K8 Summary
PREPREG:	U.S. Polymeric AS4 12k/E7K8 unidirectional tape		
FIBER:	Hercules AS4 12k	RESIN:	U.S. Polymeric E7K8
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 300 - 310°F, 120 - 130 min., 55 psi		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88
Date of prepreg manufacture		Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	303	I	291	S	304	S	310.	S
E_1^t	(Msi)	19.3	I	20.1	S	19.6	S	20.1	S
F_2^{tu}	(ksi)	5.47	S						
E_2^t	(Msi)	1.23	S						
F_1^{cu}	(ksi)	209	S	276	S	215	S	150.	S
E_1^c	(Msi)	19.0	S	17.6	S	18.5	S	18.0	S
F_2^{cu}	(ksi)	16.5	I	14.6	S	15.1	S	13.4	S
E_2^c	(Msi)								
F_{12}^{su}	(ksi)								
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.80		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.59	1.52 - 1.59	
Fiber Areal Weight	(g/m^2)	145		
Ply Thickness	(in)	0.0054	0.0054 - 0.0057	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.3(a)
C/Ep 145-UT
AS4/E7K8
Tension, 1-axis
$[0]_{10}$
75/D
Interim Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 37 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0054 in.

COMP. DENSITY: 1.53 - 1.59 g/cc
VOID CONTENT: 0.64 - 2.2 %

TEST METHOD:
ASTM D3039-76

MODULUS CALCULATION:
Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	303	293				
	Minimum	253	252				
	Maximum	345	347				
	C.V. (%)	8.26	8.94				
F_1^{tu}	B-value	(2)					
	Distribution	ANOVA					
(ksi)	C_1	26.7					
	C_2	4.40					
	No. Specimens	20					
	No. Batches	2					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	19.3	18.7				
	Minimum	18.5	17.4				
	Maximum	21.3	21.4				
E_1^t	C.V. (%)	3.79	6.10				
(Msi)	No. Specimens	20					
	No. Batches	2					

(1) As fabricated.
(2) B-values are not presented for interim data.

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Table 4.2.3(b)
C/Ep 145-UT
AS4/E7K8
Tension, 1-axis
$[0]_{10}$
75/D, -65/D, 75/0.77%
Screening Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 37 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0054 in.
COMP. DENSITY: 1.53 - 1.59 g/cc
VOID CONTENT: 0.64 - 2.2 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

			75 dry (1) 20		-65 dry (1) 20		75 0.77 (2) 20	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)		Mean			291	273	304	294
		Minimum			255	239	286	276
		Maximum			327	306	317	306
		C.V. (%)			8.93	8.90	4.16	4.22
		B-value			(3)		(3)	
		Distribution			Normal		Normal	
		C_1			291		305	
		C_2			26.0		12.7	
		No. Specimens			5		5	
		No. Batches			1		1	
ϵ_1^{tu} ($\mu\epsilon$)		Mean		13900		13500		14600
		Minimum		12500		12000		13700
		Maximum		16000		14800		15000
		C.V. (%)		11.0		8.24		3.83
		B-value		(3)		(3)		(3)
		Distribution		Normal		Normal		Normal
		C_1		13900		13500		14600
		C_2		1530		1110		561
		No. Specimens	5		5		5	
		No. Batches	1		1		1	
E_1^t (Msi)		Mean			20.1	18.8	19.6	18.9
		Minimum			19.7	18.4	19.0	18.4
		Maximum			20.6	19.3	20.1	19.4
		C.V. (%)			1.67	1.79	2.04	1.96
		No. Specimens			5		5	
		No. Batches			1		1	
ν_{12}^t		Mean		0.320				0.288
		No. Specimens	5				5	
		No. Batches	1				1	

(1) As fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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Table 4.2.3(c)
C/Ep 145-UT
AS4/E7K8
Tension, 1-axis
$[0]_{10}$
180/0.77%
Screening Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 37 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0054 in.
COMP. DENSITY: 1.53 - 1.59 g/cc
VOID CONTENT: 0.64 - 2.2 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)			180				
Moisture Content(%)			0.77				
Equilibrium at T, RH			(1)				
Source Code			20				
		Normalized	Measured				
	Mean	310	296				
	Minimum	284	274				
	Maximum	326	306				
	C.V. (%)	5.87	4.76				
F_1^{tu} (ksi)	B-value	(2)					
	Distribution	Normal					
	C_1	310					
	C_2	18.2					
	No. Specimens	5					
	No. Batches	1					
	Mean		14600				
	Minimum		13900				
	Maximum		15400				
	C.V. (%)		4.21				
ϵ_1^{tu} ($\mu\epsilon$)	B-value		(2)				
	Distribution		Normal				
	C_1		14600				
	C_2		616				
	No. Specimens	5					
	No. Batches	1					
	Mean	20.1	19.2				
	Minimum	19.1	18.5				
	Maximum	21.8	20.4				
E_1^t	C.V. (%)	5.65	4.01				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Mean		0.288				
ν_{12}^t	No. Specimens	5					
	No. Batches	1					

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are not presented for screening data.

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Table 4.2.3(d)
C/Ep 145-UT
AS4/E7K8
Tension, 2-axis
$[0]_{20}$
75/D
Interim Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 32 - 38 wt%
FIBER VOLUME: 53 - 60 %
PLY THICKNESS: 0.0057 in.

COMP. DENSITY: 1.54 - 1.59 g/cc
VOID CONTENT: 0.64 - 0.75 %

TEST METHOD:
ASTM D3039-76

MODULUS CALCULATION:
Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		20					
F_2^{tu} (ksi)	Mean	5.47					
	Minimum	4.10					
	Maximum	7.01					
	C.V. (%)	13.2					
	B-value	(2)					
	Distribution	Weibull					
	C_1	5.79					
	C_2	8.04					
	No. Specimens	20					
	No. Batches	1					
ϵ_2^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_2^t (Msi)	Mean	1.23					
	Minimum	1.16					
	Maximum	1.32					
	C.V. (%)	3.76					
	No. Specimens	20					
	No. Batches	1					

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.3(e)
C/Ep 145-UT
AS4/E7K8
Compression, 1-axis
$[0]_{10}$
75/D, -65/D, 75/0.77%
Screening Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 35 - 40 wt%
FIBER VOLUME: 51 - 57 %
PLY THICKNESS: 0.0054 in.

COMP. DENSITY: 1.52 - 1.58 g/cc
VOID CONTENT: 1.4 - 2.3 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

			75		-65		75	
Temperature(°F)			75		-65		75	
Moisture Content(%)			dry		dry		0.77	
Equilibrium at T, RH			(1)		(1)		(2)	
Source Code			20		20		20	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	245	209	276	235	215	182
		Minimum	207	176	251	213	196	166
		Maximum	269	229	299	254	238	202
		C.V. (%)	8.00	7.80	6.57	6.60	7.78	7.75
F_1^{cu}		B-value	(3)		(3)		(3)	
		Distribution	Weibull		Normal		Normal	
(ksi)		C_1	254		276		215	
		C_2	16.3		18.1		16.7	
		No. Specimens	20		5		5	
		No. Batches	1		1		1	
		Mean	11700		14400			
		Minimum	10800		13900			
		Maximum	13100		15100			
		C.V. (%)	4.81		3.89			
e_1^{cu}		B-value	(2)		(2)			
		Distribution	Normal		Normal			
($\mu\epsilon$)		C_1	11700		14400			
		C_2	564		559			
		No. Specimens	20		5			
		No. Batches	1		1			
		Mean	19.0	17.9	17.6	16.5	18.5	17.4
		Minimum	17.3	16.3	16.6	15.7	17.7	16.7
		Maximum	20.4	19.2	18.0	17.0	19.0	17.9
E_1^c		C.V. (%)	4.58	4.54	3.16	3.14	2.95	2.86
(Msi)		No. Specimens	20		5		5	
		No. Batches	1		1		1	

(1) As fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.3(l)
C/Ep 145-UT
AS4/E7K8
Compression, 1-axis
$[0]_{10}$
180/0.77%
Screening Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 35 - 40 wt%
FIBER VOLUME: 51 - 57 %
PLY THICKNESS: 0.0054 in.

COMP. DENSITY: 1.52 - 1.58 g/cc
VOID CONTENT: 1.4 - 2.3 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.77					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	150	127				
	Minimum	125	106				
	Maximum	176	150				
	C.V. (%)	14.8	15.0				
F_1^{cu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	150					
	C_2	22.2					
	No. Specimens	5					
	No. Batches	1					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	18.0	17.0				
	Minimum	17.4	16.4				
	Maximum	18.4	17.3				
E_1^c	C.V. (%)	2.46	2.41				
(Msi)	No. Specimens	5					
	No. Batches	1					

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.3(g)
C/Ep 145-UT
AS4/E7K8
Shear, 12-axis
$[(\pm 45)_2/45]_s$
75/D
Interim Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 33 - 36 wt%
FIBER VOLUME: 55 - 57 %
PLY THICKNESS: 0.0055 in.

COMP. DENSITY: 1.54 - 1.55 g/cc
VOID CONTENT: 1.9 - 2.3 %

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		20					
	Mean	16.5					
	Minimum	13.8					
	Maximum	17.0					
	C.V. (%)	6.41					
F_{12}^{su} (ksi)	B-value	(2)					
	Distribution	ANOVA					
	C_1	2.46					
	C_2	7.58					
	No. Specimens	20					
	No. Batches	2					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
γ_{12}^{su} ($\mu\epsilon$)	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean						
	Minimum						
	Maximum						
G_{12}^{s} (Msi)	C.V. (%)						
	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.3(h)
C/Ep 145-UT
AS4/E7K8
Shear, 12-axis
$[(\pm 45)_2/45]_s$
180/D, 75/0.77%, 180/0.77%
Interim Data

MATERIAL: AS4 12k/E7K8 unidirectional tape

RESIN CONTENT: 33 - 36 wt%
FIBER VOLUME: 55 - 57 %
PLY THICKNESS: 0.0055 in.

COMP. DENSITY: 1.54 - 1.55 g/cc
VOID CONTENT: 1.9 - 2.3 %

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION: Slope of initial linear portion of load-displacement curve

NORMALIZED BY: Not normalized

Temperature(°F)		180	75	180			
Moisture Content(%)		dry	0.77	0.77			
Equilibrium at T, RH		(1)	(2)	(2)			
Source Code		20	20	20			
F_{12}^{su} (ksi)	Mean	14.6	15.1	13.4			
	Minimum	14.2	13.5	13.0			
	Maximum	14.9	15.8	13.8			
	C.V. (%)	1.90	6.04	2.44			
	B-value	(3)	(3)	(3)			
	Distribution	Normal	Normal	Normal			
	C_1	14.6	15.1	13.4			
	C_2	0.277	0.905	0.328			
	No. Specimens	5	5	5			
	No. Batches	1	1	1			
γ_{12}^{su} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
G_{12}^{s} (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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4.2.4 Celion 12k/E7K8.*

MATERIAL:	Celion 12k/E7K8 unidirectional tape	**C/Ep Celion 12k/E7K8 Summary**
PREPREG:	U.S. Polymeric Celion 12k/E7K8 unidirectional tape, grade 280	
FIBER:	Celanese Celion 12k, no twist RESIN: U.S. Polymeric E7K8	
T_g(dry):	T_g(wet): T_g METHOD:	
PROCESSING:	Autoclave cure: 300 - 310°F, 120 - 130 min., 55 psi	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88
Date of prepreg manufacture		Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	293	S	281	S	300.	S	293	S
E_1^t	(Msi)	20.0	S	19.2	S	19.0	S	19.8	S
F_2^{tu}	(ksi)	6.00	S						
E_2^t	(Msi)								
F_1^{cu}	(ksi)	206	S	221	S	207	S	185	S
E_1^c	(Msi)	19.9	S	22.9	S	21.6	S	21.1	S
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)	9.9	S	12.0	S	10.0	S	10.0	S
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.8		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.59	1.59 - 1.61	
Fiber Areal Weight	(g/m^2)	280		
Ply Thickness	(in)	0.011	0.010 - 0.011	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 wt%
FIBER VOLUME: 63 - 64 %
PLY THICKNESS: 0.011 in.

COMP. DENSITY: 1.61 g/cc
VOID CONTENT: 0.53 - 1.0 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume to 60%

Table 4.2.4(a)
C/Ep 280-UT
Celion 12k/E7K8
Tension, 1-axis
$[0]_5$
75/D, -65/D, 75/0.77%
Screening Data

Temperature(°F)		75		-65		75	
Moisture Content(%)		dry		dry		0.77	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	293	310	281	302	300	314
	Minimum	265	284	268	287	292	306
	Maximum	317	332	307	387	315	330
	C.V. (%)	4.52	6.09	5.44	6.09	3.22	3.60
F_1^{tu} (ksi)	B-value	(3)		(3)		(3)	
	Distribution	Weibull		Normal		Normal	
	C_1	299		281		300	
	C_2	25.6		15.3		9.67	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean		14300		14800		
	Minimum		13500		14200		
	Maximum		14700		15800		
	C.V. (%)		3.34		3.87		
ϵ_1^{tu} ($\mu\epsilon$)	B-value		(3)		(3)		
	Distribution		Normal		Normal		
	C_1		14300		14800		
	C_2		478		573		
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean	20.0	21.1	19.2	20.6	19.0	19.9
	Minimum	18.7	20.1	18.6	20.0	18.5	19.4
	Maximum	21.9	23.0	20.3	21.8	20.0	21.0
E_1^t	C.V. (%)	4.48	4.25	3.40	3.80	3.22	3.60
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean		.286				0.292
ν_{12}^t	No. Specimens	5				5	
	No. Batches	1				1	

(1) As fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.4(b)
C/Ep 280-UT
Celion 12k/E7K8
Tension, 1-axis
$[0]_5$
180/0.77%
Screening Data

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 wt% COMP. DENSITY: 1.61 g/cc
FIBER VOLUME: 63 - 64 % VOID CONTENT: 0.53 - 1.0 %
PLY THICKNESS: 0.011 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180					
Moisture Content(%)		0.77					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured				
	Mean	293	311				
	Minimum	269	286				
	Maximum	316	335				
	C.V. (%)	6.43	7.19				
F_1^{tu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	293					
	C_2	18.9					
	No. Specimens	5					
	No. Batches	1					
	Mean		13800				
	Minimum		12300				
	Maximum		15400				
	C.V. (%)		10.4				
ϵ_1^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		13600				
	C_2		1440				
	No. Specimens	5					
	No. Batches	1					
	Mean	19.8	21.0				
	Minimum	19.4	20.6				
	Maximum	20.1	21.4				
E_1^t	C.V. (%)	1.61	1.81				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Mean		0.322				
ν_{12}^t	No. Specimens	5					
	No. Batches	1					

(1) Conditioned for 14 days at 160°F, 85% RH.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.4(c)
C/Ep 280-UT
Celion 12k/E7K8
Tension, 2-axis
$[0]_{12}$
75/D
Screening Data

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 31 - 33 wt%
FIBER VOLUME: 59 - 61 %
PLY THICKNESS: 0.011 in.
COMP. DENSITY: 1.59 - 1.60 g/cc
VOID CONTENT: 0.68 - 0.74 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		20					
F_2^{tu} (ksi)	Mean	6.00					
	Minimum	5.21					
	Maximum	6.89					
	C.V. (%)	8.79					
	B-value	(2)					
	Distribution	Weibull					
	C_1	6.24					
	C_2	12.6					
	No. Specimens	20					
	No. Batches	1					
ϵ_2^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_2^t (Msi)	Mean	1.28					
	Minimum	1.19					
	Maximum	1.36					
	C.V. (%)	4.52					
	No. Specimens	20					
	No. Batches	1					

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.4(d)
C/Ep 280-UT
Celion 12k/E7K8
Compression, 1-axis
$[0]_5$
75/D, -65/D, 75/0.77%
Screening Data

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 - 30 wt% COMP. DENSITY: 1.60 - 1.61 g/cc
FIBER VOLUME: 62 - 64 % VOID CONTENT: 0.78 - 0.79 %
PLY THICKNESS: 0.010 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		75		-65		75	
Moisture Content(%)		dry		dry		0.77	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	206	213	221	229	207	214
	Minimum	171	177	198	205	198	205
	Maximum	247	255	267	231	219	227
	C.V. (%)	8.62	8.84	12.2	13.6	5.06	5.65
F_1^{cu}	B-value	(3)		(3)		(3)	
	Distribution	Weibull		Normal		Normal	
(ksi)	C_1	214		221		207	
	C_2	12.1		27.0		10.5	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean		11200		9870		
	Minimum		10800		9210		
	Maximum		11800		10600		
	C.V. (%)		3.59		5.32		
ϵ_1^{cu}	B-value		(3)		(3)		
	Distribution		Normal		Normal		
($\mu\epsilon$)	C_1		11200		9870		
	C_2		401		526		
	No. Specimens		5		5		
	No. Batches		1		1		
	Mean	19.9	21.1	22.9	24.3	21.6	22.3
	Minimum	18.1	19.2	20.8	22.0	20.2	21.0
	Maximum	21.7	22.3	23.8	25.1	22.8	23.6
E_1^c	C.V. (%)	4.95	5.08	5.28	5.90	5.25	5.86
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	

(1) As fabricated.
(2) Conditioned for 14 days at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.4(e)
C/Ep 280-UT
Celion 12k/E7K8
Compression, 1-axis
$[0]_5$
180/0.77%
Screening Data

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 29 - 30 wt% COMP. DENSITY: 1.60 - 1.61 g/cc
FIBER VOLUME: 62 - 64 % VOID CONTENT: 0.78 - 0.79 %
PLY THICKNESS: 0.010 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume to 60%

Temperature(°F)		180			
Moisture Content(%)		0.77			
Equilibrium at T, RH		(1)			
Source Code		20			
		Normalized	Measured		
	Mean	185	192		
	Minimum	158	164		
	Maximum	220	228		
	C.V. (%)	12.9	14.5		
F_1^{cu} (ksi)	B-value	(2)			
	Distribution	Normal			
	C_1	185			
	C_2	24.0			
	No. Specimens	5			
	No. Batches	1			
	Mean				
	Minimum				
	Maximum				
	C.V. (%)				
ϵ_1^{cu} ($\mu\epsilon$)	B-value				
	Distribution				
	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Mean	21.1	22.3		
	Minimum	19.5	20.6		
	Maximum	23.1	24.5		
E_1^c (Msi)	C.V. (%)	6.80	7.63		
	No. Specimens	5			
	No. Batches	1			

(1) Conditioned at 160°F, 85% RH.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: Celion 12k/E7K8 unidirectional tape

RESIN CONTENT: 30 - 31 wt%
FIBER VOLUME: 61 - 62 %
PLY THICKNESS: 0.011 in.
COMP. DENSITY: 1.60 g/cc
VOID CONTENT: 0.41 - 0.61 %

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Table 4.2.4(f)
C/Ep 280-UT
Celion 12k/E7K8
Shear, 12-axis
[±45/45]$_s$
75/D, 180/D,
75/0.77%, 180/0.77%
Screening Data

Temperature(°F)		75	180	75	180		
Moisture Content(%)		dry	dry	0.77	0.77		
Equilibrium at T, RH		(1)	(1)	(2)	(2)		
Source Code		20	20	20	20		
F_{12}^{su} (ksi)	Mean	9.9	10.0	12.0	10.0		
	Minimum	9.3	8.1	11.3	8.2		
	Maximum	11.1	11.1	12.3	11.4		
	C.V. (%)	4.16	11.7	3.41	11.7		
	B-value	(3)	(3)	(3)	(3)		
	Distribution	Nonpara.	Normal	Normal	Normal		
	C_1		10.0	12.0	10.0		
	C_2		1.17	0.407	1.17		
	No. Specimens	20	5	5	5		
	No. Batches	1	1	1	1		
γ_{12}^{su} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
G_{12}^{s} (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) Conditioned at 160°F, 85% RH.
(3) B-values are not presented for screening data.

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4.2.5 AS4 12k/938.*

MATERIAL:	AS4 12k/938 unidirectional tape			C/Ep AS4/938 Summary
PREPREG:	Fiberite hy-E 1338H, grade 145			
FIBER:	Hercules AS4 12k, unsized, no twist	RESIN:	Fiberite 938	
T_g(dry):	°F	T_g(wet): 260 °F	T_g METHOD:	
PROCESSING:	Autoclave cure: 350 ± 10°F, 120 - 135 min., 100 ± 15 psi			

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	8/85
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture	7/85	*Date of analysis*	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	314	I	296	I				
E_1^t	(Msi)	22.4	I	19.5	I				
F_2^{tu}	(ksi)	8.96	I						
E_2^t	(Msi)	1.29	I						
F_1^{cu}	(ksi)	228	I					190	I
E_1^c	(Msi)	18.2	I					19.1	S
F_2^{cu}	(ksi)	30.4	S						
E_2^c	(Msi)								
F_{12}^{su}	(ksi)	13.0	S						
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.80	1.77 - 1.79	
Resin Density	(g/cc)	1.30	1.30	
Composite Density	(g/cc)	1.60	1.55 - 1.58	
Fiber Areal Weight	(g/m^2)	145	144 - 146	
Ply Thickness	(in)	0.0055	0.0048 - 0.0065	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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• DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.5(a)
C/Ep 145-UT
AS4/938
Tension, 1-axis
$[0]_8$
75/D, -65/D, 200/D
Interim Data

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 35 - 41 wt% COMP. DENSITY: 1.55 - 1.57 g/cc
FIBER VOLUME: 52 - 57 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: 0.0042 - 0.0052 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

		75		-65		200	
Temperature(°F)							
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	314	272	296	238	321	274
	Minimum	270	230	198	174	363	229
	Maximum	351	330	363	287	356	322
	C.V.(%)	7.45	8.79	14.4	11.0	7.79	8.10
F_1^{tu}	B-value	(3)		(3)		(3)	
	Distribution	Weibull		ANOVA		ANOVA	
(ksi)	C_1	324		49.1		26.9	
	C_2	16.5		4.64		3.78	
	No. Specimens	22		22		20	
	No. Batches	3		3		3	
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
e_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	22.4	19.4	19.5	19.0	20.4	20.8
	Minimum	18.8	17.1	18.5	16.9	18.4	18.4
	Maximum	26.9	21.0	21.5	22.0	24.0	22.4
E_1^t	C.V.(%)	9.88	4.66	4.07	5.13	7.23	6.06
(Msi)	No. Specimens	22		22		20	
	No. Batches	3		3		3	

(1) As fabricated.
(2) Gage length 2.0 inches
(3) B-values are not presented for interim data

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Table 4.2.5(b)
C/Ep 145-UT
AS4/938
Tension, 2-axis
$[90]_{16}$
75/D, 200/D
Interim Data

MATERIAL: AS4 12k/938 *unidirectional tape*

RESIN CONTENT: 35 - 40 wt% COMP. DENSITY: 1.56 - 1.58 g/cc
FIBER VOLUME: 52 - 58 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: 0.0053 - 0.0063 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Not normalized

Temperature(°F)		75.0	200			
Moisture Content(%)		dry	dry			
Equilibrium at T, RH		(1)	(1)			
Source Code		12	12			
	Mean	8.96	8.84			
	Minimum	6.50	6.85			
	Maximum	12.0	10.3			
	C.V.(%)	15.2	12.2			
F_2^{tu}	B-value	(3)	(3)			
	Distribution	Weibull	ANOVA			
(ksi)	C_1	9.54	1.18			
	C_2	7.10	3.96			
	No. Specimens	19	17			
	No. Batches	3	3			
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
e_2^{tu}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	1.29	1.23			
	Minimum	0.970	1.05			
	Maximum	1.72	1.40			
E_2^t	C.V.(%)	7.89	7.81			
(Msi)	No. Specimens	19	17			
	No. Batches	3	3			

(1) As fabricated.
(2) Gage length 2.0 inches.
(3) B-values are not presented for interim data.

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Table 4.2.5(c)
C/Ep 145-UT
AS4/938
Compression, 1-axis
$[0]_8$
75/D, 200/W
Interim Data

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 33 - 38 wt% COMP. DENSITY: 1.55 - 1.58 g/cc
FIBER VOLUME: 54 - 60 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: *0.0048 - 0.0060 in.*

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75		200		
Moisture Content(%)		dry		(1)		
Equilibrium at T, RH		(2)				
Source Code		12		12		
		Normalized	Measured	Normalized	Measured	
	Mean	228	211	190	168	
	Minimum	186	172	158	138	
	Maximum	265	251	223	194	
	C.V.(%)	9.31	8.53	8.96	9.29	
F_1^{cu}	B-value	(3)		(3)		
	Distribution	Weibull		ANOVA		
(ksi)	C_1	224		19.0		
	C_2	12.5		4.40		
	No. Specimens	25		24		
	No. Batches	3		3		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
e_1^{cu}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	18.2	18.4			
	Minimum	15.7	15.9			
	Maximum	21.0	22.5			
E_1^c	C.V.(%)	9.13	12.4			
(Msi)	No. Specimens	15				
	No. Batches	2				

(1) Specimens conditioned at 140°F and 95% relative humidity for one month.
(2) As fabricated.
(3) B-values are not presented for interim data.

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Table 4.2.5(d)
C/Ep 145-UT
AS4/938
Compression, 1-axis
$[0]_8$
200/W
Screening Data

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 33 - 38 wt% COMP. DENSITY: 1.55 - 1.58 g/cc
FIBER VOLUME: 54 - 60 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: 0.0048 - 0.0060 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		200					
Moisture Content(%)		(1)					
Equilibrium at T, RH							
Source Code		12					
		Normalized	Measured				
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
F_1^{cu}	B-value						
	Distribution						
(ksi)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	19.1	18.4				
	Minimum	16.9	16.6				
	Maximum	24.0	21.0				
E_1^c	C.V.(%)	12.8	9.10				
(Msi)	No. Specimens	13					
	No. Batches	2					

(1) Specimens conditioned at 140°F and 95% relative humidity for one month.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.5(e)
C/Ep 145-UT
AS4/938
Compression, 2-axis
$[90]_8$
75/D
Screening Data

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.56 g/cc
FIBER VOLUME: 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Not normalized

Temperature(°F)		75.0					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		12					
	Mean	30.4					
	Minimum	26.2					
	Maximum	39.7					
	C.V.(%)	16.4					
F_2^{cu}	B-value	(2)					
	Distribution	Nonpara.					
(ksi)	C_1	6					
	C_2	2.14					
	No. Specimens	10					
	No. Batches	1					
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
e_2^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean						
	Minimum						
	Maximum						
E_2^c	C.V.(%)						
(Msi)	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) B-values are not presented for screening data.

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Table 4.2.5(f)
C/Ep 145-UT
AS4/938
Shear, 12-axis
$[\pm 45]_{2s}$
200/D
Interim Data

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 35 - 37 wt% COMP. DENSITY: 1.56 - 1.58 g/cc
FIBER VOLUME: 54 - 57 % VOID CONTENT: 0.0 - <1.0 %
PLY THICKNESS: 0.0051 - 0.0063 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		200				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		12				
	Mean	13.9				
	Minimum	11.9				
	Maximum	16.0				
	C.V.(%)	7.63				
F_{12}^{su}	B-value	(2)				
	Distribution	ANOVA				
(ksi)	C_1	1.26				
	C_2	4.96				
	No. Specimens	18				
	No. Batches	3				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{12}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for interim data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.5(g)
C/Ep 145-UT
AS4/938
Shear, 12-axis
$[\pm 45]_{2s}$
75/D
Screening Data

MATERIAL: AS4 12k/938 unidirectional tape

RESIN CONTENT: 35 - 37 wt%
FIBER VOLUME: 54 - 57 %
PLY THICKNESS: 0.0051 - 0.0063 in.
COMP. DENSITY: 1.56 - 1.58 g/cc
VOID CONTENT: 0.0 - <1.0 %

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75.0				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		12				
	Mean	13.0				
	Minimum	10.8				
	Maximum	13.9				
	C.V.(%)	6.36				
F_{12}^{su}	B-value	(2)				
	Distribution	Weibull				
(ksi)	C_1	13.4				
	C_2	25.4				
	No. Specimens	13				
	No. Batches	3				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{12}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for interim data.

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4.2.6 T-300 3k/934 plain weave.*

MATERIAL:	T-300 3k/934 plain weave		C/Ep T-300/934 Summary
PREPREG:	Fiberite HMF-322/34 plain weave		
FIBER:	Toray T-300 3k	RESIN:	Fiberite 934
T_g(dry):	410°F T_g(wet):	T_g METHOD:	DSC
PROCESSING:	Autoclave cure: 355 ± 10°F, 120 - 130 min., 85-100 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture	2/84	Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW	ETW	
F_1^{tu}	(ksi)	91	I	83	I		79	I
E_1^t	(Msi)	9.1	I	10.	I		9.4	I
F_2^{tu}	(ksi)	88	I	80.	I		81	I
E_2^t	(Msi)	9.0	I	9.1	I		9.9	I
F_1^{cu}	(ksi)	95	I	104	I		44	I
E_1^c	(Msi)	8.4	I	8.2	I			
F_2^{cu}	(ksi)	90.	I	103	I		46	I
E_2^c	(Msi)	8.3	I	8.4	I			
F_{12}^{su}	(ksi)							
G_{12}^s	(Msi)							
F_{13}^{su}	(ksi)	12.0	S	11.9	S			
G_{13}^s	(Msi)							
F_{23}^{su}	(ksi)							
G_{23}^s	(Msi)							

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)		1.73 - 1.74	
Resin Density	(g/cc)	1.30		
Composite Density	(g/cc)	1.55	1.54 - 1.56	
Fiber Areal Weight	(g/m^2)	194	1.92 - 2.00	
Ply Thickness	(in)		0.0073 - 0.0084	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.6(a)
C/Ep 194-PW
T-300/934
Tension, 1-axis
$[0_f]_{12}$
75/D, -65/D
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65		
Moisture Content(%)		dry		dry		
Equilibrium at T, RH		(1)		(1)		
Source Code		12		12		
		Normalized	Measured	Normalized	Measured	
	Mean	91	94	83	85	
	Minimum	82	85	78	79	
	Maximum	99	100.	87	90	
	C.V.(%)	4.1	4.0	3.2	3.3	
F_1^{tu}	B-value	(2)		(2)		
	Distribution	Weibull		Weibull		
(ksi)	C_1	93.0		83.7		
	C_2	28.2		35.8		
	No. Specimens	20		20		
	No. Batches	4		4		
	Mean		9780		8990	
	Minimum		8880		7990	
	Maximum		11200		9800	
	C.V.(%)		5.61		6.07	
e_1^{tu}	B-value		(2)		(2)	
	Distribution		ANOVA		ANOVA	
($\mu\epsilon$)	C_1		577		592	
	C_2		3.12		3.61	
	No. Specimens	20		20		
	No. Batches	4		4		
	Mean	9.1	9.4	10.	10.	
	Minimum	8.4	8.7	8.6	9.0	
	Maximum	9.5	9.9	12.0	12.0	
E_1^t	C.V.(%)	3.3	3.6	11	10.	
(Msi)	No. Specimens	20		20		
	No. Batches	1		1		

(1) As fabricated.
(2) B-values are not presented for interim data.
(3) Width 0.5 inch, speed of testing 0.05 in./in./min, gage length below recommendation.

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Table 4.2.6(b)
C/Ep 194-PW
T-300/934
Tension, 1-axis
$[0_f]_{12}$
250/D
Screening Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.56 - 1.57 g/cc
FIBER VOLUME: 60 % VOID CONTENT: <0.5 %
PLY THICKNESS: 0.0074 - 0.0076 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		250					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
	Mean	109	113				
	Minimum	104	109				
	Maximum	114	118				
	C.V. (%)	3.54	3.42				
F_1^{tu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	86.0					
	C_2	2.86					
	No. Specimens	5					
	No. Batches	1					
	Mean		11300				
	Minimum		10900				
	Maximum		11800				
	C.V. (%)		3.11				
e_1^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		11300				
	C_2		351				
	No. Specimens	5					
	No. Batches	1					
	Mean	9.3	9.7				
	Minimum	9.1	9.4				
	Maximum	10.0	10.7				
E_1^t	C.V. (%)	4.6	5.6				
(Msi)	No. Specimens	5					
	No. Batches	1					

(1) As fabricated.
(2) B-values are not presented for screening data.
(3) Width 0.5 inch, speed of testing 0.05 in./in./min., gage length below recommendation.

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Table 4.2.6(c)
C/Ep 194-PW
T-300/934
Tension, 1-axis
$[0_f]_{12}$
160/W, 250/W
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

		160		250			
Temperature(°F)							
Moisture Content(%)		(1)		(1)			
Equilibrium at T, RH							
Source Code		12		12			
		Normalized	Measured	Normalized	Measured		
	Mean	96	98	79	82		
	Minimum	84	88	61	66		
	Maximum	104	106	95	97		
	C.V.(%)	5.7	5.1	14	11		
F_1^{tu}	B-value	(2)		(2)			
	Distribution	ANOVA		ANOVA			
(ksi)	C_1	6.0		12			
	C_2	4.8		5.3			
	No. Specimens	15		15			
	No. Batches	3		3			
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
e_1^{tu}	*B-value*						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	9.8	10.0	9.4	9.7		
	Minimum	8.1	8.6	6.8	7.1		
	Maximum	11.0	11.7	12.0	13.0		
E_1^t	C.V.(%)	8.7	8.7	17.	18		
(Msi)	No. Specimens	15		15			
	No. Batches	3		3			

(1) Immersed in water at 160°F for 14 days.
(2) B-values are not presented for interim data.
(3) Width 0.5 inch, speed of testing 0.05 in./in./min, gage length below recommendation.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.6(d)
***C/Ep* 194-PW**
T-300/934
Tension, 2-axis
$[90_t]_{12}$
75/D, -65/D
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		12		12			
		Normalized	Measured	Normalized	Measured		
	Mean	88	91	80.	82		
	Minimum	80.	82	70.	72		
	Maximum	97	99	91	95		
	C.V. (%)	5.7	5.5	6.2	6.5		
F_2^{tu}	B-value	(2)		(2)			
	Distribution	ANOVA		ANOVA			
(ksi)	C_1	5.4		5.2			
	C_2	3.5		3.3			
	No. Specimens	20		20			
	No. Batches	4		4			
	Mean		9630		9100		
	Minimum		8680		7750		
	Maximum		11100		10700		
	C.V. (%)		6.18		7.44		
ϵ_2^{tu}	B-value		(2)		(2)		
	Distribution		ANOVA		ANOVA		
($\mu\epsilon$)	C_1		616		710		
	C_2		2.82		3.08		
	No. Specimens	20		20			
	No. Batches	4		4			
	Mean	9.0	9.3	9.1	9.5		
	Minimum	8.3	8.7	8.1	8.3		
	Maximum	9.9	10.3	10.8	11.1		
E_2^t	C.V. (%)	5.0	4.8	9.3	9.2		
(Msi)	No. Specimens	20		20			
	No. Batches	4		4			

(1) As fabricated.
(2) B-values are not presented for interim data.

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Table 4.2.6(e)
C/Ep 194-PW
T-300/934
Tension, 2-axis
$[90_f]_{12}$
250/D
Screening Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.56 - 1.57 g/cc
FIBER VOLUME: 60 % VOID CONTENT: <0.5 %
PLY THICKNESS: 0.0074 - 0.0076 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		250			
Moisture Content(%)		dry			
Equilibrium at T, RH		(1)			
Source Code		12			
		Normalized	Measured		
	Mean	94	98		
	Minimum	90.	94		
	Maximum	97	101		
	C.V. (%)	2.6	2.7		
F_2^{tu}	B-value	(2)			
	Distribution	Normal			
(ksi)	C_1	93.7			
	C_2	2.47			
	No. Specimens	5			
	No. Batches	1			
	Mean		11400		
	Minimum		10400		
	Maximum		12400		
	C.V. (%)		8.59		
ϵ_2^{tu}	B-value		(2)		
	Distribution		Normal		
($\mu\epsilon$)	C_1		11400		
	C_2		981		
	No. Specimens	5			
	No. Batches	1			
	Mean	8.1	8.5		
	Minimum	8.0	8.3		
	Maximum	8.2	8.6		
E_2^t	C.V. (%)	1.1	1.5		
(Msi)	No. Specimens	5			
	No. Batches	1			

(1) As fabricated.
(2) B-values are not presented for screening data.

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Table 4.2.6(f)
C/Ep 194-PW
T-300/934
Tension, 2-axis
$[90_t]_{12}$
160/W, 250/W
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		160		250		
Moisture Content(%)		(1)		(1)		
Equilibrium at T, RH						
Source Code		12		12		
		Normalized	Measured	Normalized	Measured	
F_2^{tu} (ksi)	Mean	97	100.	81	83	
	Minimum	90.	92	73	75	
	Maximum	111	113	89	91	
	C.V. (%)	6.8	6.3	5.1	4.8	
	B-value	(2)		(2)		
	Distribution	ANOVA		ANOVA		
	C_1	7.3		4.4		
	C_2	4.8		4.5		
	No. Specimens	15		15		
	No. Batches	3		3		
e_2^{tu} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
E_2^t (Msi)	Mean	10.	10.	9.9	10.	
	Minimum	8.0	8.2	8.2	8.5	
	Maximum	11.8	12.1	11.9	12.1	
	C.V. (%)	11	11	11	11	
	No. Specimens	15		15		
	No. Batches	3		3		

(1) Immersed in water at 160°F for 14 days.
(2) B-values are not presented for interim data.

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Table 4.2.6(g)
C/Ep 194-PW
T-300/934
Compression, 1-axis
$[0_f]_{12}$
75/D, -65/D, 250/D
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88 Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65		250	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean	95	98	104	108		
	Minimum	83	87	87	90.		
	Maximum	120	125	133	139		
	C.V.(%)	10.	10.	13	14		
	B-value	(2)		(2)			
	Distribution	ANOVA		ANOVA			
	C_1	10.		15			
	C_2	3.9		3.7			
	No. Specimens	20		20			
	No. Batches	4		4			
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^c (4) (Msi)	Mean	8.4	8.8	8.2	8.6	8.4	8.9
	Minimum	7.7	8.0	7.4	7.8	7.9	8.1
	Maximum	9.0	9.4	8.9	9.7	10.0	10.1
	C.V.(%)	5.1	5.3	5.1	5.7	6.3	6.4
	No. Specimens	20		20		19	
	No. Batches	4		4		4	

(1) As fabricated.
(2) B-values are not presented for interim data.
(3) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.
(4) Specimen thickness of 0.09 - 0.10 inch is less than nominal 0.12 inch thickness per method.

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Table 4.2.6(h)
C/Ep 194-PW
T-300/934
Compression, 1-axis
$[0_f]_{12}$
250/D
Screening Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.56 - 1.57 g/cc
FIBER VOLUME: 60 % VOID CONTENT: <0.5 %
PLY THICKNESS: 0.0074 - 0.0076 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		250					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
	Mean	100.	105				
	Minimum	94	98				
	Maximum	107	111				
	C.V. (%)	5.6	5.1				
F_1^{cu} (3)	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	100.					
	C_2	5.64					
	No. Specimens	5					
	No. Batches	1					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean						
	Minimum						
	Maximum						
E_1^c	C.V. (%)						
(Msi)	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) B-values are not presented for screening data.
(3) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.

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Table 4.2.6(i)
C/Ep 194-PW
T-300/934
Compression, 1-axis
$[0_f]_{12}$
160/W, 250/W
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		160 (1)		250 (1)			
Moisture Content(%)							
Equilibrium at T, RH							
Source Code		12		12			
		Normalized	Measured	Normalized	Measured		
	Mean	74	76	44	46		
	Minimum	67	68	40	41		
	Maximum	81	84	49	51		
	C.V.(%)	6.9	5.6	6.2	6.2		
F_1^{cu} (3) (ksi)	B-value	(2)		(2)			
	Distribution	ANOVA		Weibull			
	C_1	5.6		45.4			
	C_2	4.9		17.4			
	No. Specimens	15		15			
	No. Batches	3		3			
e_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^c (Msi)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	No. Specimens						
	No. Batches						

(1) Immersed in water at 160°F for 14 days.
(2) B-values are not presented for interim data.
(3) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.

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Table 4.2.6(j)
C/Ep 194-PW
T-300/934
Compression, 2-axis
$[90_f]_{12}$
75/D, -65/D, 250/D
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

		75		-65		250	
Temperature(°F)							
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_2^{cu} (3) (ksi)	Mean	90.	93	103	106		
	Minimum	81	85	94	98		
	Maximum	100.	104	116	121		
	C.V. (%)	5.9	6.0	6.2	6.1		
	B-value	(2)		(2)			
	Distribution	ANOVA		Normal			
	C_1	5.6		103			
	C_2	3.2		6.18			
	No. Specimens	20		20			
	No. Batches	4		4			
e_2^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_2^c (4) (Msi)	Mean	8.3	8.6	8.4	8.8	8.8	9.0
	Minimum	7.4	7.7	7.5	7.7	7.9	8.1
	Maximum	9.3	9.5	9.0	9.4	10.2	10.6
	C.V. (%)	7.0	6.6	5.1	5.5	8.4	8.9
	No. Specimens	20		20		20	
	No. Batches	4		4		4	

(1) As fabricated.
(2) B-values are not presented for interim data.
(3) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.
(4) Specimen thickness of 0.09 - 0.10 inch is less than nominal 0.120 inch thickness per method.

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Table 4.2.6(k)
C/Ep 194-PW
T-300/934
Compression, 2-axis
$[90_t]_{12}$
250/D
Screening Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.56 - 1.57 g/cc
FIBER VOLUME: 60 % VOID CONTENT: <0.5 %
PLY THICKNESS: 0.0074 - 0.0076 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		250					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
F_2^{cu} (3) (ksi)	Mean	82	85				
	Minimum	77	81				
	Maximum	84	88				
	C.V. (%)	3.4	3.4				
	B-value	(2)					
	Distribution	Normal					
	C_1	81.7					
	C_2	2.74					
	No. Specimens	5					
	No. Batches	1					
e_2^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_2^c (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) B-values are not presented for screening data.
(3) Tab thickness of 0.112 - 0.120 inch is larger than 0.070 inch nominal thickness per method.

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Table 4.2.6(l)
C/Ep 194-PW
T-300/934
Compression, 2-axis
$[90_f]_{12}$
75/D, -65/D, 250/D
Interim Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		160		250			
Moisture Content(%)		(1)		(1)			
Equilibrium at T, RH							
Source Code		12		12			
		Normalized	Measured	Normalized	Measured		
F_2^{cu} (3) (ksi)	Mean	75	77	46	47		
	Minimum	63	66	38	39		
	Maximum	81	83	59	60		
	C.V. (%)	7.2	6.5	11	11		
	B-value	(2)		(2)			
	Distribution	ANOVA		ANOVA			
	C_1	6.0		5.9			
	C_2	5.0		5.1			
	No. Specimens	15		15			
	No. Batches	3		3			
ϵ_2^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_2^c (Msi)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	No. Specimens						
	No. Batches						

(1) Immersed in water at 160°F for 14 days.
(2) B-values are not presented for interim data.
(3) Tab thickness of 0.112 - 0.120 inch is larger then 0.070 nominal thickness per method.

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Table 4.2.6(m)
C/Ep 194-PW
T-300/934
Shear, 13-axis
$[0_f]_{12}$
75/D, -65/D, 250/D
Screening Data

MATERIAL: T-300 3k/934 plain weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.54 - 1.57 g/cc
FIBER VOLUME: 58 - 60 % VOID CONTENT: <0.5 - 1.2 %
PLY THICKNESS: 0.0074 - 0.0082 in.

TEST METHOD: ASTM D2344 (1)

MODULUS CALCULATION: Chord between 20 and 40% of typical ultimate load

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75	-65	250		
Moisture Content(%)		dry	dry	dry		
Equilibrium at T, RH		(2)	(2)	(2)		
Source Code		12	12	12		
	Mean	12.0	11.9	9.2		
	Minimum	10.5	10.0	9.1		
	Maximum	13.4	13.9	9.5		
	C.V.(%)	6.89	8.38	2.1		
F_{13}^{su}	B-value	(3)	(3)	(3)		
	Distribution	ANOVA	ANOVA	Normal		
(ksi)	C_1	1.07	0.901	9.2		
	C_2	3.41	3.71	0.20		
	No. Specimens	20	20	5		
	No. Batches	4	4	1		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) Length-to-thickness ratio is approximately 11.
(2) As fabricated.
(3) B-values are not presented for screening data.

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4.2.7 Celion 12k/938.*

				C/Ep Celion 12k/938 Summary
MATERIAL:	Celion 12k/938 unidirectional tape			
PREPREG:	Fiberite HY-E 1638			
FIBER:	Celanese Celion 12k, EP06, no twist	RESIN:	Fiberite 938	
T_g(dry):	T_g(wet):	T_g METHOD:		
PROCESSING:	Autoclave cure: 355 ± 10°F, 120 - 130 min., 85 - 100 psig			

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Date of fiber manufacture	5/85	Date of testing	7/85
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture		Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW	ETW	
F_1^{tu}	(ksi)	273	I	262	I		277	I
E_1^t	(Msi)	19.7	I	19.0	I		18.9	I
F_2^{tu}	(ksi)	9.6	I	9.5	I		5.8	I
E_2^t	(Msi)	1.35	I	1.35	I		1.19	I
F_1^{cu}	(ksi)	201	I	240.	I		185	I
E_1^c	(Msi)	17.2	I	18.8	I		18.2	I
F_2^{cu}	(ksi)							
E_2^c	(Msi)							
F_{12}^{su}	(ksi)	14	I	16	I		14	I
G_{12}^s	(Msi)							
F_{13}^{su}	(ksi)	18.3	S					
G_{13}^s	(Msi)							
F_{23}^{su}	(ksi)							
G_{23}^s	(Msi)							

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.78		
Resin Density	(g/cc)	1.30		
Composite Density	(g/cc)		1.54 - 1.61	
Fiber Areal Weight	(g/m^2)	145	144 - 147	
Ply Thickness	(in)		0.0040 - 0.0073	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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Table 4.2.7(a)
C/Ep 145-UT
Celion 12k/938
Tension, 1-axis
$[0]_7$
75/D, -65/D, 250/D
Interim Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 36 wt%
FIBER VOLUME: 56 - 65 %
PLY THICKNESS: 0.0040 - 0.0063 in.

COMP. DENSITY: 1.55 - 1.61 g/cc
VOID CONTENT: <1.1 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Secant at 25% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75		-67		250	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		12		12		12	
		Normalized	*Measured*	*Normalized*	*Measured*	*Normalized*	*Measured*
F_1^{tu} (ksi)	Mean	273	270	262	278	309	319
	Minimum	223	207	235	254	295	306
	Maximum	324	319	290	303	328	337
	C.V.(%)	7.56	9.81	7.67	6.25	3.00	2.90
	B-value	(2)		(2)		(2)	
	Distribution	ANOVA		ANOVA		Weibull	
	C_1	21.0		25.1		314	
	C_2	2.42		18.0		34.5	
	No. Specimens	102		10		15	
	No. Batches	3		2		3	
e_1^{tu} ($\mu\epsilon$)	Mean		13100		12800		14800
	Minimum		10600		11500		12900
	Maximum		14800		14000		16100
	C.V.(%)		6.95		6.72		5.81
	B-value		(2)		(2)		(2)
	Distribution		ANOVA		ANOVA		Weibull
	C_1		946		1060		15100
	C_2		3.14		17.2		21.4
	No. Specimens	102		10		15	
	No. Batches	3		2		3	
E_1^t (Msi)	Mean	19.7	19.5	19.0	20.2	20.1	20.7
	Minimum	16.9	16.5	17.3	18.1	16.9	17.9
	Maximum	23.1	21.8	20.3	22.0	23.4	23.4
	C.V.(%)	5.22	5.59	4.94	5.94	9.12	7.49
	No. Specimens	102		10		15	
	No. Batches	3		2		3	
ν_{12}^t	Mean		0.317		0.279		0.280
	No. Specimens	102		10		10	
	No. Batches	3		2		2	

(1) As fabricated.
(2) B-values are not presented for interim data.

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Table 4.2.7(b)
C/Ep 145-UT
Celion 12k/938
Tension, 1-axis
$[0]_7$
180/1.1%
Interim Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 36 wt%
FIBER VOLUME: 56 - 64 %
PLY THICKNESS: 0.0044 - 0.0063 in.
COMP. DENSITY: 1.55 - 1.59 g/cc
VOID CONTENT: <1.4 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Secant at 25% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180					
Moisture Content(%)		1.1					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
	Mean	277	282				
	Minimum	236	219				
	Maximum	307	328				
	C.V.(%)	8.89	14.3				
F_1^{tu}	B-value	(3)					
	Distribution	ANOVA					
(ksi)	C_1	27.7					
	C_2	5.36					
	No. Specimens	15					
	No. Batches	3					
	Mean		14000				
	Minimum		11800				
	Maximum		15700				
	C.V.(%)		8.13				
ϵ_1^{tu}	B-value		(3)				
	Distribution		ANOVA				
($\mu\epsilon$)	C_1		1180				
	C_2		3.36				
	No. Specimens	15					
	No. Batches	3					
	Mean	18.9	19.2				
	Minimum	17.7	16.4				
	Maximum	20.5	21.9				
E_1^t	C.V.(%)	4.81	9.74				
(Msi)	No. Specimens	15					
	No. Batches	3					
	Mean		0.345				
ν_{12}^t (2)	No. Specimens	14					
	No. Batches	3					

(1) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(2) Poisson's ratio measured at 25% of typical ultimate load.
(3) B-values are not presented for interim data.

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Table 4.2.7(c)
C/Ep 145-UT
Celion 12k/938
Tension, 2-axis
$[0]_{20}$
75/D, -65/D, 250/D, 180/1.1%
Interim Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 32 - 37 wt% COMP. DENSITY: 1.55 - 1.58 g/cc
FIBER VOLUME: 55 - 60 % *VOID CONTENT:* <1.3 %
PLY THICKNESS: 0.0053 - 0.0064 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Secant at 25% of typical ultimate load

NORMALIZED BY: Not normalized

Temperature(°F)		75	-67	250	180		
Moisture Content(%)		dry	dry	dry	1.1		
Equilibrium at T, RH		(1)	(1)	(1)	(2)		
Source Code		12	12	12	12		
	Mean	9.6	9.5	8.8	5.8		
	Minimum	7.5	8.5	7.1	5.0		
	Maximum	13.9	10.4	10.7	6.6		
	C.V.(%)	13	6.6	11	8.4		
F_2^{tu}	B-value	(3)	(3)	(3)	(3)		
	Distribution	ANOVA	Weibull	Weibull	ANOVA		
(ksi)	C_1	1.3	9.8	9.2	0.54		
	C_2	2.7	18	10.	5.1		
	No. Specimens	101	15	10	15		
	No. Batches	3	3	2	3		
	Mean	7200	6700	7620	4900		
	Minimum	1300	5500	6900	4200		
	Maximum	9500	7900	9300	5800		
	C.V.(%)	15	9.2	9.5	8.6		
e_1^{tu}	B-value	(3)	(3)	(3)	(3)		
	Distribution	Nonpara.	Weibull	Normal	Weibull		
($\mu\epsilon$)	C_1	5	7000	7600	5100		
	C_2		12	720	12		
	No. Specimens	97	15	10	15		
	No. Batches	3	3	2	3		
	Mean	1.35	1.35	1.22	1.19		
	Minimum	1.14	1.25	0.94	1.03		
	Maximum	1.82	1.51	1.52	1.36		
E_2^t	C.V.(%)	9.29	4.96	12.5	8.65		
(Msi)	No. Specimens	101	15	10	15		
	No. Batches	3	3	2	3		

(1) As fabricated.
(2) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(3) B-values are not presented for interim data.

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Table 4.2.7(d)
C/Ep 145-UT
Celion 12k/938
Compression, 1-axis
$[0]_7$
75/D, -65/D, 250/D
Interim Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 26 - 35 wt% COMP. DENSITY: 1.56 - 1.61 g/cc
FIBER VOLUME: 57 - 67 % VOID CONTENT: <1.5 %
PLY THICKNESS: 0.0046 - 0.0073 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord modulus between 20% and 40% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		75		-67		250	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		12		12		12	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	201	198	240	240	195	201
	Minimum	166	172	204	216	180	179
	Maximum	255	246	286	276	214	229
	C.V.(%)	9.88	8.99	11.3	8.25	5.48	7.26
F_1^{cu}	B-value	(2)		(2)		(2)	
	Distribution	ANOVA		ANOVA		ANOVA	
(ksi)	C_1	21.4		31.1		11.9	
	C_2	3.93		5.59		5.07	
	No. Specimens	102		15		15	
	No. Batches	3		3		3	
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
e_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	17.2	18.2	18.8	19.1	18.1	18.1
	Minimum	14.7	15.0	16.6	16.6	17.1	16.3
	Maximum	21.0	21.5	21.7	22.5	19.1	20.3
E_1^c	C.V.(%)	6.87	7.64	7.14	9.74	3.73	7.07
(Msi)	No. Specimens	97		15		15	
	No. Batches	3		3		3	

(1) As fabricated.
(2) B-values are not presented for interim data.

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Table 4.2.7(e)
C/Ep 145-UT
Celion 12k/938
Compression, 1-axis
$[0]_7$
180/1.1%
Interim Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 34 wt%
FIBER VOLUME: 58 - 65 %
PLY THICKNESS: 0.0044 - 0.0073 in.
COMP. DENSITY: 1.58 - 1.60 g/cc
VOID CONTENT: <1.0 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Chord modulus between 20% and 40% of typical ultimate load

NORMALIZED BY: Fiber volume fraction to 60%

Temperature(°F)		180					
Moisture Content(%)		1.1					
Equilibrium at T, RH		(1)					
Source Code		12					
		Normalized	Measured				
	Mean	185	188				
	Minimum	157	160				
	Maximum	206	217				
	C.V.(%)	7.40	7.55				
F_1^{cu}	B-value	(2)					
	Distribution	Weibull					
(ksi)	C_1	191					
	C_2	16.3					
	No. Specimens		15				
	No. Batches		3				
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
e_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	18.2	19.2				
	Minimum	15.7	15.8				
	Maximum	22.3	23.7				
E_1^c	C.V.(%)	8.88	10.5				
(Msi)	No. Specimens		15				
	No. Batches		3				

(1) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(2) B-values are not presented for interim data.

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Table 4.2.7(f)
C/Ep 145-UT
Celion 12k/938
Shear, 12-axis
$[\pm 45]_{2s}$
75/D, -65/D, 250/D, 180/1.1%
Interim Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 28 - 34 wt%
FIBER VOLUME: 58 - 65 %
PLY THICKNESS: 0.0044 - 0.0064 in.

COMP. DENSITY: 1.57 - 1.61 g/cc
VOID CONTENT: <1.4 %

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	-67	250	180	
Moisture Content(%)		dry	dry	dry	1.1	
Equilibrium at T, RH		(1)	(1)	(1)	(2)	
Source Code		12	12	12	12	
F_{12}^{su} (ksi)	Mean	14	16	14	14	
	Minimum	11	14	13	13	
	Maximum	16	18	15	14	
	C.V.(%)	7.3	10.	6.1	3.6	
	B-value	(3)	(3)	(3)	(3)	
	Distribution	ANOVA	ANOVA	Weibull	ANOVA	
	C_1	1.1	1.8	14	0.53	
	C_2	4.4	5.8	19	4.6	
	No. Specimens	102	14	14	15	
	No. Batches	3	3	3	3	
γ_{12}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
G_{12}^{s} (Msi)	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) Conditioned at 160°F, 88% R.H. until weight gain was between 1.0 and 1.2%.
(3) B-values are not presented for interim data.

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Table 4.2.7(g)
C/Ep 145-UT
Celion 12k/938
Shear, 13-axis
$[0]_{14}$
75/D
Screening Data

MATERIAL: Celion 12k/938 unidirectional tape

RESIN CONTENT: 31 - 40 wt%
FIBER VOLUME: 52 - 62 %
PLY THICKNESS: 0.0051 - 0.0064 in.
COMP. DENSITY: 1.54 - 1.59 g/cc
VOID CONTENT: <1.0 %

TEST METHOD: ASTM D2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		12				
	Mean	18.3				
	Minimum	16.6				
	Maximum	19.7				
	C.V.(%)	3.29				
F_{13}^{su}	B-value	(2)				
	Distribution	ANOVA				
(ksi)	C_1	0.619				
	C_2	2.76				
	No. Specimens	102				
	No. Batches	3				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated
(2) B-values are not presented for screening data.

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4.2.8 AS4 12k/3502.*

				C/Ep AS4/3502 Summary
MATERIAL:	AS4 12k/3502 unidirectional tape			
PREPREG:	Hercules AS4/3502 unidirectional tape			
FIBER:	Hercules AS4 12k, surface-treated	RESIN:	Hercules 3502	
T_g(dry):	460°F	T_g(wet):	T_g METHOD: TMA	
PROCESSING:	Autoclave cure: 275°F, 45 min., 350°F, 2 hours, 85 psig; Postcure: 400°F, 4 hours			

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	12/80 - 2/82	Date of testing	
Date of resin manufacture		Date of data submittal	6/90
Date of prepreg manufacture	12/80 - 2/82	Date of analysis	1/93
Data of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	253	I					251	I
E_1^t	(Msi)	18.9	I					19.0	I
F_2^{tu}	(ksi)	8.04	I			3.27	I	3.29	I
E_2^t	(Msi)	1.5	I			1.04	I	1.04	I
F_1^{cu}	(ksi)			226	I			176	I
E_1^c	(Msi)			19.3	I			19.6	I
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)								
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.79	1.78 - 1.81	
Resin Density	(g/cc)	1.26		
Composite Density	(g/cc)	1.58		
Fiber Areal Weight	(g/m^2)			
Ply Thickness	(in)		0.0047 - 0.0062	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.8(a)
C/Ep -UT
AS4/3502
Tension, 1-axis
$[0]_6$
75/D, 265/D, 265/W
Interim Data

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 25 - 29 wt% COMP. DENSITY: 1.59 - 1.62 g/cc
FIBER VOLUME: 63 - 68 % VOID CONTENT:
PLY THICKNESS: 0.0055 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75		265		265	
Moisture Content(%)		dry		dry		wet	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	253	275	269	292	251	273
	Minimum	212	226	148	165	183	196
	Maximum	294	323	314	358	287	315
	C.V.(%)	8.35	9.49	15.2	16.5	9.09	10.4
F_1^{tu}	B-value	(3)		(3)		(3)	
	Distribution	ANOVA		ANOVA		ANOVA	
(ksi)	C_1	21.5		42.0		24.0	
	C_2	2.20		2.62		2.83	
	No. Specimens	30		20		25	
	No. Batches	5		4		5	
	Mean		12400		13900		12400
	Minimum		10200		10400		9220
	Maximum		14400		15700		13900
	C.V.(%)		8.65		12.0		8.95
ϵ_1^{tu}	B-value		(3)		(3)		(3)
	Distribution		ANOVA		ANOVA		ANOVA
($\mu\epsilon$)	C_1		1120		1850		1170
	C_2		2.62		3.92		2.87
	No. Specimens	30		20		25	
	No. Batches	5		4		5	
	Mean	18.7	20.4	18.4	20.0	19.0	20.6
	Minimum	17.3	18.9	17.4	19.1	18.0	19.2
	Maximum	20.2	22.2	19.7	20.8	19.7	22.1
E_1^t	C.V.(%)	3.88	3.37	3.52	2.59	3.53	3.22
(Msi)	No. Specimens	29		20		25	
	No. Batches	5		4		5	
	Mean		0.340		0.356		0.280
ν_{12}^t	No. Specimens	30		20		25	
	No. Batches	5		4		5	

(1) Conditioned at 180°F, ambient relative humidity for 2 days
(2) Conditioned at 180°F, 75% relative humidity for 10 days
(3) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.8(b)
C/Ep -UT
AS4/3502
Tension, 2-axis
$[90]_{16}$
75/D, 75/W, 265/W
Interim Data

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 25 - 29 wt% COMP. DENSITY: 1.59 - 1.62 g/cc
FIBER VOLUME: 63 - 68 % VOID CONTENT:
PLY THICKNESS: 0.0055 - 0.0059 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Not normalized

Temperature(°F)		75	75	265		
Moisture Content(%)		dry	wet	wet		
Equilibrium at T, RH		(1)	(2)	(2)		
Source Code		26	26	26		
	Mean	8.04	3.27	3.29		
	Minimum	5.93	2.54	2.62		
	Maximum	10.6	4.15	4.15		
	C.V.(%)	13.5	16.3	13.0		
F_2^{tu}	B-value	(3)	(3)	(3)		
	Distribution	ANOVA	ANOVA	ANOVA		
(ksi)	C_1	1.11	0.560	0.452		
	C_2	2.36	3.79	3.16		
	No. Specimens	30	15	20		
	No. Batches	5	3	4		
	Mean	5500	3320	3440		
	Minimum	4000	2750	2840		
	Maximum	7390	4200	4200		
	C.V.(%)	13.7	13.3	12.1		
e_2^{tu}	B-value	(3)	(3)	(3)		
	Distribution	Weibull	ANOVA	ANOVA		
($\mu\epsilon$)	C_1	5820	506	456		
	C_2	7.67	5.66	3.79		
	No. Specimens	30	15	20		
	No. Batches	5	3	4		
	Mean	1.50	1.04	1.04		
	Minimum	1.43	0.95	0.95		
	Maximum	1.58	1.10	1.10		
E_2^t	C.V.(%)	2.76	5.1	4.3		
(Msi)	No. Specimens	30	15	20		
	No. Batches	5	3	4		

(1) Conditioned at 180°F, ambient relative humidity for 2 days
(2) Conditioned at 180°F, 75% relative humidity for 63 days
(3) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: AS4 12k/3502 unidirectional tape

RESIN CONTENT: 25 - 29 wt% COMP. DENSITY: 1.59 - 1.62 g/cc
FIBER VOLUME: 63 - 68 % VOID CONTENT:
PLY THICKNESS: 0.0047 - 0.0062 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3410C

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Table 4.2.8(c)
C/Ep -UT
AS4/3502
Compression, 1-axis
$[0]_8$
-65/D, 265/D, 265/W
Interim Data

Temperature(°F)		-65		265		265	
Moisture Content(%)		dry		dry		wet	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	226	225	228	249	176	192
	Minimum	173	180.	142	150.	139	146
	Maximum	307	283	275	292	208	228
	C.V.(%)	16.8	14.0	15.0	15.1	11.5	13.3
F_1^{cu}	B-value	(3)		(3)		(3)	
	Distribution	Weibull		Weibull		Weibull	
(ksi)	C_1	242		241		184	
	C_2	6.23		8.66		10.6	
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Mean		16200		13400		10500
	Minimum		11100		7370		7770
	Maximum		21200		16000		12800
	C.V.(%)		17.4		16.2		14.1
e_1^{cu}	B-value		(3)		(3)		(3)
	Distribution		Weibull		Weibull		Weibull
($\mu\epsilon$)	C_1		17400		14200		11100
	C_2		6.39		8.53		8.71
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Mean	19.3	21.1	21.2	23.2	19.6	21.4
	Minimum	17.1	19.3	17.1	19.3	18.5	20.5
	Maximum	21.8	23.7	23.1	26.3	20.6	22.5
E_1^c	C.V.(%)	6.63	7.30	9.53	9.70	3.85	3.70
(Msi)	No. Specimens	15		15		15	
	No. Batches	3		3		3	

(1) Conditioned at 180°F, ambient relative humidity for 2 days
(2) Conditioned at 150°F, 98% relative humidity for 14 days
(3) B-values data are not presented for interim data.

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4.2.9 Celion 3000/E7K8 plain weave.*

MATERIAL:	Celion 3000/E7K8 plain weave		C/Ep Celion 3000/E7K8 Summary
PREPREG:	U.S. Polymeric Celion 3000/E7K8 Plain Weave, Grade 195		
FIBER:	Celanese Celion 3000	RESIN: U.S. Polymeric E7K8	
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave: 310°F, 2 hours, 85 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88
Date of prepreg manufacture	2/86 - 3/86	Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	132	S	110	S	125	S	123	S
E_1^t	(Msi)	9.67	S	9.98	S	9.23	S	9.55	S
F_2^{tu}	(ksi)	128	S	113	S	119	S	130	S
E_2^t	(Msi)	9.50	S	9.51	S	9.08	S	9.35	S
F_1^{cu}	(ksi)	104	S	121	S	94.9	S	78.9	S
E_1^c	(Msi)	9.88	S	9.83	S	9.39	S	8.97	S
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)								
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)	10.3	S	11.6	S	9.81	S	6.92	S
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.8		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.54	1.37 - 1.54	
Fiber Areal Weight	(g/m^2)	195		
Ply Thickness	(in)	0.0075	0.0078 - 0.011	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(a)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{10}$
75/D, -65/D
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 37 - 38 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 55 - 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0085 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	132	128	110	106		
	Minimum	120	115	101	98.4		
	Maximum	143	140	118	113		
	C.V.(%)	4.7	5.8	6.2	5.4		
F_1^{tu}	B-value	(2)		(2)			
	Distribution	Weibull		Normal			
(ksi)	C_1	135		110			
	C_2	25.7		6.88			
	No. Specimens	20		5			
	No. Batches	1		1			
	Mean		13700		11000		
	Minimum		12300		10200		
	Maximum		14800		11600		
	C.V.(%)		4.5		5.4		
e_1^{tu}	B-value		(2)		(2)		
	Distribution		Weibull		Normal		
($\mu\epsilon$)	C_1		14000		11000		
	C_2		26.8		592		
	No. Specimens	20		5			
	No. Batches	1		1			
	Mean	9.67	9.38	9.98	9.66		
	Minimum	9.49	8.85	9.82	9.46		
	Maximum	9.98	9.74	10.0	9.90		
E_1^t	C.V.(%)	1.2	2.5	1.0	1.8		
(Msi)	No. Specimens	20		5			
	No. Batches	1		1			
	Mean		0.0580				
ν_{12}^t	No. Specimens	5					
	No. Batches	1					

(1) As fabricated
(2) *B-values are not presented for screening data*

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(b)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{10}$
75/W, 180/W
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 55 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0081 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180			
Moisture Content(%)		wet		wet			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	125	122	123	120		
	Minimum	111	105	114	112		
	Maximum	130	129	131	127		
	C.V.(%)	6.3	8.1	6.5	6.3		
F_1^{tu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
(ksi)	C_1	125		123			
	C_2	7.93		7.99			
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean		13700		12800		
	Minimum		12100		11200		
	Maximum		14300		14100		
	C.V.(%)		6.9		9.6		
ϵ_1^{tu}	B-value		(2)		(2)		
	Distribution		Normal		Normal		
($\mu\epsilon$)	C_1		13700		12800		
	C_2		939		1230		
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean	9.23	9.01	9.55	9.33		
	Minimum	8.93	8.81	9.37	9.15		
	Maximum	9.53	9.20	9.84	9.63		
E_1^t	C.V.(%)	2.5	1.7	1.9	2.0		
(Msi)	No. Specimens	5		5			
	No. Batches	1		1			
	Mean		0.0620		0.0560		
ν_{12}^t	No. Specimens	5		5			
	No. Batches	1		1			

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(c)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{12}$
75/D, -65/D
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 39 - 44 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 51 - 54 % VOID CONTENT: 0.04 - 0.5 %
PLY THICKNESS: 0.0079 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65		
Moisture Content(%)		dry		dry		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	132	122	122	115	
	Minimum	106	100	117	111	
	Maximum	147	136	126	123	
	C.V.(%)	7.5	7.5	2.8	4.3	
F_1^{tu}	B-value	(2)		(2)		
	Distribution	Weibull		Normal		
(ksi)	C_1	136		122		
	C_2	16.4		3.44		
	No. Specimens	20		5		
	No. Batches	1		1		
	Mean		14100			
	Minimum		13600			
	Maximum		14600			
	C.V.(%)		2.6			
e_1^{tu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		14100			
	C_2		371			
	No. Specimens	5				
	No. Batches	1				
	Mean	9.96	9.21	9.29	8.82	
	Minimum	9.30	8.74	8.95	8.51	
	Maximum	9.98	9.78	9.66	9.41	
E_1^t	C.V.(%)	1.2	2.5	2.8	4.0	
(Msi)	No. Specimens	20		5		
	No. Batches	1		1		

(1) As fabricated
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(d)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 1-axis
$[0_f]_{12}$
75/W, 180/W
Screening Data

MATERIAL: *Celion 3000/E7K8 Plain Weave*

RESIN CONTENT: 42 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 51 % VOID CONTENT: 0.48 %
PLY THICKNESS: 0.0081 - 0.0083 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180			
Moisture Content(%)		wet		wet			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	145	129	148	133		
	Minimum	143	125	139	124		
	Maximum	148	131	154	142		
	C.V.(%)	1.6	1.8	4.0	5.6		
F_1^{tu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
(ksi)	C_1	145		149			
	C_2	2.23		5.94			
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean		13400				
	Minimum		12300				
	Maximum		14300				
	C.V.(%)		5.30				
e_1^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		13400				
	C_2		713				
	No. Specimens	5					
	No. Batches	1					
	Mean	10.6	9.42	10.3	9.21		
	Minimum	10.1	8.79	10.1	8.91		
	Maximum	11.4	10.0	10.5	9.53		
E_1^t	C.V.(%)	4.9	5.0	1.3	2.7		
(Msi)	No. Specimens	5		5			
	No. Batches	1		1			
	Mean		0.0560		0.0560		
ν_{12}^t	No. Specimens	5		5			
	No. Batches	1		1			

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(e)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 2-axis
$[90_t]_{10}$
75/D, -65/D
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 56 % *VOID CONTENT:* 0.0 %
PLY THICKNESS: 0.0078 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	128	127	113	111		
	Minimum	120	115	101	100		
	Maximum	137	134	125	122		
	C.V.(%)	3.6	3.7	9.1	8.9		
F_2^{tu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
(ksi)	C_1	128		113			
	C_2	4.64		10.3			
	No. Specimens	20		5			
	No. Batches	1		1			
	Mean		13400		11700		
	Minimum		12600		10700		
	Maximum		14200		12700		
	C.V.(%)		3.5		7.7		
ϵ_2^{tu}	B-value		(2)		(2)		
	Distribution		Weibull		Normal		
($\mu\epsilon$)	C_1		13600		11700		
	C_2		32.5		902		
	No. Specimens	20		5			
	No. Batches	1		1			
	Mean	9.50	9.37	9.51	9.34		
	Minimum	9.36	9.04	9.29	9.20		
	Maximum	9.69	9.71	9.65	9.68		
E_2^t	C.V.(%)	0.98	1.8	1.6	2.1		
(Msi)	No. Specimens	20		5			
	No. Batches	1		1			

(1) As fabricated
(2) B-values are not presented for screening data

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Table 4.2.9(f)
C/Ep 195-PW
Celion 3000/E7K8
Tension, 2-axis
$[90_f]_{10}$
75/W, 180/W
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0078 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

			75 wet (1) 20		180 wet (1) 20			
			Normalized	Measured	Normalized	Measured		
		Mean	119	117	130	128		
		Minimum	105	104	129	125		
		Maximum	130	126	132	131		
		C.V.(%)	7.8	7.3	0.89	1.8		
F_2^{tu}		B-value	(2)		(2)			
		Distribution	Normal		Normal			
(ksi)		C_1	119		130			
		C_2	9.35		1.16			
		No. Specimens	5		5			
		No. Batches	1		1			
		Mean		13100		14200		
		Minimum		11400		13700		
		Maximum		14400		14800		
		C.V.(%)		8.7		3.5		
ϵ_2^{tu}		B-value		(2)		(2)		
		Distribution		Normal		Normal		
($\mu\epsilon$)		C_1		13100		14200		
		C_2		1135		490		
		No. Specimens	5		5			
		No. Batches	1		1			
		Mean	9.08	8.92	9.35	9.18		
		Minimum	8.98	8.73	9.26	8.96		
		Maximum	9.21	9.14	9.48	9.38		
E_2^t		C.V.(%)	1.2	1.6	1.2	1.8		
(Msi)		No. Specimens	5		5			
		No. Batches	1		1			

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(g)
***C/Ep* 195-PW**
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{10}$
75/D, -65/D, 180/D
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 - 40 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 53 - 55 % VOID CONTENT: 0.0 - 0.75 %
PLY THICKNESS: 0.0079 - 0.0084 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: *Specimen thickness and batch fiber volume to 57%*

Temperature(°F)		75		-65		180	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	104	101	121	118	97.4	94.5
	Minimum	90.5	87.7	113	111	87.5	85.1
	Maximum	122	120	132	126	105	100
	C.V.(%)	8.3	8.7	5.9	4.7	7.2	7.1
	B-value	(2)		(2)		(2)	
	Distribution	Weibull		Normal		Normal	
	C_1	108		121		97.4	
	C_2	13.0		7.19		7.00	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
e_1^{tu} ($\mu\epsilon$)	Mean	10900		12200		10400	
	Minimum	10500		12000		10200	
	Maximum	11200		12300		10800	
	C.V.(%)	2.2		1.0		2.3	
	B-value	(2)		(2)		(2)	
	Distribution	Weibull		Normal		Normal	
	C_1	11000		12000		10400	
	C_2	54.2		122		239	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
E_1^c (Msi)	Mean	9.88	9.02	9.83	9.33	9.45	9.16
	Minimum	9.56	8.65	9.75	9.20	9.14	8.89
	Maximum	10.3	9.29	9.95	9.48	9.66	9.37
	C.V.(%)	2.3	2.0	1.0	1.1	2.3	2.0
	No. Specimens	20		5		5	
	No. Batches	1		1		1	

(1) As fabricated
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.9(h)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{10}$
75/W, 180/W
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 - 37 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 54 - 56 % VOID CONTENT: 0.0 - 0.70 %
PLY THICKNESS: 0.0073 - 0.0086 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180			
Moisture Content(%)		wet		wet			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	94.9	92.6	78.9	77.6		
	Minimum	89.7	88.2	72.7	70.5		
	Maximum	102	98.8	83.2	82.3		
	C.V.(%)	5.5	4.9	5.7	6.0		
F_1^{cu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
(ksi)	C_1	94.9		78.9			
	C_2	5.47		4.53			
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean		9800		8130		
	Minimum		8970		7620		
	Maximum		10400		8600		
	C.V.(%)		6.0		4.4		
e_1^{cu}	B-value		(2)		(2)		
	Distribution		Normal		Normal		
($\mu\epsilon$)	C_1		9799		8128		
	C_2		590		356		
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean	9.39	8.92	8.97	8.52		
	Minimum	8.80	8.12	8.45	8.18		
	Maximum	10.2	9.79	9.54	8.80		
E_1^c	C.V.(%)	6.3	6.8	4.4	3.5		
(Msi)	No. Specimens	5		5			
	No. Batches	1		1			

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are not presented for screening data

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Table 4.2.9(i)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/D, -65/D, 180/D
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 38 - 40 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 52 - 54 % *VOID CONTENT:* 0.0 %
PLY THICKNESS: 0.0078 - 0.0084 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65		180	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	114	107	133	122	103	97.6
	Minimum	86.4	84.4	127	116	96.0	89.2
	Maximum	128	121	139	129	114	107
	C.V.(%)	9.5	9.1	3.9	4.6	6.8	7.2
F_1^{cu}	B-value	(2)		(2)		(2)	
	Distribution	Weibull		Normal		Normal	
(ksi)	C_1	118		133		103	
	C_2	13.8		5.22		6.99	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean	13500					
	Minimum	13000					
	Maximum	13700					
	C.V.(%)	1.6					
e_1^{cu}	B-value	(2)					
	Distribution	Nonpara.					
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens	20					
	No. Batches	1					
	Mean	8.22	7.80	8.45	7.71	8.40	7.67
	Minimum	8.07	7.51	8.27	7.43	8.20	7.58
	Maximum	8.50	8.05	8.73	8.09	8.54	7.84
E_1^c	C.V.(%)	1.6	2.2	2.3	3.4	1.5	1.4
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	

(1) As fabricated
(2) B-values are not presented for screening data

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Table 4.2.9(j)
C/Ep 195-PW
Celion 3000/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/W, 180/W
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 38 - 40 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 52 - 54 % VOID CONTENT: 0.0 - 0.04 %
PLY THICKNESS: 0.0080 - 0.0084 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	96.1	90.7	80.2	75.7	
	Minimum	83.9	78.4	74.4	72.2	
	Maximum	107	101	83.3	79.9	
	C.V.(%)	9.3	9.4	4.7	4.4	
F_1^{cu}	B-value	(2)		(2)		
	Distribution	Normal		Normal		
(ksi)	C_1	96.1		80.2		
	C_2	8.91		3.73		
	No. Specimens	5		5		
	No. Batches	1		1		
	Mean		10700			
	Minimum		10600			
	Maximum		11000			
	C.V.(%)		1.5			
ϵ_1^{cu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		10700			
	C_2		164			
	No. Specimens	5				
	No. Batches	1				
	Mean	9.08	8.30	9.36	8.54	
	Minimum	8.84	7.91	9.14	8.20	
	Maximum	9.17	8.62	9.57	8.84	
E_1^c	C.V.(%)	1.5	3.2	2.0	2.9	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		

(1) Conditioned at 160°F, 85% relative humidity for 7 days
(2) B-values are not presented for screening data

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Table 4.2.9(k)
C/Ep 195-PW
Celion 3000/E7K8
Shear, 13-axis
$[0_f]_{14}$
75/D, -65/D, 180/D, 75/W, 180/W
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 36 - 39 wt%
FIBER VOLUME: 54 - 56 %
PLY THICKNESS: 0.0079 - 0.0081 in.

COMP. DENSITY: 1.55 g/cc
VOID CONTENT: 0.0 - 0.75 %

TEST METHOD: ASTM D2344-68

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	75	180
Moisture Content(%)		dry	dry	dry	wet	wet
Equilibrium at T, RH		(1)	(1)	(1)	(2)	(2)
Source Code		20	20	20	20	20
	Mean	10.3	11.6	9.70	9.81	6.92
	Minimum	9.43	10.7	9.34	9.24	6.60
	Maximum	11.4	13.6	9.94	10.4	7.22
	C.V.(%)	4.3	10.8	3.0	5.8	3.4
F^{su}_{13} (ksi)	B-value	(3)	(3)	(3)	(3)	(3)
	Distribution	Normal	Normal	Normal	Normal	Normal
	C_1	10.5	10.2	9.70	11.6	8.72
	C_2	22.7	0.395	0.293	1.25	0.247
	No. Specimens	20	5	5	5	5
	No. Batches	1	1	1	1	1
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ^{su}_{13} ($\mu\epsilon$)	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G^{s}_{13}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated
(2) Conditioned at 160°F, 85% relative humidity for 7 days
(3) B-values are not presented for screening data

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Table 4.2.9(l)
C/Ep 195-PW
Celion 3000/E7K8
Shear, 13-axis
$[0_f]_{12}$
75/D, -65/D, 180/D, 75/W, 180/W
Screening Data

MATERIAL: Celion 3000/E7K8 Plain Weave

RESIN CONTENT: 39 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 54 % VOID CONTENT: 0.29 %
PLY THICKNESS: 0.0080 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344-68

NORMALIZED BY: Not normalized

Temperature(°F)		75	-65	180	75	180
Moisture Content(%)		dry	dry	dry	wet	wet
Equilibrium at T, RH		(1)	(1)	(1)	(2)	(2)
Source Code		20	20	20	20	20
	Mean	9.76	10.2	9.72	9.72	8.72
	Minimum	9.00	9.54	8.76	8.76	8.35
	Maximum	10.7	10.5	10.3	10.3	9.00
	C.V.(%)	4.8	3.9	6.1	6.1	2.8
F^{su}_{13}	B-value	(3)	(3)	(3)	(3)	(3)
	Distribution	Normal	Normal	Normal	Normal	Normal
(ksi)	C_1	9.76	10.17	9.72	9.72	8.72
	C_2	0.470	0.395	0.591	0.591	0.247
	No. Specimens	20	5	5	5	5
	No. Batches	1	1	1	1	1
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ^{su}_{13}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G^{s}_{13}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated
(2) Conditioned at 160°F, 85% relative humidity for 7 days
(3) B-values are not presented for screening data

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4.2.10 HITEX 33 6k/E7K8 plain weave. *

MATERIAL:	HITEX 33 6k/E7K8 plain weave		C/Ep HITEX 33/E7K8 Summary
PREPREG:	U.S. Polymeric Hitex 33 6k/E7K8 Plain weave		
FIBER:	Hitco HITEX 33 6k G'	RESIN:	U.S. Polymeric E7K8
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave: 310°F, 2 hours, 85 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)								
E_1^t	(Msi)								
F_2^{tu}	(ksi)	131	S	126	S	134	S	138	S
E_2^t	(Msi)	8.65	S	8.10	S	9.61	S	9.91	S
F_1^{cu}	(ksi)	136	S	155	S				
E_1^c	(Msi)	9.11	S	10.1	S				
F_2^{cu}	(ksi)	104	S	128	S	99.2	S	84.0	S
E_2^c	(Msi)	8.92	S	9.49	S	9.30	S	8.96	S
F_{12}^{su}	(ksi)								
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)	8.67	S	8.83	S	9.40	S	8.35	S
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.77		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.56		
Fiber Areal Weight	(g/m^2)	195		
Ply Thickness	(in)	0.0085	0.0077 - 0.0099	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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Table 4.2.10(a)
C/Ep 195-PW
HITEX 33/E7K8
Tension, 2-axis
$[0_f]_{12}$
75/D, -65/D, 75/W
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 37 - 41 wt% COMP. DENSITY: 1.53 - 1.55 g/cc
FIBER VOLUME: 51 - 55 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0087 - 0.0098 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65		75	
Moisture Content(%)		dry		dry		wet	
Equilibrium at T, RH		(1)		(1)		(2)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	131	124	126	122	134	119
	Minimum	120	103	122	111	130	114
	Maximum	139	136	131	131	137	125
	C.V. (%)	4.3	6.8	3.1	6.7	2.8	3.8
F_2^{tu} (ksi)	B-value	(3)		(3)		(3)	
	Distribution	Weibull		Normal		Normal	
	C_1	134		126		134	
	C_2	28.2		3.88		3.69	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean		14300		15600		10500
	Minimum		13700		14600		9930
	Maximum		14900		16500		10800
	C.V. (%)		3.8		4.4		3.2
ϵ_2^{tu} ($\mu\epsilon$)	B-value		(3)		(3)		(3)
	Distribution		Normal		Normal		Normal
	C_1		14400		15600		10500
	C_2		541		687		335
	No. Specimens	5		5		5	
	No. Batches	1		1		1	
	Mean	8.65	8.14	8.10	7.82	9.61	8.55
	Minimum	8.01	7.52	7.73	7.54	9.26	8.20
	Maximum	9.65	8.62	8.29	8.26	9.94	9.13
E_2^t	C.V. (%)	6.2	3.1	2.7	3.4	2.8	4.1
(Msi)	No. Specimens	20		5		5	
	No. Batches	1		1		1	
	Mean		0.0460				0.0540
ν_{21}^t	No. Specimens	5				5	
	No. Batches	1				1	

(1) As fabricated
(2) Conditioned at 160°F, 85% relative humidity for 14 days
(3) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 41 wt% COMP. DENSITY: 1.53 g/cc
FIBER VOLUME: 51 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0089 - 0.0094 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: *Specimen thickness and batch fiber volume to 57%*

Table 4.2.10(b)
C/Ep 195-PW
HITEX 33/E7K8
Tension, 2-axis
$[0_f]_{12}$
180/W
Screening Data

Temperature(°F)		180					
Moisture Content(%)		wet					
Equilibrium at T, RH		(1)					
Source Code		20					
		Normalized	Measured	Normalized	Measured		
	Mean	138	122				
	Minimum	120	107				
	Maximum	155	135				
	C.V. (%)	10.2	9.1				
F_2^{tu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	138					
	C_2	14.1					
	No. Specimens	5					
	No. Batches	1					
	Mean		10400				
	Minimum		9840				
	Maximum		10800				
	C.V. (%)		3.6				
e_2^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		10400				
	C_2		372				
	No. Specimens	5					
	No. Batches	1					
	Mean	9.91	8.80				
	Minimum	9.11	8.23				
	Maximum	10.7	9.23				
E_2^t	C.V. (%)	7.2	5.3				
(Msi)	No. Specimens	5					
	No. Batches	1					
	Mean		0.0700				
ν_{21}^t	No. Specimens	5					
	No. Batches	1					

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are not presented for screening data

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Table 4.2.10(c)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/D, -65/D, 180/D
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cc
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0079 - 0.0099 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

			75.0		-65.0		180.0	
Temperature(°F)			75.0		-65.0		180.0	
Moisture Content(%)			dry		dry		dry	
Equilibrium at T, RH			(1)		(1)		(1)	
Source Code			20		20		20	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	136	112	155	128	130	107
		Minimum	111	98.4	147	118	118	94.9
		Maximum	158	128	164	139	139	117
		C.V.(%)	8.4	7.5	5.5	7.5	6.3	7.8
F_1^{cu}		B-value	(2)		(2)		(2)	
		Distribution	Weibull		Normal		Normal	
(ksi)		C_1	141		155		130	
		C_2	13.3		8.51		8.21	
		No. Specimens	20		5		5	
		No. Batches	1		1		1	
		Mean	14400					
		Minimum	13700					
		Maximum	15200					
		C.V.(%)	3.1					
ϵ_1^{cu}		B-value	(2)					
		Distribution	Weibull					
($\mu\epsilon$)		C_1	14600					
		C_2	34.7					
		No. Specimens	20					
		No. Batches	1					
		Mean	9.11	7.53	10.1	8.30	9.37	7.75
		Minimum	8.64	6.83	9.72	7.74	9.15	7.38
		Maximum	9.63	8.17	10.8	8.76	9.66	8.66
E_1^c		C.V.(%)	3.0	5.2	4.0	5.1	2.4	7.1
(Msi)		No. Specimens	20		5		5	
		No. Batches	1		1		1	

(1) As fabricated
(2) B-values are not presented for screening data

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Table 4.2.10(d)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/W, 180/W
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cc
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0081 - 0.0098 in.

TEST METHOD: *MODULUS CALCULATION:*
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75.0		180.0		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	133	110	68.5	56.4	
	Minimum	130	100	54.2	46.7	
	Maximum	139	116	75.8	62.2	
	C.V.(%)	2.8	5.8	13.6	12.0	
F_1^{cu}	B-value	(2)		(2)		
	Distribution	Normal		Normal		
(ksi)	C_1	133		68.5		
	C_2	3.71		9.31		
	No. Specimens	5		5		
	No. Batches	1		1		
	Mean		14600			
	Minimum		14000			
	Maximum		15400			
	C.V.(%)		3.6			
e_1^{cu}	B-value		(2)			
	Distribution		Normal			
($\mu\epsilon$)	C_1		14600			
	C_2		525			
	No. Specimens	5				
	No. Batches	1				
	Mean	8.78	7.24	9.43	7.78	
	Minimum	8.41	7.04	9.32	7.69	
	Maximum	9.07	7.51	9.64	7.89	
E_1^c	C.V.(%)	3.2	2.5	1.4	9.5	
(Msi)	No. Specimens	5		5		
	No. Batches	1		1		

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are not presented for screening data

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Table 4.2.10(e)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_f]_6$
75/D, -65/D, 180/D
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.53 g/cc
FIBER VOLUME: 51 - 52 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0083 - 0.0087 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		-65		180	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		20		20		20	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_2^{cu} (ksi)	Mean	104	92.4	128	114	99.4	88.6
	Minimum	77.9	70.4	111	98.8	86.4	77.0
	Maximum	125	109	138	123	113	101
	C.V. (%)	13.1	12.6	8.0	8.1	12.0	12.0
	B-value	(2)		(2)		(2)	
	Distribution	Weibull		Normal		Normal	
	C_1	110		128		99.4	
	C_2	9.70		10.3		11.9	
	No. Specimens	20		5		5	
	No. Batches	1		1		1	
ϵ_2^{cu} ($\mu\epsilon$)	Mean		10900				
	Minimum		10400				
	Maximum		11400				
	C.V. (%)		2.4				
	B-value		(2)				
	Distribution		Weibull				
	C_1		11050				
	C_2		46.54				
	No. Specimens	20					
	No. Batches	1					
E_2^c (Msi)	Mean	8.92	8.21	9.49	8.74	9.07	8.35
	Minimum	8.50	7.78	9.36	8.65	8.95	8.20
	Maximum	9.40	8.77	9.58	8.93	9.18	8.52
	C.V. (%)	2.5	3.4	0.9	1.3	1.3	1.7
	No. Specimens	20		5		5	
	No. Batches	1		1		1	

(1) As fabricated
(2) B-values are not presented for screening data

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Table 4.2.10(f)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_f]_6$
75/W, 180/W
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.53 g/cc
FIBER VOLUME: 51 - 52 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0080 - 0.0083 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: *Specimen thickness and batch fiber volume to 57%*

Temperature(°F)		75		180			
Moisture Content(%)		wet		wet			
Equilibrium at T, RH		(1)		(1)			
Source Code		20		20			
		Normalized	Measured	Normalized	Measured		
	Mean	99.2	88.5	84.0	74.9		
	Minimum	80.9	72.2	74.2	66.1		
	Maximum	112	100	88.8	79.2		
	C.V. (%)	12.1	12.1	7.0	6.9		
F_2^{cu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
(ksi)	C_1	99.2		84.0			
	C_2	12.0		5.8			
	No. Specimens	5		5			
	No. Batches	1		1			
	Mean		10200				
	Minimum		9910				
	Maximum		10900				
	C.V. (%)		3.7				
e_2^{cu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		10200				
	C_2		381				
	No. Specimens	5					
	No. Batches	1					
	Mean	9.30	8.56	8.96	8.25		
	Minimum	8.74	7.98	8.69	8.03		
	Maximum	9.56	8.78	9.31	8.43		
E_2^c	C.V. (%)	3.5	3.9	2.9	2.0		
(Msi)	No. Specimens	5		5			
	No. Batches	1		1			

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are not presented for screening data

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Table 4.2.10(g)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_t]_{12}$
75/D, -65/D, 180/D
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cc
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0080 - 0.0097 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

			75		-65		180	
Temperature(°F)			75		-65		180	
Moisture Content(%)			dry		dry		dry	
Equilibrium at T, RH			(1)		(1)		(1)	
Source Code			20		20		20	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	132	110	147	122	132	110
		Minimum	114	97.9	138	115	128	106
		Maximum	145	118	161	127	146	117
		C.V. (%)	5.7	5.3	6.0	4.1	5.9	4.7
F_2^{cu}		B-value	(2)		(2)		(2)	
		Distribution	Weibull		Normal		Normal	
(ksi)		C_1	136		147		132	
		C_2	21.6		8.78		7.73	
		No. Specimens	20		5		5	
		No. Batches	1		1		1	
		Mean		14100				
		Minimum		13400				
		Maximum		14700				
		C.V. (%)		2.6				
ϵ_2^{cu}		B-value		(2)				
		Distribution		Weibull				
($\mu\epsilon$)		C_1		14300				
		C_2		46.4				
		No. Specimens	20					
		No. Batches	1					
		Mean	8.74	7.27	9.09	7.54	9.11	7.57
		Minimum	8.41	6.70	8.12	7.07	8.61	7.41
		Maximum	9.20	8.06	10.1	7.90	9.49	7.71
E_2^c		C.V. (%)	2.6	4.7	9.1	5.6	3.8	1.5
(Msi)		No. Specimens	20		5		5	
		No. Batches	1		1		1	

(1) As fabricated
(2) B-values are not presented for screening data

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Table 4.2.10(h)
C/Ep 195-PW
HITEX 33/E7K8
Compression, 2-axis
$[90_f]_{12}$
75/W, 180/W
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 45 wt% COMP. DENSITY: 1.51 g/cc
FIBER VOLUME: 47 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0080 - 0.0097 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		180		
Moisture Content(%)		wet		wet		
Equilibrium at T, RH		(1)		(1)		
Source Code		20		20		
		Normalized	Measured	Normalized	Measured	
	Mean	117	97.4	61.1	50.8	
	Minimum	107	88.4	52.2	44.1	
	Maximum	132	105	66.4	57.2	
	C.V. (%)	9.1	6.9	9.9	9.9	
F_2^{cu} (ksi)	B-value	(2)		(2)		
	Distribution	Normal		Normal		
	C_1	117		61.1		
	C_2	10.6		6.04		
	No. Specimens	5		5		
	No. Batches	1		1		
	Mean		13500			
	Minimum		12700			
	Maximum		14200			
	C.V. (%)		4.2			
ϵ_2^{cu} ($\mu\epsilon$)	B-value		(2)			
	Distribution		Normal			
	C_1		13500			
	C_2		564			
	No. Specimens	5				
	No. Batches	1				
	Mean	8.99	7.48	9.26	7.71	
	Minimum	8.48	7.08	8.76	7.32	
	Maximum	9.54	7.8	9.69	8.39	
E_2^c (Msi)	C.V. (%)	4.5	4.0	4.0	6.2	
	No. Specimens	5		5		
	No. Batches	1		1		

(1) Conditioned at 160°F, 85% relative humidity for 14 days
(2) B-values are not presented for screening data

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Table 4.2.10(i)
C/Ep 195-PW
HITEX 33/E7K8
Shear, 13-axis
$[90_t]_6$
75/D, -65/D, 180/D
Screening Data

MATERIAL: HITEX 33/E7K8 Plain weave

RESIN CONTENT: 44 wt% COMP. DENSITY: 1.51 g/cc
FIBER VOLUME: 48 % VOID CONTENT: 0.18 %
PLY THICKNESS: 0.0077 - 0.0093 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344-76

NORMALIZED BY: Not normalized

Temperature(°F)		75.0	-65.0	75.0	180.0	
Moisture Content(%)		dry	dry	wet	wet	
Equilibrium at T, RH		(1)	(1)	(2)	(2)	
Source Code		20	20	20	20	
	Mean	8.67	8.83	9.40	8.35	
	Minimum	7.77	8.14	9.20	7.83	
	Maximum	9.40	9.37	9.73	8.80	
	C.V.(%)	5.0	6.3	2.1	4.5	
F_{13}^{su}	B-value	(3)	(3)	(3)	(3)	
	Distribution	Weibull	Normal	Normal	Normal	
(ksi)	C_1	8.86	8.83	9.40	8.35	
	C_2	23.6	0.554	0.202	0.379	
	No. Specimens	20	5	5	5	
	No. Batches	1	1	1	1	
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{su}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated
(2) Conditioned at 160°F, 85% relative humidity for 14 days
(3) B-values are not presented for screening data

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4.2.11 AS4/E7K8 plain weave. *

MATERIAL:	AS4 3k/E7K8 plain weave		C/Ep AS4/E7K8 Summary
PREPREG:	U.S. Polymeric AS4/E7K8 Plain weave		
FIBER:	Hercules AS4 3k	RESIN:	U.S. Polymeric E7K8
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave: 290°F, 2 hours, 85 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	1/88, 6/90
Date of prepreg manufacture	2/86 - 7/89	Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD	RTW	ETW
F_1^{tu}	(ksi)					
E_1^t	(Msi)					
F_2^{tu}	(ksi)					
E_2^t	(Msi)					
F_1^{cu}	(ksi)	111	I			
E_1^c	(Msi)	9.02	I			
F_2^{cu}	(ksi)					
E_2^c	(Msi)					
F_{12}^{su}	(ksi)					
G_{12}^s	(Msi)					
F_{13}^{su}	(ksi)	9.68	S			
G_{13}^s	(Msi)					
F_{23}^{su}	(ksi)					
G_{23}^s	(Msi)					

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.77		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.56		
Fiber Areal Weight	(g/m^2)	195		
Ply Thickness	(in)	0.0087	0.0074 - 0.0088	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.11(a)
C/Ep 195-PW
AS4/E7K8
Compression, 1-axis
$[0_f]_{12}$
75/D
Interim Data

MATERIAL: *AS4 3k/E7K8 Plain weave*

RESIN CONTENT: 37 - 48 wt%
FIBER VOLUME: 48 - 55 %
PLY THICKNESS: *0.0074 - 0.0085 in.*
COMP. DENSITY: 1.52 - 1.54 g/cc
VOID CONTENT: 0.0 - 1.9 %

TEST METHOD: SACMA 1-88

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)	75		
Moisture Content(%)	dry		
Equilibrium at T, RH	(1)		
Source Code	20,27		

		Normalized	Measured				
	Mean	111	98.7				
	Minimum	64.4	58.0				
	Maximum	138	122				
	C.V. (%)	11.7	11.4				
F_1^{cu}	B-value	(2)					
	Distribution	ANOVA					
(ksi)	C_1	13.3					
	C_2	1.81					
	No. Specimens	206					
	No. Batches	18					
	Mean		11600				
	Minimum		8820				
	Maximum		15000				
	C.V. (%)		14.5				
ϵ_1^{cu}	B-value		(2)				
	Distribution		ANOVA				
($\mu\epsilon$)	C_1		1730				
	C_2		1.97				
	No. Specimens	190					
	No. Batches	17					
	Mean	9.02					
	Minimum	7.87					
	Maximum	10.5					
E_1^c	C.V. (%)	5.24					
(Msi)	No. Specimens	210					
	No. Batches	18					

(1) As fabricated.
(2) B-values are not presented for interim data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.11(b)
C/EP 195-PW
AS4/E7K8
Shear, 13-axis
$[0_f]_{12}$
75/D
Screening Data

MATERIAL: AS4/E7K8 Plain weave

RESIN CONTENT: 38 - 48 wt% COMP. DENSITY: 1.52 - 1.54 g/cc
FIBER VOLUME: 48 - 55 % VOID CONTENT: 0.0 - 1.9 %
PLY THICKNESS: 0.0074 - 0.0085 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344 - 84

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		20,27				
F^{su}_{13} (ksi)	Mean	9.68				
	Minimum	7.53				
	Maximum	14.2				
	C.V. (%)	12.0				
	B-value	(2)				
	Distribution	ANOVA				
	C_1	1.20				
	C_2	1.95				
	No. Specimens	170				
	No. Batches	16				
γ^{su}_{13} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
G^{s}_{13} (Msi)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for screening data.

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4.2.12 AS4/3501-6 (bleed).*

MATERIAL:	AS4/3501-6 unidirectional tape		C/Ep AS4/3501-6 Summary
PREPREG:	Hercules AS4/3501-6 unidirectional tape		
FIBER:	Hercules AS4	RESIN:	Hercules 3501-6
T_g(dry):	390°F T_g(wet):	T_g METHOD:	TMA
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10 min., 100 ± 10 psig, bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/90
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	291	I						
E_1^t	(Msi)	19.6	I						
F_2^{tu}	(ksi)	7.78	S						
E_2^t	(Msi)	1.48	S						
F_1^{cu}	(ksi)	210.	I			202	S	169	S
E_1^c	(Msi)	17.8	S			17.4	S	17.7	S
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)								
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)	17.3	S			17.3	S	8.98	S
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.8		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.59		
Fiber Areal Weight	(g/m^2)	145		
Ply Thickness	(in)		0.0041 - 0.0059	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(a)
C/Ep 145-UT
AS4/3501-6
Tension, 1-axis
$[0]_8$
75/D
Interim Data

MATERIAL: AS4/3501-6 (bleed)

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.56 g/cc
FIBER VOLUME: 54 - 55 % VOID CONTENT:
PLY THICKNESS: 0.0048 - 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		26					
		Normalized	Measured				
	Mean	291	295				
	Minimum	263	271				
	Maximum	326	326				
	C.V. (%)	6.09	5.05				
F_1^{tu}	B-value	(2)					
(ksi)	Distribution	Weibull					
	C_1	300					
	C_2	18.4					
	No. Specimens	21					
	No. Batches	7					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{tu}	B-value						
($\mu\epsilon$)	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	19.6	19.9				
	Minimum	18.0	18.3				
	Maximum	21.1	22.6				
E_1^t	C.V. (%)	3.73	6.48				
(Msi)	No. Specimens	21					
	No. Batches	7					

(1) As fabricated.
(2) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(b)
C/Ep 145-UT
AS4/3501-6
Tension, 2-axis
$[90]_8$
75/D
Screening Data

MATERIAL: AS4/3501-6 (bleed)

RESIN CONTENT: 28 - 29 wt% COMP. DENSITY: 1.60 - 1.61 g/cc
FIBER VOLUME: 63 - 64 % VOID CONTENT:
PLY THICKNESS: 0.0048 - 0.0057 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		26				
F_2^{tu} (ksi)	Mean	7.78				
	Minimum	7.00				
	Maximum	9.50				
	C.V. (%)	12.1				
	B-value	(2)				
	Distribution	Normal				
	C_1	7.78				
	C_2	0.941				
	No. Specimens	6				
	No. Batches	2				
ϵ_2^{tu} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
E_2^t (Msi)	Mean	1.48				
	Minimum	1.40				
	Maximum	1.50				
	C.V. (%)	2.75				
	No. Specimens	6				
	No. Batches	2				

(1) As fabricated.
(2) B-values are not presented for screening data.

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• ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(c)
C/Ep 145-UT
AS4/3501-6
Compression, 1-axis
$[0]_8$
75/D, 200/D
Interim Data

MATERIAL: AS4/3501-6 (bleed)

RESIN CONTENT: 28 - 34 wt% COMP. DENSITY: 1.58 - 1.61 g/cc
FIBER VOLUME: 58 - 65 % VOID CONTENT:
PLY THICKNESS: 0.0041 - 0.0055 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75		200			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		26		26			
		Normalized	Measured	Normalized	Measured		
	Mean	210	214	196	201		
	Minimum	144	161	148	165		
	Maximum	269	260	242	237		
	C.V. (%)	16.0	13.5	13.6	10.7		
F_1^{cu}	B-value	(2)		(2)			
	Distribution	ANOVA		ANOVA			
(ksi)	C_1	34.7	27.7				
	C_2	2.39	2.52				
	No. Specimens	26		27			
	No. Batches	7		7			
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
e_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean			16.3	17.4		
	Minimum			13.0	14.3		
	Maximum			18.7	19.6		
E_1^c	C.V. (%)			10.7	10.1		
(Msi)	No. Specimens			15			
	No. Batches			3			

(1) As fabricated.
(2) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(d)
C/Ep 145-UT
AS4/3501-6
Compression, 1-axis
$[0]_8$
75/D, 75/W, 200/W
Screening Data

MATERIAL: AS4/3501-6 (bleed)

RESIN CONTENT: 28 - 34 wt% COMP. DENSITY: 1.58 - 1.61 g/cc
FIBER VOLUME: 58 - 65 % VOID CONTENT:
PLY THICKNESS: 0.0041 - 0.0055 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75		75		200	
Moisture Content(%)		dry		wet		wet	
Equilibrium at T, RH		(1)		(2)		(1)	
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{cu} (ksi)	Mean			202	213	169	179
	Minimum			165	179	100	107
	Maximum			274	266	212	226
	C.V. (%)			18.0	14.1	22.2	22.9
	B-value			(3)		(3)	
	Distribution			Weibull		ANOVA	
	C_1			217		41.7	
	C_2			5.89		5.28	
	No. Specimens			10		10	
	No. Batches			2		3	
ϵ_1^{cu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^c (Msi)	Mean	17.8	18.8	17.4	18.5	17.7	18.7
	Minimum	15.1	16.4	15.6	17.1	12.1	13.4
	Maximum	20.3	20.0	20.3	20.6	27.2	25.5
	C.V. (%)	7.50	7.18	9.14	5.84	21.6	15.8
	No. Specimens	14		10		10	
	No. Batches	3		2		3	

(1) As fabricated.
(2) Conditioned at 140°F, 95% relative humidity 30 days.
(3) B-values are not presented for screening data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.12(e)
C/Ep 145-UT
AS4/3501-6
Shear, 13-axis
$[0]_8$
75/D, 200/D, 75 W, 200/W
Screening Data

MATERIAL: AS4/3501-6 (bleed)

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 58 - 62 %
PLY THICKNESS: 0.0047 - 0.0055 in.

COMP. DENSITY: 1.58 - 1.60 g/cc
VOID CONTENT:

TEST METHOD:
ASTM D2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	200	75	200	
Moisture Content(%)		dry	dry	wet	wet	
Equilibrium at T, RH		(1)	(1)	(2)	(2)	
Source Code		26	26	26	26	
F_{13}^{su} (ksi)	Mean	17.3	13.0	13.9	9.0	
	Minimum	14.1	11.1	13.1	8.3	
	Maximum	19.4	14.9	15.5	10.1	
	C.V. (%)	7.63	11.6	6.13	6.4	
	B-value	(3)	(3)	(3)	(3)	
	Distribution	ANOVA	ANOVA	Normal	Normal	
	C_1	1.38	1.59	13.9	9.0	
	C_2	2.62	2.77	0.852	0.58	
	No. Specimens	21	21	6	9	
	No. Batches	7	7	2	3	
γ_{13}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
G_{13}^{s} (Msi)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) Conditioned at 140°F, 95% relative humidity for 30 days.
(3) B-values are not presented for screening data.

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4.2.13 AS4/3501-6 (no bleed).*

MATERIAL:	AS4/3501-6 unidirectional tape		C/Ep AS4/3501-6 Summary
PREPREG:	Hercules AS4/3501-6 unidirectional tape		
FIBER:	Hercules AS4, unsized	RESIN:	Hercules 3501-6 TMA
T_g(dry):	390°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10 min., 100 ± 10 psig, no bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture	~12/82 - 8/89	Date of testing	~6/83 - ~4/91
Date of resin manufacture		Date of data submittal	6/90
Date of prepreg manufacture	1/83 - 11/89	Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW	ETW	
F_1^{tu}	(ksi)	290	I	261	S			
E_1^t	(Msi)	18.9	I	21.1	S			
F_2^{tu}	(ksi)	7.97	S					
E_2^t	(Msi)	1.22	S					
F_1^{cu}	(ksi)	233	I				191	S
E_1^c	(Msi)	18.8	I				18.3	S
F_2^{cu}	(ksi)							
E_2^c	(Msi)							
F_{12}^{su}	(ksi)							
G_{12}^s	(Msi)							
F_{13}^{su}	(ksi)	17.9	S					
G_{13}^s	(Msi)							
F_{23}^{su}	(ksi)							
G_{23}^s	(Msi)							

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.8		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.59		
Fiber Areal Weight	(g/m^2)	145	142 - 149	
Ply Thickness	(in)		0.0055 - 0.0062	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL:	AS4/3501-6 (no bleed)		
RESIN CONTENT:	36 - 39 wt%	COMP. DENSITY:	1.55 - 1.57 g/cc
FIBER VOLUME:	52 - 56 %	VOID CONTENT:	
PLY THICKNESS:	0.0055 - 0.0060 in.		
TEST METHOD:		MODULUS CALCULATION:	
ASTM D3039-76		Initial tangent	

Table 4.2.13(a)
C/Ep 145-UT
AS4/3501-6
Tension, 1-axis
$[0]_8$
75/D
Interim Data

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		26					
		Normalized	Measured				
	Mean	290	262				
	Minimum	262	235				
	Maximum	322	286				
	C.V. (%)	5.62	5.38				
F_1^{tu}	B-value	(2)					
	Distribution	ANOVA					
(ksi)	C_1	16.5					
	C_2	2.05					
	No. Specimens	30					
	No. Batches	10					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
e_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	18.9	17.1				
	Minimum	17.0	15.5				
	Maximum	20.3	17.9				
E_1^t	C.V. (%)	4.00	3.20				
(Msi)	No. Specimens	30					
	No. Batches	10					

(1) As fabricated.
(2) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.56 g/cc
FIBER VOLUME: 54 - 55 % VOID CONTENT:
PLY THICKNESS: 0.0060 - 0.0062 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Table 4.2.13(b)
C/Ep 145-UT
AS4/3501-6
Tension, 1-axis
$[0]_8$
-65/D, 200/D
Screening Data

		-65 dry (1) 26		200 dry (1) 26			
		Normalized	Measured	Normalized	Measured		
F_1^{tu} (ksi)	Mean	261	237	315	286		
	Minimum	207	187	278	247		
	Maximum	300	274	330	297		
	C.V. (%)	12.4	12.8	4.89	5.59		
	B-value	(2)		(2)			
	Distribution	ANOVA		Nonpara.			
	C_1	34.9		6			
	C_2	4.69		2.25			
	No. Specimens	9		9			
	No. Batches	3		3			
ϵ_1^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^t (Msi)	Mean	21.1	19.2	20.8	18.9		
	Minimum	19.7	17.7	19.4	17.4		
	Maximum	22.3	21.4	22.0	20.2		
	C.V. (%)	4.60	5.78	4.72	4.70		
	No. Specimens	9		9			
	No. Batches	3		3			

(1) As fabricated.
(2) B-values are not presented for screening data.

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• ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(c)
C/Ep 145-UT
AS4/3501-6
Tension, 2-axis
$[90]_8$
75/D
Screening Data

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.56 g/cc
FIBER VOLUME: 54 - 55 % VOID CONTENT:
PLY THICKNESS: 0.0060 - 0.0062 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76 Initial tangent

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		26				
F_2^{tu} (ksi)	Mean	8.0				
	Minimum	6.8				
	Maximum	9.3				
	C.V. (%)	10.				
	B-value	(2)				
	Distribution	Normal				
	C_1	8.0				
	C_2	0.81				
	No. Specimens	9				
	No. Batches	3				
ϵ_2^{tu} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
E_2^t (Msi)	Mean	1.2				
	Minimum	1.1				
	Maximum	1.4				
	C.V. (%)	8.9				
	No. Specimens	9				
	No. Batches	3				

(1) As fabricated.
(2) B-values are not presented for screening data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(d)
C/Ep 145-UT
AS4/3501-6
Compression, 1-axis
$[0]_8$
75/D, 200/D, 200/W
Interim Data

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 36 - 39 wt% COMP. DENSITY: 1.55 - 1.57 g/cc
FIBER VOLUME: 52 - 56 % VOID CONTENT:
PLY THICKNESS: 0.0056 - 0.0060 in.

TEST METHOD: SACMA 1-88

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Specimen thickness and batch fiber volume to 60%

Temperature(°F)		75 dry		200 dry		200 wet	
Moisture Content(%)		(1)		(1)		(2)	
Equilibrium at T, RH		26		26		26	
Source Code							
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	233	211	213	193	191	173
	Minimum	200	186	174	157	142	128
	Maximum	260	234	267	243	220	201
	C.V. (%)	6.39	6.16	9.74	10.0	11.0	11.4
F_1^{cu}	B-value	(3)		(3)		(3)	
	Distribution	ANOVA		ANOVA		ANOVA	
(ksi)	C_1	15.2		21.0		22.4	
	C_2	2.21		2.00		4.17	
	No. Specimens	30		30		15	
	No. Batches	8		10		3	
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
e_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	18.8	17.0			18.3	16.6
	Minimum	17.9	16.2			17.5	15.7
	Maximum	19.7	17.8			19.1	17.3
E_1^c	C.V. (%)	3.21	3.53			2.62	3.16
(Msi)	No. Specimens	15				15	
	No. Batches	3				3	

(1) As fabricated.
(2) Conditioned at 140°F, 95% relative humidity for 30 days.
(3) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.13(e)
C/Ep 145-UT
AS4/3501-6
Shear, 13-axis
$[0]_8$
75/D, 200/D
Screening Data

MATERIAL: AS4/3501-6 (no bleed)

RESIN CONTENT: 36 - 39 wt%
FIBER VOLUME: 52 - 56 %
PLY THICKNESS: 0.0057 - 0.0063 in.

COMP. DENSITY: 1.55 - 1.57 g/cc
VOID CONTENT:

TEST METHOD: ASTM D2344

MODULUS CALCULATION: Initial tangent

NORMALIZED BY: Not normalized

Temperature(°F)		75	200			
Moisture Content(%)		dry	dry			
Equilibrium at T, RH		(1)	(1)			
Source Code		26	26			
F_{13}^{su} (ksi)	Mean	17.9	14.0			
	Minimum	16.5	12.9			
	Maximum	19.0	15.4			
	C.V. (%)	4.46	4.73			
	B-value	(2)	(2)			
	Distribution	ANOVA	ANOVA			
	C_1	0.824	0.683			
	C_2	2.36	2.34			
	No. Specimens	30	30			
	No. Batches	8	10			
γ_{13}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
G_{13}^{s} (Msi)	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for screening data.

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4.2.14 AS4/3501-6 Plain weave.*

MATERIAL:	AS4 3k/3501-6 Plain weave		C/Ep AS4/3501-6 Summary
PREPREG:	Hercules AW193P		
FIBER:	Hercules AS4 3k W	RESIN:	Hercules 3501-6
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10°F, 100 ± 10 psig, no bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	124	S	112	S	68.5	S		
E_1^t	(Msi)	9.77	S	10.5	S	6.02	S		
F_2^{tu}	(ksi)								
E_2^t	(Msi)								
F_1^{cu}	(ksi)	130.	I			112	I	58.7	I
E_1^c	(Msi)	9.22	I			9.35	I	9.05	I
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)								
G_{12}^s	(Msi)								
F_{13}^{su}	(ksi)	10.9	S			10.9	S	5.3	S
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.80		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.58	1.54 - 1.56	
Fiber Areal Weight	(g/m^2)	193	193	
Ply Thickness	(in)	0.0070	0.0074 - 0.0086	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 38 wt% COMP. DENSITY: 1.56 g/cc
FIBER VOLUME: 53 - 54 % VOID CONTENT:
PLY THICKNESS: 0.0074 - 0.0080 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Table 4.2.14(a)
C/Ep 193-PW
AS4/3501-6
Tension, 1-axis
$[0_f]_8$
75/D, -65/D, 200/D
Screening Data

Temperature(°F)		75		-65		200	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	124	117	112	105	126	119
	Minimum	117	111	103	98.1	116	108
	Maximum	133	124	120	112	133	126
	C.V. (%)	4.18	3.56	4.63	4.00	4.79	5.88
	B-value	(2)		(2)		(2)	
	Distribution	Normal		Normal		Normal	
	C_1	124		112		126	
	C_2	5.17		5.17		6.05	
	No. Specimens	9		9		9	
	No. Batches	3		3		3	
e_1^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^t (Msi)	Mean	9.77	9.21	10.5	9.89	10.1	9.50
	Minimum	9.41	8.80	9.72	9.10	7.10	6.70
	Maximum	10.2	9.50	11.1	10.4	10.7	10.1
	C.V. (%)	2.99	2.45	4.58	4.21	11.2	11.2
	No. Specimens	9		9		9	
	No. Batches	3		3		3	

(1) *As fabricated.*
(2) B-values are not presented for screening data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.14(b)
C/Ep 193-PW
AS4/3501-6
Compression, 1-axis
$[0_f]_{14}$
75/D, 200/D, -75/W
Interim Data

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.54 - 1.55 g/cc
FIBER VOLUME: 51 - 52 % VOID CONTENT:
PLY THICKNESS: 0.0081 - 0.0086 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75		200		75	
Moisture Content(%)		(1)		(1)		(2)	
Equilibrium at T, RH		dry		dry		wet	
Source Code		26		26		26	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	130	117	108	97.3	112	101
	Minimum	115	104	92.8	83.0	99.6	88.0
	Maximum	140	127	121	109	122	109
	C.V. (%)	6.45	6.49	7.44	7.71	5.56	5.65
F_1^{cu}	B-value	(3)		(3)		(3)	
	Distribution	Nonpara.		Weibull		ANOVA	
(ksi)	C_1	8		112		6.83	
	C_2	1.54		15.1		4.85	
	No. Specimens	15		15		15	
	No. Batches	3		3		3	
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	9.22	8.29	9.79	8.79	9.35	8.41
	Minimum	8.53	7.70	9.22	8.40	8.84	8.10
	Maximum	9.84	8.80	10.2	9.10	9.87	8.80
E_1^c	C.V. (%)	3.40	4.28	3.26	2.50	2.98	2.44
(Msi)	No. Specimens	15		15		15	
	No. Batches	3		3		3	

(1) As fabricated.
(2) Conditioned at 140°F, 95% relative humidity for 30 days.
(3) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.14(c)
C/Ep 193-PW
AS4/3501-6
Compression, 1-axis
$[0_f]_{14}$
200/W
Interim Data

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.54 - 1.55 g/cc
FIBER VOLUME: 51 - 52 % VOID CONTENT:
PLY THICKNESS: 0.0081 - 0.0086 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		200					
Moisture Content(%)		(1)					
Equilibrium at T, RH		wet					
Source Code		26					
		Normalized	Measured				
	Mean	58.7	52.7				
	Minimum	51.7	46.2				
	Maximum	65.4	59.7				
	C.V. (%)	7.27	7.58				
F_1^{cu} (ksi)	B-value	(2)					
	Distribution	Weibull					
	C_1	60.6					
	C_2	15.6					
	No. Specimens	15					
	No. Batches	3					
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
e_1^{cu} ($\mu\epsilon$)	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	9.05	8.14				
	Minimum	8.73	7.80				
	Maximum	9.42	8.50				
E_1^c	C.V. (%)	2.40	2.89				
(Msi)	No. Specimens	15					
	No. Batches	3					

(1) Conditioned at 140°F, 95% relative humidity for 30 days.
(2) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.14(d)
C/Ep 193-PW
AS4/3501-6
Shear, 13-axis
$[0_f]_{14}$
200/W
Interim Data

MATERIAL: AS4 3k/3501-6 Plain weave

RESIN CONTENT: 39 - 41 wt% COMP. DENSITY: 1.54 - 1.55 g/cc
FIBER VOLUME: 51 - 52 % VOID CONTENT:
PLY THICKNESS: 0.0077 - 0.0082 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344

NORMALIZED BY: Not normalized

Temperature(°F)		75	200	75	200	
Moisture Content(%)		dry	dry	wet	wet	
Equilibrium at T, RH		(1)	(1)	(2)	(2)	
Source Code		26	26	26	26	
	Mean	10.9	8.4	10.9	5.3	
	Minimum	9.7	8.1	10.0	5.2	
	Maximum	11.9	8.8	11.4	5.5	
	C.V. (%)	6.09	2.5	3.47	2.3	
F_{13}^{su}	B-value	(3)	(3)	(3)	(3)	
(ksi)	Distribution	Weibull	Normal	Weibull	Nonpara.	
	C_1	11.2	8.4	11.0	7	
	C_2	20.1	0.21	35.4	1.81	
	No. Specimens	15	9	15	12	
	No. Batches	3	3	3	3	
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
γ_{13}^{su}	B-value					
($\mu\epsilon$)	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V. (%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) Conditioned at 140°F, 95% relative humidity for 30 days.
(3) B-values are not presented for screening data.

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4.2.15 AS4/3501-6S 5-harness satin weave.*

MATERIAL:	AS4 3k/3501-6S 5-harness satin weave		C/Ep AS4/3501-6S Summary
PREPREG:	Hercules AW280-5HS		
FIBER:	Hercules AS4 3k W	RESIN:	Hercules 3501-6S
T_g(dry):	T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 240 ± 10°F, 60 min., 85 psig; 350 ± 10°F, 120 ± 10°F, 100 ± 10 psig, no bleed		

* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	6/88
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD	RTW	ETW
F_1^{tu}	(ksi)	112	I			
E_1^t	(Msi)	9.73	I			
F_2^{tu}	(ksi)					
E_2^t	(Msi)					
F_1^{cu}	(ksi)	124	I			
E_1^c	(Msi)					
F_2^{cu}	(ksi)					
E_2^c	(Msi)					
F_{12}^{su}	(ksi)					
G_{12}^s	(Msi)					
F_{13}^{su}	(ksi)	11.0	S			
G_{13}^s	(Msi)					
F_{23}^{su}	(ksi)					
G_{23}^s	(Msi)					

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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• ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.80		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.58	1.58 - 1.59	
Fiber Areal Weight	(g/m^2)	280	279 - 284	
Ply Thickness	(in)		0.0106 - 0.0107	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.15(a)
C/Ep 280-5HS
AS4/3501-6S
Tension, 1-axis
$[0_f]_8$
75/D
Interim Data

MATERIAL: AS4 3k/3501-6S 5-harness satin weave

RESIN CONTENT: 33 - 35 wt%
FIBER VOLUME: 57 - 60 %
PLY THICKNESS: 0.0106 - 0.0107 in.

COMP. DENSITY: 1.58 - 1.59 g/cc
VOID CONTENT:

TEST METHOD:
ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Temperature(°F)		75					
Moisture Content(%)		dry					
Equilibrium at T, RH		(1)					
Source Code		26					
		Normalized	Measured				
F_1^{tu} (ksi)	Mean	112	115				
	Minimum	97.6	100				
	Maximum	123	126				
	C.V. (%)	5.78	5.55				
	B-value	(2)					
	Distribution	ANOVA					
	C_1	6.63					
	C_2	2.26					
	No. Specimens	30					
	No. Batches	10					
ϵ_1^{tu} ($\mu\epsilon$)	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
E_1^t (Msi)	Mean	9.73	10.0				
	Minimum	8.93	9.20				
	Maximum	10.1	10.3				
	C.V. (%)	2.48	2.31				
	No. Specimens	30					
	No. Batches	10					

(1) As fabricated.
(2) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: AS4 3k/3501-6S 5-harness satin weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.58 - 1.59 g/cc
FIBER VOLUME: 57 - 60 % VOID CONTENT:
PLY THICKNESS: 0.0106 - 0.0107 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Specimen thickness and batch fiber volume to 57%

Table 4.2.15(b)
C/Ep 280-5HS
AS4/3501-6S
Compression, 1-axis
$[0_f]_6$
75/D, 200/D
Interim Data

Temperature(°F)		75		200			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		26		26			
		Normalized	Measured	Normalized	Measured		
	Mean	124	128	110	113		
	Minimum	108	111	96.1	99.0		
	Maximum	144	148	122	125		
	C.V. (%)	6.73	6.74	6.31	6.24		
F_1^{cu}	B-value	(2)		(2)			
	Distribution	Weibull		ANOVA			
(ksi)	C_1	128		7.04			
	C_2	15.4		2.10			
	No. Specimens	30		30			
	No. Batches	10		10			
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
ϵ_1^{cu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean						
	Minimum						
	Maximum						
E_1^c	C.V. (%)						
(Msi)	No. Specimens						
	No. Batches						

(1) As fabricated.
(2) B-values are not presented for interim data.

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* ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.2.15(c)
C/Ep 280-5HS
AS4/3501-6S
Shear, 13-axis
$[0_f]_8$
75/D, 200/D
Screening Data

MATERIAL: AS4 3k/3501-6S 5-harness satin weave

RESIN CONTENT: 33 - 35 wt% COMP. DENSITY: 1.58 - 1.59 g/cc
FIBER VOLUME: 57 - 60 % VOID CONTENT:
PLY THICKNESS: 0.0106 - 0.0107 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344

NORMALIZED BY: Not normalized

Temperature(°F)		75	200			
Moisture Content(%)		dry	dry			
Equilibrium at T, RH		(1)	(1)			
Source Code		26	26			
	Mean	11.0	9.53			
	Minimum	9.00	8.40			
	Maximum	13.2	10.8			
	C.V. (%)	10.8	6.70			
F_{13}^{su}	B-value	(2)	(2)			
(ksi)	Distribution	ANOVA	ANOVA			
	C_1	1.22	0.66			
	C_2	2.18	2.32			
	No. Specimens	30	30			
	No. Batches	10	10			
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
γ_{13}^{su}	B-value					
($\mu\epsilon$)	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V. (%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for screening data.

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4.3 CARBON - POLYESTER COMPOSITES.

4.4 CARBON - BISMALEIMIDE COMPOSITES.

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4.4.1 T-300 3k/F650.*

				C/BMI T-300/F650 Summary
MATERIAL:	T-300 3k/F650 unidirectional tape			
PREPREG:	Hexcel T3T190/F652 unidirectional tape			
FIBER:	Toray T-300 3k	RESIN:	Hexcel F650	
T_g(dry):	600°F T_g(wet): °F	T_g METHOD:		
PROCESSING:	Autoclave cure: 375°F, 4 hours, 85 psig; Postcure: 475°F, 4 hours, free-standing oven			

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW	ETW
F_1^{tu}	(ksi)	248	S	194	S		
E_1^t	(Msi)	18.9	S				
F_2^{tu}	(ksi)						
E_2^t	(Msi)						
F_1^{cu}	(ksi)						
E_1^c	(Msi)						
F_2^{cu}	(ksi)						
E_2^c	(Msi)						
F_{12}^{su}	(ksi)						
G_{12}^s	(Msi)						
F_{13}^{su}	(ksi)	14.1	S				
G_{13}^s	(Msi)						
F_{23}^{su}	(ksi)						
G_{23}^s	(Msi)						

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.76		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.56	1.57	
Fiber Areal Weight	(g/m^2)	189		
Ply Thickness	(in)	0.0070		

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.1(a)
C/BMI 189-UT
T-300/F650
Tension, 1-axis
$[0]_8$
75/D, -67/D, 400/D
Screening Data

MATERIAL: T-300 3k/F650

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.57 g/cc
FIBER VOLUME: 61 % VOID CONTENT:
PLY THICKNESS: 0.0070 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Batch fiber volume fraction to 60%

			75		-67		400	
Temperature(°F)			75		-67		400	
Moisture Content(%)			dry		dry		dry	
Equilibrium at T, RH			(1)		(1)		(1)	
Source Code			21		21		21	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	248	252	194	197	229	233
		Minimum	216	220	167	170	216	220
		Maximum	293	298	212	216	243	247
		C.V.(%)	7.14	7.15	8.68	8.68	3.97	3.97
F_1^{tu}		B-value	(2)		(2)		(2)	
		Distribution	Normal		Normal		Normal	
(ksi)		C_1	248		194		230.	
		C_2	17.7		16.8		11.1	
		No. Specimens	15		5		7	
		No. Batches	1		1		1	
		Mean						
		Minimum						
		Maximum						
		C.V.(%)						
ϵ_1^{tu}		B-value						
		Distribution						
($\mu\epsilon$)		C_1						
		C_2						
		No. Specimens						
		No. Batches						
		Mean	18.9	19.2			19.1	19.4
		Minimum	16.5	16.8			16.8	17.1
		Maximum	20.3	20.6			21.0	21.4
E_1^t		C.V.(%)	5.58	5.49			7.26	7.23
(Msi)		No. Specimens	15				9	
		No. Batches	1				1	

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.1(b)
C/BMI 189-UT
T-300/F650
Shear, 13-axis
$[0]_{34}$
75/D, 400/D
Screening Data

MATERIAL: T-300 3k/F650

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.57 g/cc
FIBER VOLUME: 61 % VOID CONTENT:
PLY THICKNESS: 0.0070 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344

NORMALIZED BY: Not normalized

Temperature(°F)		75	400			
Moisture Content(%)		dry	dry			
Equilibrium at T, RH		(1)	(1)			
Source Code		21	21			
	Mean	14.1	9.39			
	Minimum	13.5	8.77			
	Maximum	15.0	10.1			
	C.V.(%)	3.04	4.25			
F_{13}^{su}	B-value	(2)	(2)			
	Distribution	Weibull	Weibull			
(ksi)	C_1	14.3	9.59			
	C_2	32.3	24.6			
	No. Specimens	15	15			
	No. Batches	1	1			
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for screening data.

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4.4.2 T-300 3k/F650 8-harness satin weave. *

MATERIAL:	T-300 3k/F650 8-harness satin weave		C/Ep T-300/F650 Summary
PREPREG:	Hexcel F3T584/F650 8-harness satin weave		
FIBER:	Toray T-300 3k	RESIN: Hexcel F650	
T_g(dry):	600°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 375°F, 4 hours, 85 psig; Postcure: 475°F, 4 hours, free-standing oven		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD	RTW	ETW
F_1^{tu}	(ksi)					
E_1^t	(Msi)					
F_2^{tu}	(ksi)					
E_2^t	(Msi)					
F_1^{cu}	(ksi)					
E_1^c	(Msi)					
F_2^{cu}	(ksi)					
E_2^c	(Msi)					
F_{12}^{su}	(ksi)	9.77	S			
G_{12}^s	(Msi)	0.69	S			
F_{13}^{su}	(ksi)	5.83	S			
G_{13}^s	(Msi)					
F_{23}^{su}	(ksi)					
G_{23}^s	(Msi)					

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.75		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.54		
Fiber Areal Weight	(g/m^2)	370		
Ply Thickness	(in)	0.015		

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.2(a)
C/BMI 370-8HS
T-300/F650
Shear, 12-axis
$[\pm 45_t]_{4s}$
75/D
Screening Data

MATERIAL: T-300 3k/F650

RESIN CONTENT: 40 wt%
FIBER VOLUME: 52 %
PLY THICKNESS: 0.015 - 0.015 in.

COMP. DENSITY: 1.51 g/cc
VOID CONTENT:

TEST METHOD:
ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		21				
	Mean	9.77				
	Minimum	8.57				
	Maximum	11.1				
	C.V.(%)	8.78				
F_{12}^{su}	B-value	(2)				
	Distribution	Weibull				
(ksi)	C_1	10.2				
	C_2	12.9				
	No. Specimens	15				
	No. Batches	1				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	0.69				
	Minimum	0.59				
	Maximum	0.81				
G_{12}^{s}	C.V.(%)	10.				
(Msi)	No. Specimens	14				
	No. Batches	1				

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.2(b)
C/BMI 370-8HS
T-300/F650
Shear, 13-axis
$[0_f]_8$
75/D, 350/D, 450/D
Screening Data

MATERIAL: T-300 3k/F650

RESIN CONTENT: 40 wt%
FIBER VOLUME: 52 %
PLY THICKNESS: 0.015 - 0.015 in.

COMP. DENSITY: 1.51 g/cc
VOID CONTENT:

TEST METHOD:
ASTM D2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75	350	450		
Moisture Content(%)		dry	dry	dry		
Equilibrium at T, RH		(1)	(1)	(1)		
Source Code		21	21	21		
	Mean	5.83	5.59	5.80		
	Minimum	4.75	4.93	5.23		
	Maximum	8.06	6.44	6.57		
	C.V.(%)	15.0	10.9	6.81		
F_{13}^{su}	B-value	(2)	(2)	(2)		
	Distribution	Nonpara.	Weibull	Weibull		
(ksi)	C_1	8	5.86	5.98		
	C_2	1.54	11.0	15.5		
	No. Specimens	15	10	10		
	No. Batches	1	1	1		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for screening data.

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4.4.3 T-300 3k/F652 8-harness satin weave. *

MATERIAL:	T-300 3k/F652 8-harness satin weave		C/Ep T-300/F652 Summary
PREPREG:	Hexcel F3G584/F652 8-harness satin weave		
FIBER:	Amoco Thornel T-300	RESIN: Hexcel F652	
T_g(dry):	600°F T_g(wet):	T_g METHOD:	
PROCESSING:	Press cured: 400°F, 2.5 hours, 125 psig; Postcure: 550°F, 4 hours		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD	RTW	ETW
F_1^{tu}	(ksi)	73.6	S			
E_1^t	(Msi)	9.71	S			
F_2^{tu}	(ksi)					
E_2^t	(Msi)					
F_1^{cu}	(ksi)	61.4	S			
E_1^c	(Msi)	8.41	S			
F_2^{cu}	(ksi)					
E_2^c	(Msi)					
F_{12}^{su}	(ksi)	5.97	S			
G_{12}^s	(Msi)					
F_{13}^{su}	(ksi)					
G_{13}^s	(Msi)					
F_{23}^{su}	(ksi)					
G_{23}^s	(Msi)					

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.76		
Resin Density	(g/cc)	1.26		
Composite Density	(g/cc)	1.55	1.57	
Fiber Areal Weight	(g/m^2)	367		
Ply Thickness	(in)	.00124		

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.3(a)
C/Ep 367-8HS
T-300/F652
Tension, 1-axis
$[0_f]_{10}$
70/D
Screening Data

MATERIAL: T-300 3k/F652 8-harness satin weave

RESIN CONTENT: 27.2 wt% COMP. DENSITY: 1.57 - g/cc
FIBER VOLUME: 64.8 % VOID CONTENT:
PLY THICKNESS: 0.0012 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Batch fiber volume to 57%

Temperature(°F)		70			
Moisture Content(%)		dry			
Equilibrium at T, RH		(1)			
Source Code		21			
		Normalized	Measured		
	Mean	73.6	84.0		
	Minimum	58.8	67.1		
	Maximum	84.3	96.1		
	C.V.(%)	10.1	10.0		
F_1^{tu}	B-value	(2)			
	Distribution	Weibull			
(ksi)	C_1	76.8			
	C_2	12.3			
	No. Specimens	15			
	No. Batches	1			
	Mean				
	Minimum				
	Maximum				
	C.V.(%)				
ϵ_1^{tu}	B-value				
	Distribution				
($\mu\epsilon$)	C_1				
	C_2				
	No. Specimens				
	No. Batches				
	Mean	9.71	11.1		
	Minimum	8.94	10.2		
	Maximum	10.2	11.6		
E_1^t	C.V.(%)	4.36	4.28		
(Msi)	No. Specimens	15			
	No. Batches	1			

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.3(b)
C/Ep 367-8HS
T-300/F652
Shear, 13-axis
$[0_f]_{10}$
70/D, 600/D
Screening Data

MATERIAL: T-300 3k/F652 8-harness satin weave

RESIN CONTENT: 27.2 wt% COMP. DENSITY: 1.57 - g/cc
FIBER VOLUME: 64.8 % VOID CONTENT:
PLY THICKNESS: 0.0012 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344

NORMALIZED BY: Not normalized

Temperature(°F)		70	600			
Moisture Content(%)		dry	dry			
Equilibrium at T, RH		(1)	(1)			
Source Code		21	21			
	Mean	6.97	4.59			
	Minimum	6.13	4.29			
	Maximum	6.64	4.82			
	C.V.(%)	8.17	3.60			
F_{13}^{su}	B-value	(2)	(2)			
	Distribution	Weibull	Weibull			
(ksi)	C_1	6.18	4.66			
	C_2	14.8	36.8			
	No. Specimens	15	15			
	No. Batches	1	1			
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated.
(2) B-values are not presented for screening data.

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4.4.4 AS4/5250-3.*

MATERIAL:	AS4/5250-3 unidirectional tape	C/Ep AS4/5250-3 Summary
PREPREG:	Narmco AS4/5250-3 unidirectional tape, grade 147	
FIBER:	Hercules AS4 RESIN: Narmco 5250-3	
T_g(dry):	642°F T_g(wet): 561°F T_g METHOD: DMA	
PROCESSING:	Autoclave cure: 250°F, 60 min.; 350°F, 360 min., 85 psi; Postcure: 475°F, 6 hours	

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	12/88
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	252	S	270.	S	268	S	235	S
E_1^t	(Msi)	15.9	S	16.4	S	16.6	S	16.7	S
F_2^{tu}	(ksi)	4.61	S	4.98	S				
E_2^t	(Msi)	1.24	S	1.40	S				
F_1^{cu}	(ksi)	175	S	198	S	194	S	127	S
E_1^c	(Msi)	17.0	S	15.5	S	18.5	S	18.1	S
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)	9.61	S	10.1	S	12.5	S	9.81	S
G_{12}^s	(Msi)	0.77	S	0.84	S	0.79	S	0.49	S
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.80		
Resin Density	(g/cc)	1.25		
Composite Density	(g/cc)	1.58	1.52 - 1.63	
Fiber Areal Weight	(g/m^2)	147	132 - 165	ASTM D3529
Ply Thickness	(in)	0.0051 - 0.0059	0.0050 - 0.0062	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(a)
C/Ep 147-UT
AS4/5250-3
Tension, 1-axis
$[0]_8$
72/D, -67/D, 350/D
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 26 - 28 wt% COMP. DENSITY: 1.58 - 1.61 g/cc
FIBER VOLUME: 63 - 66 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		72		-67		350	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		(2)		(2)		(2)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	252	291	270	311	266	308
	Minimum	223	255	249	285	241	276
	Maximum	275	322	288	332	283	325
	C.V.(%)	7.63	8.48	6.12	6.48	6.87	7.54
F_1^{tu}	B-value	(3)		(3)		(3)	
	Distribution	Normal		Normal		Normal	
(ksi)	C_1	252		270.		266	
	C_2	19.2		16.5		18.3	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Mean		17100		15800		15900
	Minimum		14900		14100		14800
	Maximum		20000		18000		17100
	C.V.(%)		13.3		9.6		4.98
e_1^{tu}	B-value		(3)		(3)		(3)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		17100		15800		15900
	C_2		2270		1520		789
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Mean	15.9	18.3	16.4	18.9	16.4	19.0
	Minimum	15.3	17.7	15.9	18.5	15.8	18.2
	Maximum	16.4	18.9	16.8	19.4	16.7	19.5
E_1^t	C.V.(%)	3.04	2.51	2.23	1.91	2.07	2.85
(Msi)	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Mean		0.300		0.295		0.302
ν_{12}^t	No. Specimens	6		6		6	
	No. Batches	1		1		1	

(1) As fabricated
(2) Reference 4.4.4
(3) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(b)
C/Ep 147-UT
AS4/5250-3
Tension, 1-axis
$[0]_8$
450/D, 74/0.70%, 350/0.73%
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 26 - 28 wt% COMP. DENSITY: 1.61 - 1.63 g/cc
FIBER VOLUME: 63 - 67 % VOID CONTENT: 0.0 - 0.9 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

			450.		74		350.	
Temperature(°F)			450.		74		350.	
Moisture Content(%)			dry		0.70		0.73	
Equilibrium at T, RH			(1)		160°F, 95%		(2)	
Source Code			25		25		25	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	253	292	268	312	249	287
		Minimum	208	237	235	268	232	264
		Maximum	269	314	293	347	261	305
		C.V.(%)	8.87	9.64	7.74	8.99	4.50	5.42
F_1^{tu}		B-value	(4)		(4)		(4)	
		Distribution	Nonpara.		Normal		Normal	
(ksi)		C_1			268		249	
		C_2			20.7		11.2	
		No. Specimens	6		6		5	
		No. Batches	1		1		1	
		Mean		13900		15200		14900
		Minimum		11700		13500		13200
		Maximum		15000		16600		15500
		C.V.(%)		8.14		7.14		6.46
e_1^{tu}		B-value		(4)		(4)		(4)
		Distribution		Normal		Normal		Normal
($\mu\epsilon$)		C_1		13900		15200		14900
		C_2		1130		1080		961
		No. Specimens	6		6		6	
		No. Batches	1		1		1	
		Mean	16.5	19.0	16.6	19.3	15.9	18.4
		Minimum	15.7	18.1	16.2	18.9	15.4	17.8
		Maximum	16.9	19.7	17.3	19.9	16.4	19.1
E_1^t		C.V.(%)	3.43	3.56	2.36	1.82	2.41	2.71
(Msi)		No. Specimens	6		6		5	
		No. Batches	1		1		1	
		Mean		0.295		0.335		0.368
ν_{12}^t		No. Specimens	6		6		5	
		No. Batches	1		1		1	

(1) As fabricated
(2) Conditioned at 160°F, 95% relative humidity for 29 days (75% saturation)
(3) Reference 4.4.4
(4) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(c)
C/Ep 147-UT
AS4/5250-3
Tension, 1-axis
$[0]_8$
350/1.0%
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 26 - 28 wt% COMP. DENSITY: 1.61 g/cc
FIBER VOLUME: 63 - 66 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)	350.		
Moisture Content(%)	1.0		
Equilibrium at T, RH	160°F, 95%		
Source Code	(1)		

		Normalized	Measured				
	Mean	235	270				
	Minimum	176	202				
	Maximum	259	296				
	C.V.(%)	12.8	13.0				
F_1^{tu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	235					
	C_2	29.9					
	No. Specimens	6					
	No. Batches	1					
	Mean		14400				
	Minimum		9950				
	Maximum		16200				
	C.V.(%)		16.0				
e_1^{tu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		14400				
	C_2		2300				
	No. Specimens	6					
	No. Batches	1					
	Mean	16.7	19.2				
	Minimum	15.5	17.7				
	Maximum	18.4	21.2				
E_1^t	C.V.(%)	6.43	6.26				
(Msi)	No. Specimens	6					
	No. Batches	1					
	Mean		0.363				
ν_{12}^t	No. Specimens	4					
	No. Batches	1					

(1) Reference 4.4.4
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(d)
C/Ep 147-UT
AS4/5250-3
Tension, 2-axis
$[90]_8$
72/D, -67/D, 350/D, 450/D
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 27 - 40 wt% COMP. DENSITY: 1.52 - 1.61 g/cc
FIBER VOLUME: 51 - 65 % VOID CONTENT: 0.1 - 0.8 %
PLY THICKNESS: 0.0051 - 0.0059 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Not normalized

Temperature(°F)		72	-67	350	450		
Moisture Content(%)		dry	dry	dry	dry		
Equilibrium at T, RH		(1)	(1)	(1)	(1)		
Source Code		25	25	25	25		
	Mean	4.61	4.98	4.63	4.54		
	Minimum	3.52	4.68	3.43	4.13		
	Maximum	5.65	5.94	5.33	5.19		
	C.V.(%)	18.4	9.69	13.7	9.20		
F_2^{tu}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Normal	Nonpara.	Normal	Normal		
(ksi)	C_1	4.61		4.63	4.54		
	C_2	0.847		0.637	0.417		
	No. Specimens	6	6	6	6		
	No. Batches	1	1	1	1		
	Mean	3540	3580	4680	4330		
	Minimum	2000	3180	3300	3600		
	Maximum	4900	4740	6000	5600		
	C.V.(%)	26.9	16.5	19.0	18.0		
e_2^{tu}	B-value	(2)	(2)	(2)	(2)		
	Distribution	Normal	Lognormal	Normal	Normal		
($\mu\epsilon$)	C_1	3540	8.17	4680	4330		
	C_2	955	0.149	889	782		
	No. Specimens	6	6	6	6		
	No. Batches	1	1	1	1		
	Mean	1.24	1.40	1.04	1.08		
	Minimum	1.17	1.26	0.940	0.930		
	Maximum	1.35	1.47	1.16	1.26		
E_2^t	C.V.(%)	5.90	5.50	8.50	10.3		
(Msi)	No. Specimens	6	6	5	6		
	No. Batches	1	1	1	1		

(1) As fabricated
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(e)
C/Ep 147-UT
AS4/5250-3
Compression, 1-axis
$[0]_8$
72/D, -67/D, 350/D
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 36 - 38 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 53 - 56 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0057 - 0.0062 in.

TEST METHOD: ASTM D3410A-87

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

			74		-67		350.	
Temperature(°F)								
Moisture Content(%)			dry		dry		dry	
Equilibrium at T, RH			(1)		(1)		(1)	
Source Code			(2)		(2)		(2)	
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	175	158	198	179	174	148
		Minimum	122	110	176	160	141	127
		Maximum	203	184	222	201	235	185
		C.V.(%)	15.9	15.9	8.0	8.0	23.6	15.9
F_1^{cu}		B-value	(3)		(3)		(3)	
		Distribution	Normal		Normal		Normal	
(ksi)		C_1	175		198		174	
		C_2	27.7		15.8		41.1	
		No. Specimens	6		6		6	
		No. Batches	1		1		1	
		Mean	12100		19800		15300	
		Minimum	8000		8360		10200	
		Maximum	22700		26700		18400	
		C.V.(%)	46.2		43.9		18.1	
e_1^{cu}		B-value	(3)		(3)		(3)	
		Distribution	Normal		Normal		Normal	
($\mu\epsilon$)		C_1	12100		19800		15300	
		C_2	5570		8710		2770	
		No. Specimens	6		6		6	
		No. Batches	1		1		1	
		Mean	17.0	15.4	15.5	14.0	17.4	14.9
		Minimum	14.1	12.8	13.9	12.6	15.2	13.8
		Maximum	22.7	20.5	18.5	16.7	21.9	17.2
E_1^c		C.V.(%)	20.1	20.0	10.7	10.6	14.7	8.55
		No. Specimens	6		6		6	
(Msi)		No. Batches	1		1		1	

(1) As fabricated
(2) Reference 4.4.4
(3) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(f)
C/Ep 147-UT
AS4/5250-3
Compression, 1-axis
$[0]_8$
450/D, 74/0.82%, 350/0.79%
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 36 - 38 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 53 - 56 % VOID CONTENT: 0.1 - 0.9 %
PLY THICKNESS: 0.0057 - 0.0062 in.

TEST METHOD: ASTM D3410A-87

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		450.		74		350.	
Moisture Content(%)		dry		0.82		0.79	
Equilibrium at T, RH		(1)		160°F, 95%		(2)	
Source Code		(3)		(3)		(3)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	153	131	194	176	153	139
	Minimum	119	108	175	159	113	102
	Maximum	207	163	216	195	173	157
	C.V.(%)	21.2	15.1	8.6	8.63	15.5	15.5
F_1^{cu}	B-value	(4)		(4)		(4)	
	Distribution	Normal		Normal		Normal	
(ksi)	C_1	153		194		153	
	C_2	32.4		16.7		23.8	
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Mean	8480		15900		12600	
	Minimum	2900		10600		6400	
	Maximum	14600		22900		1600	
	C.V.(%)	44.7		32.5		30.2	
e_1^{cu}	B-value	(4)		(4)		(4)	
	Distribution	Normal		Normal		Normal	
($\mu\epsilon$)	C_1	8480		15900		12600	
	C_2	3790		5170		3810	
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Mean	18.2	15.6	18.5	16.8	16.1	14.6
	Minimum	14.0	12.6	16.4	14.9	14.3	12.9
	Maximum	21.7	17.1	21.5	19.5	18.2	16.5
E_1^c	C.V.(%)	16.0	10.4	9.42	9.39	9.78	9.75
(Msi)	No. Specimens	6		6		5	
	No. Batches	1		1		1	

(1) As fabricated
(2) Conditioned at 160°F, 95% relative humidity for 7 days (75% saturation)
(3) Reference 4.4.4
(4) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(g)
C/Ep 147-UT
AS4/5250-3
Compression, 1-axis
$[0]_8$
350/1.0%
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 36 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 56 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0050 - 0.0053 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		350.					
Moisture Content(%)		1.0					
Equilibrium at T, RH		160°F, 95%					
Source Code		(1)					
		Normalized	Measured				
	Mean	127	115				
	Minimum	108	97.9				
	Maximum	152	138				
	C.V.(%)	11.4	11.4				
F_1^{cu}	B-value	(2)					
	Distribution	Normal					
(ksi)	C_1	127					
	C_2	14.4					
	No. Specimens	6					
	No. Batches	1					
	Mean		8120				
	Minimum		6600				
	Maximum		9180				
	C.V(%)		11.5				
ϵ_1^{cu}	B-value		(2)				
	Distribution		Normal				
($\mu\epsilon$)	C_1		8120				
	C_2		934				
	No. Specimens	6					
	No. Batches	1					
	Mean	18.1	16.4				
	Minimum	16.6	15.0				
	Maximum	20.7	18.7				
E_1^c	C.V.(%)	7.93	7.89				
(Msi)	No. Specimens	6					
	No. Batches	1					

(1) Reference 4.4.4
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 28 - 32 wt% COMP. DENSITY: 1.58 - 1.61 g/cc
FIBER VOLUME: 59 - 63 % VOID CONTENT: 0.0 - 1.2 %
PLY THICKNESS: 0.0055 - 0.0058 in.

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Table 4.4.4(h)
C/Ep 147-UT
AS4/5250-3
Shear, 12-axis
$[\pm 45]_{4s}$
72/D, -67/D, 350/D, 450/D
Screening Data

Temperature(°F)		72	-67	350.	450.	
Moisture Content(%)		dry	dry	dry	dry	
Equilibrium at T, RH		(1)	(1)	(1)	(1)	
Source Code		(2)	(2)	(2)	(2)	
F_{12}^{su} (ksi)	Mean	9.61	10.1	10.4	9.01	
	Minimum	8.49	9.67	9.55	8.44	
	Maximum	10.4	10.5	11.0	9.47	
	C.V.(%)	6.95	3.50	5.31	4.87	
	B-value	(3)	(3)	(3)	(3)	
	Distribution	Normal	Normal	Normal	Normal	
	C_1	9.61	10.1	10.4	9.01	
	C_2	0.668	0.352	0.553	0.439	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
γ_{12}^{su} ($\mu\epsilon$)	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
G_{12}^{s} (Msi)	Mean	0.77	0.84	0.66	0.62	
	Minimum	0.71	0.78	0.62	0.50	
	Maximum	0.83	0.86	0.72	0.69	
	C.V.(%)	5.6	3.6	5.3	12.	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	

(1) As fabricated
(2) Reference 4.4.4
(3) *B-values are not presented for screening data*

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.4.4(i)
C/Ep 147-UT
AS4/5250-3
Shear, 12-axis
$[\pm 45]_{4s}$
74/0.55%, 350/0.55%, 350/1.1%
Screening Data

MATERIAL: AS4/5250-3 unidirectional material

RESIN CONTENT: 28 - 32 wt% COMP. DENSITY: 1.58 - 1.61 g/cc
FIBER VOLUME: 59 - 63 % VOID CONTENT: 0.0 - 1.2 %
PLY THICKNESS: 0.0055 - 0.0058 in.

TEST METHOD: ASTM D3518-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		74	350.	350.		
Moisture Content(%)		0.55	0.55	1.1		
Equilibrium at T, RH		160°F, 95%	(1)	160°F, 95%		
Source Code		(2)	(2)	(2)		
	Mean	12.5	8.70	9.81		
	Minimum	11.3	8.24	8.13		
	Maximum	13.2	8.95	10.6		
	C.V.(%)	5.26	3.42	9.27		
F_{12}^{su} (ksi)	B-value	(3)	(3)	(3)		
	Distribution	Normal	Normal	Normal		
	C_1	12.5	8.70	9.81		
	C_2	0.656	0.298	0.909		
	No. Specimens	6	5	6		
	No. Batches	1	1	1		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su} ($\mu\epsilon$)	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	0.79	0.46	0.49		
	Minimum	0.77	0.43	0.40		
	Maximum	0.81	0.48	0.56		
G_{12}^{s} (Msi)	C.V.(%)	1.9	4.0	14		
	No. Specimens	6	6	4		
	No. Batches	1	1	1		

(1) Conditioned at 160°F, 95% relative humidity for 3 days (75% saturation)
(2) Reference 4.4.4
(3) B-values are not presented for screening data

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4.5 CARBON - POLYIMIDE COMPOSITES.

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4.5.1 Celion 3000/F670 8-harness satin weave. *

MATERIAL:	Celion 3000/F670 8-harness satin weave		C/PI Celion 3000/F670 Summary
PREPREG:	Hexcel F3L584/F670 8-harness satin weave		
FIBER:	Celanese Celion 3000	RESIN: Hexcel F670 (PMR-15)	
T_g(dry):	635°F T_g(wet):	T_g METHOD:	
PROCESSING:	Autoclave cure: 440°F, 2 hours; 600°F, 3 Hours, 200 psig; Postcure		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	8/87
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture	2/87 - 5/87	Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD	RTW	ETW
F_1^{tu}	(ksi)	132	S			
E_1^t	(Msi)	9.06	S			
F_2^{tu}	(ksi)	107	S			
E_2^t	(Msi)	8.43	S			
F_1^{cu}	(ksi)	99.4	S			
E_1^c	(Msi)	8.60	S			
F_2^{cu}	(ksi)	78.9	S			
E_2^c	(Msi)	8.08	S			
F_{12}^{su}	(ksi)					
G_{12}^s	(Msi)					
F_{13}^{su}	(ksi)	10.9	S			
G_{13}^s	(Msi)					
F_{23}^{su}	(ksi)	11.1	S			
G_{23}^s	(Msi)					

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.8		
Resin Density	(g/cc)	1.32		
Composite Density	(g/cc)	1.59	1.59 - 1.63	
Fiber Areal Weight	(g/m^2)	384		
Ply Thickness	(in)		0.0132 - 0.0144	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(a)
C/PI 384-8HS
Celion 3000/F670
Tension, 1-axis
$[0_f]_8$
75/D, 550/D
Screening Data

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt% COMP. DENSITY: 1.59 - 1.63 g/cc
FIBER VOLUME: 57 - 64 % VOID CONTENT: 0.0 - 0.62 %
PLY THICKNESS: 0.0132 - 0.0144 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F)		75		550			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		22		22			
		Normalized	Measured	Normalized	Measured		
	Mean	132	136	116	120		
	Minimum	127	131	95.4	98.7		
	Maximum	140	144	129	134		
	C.V.(%)	2.75	2.76	7.94	7.95		
F_1^{tu}	B-value	(2)		(2)			
	Distribution	Normal		Normal			
(ksi)	C_1	132		116			
	C_2	3.63		9.18			
	No. Specimens	9		9			
	No. Batches	3		3			
	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
e_1^{tu}	B-value						
	Distribution						
($\mu\epsilon$)	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	9.03	9.35	8.67	8.98		
	Minimum	8.66	8.96	8.50	8.80		
	Maximum	9.35	9.68	9.07	9.39		
E_1^t	C.V.(%)	3.22	3.23	2.54	2.55		
(Msi)	No. Specimens	9		9			
	No. Batches	3		3			

(1) As fabricated.
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(b)
C/PI 384-8HS
Celion 3000/F670
Tension, 2-axis
$[0_f]_8$
75/D, 550/D
Screening Data

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 57 - 64 %
PLY THICKNESS: 0.0132 - 0.0144 in.

COMP. DENSITY: 1.59 - 1.63 g/cc
VOID CONTENT: 0.0 - 0.62 %

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F)		75		550		
Moisture Content(%)		dry		dry		
Equilibrium at T, RH		(1)		(1)		
Source Code		22		22		
		Normalized	Measured	Normalized	Measured	
	Mean	107	111	90.4	93.5	
	Minimum	85.6	88.6	61.9	64.1	
	Maximum	129	133	123	127	
	C.V. (%)	15.7	15.7	23.8	23.8	
F_y^{tu}	B-value	(2)		(2)		
(ksi)	Distribution	ANOVA		ANOVA		
	C_1	19.3		24.7		
	C_2	6.09		6.02		
	No. Specimens	9		9		
	No. Batches	3		3		
	Mean					
	Minimum					
	Maximum					
	C.V. (%)					
ϵ_y^{tu}	B-value					
($\mu\epsilon$)	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	8.43	8.73	8.23	8.52	
	Minimum	7.43	7.69	7.58	7.85	
	Maximum	9.33	9.66	8.84	9.15	
E_y^t	C.V. (%)	7.45	7.46	5.49	5.48	
(Msi)	No. Specimens	9		9		
	No. Batches	3		3		

(1) As fabricated.
(2) B-values are not presented for screening data.

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Check the source to verify that this is the current version before use.

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(c)
C/PI 384-8HS
Celion 3000/F670
Compression, 1-axis
$[0_t]_8$
75/D, 550/D
Screening Data

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt% COMP. DENSITY: 1.59 - 1.63 g/cc
FIBER VOLUME: 57 - 64 % VOID CONTENT: 0.0 - 0.62 %
PLY THICKNESS: 0.0132 - 0.0144 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F)		75		550		
Moisture Content(%)		dry		dry		
Equilibrium at T, RH		(1)		(1)		
Source Code		22		22		
		Normalized	Measured	Normalized	Measured	
	Mean	99.4	103	66.0	68.3	
	Minimum	87.9	91.3	59.0	61.1	
	Maximum	118	122	71.7	74.2	
	C.V.(%)	9.33	9.33	6.60	6.59	
F_1^{cu}	B-value	(2)		(2)		
	Distribution	ANOVA		Normal		
(ksi)	C_1	10.2		66.0		
	C_2	5.28		4.36		
	No. Specimens	9		9		
	No. Batches	3		3		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
e_1^{cu}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	8.61	8.92	8.09	8.38	
	Minimum	8.40	8.69	7.26	7.51	
	Maximum	9.09	9.41	8.78	9.09	
E_1^c	C.V.(%)	2.54	2.54	5.19	5.21	
(Msi)	No. Specimens	9		9		
	No. Batches	3		3		

(1) As fabricated
(2) B-values are not presented for screening data.

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Check the source to verify that this is the current version before use.

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(d)
C/PI 384-8HS
Celion 3000/F670
Compression, 2-axis
$[0_f]_8$
75/D, 550/D
Screening Data

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt% COMP. DENSITY: 1.59 - 1.63 g/cc
FIBER VOLUME: 57 - 64 % VOID CONTENT: 0.0 - 0.62 %
PLY THICKNESS: 0.0132 - 0.0144 in.

TEST METHOD: MODULUS CALCULATION:
SACMA 1-88

NORMALIZED BY: Fiber volume fraction to 57%

Temperature(°F)		75		550			
Moisture Content(%)		dry		dry			
Equilibrium at T, RH		(1)		(1)			
Source Code		22		22			
		Normalized	Measured	Normalized	Measured		
	Mean	78.9	81.7	54.2	56.1		
	Minimum	76.1	78.8	52.4	54.2		
	Maximum	80.7	83.5	56.6	58.6		
	C.V. (%)	3.10	3.10	4.02	4.03		
F_2^{cu} (ksi)	B-value	(2)		(2)			
	Distribution						
	C_1						
	C_2						
	No. Specimens	3		3			
	No. Batches	1		1			
	Mean						
	Minimum						
	Maximum						
	C.V. (%)						
e_2^{cu} ($\mu\epsilon$)	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
	Mean	8.08	8.37	7.67	7.94		
	Minimum	8.03	8.31	7.59	7.86		
	Maximum	8.14	8.43	7.77	8.04		
E_2^c	C.V. (%)	0.681	0.720	1.19	1.15		
(Msi)	No. Specimens	3		3			
	No. Batches	1		1			

(1) As fabricated.
(2) Insufficient observations to complete the statistical calculations.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.5.1(e)
C/PI 384-8HS
Celion 3000/F670
Shear, 13-axis
$[0_f]_8$
75/D
Screening Data

MATERIAL: Celion 3000/F670 8-harness satin weave

RESIN CONTENT: 30 - 34 wt%
FIBER VOLUME: 57 - 64 %
PLY THICKNESS: 0.0132 - 0.0144 in.

COMP. DENSITY: 1.59 - 1.63 g/cc
VOID CONTENT: 0.0 - 0.62 %

TEST METHOD: ASTM D2344

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		22				
	Mean	10.9				
	Minimum	9.70				
	Maximum	12.0				
	C.V.(%)	6.15				
F_{13}^{su}	B-value	(2)				
	Distribution	ANOVA				
(ksi)	C_1	0.722				
	C_2	4.78				
	No. Specimens	9				
	No. Batches	3				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{13}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{13}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated
(2) B-values are not presented for screening data.

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

MATERIAL:	Celion 3000/F670 8-harness satin weave		
RESIN CONTENT:	30 - 34 wt%	COMP. DENSITY:	1.59 - 1.63 g/cc
FIBER VOLUME:	57 - 64 %	VOID CONTENT:	0.0 - 0.62 %
PLY THICKNESS:	0.0132 - 0.0144 in.		
TEST METHOD:		MODULUS CALCULATION:	
ASTM D2344			
NORMALIZED BY:	*Not normalized*		

Table 4.5.1(f)
C/PI 384-8HS
Celion 3000/F670
Shear, 23-axis
$[0_f]_8$
75/D
Screening Data

Temperature(°F)		75				
Moisture Content(%)		dry				
Equilibrium at T, RH		(1)				
Source Code		22				
	Mean	11.1				
	Minimum	10.4				
	Maximum	11.7				
	C.V.(%)	5.88				
F_{23}^{su} (ksi)	B-value	(2)				
	Distribution					
	C_1					
	C_2					
	No. Specimens	3				
	No. Batches	1				
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{23}^{su} ($\mu\epsilon$)	B-value					
	Distribution					
	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean					
	Minimum					
	Maximum					
G_{23}^{s}	C.V.(%)					
(Msi)	No. Specimens					
	No. Batches					

(1) As fabricated
(2) Insufficient observations to complete the statistical calculations.

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4.6 CARBON - PHENOLIC COMPOSITES.

4.7 CARBON - SILICONE COMPOSITES.

4.8 CARBON - POLYBENZIMIDAZOLE COMPOSITES.

4.9 CARBON - PEEK COMPOSITES.

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4.9.1 IM-6 12k/APC-2.*

MATERIAL:	IM-6 12k/APC-2 unidirectional tape		C/PEEK IM-6/APC-2 Summary
PREPREG:	Fiberite IM-6/APC-2 unidirectional tape		
FIBER:	Hercules IM-6 12k	RESIN: Fiberite APC-2	
T_g(dry):	291°F T_g(wet): 309°F	T_g METHOD: DMA	
PROCESSING:	Autoclave cure: 720°F, 30 - 45 min., 60 psig		

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	12/88
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD		CTD		RTW		ETW	
F_1^{tu}	(ksi)	350.	S	376	S	352	S	364	S
E_1^t	(Msi)	21.6	S	22.0	S	21.2	S	21.2	S
F_2^{tu}	(ksi)	9.41	S	9.67	S				
E_2^t	(Msi)	1.28	S	1.41	S				
F_1^{cu}	(ksi)	167	S	156	S	174	S	154	S
E_1^c	(Msi)	19.4	S	20.4	S	21.4	S	20.3	S
F_2^{cu}	(ksi)								
E_2^c	(Msi)								
F_{12}^{su}	(ksi)	23.9	S	25.4	S	23.0	S	20.0	S
G_{12}^s	(Msi)	0.78	S	0.91	S	0.79	S	0.71	S
F_{13}^{su}	(ksi)								
G_{13}^s	(Msi)								
F_{23}^{su}	(ksi)								
G_{23}^s	(Msi)								

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	1.73		
Resin Density	(g/cc)	1.28		
Composite Density	(g/cc)	1.55	1.54 - 1.58	ASTM D792
Fiber Areal Weight	(g/m^2)			
Ply Thickness	(in)	0.0054	0.0052 - 0.0058	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(a)
C/PEEK -UT
IM-6/APC-2
Tension, 1-axis
$[0]_8$
74/D, -67/D, 180/D
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 61 - 62 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0053 - 0.0054 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

			74 dry (1) (2)		-67 dry (1) (2)		180. dry (1) (2)	
Temperature(°F)								
Moisture Content(%)								
Equilibrium at T, RH								
Source Code								
			Normalized	Measured	Normalized	Measured	Normalized	Measured
		Mean	350.	370	376	398	327	344
		Minimum	266	282	326	345	234	248
		Maximum	426	455	412	439	402	421
		C.V.(%)	15.9	16.0	8.69	8.93	17.3	16.8
F_1^{tu}		B-value	(3)		(3)		(3)	
		Distribution	Normal		Normal		Normal	
(ksi)		C_1	350		376		327	
		C_2	55.5		32.7		56.4	
		No. Specimens	6		6		6	
		No. Batches	1		1		1	
		Mean		13600		15900		14100
		Minimum		8100		13500		10400
		Maximum		17500		17200		16800
		C.V.(%)		24.6		9.23		14.9
e_1^{tu}		B-value		(3)		(3)		(3)
		Distribution		Normal		Normal		Normal
($\mu\epsilon$)		C_1		13600		15900		14100
		C_2		3350		1470		2100
		No. Specimens	6		6		6	
		No. Batches	1		1		1	
		Mean	21.6	22.9	22.0	23.3	23.2	24.4
		Minimum	21.3	22.4	20.9	22.2	22.3	23.6
		Maximum	22.0	23.3	23.2	24.5	23.7	25.0
E_1^t		C.V.(%)	1.41	1.58	3.35	3.26	2.24	2.17
(Msi)		No. Specimens	6		6		6	
		No. Batches	1		1		1	
		Mean		0.342		0.357		0.355
ν_{12}^t		No. Specimens	6		6		6	
		No. Batches	1		1		1	

(1) As fabricated
(2) Reference 4.9.1
(3) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(b)
C/PEEK -UT
IM-6/APC-2
Tension, 1-axis
$[0]_8$
250/D, 74/0.13%, 180/0.11%
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 61 - 62 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0053 - 0.0054 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		250.		74		180.	
Moisture Content(%)		dry		0.13		0.11	
Equilibrium at T, RH		(1)		160°F, 95%		(2)	
Source Code		(3)		(3)		(3)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
F_1^{tu} (ksi)	Mean	304	322	352	271	369	390
	Minimum	253	269	271	286	303	320
	Maximum	341	363	415	434	403	425
	C.V.(%)	11.4	11.4	14.6	14.2	12.3	12.2
	B-value	(4)		(4)		(4)	
	Distribution	Normal		Normal		Normal	
	C_1	304		352		369	
	C_2	34.7		51.4		45.3	
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
e_1^{tu} ($\mu\epsilon$)	Mean		14800		18100		16300
	Minimum		12500		15700		14400
	Maximum		16400		20800		17200
	C.V.(%)		11.8		10.8		6.70
	B-value		(4)		(4)		(4)
	Distribution		Normal		Normal		Normal
	C_1		14800		18100		16300
	C_2		1760		1960		1090
	No. Specimens		6		6		5
	No. Batches		1		1		1
E_1^t (Msi)	Mean	21.4	22.7	21.2	22.3	21.8	23.0
	Minimum	20.5	21.9	20.4	21.6	20.9	22.1
	Maximum	22.1	23.4	22.0	23.0	22.2	23.5
	C.V.(%)	2.70	2.42	3.15	3.04	2.42	2.42
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
ν_{12}^t	Mean		0.338		0.372		0.366
	No. Specimens	6		6		5	
	No. Batches	1		1		1	

(1) *As fabricated*
(2) Conditioned at 160°F, 96% relative humidity for 3 days (75% saturation)
(3) Reference 4.9.1
(4) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(c)
C/PEEK -UT
IM-6/APC-2
Tension, 1-axis
$[0]_8$
180/0.14%
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 61 - 62 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0053 - 0.0054 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3039-76

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		180.			
Moisture Content(%)		0.14%			
Equilibrium at T, RH		160°F, 95%			
Source Code		(1)			
		Normalized	Measured		
	Mean	364	385		
	Minimum	325	344		
	Maximum	411	436		
	C.V.(%)	10.2	10.1		
F_1^{tu} (ksi)	B-value	(2)			
	Distribution	Normal			
	C_1	364			
	C_2	37.2			
	No. Specimens	6			
	No. Batches	1			
	Mean		15400		
	Minimum		13600		
	Maximum		17200		
	C.V.(%)		9.24		
e_1^{tu} ($\mu\epsilon$)	B-value		(2)		
	Distribution		Normal		
	C_1		15400		
	C_2		1420		
	No. Specimens	6			
	No. Batches	1			
E_1^t (Msi)	Mean	21.2	22.4		
	Minimum	20.5	21.8		
	Maximum	22.2	23.2		
	C.V.(%)	3.14	2.77		
	No. Specimens	6			
	No. Batches	1			
ν_{12}^t	Mean		0.332		
	No. Specimens	6			
	No. Batches	1			

(1) Reference 4.9.1
(2) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(d)
C/PEEK -UT
IM-6/APC-2
Tension, 2-axis
$[90]_{16}$
74/D, -67/D, 180/D, 250/D
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 31 - 34 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: ASTM D3039-76

MODULUS CALCULATION:

NORMALIZED BY: Not normalized

Temperature(°F)		74	-67	180.	250.	
Moisture Content(%)		dry	dry	dry	dry	
Equilibrium at T, RH		(1)	(1)	(1)	(1)	
Source Code		(2)	(2)	(2)	(2)	
	Mean	9.41	9.67	11.1	9.07	
	Minimum	8.53	8.72	10.0	7.30	
	Maximum	10.6	10.7	12.2	9.72	
	C.V.(%)	9.35	6.52	8.87	10.1	
F_2^{tu}	B-value	(3)	(3)	(3)	(3)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	9.41	9.67	11.1	9.07	
	C_2	0.880	0.631	0.985	0.916	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Mean	7610	7120	10900	12300	
	Minimum	6650	6450	8850	8510	
	Maximum	8830	8180	14900	23600	
	C.V.(%)	11.2	8.15	20.0	45.5	
e_2^{tu}	B-value	(3)	(3)	(3)	(3)	
	Distribution	Normal	Normal	Normal	Nonpara.	
($\mu\epsilon$)	C_1	7610	7120	10900		
	C_2	850	581	2180		
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Mean	1.28	1.41	1.22	1.32	
	Minimum	1.24	1.35	1.17	1.27	
	Maximum	1.36	1.46	1.25	1.38	
E_2^t	C.V.(%)	3.33	3.32	2.13	3.44	
(Msi)	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	

(1) As fabricated
(2) Reference 4.9.1
(3) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(e)
C/PEEK -UT
IM-6/APC-2
Compression, 1-axis
$[0]_{16}$
74/D, -67/D, 180/D
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		72.0		-67		180.	
Moisture Content(%)		dry		dry		dry	
Equilibrium at T, RH		(1)		(1)		(1)	
Source Code		(2)		(2)		(2)	
		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	167	169	156	160	156	155
	Minimum	139	144	115	118	103	96.7
	Maximum	197	200.	179	181	195	190.
	C.V.(%)	13.3	13.3	16.0	15.6	20.2	20.4
F_1^{cu}	B-value	(3)		(3)		(3)	
	Distribution	Normal		Normal		Normal	
(ksi)	C_1	167		156		156	
	C_2	22.1		25.0		31.5	
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Mean		8790		7910		8010
	Minimum		7780		4510		5950
	Maximum		10500		9630		9350
	C.V.(%)		11.8		24.7		14.9
e_1^{cu}	B-value		(3)		(3)		(3)
	Distribution		Normal		Normal		Normal
($\mu\epsilon$)	C_1		8790		7910		8010
	C_2		1040		1950		1200
	No. Specimens	6		6		6	
	No. Batches	1		1		1	
	Mean	19.4	19.7	20.4	20.9	21.4	21.2
	Minimum	17.6	18.1	16.9	17.3	17.0	16.0
	Maximum	20.9	21.2	24.0	24.8	27.5	26.7
E_1^c	C.V.(%)	6.54	7.17	12.2	12.6	16.1	16.1
(Msi)	No. Specimens	6		6		6	
	No. Batches	1		1		1	

(1) As fabricated
(2) Reference 4.9.1
(3) B-values are not presented for screening data

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Table 4.9.1(f)
C/PEEK -UT
IM-6/APC-2
Compression, 1-axis
$[0]_{16}$
250/D, 74/0.12%, 180/0.097%
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: ASTM D3410A-87

MODULUS CALCULATION:

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

		250. dry (1) (3)		74 0.12 160°F, 95% (3)		180. 0.097 (2) (3)	
Temperature(°F) / Moisture Content(%) / Equilibrium at T, RH / Source Code		Normalized	Measured	Normalized	Measured	Normalized	Measured
	Mean	129	126	174	176	162	160
	Minimum	70.0	71.5	141	144	156	146
	Maximum	154	145	186	192	168	169
	C.V.(%)	23.6	21.8	9.6	9.7	3.25	5.36
F_1^{cu} (ksi)	B-value	(4)		(4)		(4)	
	Distribution	Normal		Normal		Normal	
	C_1	129		174		162	
	C_2	30.5		16.7		5.26	
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Mean	6860		8690		8310	
	Minimum	3380		6950		7500	
	Maximum	8990		12100		9390	
	C.V.(%)	28.7		23.5		8.94	
ϵ_1^{cu} ($\mu\epsilon$)	B-value	(4)		(4)		(4)	
	Distribution	Normal		Normal		Normal	
	C_1	6860		8690		8310	
	C_2	1970		2050		743	
	No. Specimens	6		6		5	
	No. Batches	1		1		1	
	Mean	21.2	20.7	21.4	21.6	19.5	19.3
	Minimum	19.6	19.0	18.8	19.3	18.7	18.6
	Maximum	24.7	23.2	23.9	23.9	20.0	20.7
E_1^c	C.V.(%)	8.47	7.37	8.60	7.38	2.91	4.42
(Msi)	No. Specimens	6		6		5	
	No. Batches	1		1		1	

(1) As fabricated
(2) Conditioned at 160°F, 95% relative humidity for 10 days (75% saturation)
(3) Reference 4.9.1
(4) B-values are not presented for screening data

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Table 4.9.1(g)
C/PEEK -UT
IM-6/APC-2
Compression, 1-axis
$[0]_{16}$
180/0.11%
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 60 - 62 % VOID CONTENT: 0.0 %
PLY THICKNESS: 0.0054 - 0.0058 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3410A-87

NORMALIZED BY: Ply thickness to 0.0055 in. and batch fiber volume to 60%

Temperature(°F)		180			
Moisture Content(%)		0.11			
Equilibrium at T, RH		160°F, 95%			
Source Code		(1)			
		Normalized	Measured		
	Mean	154	151		
	Minimum	105	98.5		
	Maximum	189	183		
	C.V.(%)	18.2	19.3		
F_1^{cu} (ksi)	B-value	(2)			
	Distribution	Normal			
	C_1	154			
	C_2	28.0			
	No. Specimens	6			
	No. Batches	1			
	Mean		8180		
	Minimum		6580		
	Maximum		9500		
	C.V.(%)		13.0		
ϵ_1^{cu} ($\mu\epsilon$)	B-value		(2)		
	Distribution		Normal		
	C_1		8180		
	C_2		1070		
	No. Specimens	6			
	No. Batches	1			
E_1^c (Msi)	Mean	20.3	19.8		
	Minimum	15.6	15.7		
	Maximum	25.3	24.6		
	C.V.(%)	18.4	17.6		
	No. Specimens	6			
	No. Batches	1			

(1) Reference 4.9.1
(2) B-values are not presented for screening data

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Table 4.9.1(h)
C/PEEK -UT
IM-6/APC-2
Shear, 12-axis
$[\pm 45]_{4s}$
74/D, -67/D, 180/D, 250/D
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 31 - 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 61 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0052 - 0.0056 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		74	-67	180.	250.	
Moisture Content(%)		dry	dry	dry	dry	
Equilibrium at T, RH		(1)	(1)	(1)	(1)	
Source Code		(2)	(2)	(2)	(2)	
	Mean	23.9	25.4	22.4	19.8	
	Minimum	18.9	18.1	17.2	14.2	
	Maximum	27.8	29.0	25.3	23.1	
	C.V.(%)	14.8	14.8	15.6	15.1	
F_{12}^{su}	B-value	(3)	(3)	(3)	(3)	
	Distribution	Normal	Normal	Normal	Normal	
(ksi)	C_1	23.9	25.4	22.4	19.8	
	C_2	3.53	3.77	3.49	2.98	
	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	0.78	0.91	0.78	0.71	
	Minimum	0.73	0.83	0.72	0.63	
	Maximum	0.83	0.96	0.86	0.79	
G_{12}^{s}	C.V.(%)	5.5	5.5	6.2	9.3	
(Msi)	No. Specimens	6	6	6	6	
	No. Batches	1	1	1	1	

(1) As fabricated
(2) Reference 4.9.1.
(3) B-values are not presented for screening data

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Table 4.9.1(i)
C/PEEK -UT
IM-6/APC-2
Shear, 12-axis
$[\pm 45]_{4s}$
74/0.21%, 180/0.17%, 180/0.20%
Screening Data

MATERIAL: IM-6 12k/APC-2 unidirectional tape

RESIN CONTENT: 31 - 32 wt% COMP. DENSITY: 1.55 g/cc
FIBER VOLUME: 61 % VOID CONTENT: 0.0 - 0.2 %
PLY THICKNESS: 0.0052 - 0.0056 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D3518-76

NORMALIZED BY: Not normalized

Temperature(°F)		74	180.	180.		
Moisture Content(%)		0.21	0.17	0.20		
Equilibrium at T, RH		160°F, 95%	(1)	160°F, 95%		
Source Code		(2)	(2)	(2)		
	Mean	23.0	23.3	20.0		
	Minimum	16.2	21.8	14.5		
	Maximum	26.7	24.0	26.1		
	C.V.(%)	15.4	3.85	22.4		
F_{12}^{su}	B-value	(3)	(3)	(3)		
	Distribution	Normal	Normal	Normal		
(ksi)	C_1	23.0	23.3	20.0		
	C_2	3.55	0.897	4.48		
	No. Specimens	6	5	6		
	No. Batches	1	1	1		
	Mean					
	Minimum					
	Maximum					
	C.V.(%)					
γ_{12}^{su}	B-value					
	Distribution					
($\mu\epsilon$)	C_1					
	C_2					
	No. Specimens					
	No. Batches					
	Mean	0.79	0.76	0.71		
	Minimum	0.65	0.74	0.64		
	Maximum	0.89	0.78	0.78		
G_{12}^{s}	C.V.(%)	10.	2.7	9.0		
(Msi)	No. Specimens	6	4	6		
	No. Batches	1	1	1		

(1) Conditioned at 160°F, 95% relative humidity for 27 days (75% saturation)
(2) Reference 4.9.1
(3) B-values are not presented for screening data

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REFERENCES

4.4.4 Rondeau, R.A., Askins, D. R., and Sjoblom, P., "Development of Engineering Data on New Aerospace Materials," University of Dayton Research Institute, UDR-TR-88-88, AFWAL-TR-88-4217, December 1988, Distribution authorized to DoD and DoD contractors only; critical technology; September 1988. Other requests for this document shall be referred to AFWAL/MLSE, OH 45433-6533.

4.9.1 Rondeau, R.A., Askins, D. R., and Sjoblom, P., "Development of Engineering Data on New Aerospace Materials," University of Dayton Research Institute, UDR-TR-88-88, AFWAL-TR-88-4217, December 1988, Distribution authorized to DoD and DoD contractors only; critical technology; September 1988. Other requests for this document shall be referred to AFWAL/MLSE, OH 45433-6533.

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5.6 ARAMID - PHENOLIC COMPOSITES.

5.7 ARAMID - SILICONE COMPOSITES.

5.8 ARAMID - POLYBENZIMIDAZOLE COMPOSITES.

5.9 ARAMID - PEEK COMPOSITES.

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7.6 BORON - PHENOLIC COMPOSITES.

7.7 BORON - SILICONE COMPOSITES.

7.8 BORON - POLYBENZIMIDAZOLE COMPOSITES.

7.9 BORON - PEEK COMPOSITES.

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8.9 ALUMINA - PEEK COMPOSITES.

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10.1 INTRODUCTION.

10.2 QUARTZ - EPOXY COMPOSITES.

10.3 QUARTZ - POLYESTER COMPOSITES.

10.4 QUARTZ - BISMALEIMIDE COMPOSITES.

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10.4.1 Astroquartz II/F650 8-harness satin weave.*

MATERIAL:	Astroquartz II/F650 8-harness satin weave
PREPREG:	Hexcel AQII581/F650 8-harness satin weave
FIBER:	J.P. Stevens Astroquartz II RESIN: Hexcel F650
T_g(dry):	600°F T_g(wet): T_g METHOD:
PROCESSING:	Autoclave cure: 375°F, 4 hours, 85 psig; Postcure: 475°F, 4 hours

Q/BMI Astroquartz II/F650 Summary

* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

Date of fiber manufacture		Date of testing	
Date of resin manufacture		Date of data submittal	4/89
Date of prepreg manufacture		Date of analysis	1/93
Date of composite manufacture			

LAMINA MEAN MECHANICAL PROPERTY SUMMARY

		RTD	CTD	RTW	ETW
F_1^{tu}	(ksi)				
E_1^t	(Msi)				
F_2^{tu}	(ksi)				
E_2^t	(Msi)				
F_1^{cu}	(ksi)				
E_1^c	(Msi)				
F_2^{cu}	(ksi)				
E_2^c	(Msi)				
F_{12}^{su}	(ksi)				
G_{12}^s	(Msi)				
F_{13}^{su}	(ksi)	6.41 S			
G_{13}^s	(Msi)				
F_{23}^{su}	(ksi)				
G_{23}^s	(Msi)				

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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* DATA WERE SUBMITTED BEFORE THE ESTABLISHMENT OF DATA DOCUMENTATION REQUIREMENTS (JUNE 1989). ALL DOCUMENTATION PRESENTLY REQUIRED WERE NOT SUPPLIED FOR THIS MATERIAL.

		Nominal	As Submitted	Test Method
Fiber Density	(g/cc)	2.17		
Resin Density	(g/cc)	1.27		
Composite Density	(g/cc)	1.78	1.73	
Fiber Areal Weight	(g/m^2)	285		
Ply Thickness	(in)	0.0100	0.010	

LAMINA MEAN THERMOPHYSICAL PROPERTY SUMMARY

	RTD	CTD	RTW	ETW

Classes of data: FA - Fully approved, I - Interim, S - Screening (See Volume 1, Section 8.2.1)

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Table 10.4.1(a)
Q/BMI 285-8HS
Astroquartz II/F650
Tension, 1-axis
$[0_f]_{12}$
75/D, 450/D
Screening Data

MATERIAL: Astroquartz II/F650 8-harness satin weave

RESIN CONTENT: 37 wt% COMP. DENSITY: 1.73 g/cc
FIBER VOLUME: 51 % VOID CONTENT:
PLY THICKNESS: 0.010 in.

TEST METHOD: MODULUS CALCULATION:
ASTM D2344

NORMALIZED BY: Not normalized

Temperature(°F)		75	450				
Moisture Content(%)		dry	dry				
Equilibrium at T, RH		(1)	(1)				
Source Code		21	21				
F_{13}^{su} (ksi)	Mean	6.41	6.56				
	Minimum	6.31	6.43				
	Maximum	6.50	6.72				
	C.V.(%)	1.06	1.69				
	B-value	(2)	(2)				
	Distribution	Normal	Normal				
	C_1	6.41	6.56				
	C_2	0.068	0.111				
	No. Specimens	5	5				
	No. Batches	1	1				
γ_{13}^{su} (μin/in)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	B-value						
	Distribution						
	C_1						
	C_2						
	No. Specimens						
	No. Batches						
G_{13}^{is} (Msi)	Mean						
	Minimum						
	Maximum						
	C.V.(%)						
	No. Specimens						
	No. Batches						

(1) As fabricated
(2) B-values for screening data are not presented.

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10.5 QUARTZ - POLYIMIDE COMPOSITES.

10.6 QUARTZ - PHENOLIC COMPOSITES.

10.7 QUARTZ - SILICONE COMPOSITES.

10.8 QUARTZ - POLYBENZIMIDAZOLE COMPOSITES.

10.9 QUARTZ - PEEK COMPOSITES.

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Appendix A1 MIL-HDBK-17A DATA

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A1.1 GENERAL INFORMATION. The data on polymer-matrix composite materials which were presented in MIL-HDBK-17A, dated January 1971, are presented in this appendix. MIL-HDBK-17 A has been superseded so these data are presented here so they can be referenced in a current publication. However, these data do not meet the data requirements in Volume 1.
The materials which were included in MIL-HDBK-17A are listed in Table A1. Of the sixteen materials, six are still available, five are no longer available, and the availability of the other five materials could not be determined. The data from the six available materials are provided in this appendix. The data from the remaining materials may be added as availability of the material or usefulness of the data is determined. Note that Narmco 5505 has been licensed to AVCO and those data are presented herein as AVCO 5505.

The table and figure numbers used in this appendix are similar to those in MIL-HDBK-17A. The chapter identification has been changed from 4 to A1 but the rest of all figure and table numbers has

TABLE A1 *Materials from MIL-HDBK-17A*

Available:
U.S. Polymeric E-720E/7781 (ECDE-1/0-550) Fiberglass Epoxy
Hexcel F-161/7743(550) Fiberglass Epoxy
Hexcel F-161/7781(ECDE-1/0-550) Fiberglass Epoxy
Narmco N588/7781 (ECDE-1/0-550) Fiberglass Epoxy
Narmco 506/7781 (ECDE-1/0-A1100) Fiberglass Phenolic
AVCO 5505 Boron Epoxy
Not available:
U.S. Polymeric E-779/7743 (Volan) Fiberglass Epoxy
3M XP251S Fiberglass Epoxy
U.S. Polymeric S-860/1581 (ECG-1/2-112) Neutral pH Fiberglass Silicone
U.S. Polymeric P670A/7781 (ECDE-1/0) Fiberglass Modified DAP Polyester
SP272 Boron Epoxy
Availability unknown:
Bloomingdale BP915/7781 (ECDE-1/0-550) Fiberglass Epoxy
Bloomingdale BP911/7781 (ECDE-1/0 Volan) Fiberglass Epoxy
Cordo E293/7781 (ECDE-1/0-550) Fiberglass Epoxy
Styrene-Alkyd Polyester/7781 Fiberglass
Cordo IFRR/7781 (ECDE-1/0) Fiberglass Modified DAP Polyester

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not been changed. For example, Table A1.40 presents is the same as Table 4.40 in MIL-HDBK-17A. The MIL-HDBK-17A text describing the test program and methods is reproduced in Sections A1.2 through A1.4.

A1.2 INTRODUCTION. The laminate properties presented in this chapter have been generated in test programs conducted at the U.S. Forest Products Laboratory and elsewhere (Reference A1.2).[1] Properties are given for fiberglass with epoxy, phenolic, silicone and polyester resins and for boron with epoxy. Additional information on these and other material combinations will be issued as supplements or revisions of the present handbook edition.

A1.3 HANDBOOK TEST PROGRAM.

A1.3.1 Objectives. The objectives of the handbook test program are to obtain statistically significant data for materials currently in use and to determine the degree of reproducibility attained in their fabrication. A minimum requirement is that test results include data from three sets of panels which are representative of the manufacturing procedures employed by three different fabricators. The properties listed in the charts and tables of this chapter represent test results from only one set of panels for each material system. Properties are therefore not given minimum values and are considered to be "typical" for each material. When the minimum number of tests has been completed for a material, its properties will be assigned values on a B-basis; that is, the value above which 90 percent of the population of values is expected to fall with a confidence of 95 percent.

A1.3.2 Preimpregnated materials. All test panels are fabricated from prepregs. Emphasis is placed on materials for use as facings in sandwich type structures. The prepregs for facings are normally processed to conform with two methods of sandwich fabrication. These are the laminate grades for two-step sandwich constructions and the controlled flow adhesive grades for one-step sandwich constructions. Only laminates simulating precured facings, that is, for use in two-step sandwiches, have been subjected to the narrow coupon tests listed in this chapter. The controlled flow adhesive prepregs are best tested as sandwich panels, and such testing is not at present included in the handbook program.

The prepreg materials comply with the specifications established by the individual fabricators. In general, the materials are autoclave molding grades with flows controlled to attain minimum bleedout and optimum bonding of the plies. When possible handling characteristics are specified consistent with the objectives of collimated plies in the laminate and the retention of fiber orientation during lay-up and cure.

Imposed tolerances on the gravimetric resin content of the prepregs are dependent on the type of reinforcement. For bidirectional woven broadgoods such as style 7781 fabric, the resin fraction is specified as not varying by more than two percent from the assigned devolatilized resin content. For directionally woven broadgoods such as style 7743 fabric, and nonwoven parallel fiber tapes such as XP251S, variation from the assigned devolatilized resin content is not to exceed three percent.

A1.3.3 Test panels. A minimum size of the test panels has been established as two feet parallel to the warp direction by three feet parallel to the width for woven fabrics. For the non-woven

[1]Exceptions are the data for fiberglass-polyester laminates, taken from earlier sources, and the data for boron-epoxy panels which were compiled under special contract and published separately (Reference A1.2).

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laminates, including unidirectional, crossplied and quasi-isotropic configurations, the three foot dimension is parallel to the fiber direction in the outer plies.

It is desirable that the test laminates be fabricated so that fiber alignment and orthotropy are maintained and that they are symmetrically balanced. Such conditions are generally attained in the test panels and they are designated in the following data summary tables as balanced and parallel. One set of panels (Table A1.1) is not balanced. In this case the laminates are parallel plied.

A1.3.4 Test procedures. Conventional uniaxial tests are conducted at constant crosshead rates. The direction parallel to the warp of woven fabrics is designated as the 0° or 1-direction. The direction perpendicular to the 0° direction is designated as the 90° or 2-direction. For non-woven unidirectional laminates, the 0° direction corresponds to the fiber direction. For crossplied and quasi-isotropic laminates, the 0° direction corresponds to the fiber direction in the outer plies.

A1.3.4.1 *Tensile tests.* Tensile tests for woven fabric laminates have been conducted initially using the method of ASTM D 638 and Type I specimens (Reference 1.3.4.1(a)). Later tests are conducted with a modified specimen (Reference A1.2) and the method is designated as MIL-HDBK-17 tensile test. Tab ended specimens are used to test the 0° tensile properties of the non-woven unidirectional laminates (Reference A1.3.4.1(b)).

A1.3.4.2 *Compression tests.* Compression tests have been conducted with the end clamped and jig stabilized ASTM D 695 specimen (Reference A1.3.4.2) and with the MIL-HDBK-17 compression specimen (Reference A1.2) in which the specimen and fixture have been modified.

A1.3.4.3 *Shear tests.* The picture frame method (Reference A1.2) has been used to determine the 0° - 90° shear properties of one material system at three resin fractions (Figure A1.6.3). In these tests it is assumed that 88 percent of the load is reacted by the specimen, while the pins in the fixture react the remainder. The other materials are tested by a modified rail shear method (Reference A1.3.4.3).

A1.3.4.4 *Interlaminar shear.* Interlaminar shear properties are determined by the short beam test method (Reference A1.3.4.1(b)), or by the method of ASTM D 2733-68T when indicated (Reference 1.3.4.4).

A1.3.4.5 *Flexural tests.* Flexural properties are determined by the method of ASTM D 790 (Reference 1.3.4.5).

A1.3.4.6 *Bearing strength.* Bearing strengths are determined by the method of ASTM D 953 (Reference 1.3.4.6).

A1.3.5 Dry conditioning. Specimens are dry conditioned by allowing them to attain equilibrium at 70°F to 75°F and 45 percent to 55 percent relative humidity for a minimum of ten days. When tested at other than room temperature, the dry conditioned specimens are soaked at the test temperature for one-half hour prior to applying load.

A1.3.6 Wet conditioning. Specimens are wet conditioned at 125°F and 95 percent to 100 percent relative humidity for 1000 hours (42 days). When tested at temperatures below freezing, the wet conditioned specimens are cycled four times from the wet condition at 125°F to the sub-freezing test temperature; the dwell time at each temperature being one-half hour. Wet specimens tested at 160°F are soaked for one-half hour at this temperature immediately prior to testing. Some materials are shown as being tested at 220°F after wet conditioning. Such testing has been discontinued since these results appear inconclusive.

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A1.3.7 Test schedule. The 0° and 90° tension and compression properties are determined at three reference temperatures, 65°F, 70°F - 75°F and 160°F, for both dry and wet conditioned specimens. Dry conditioned specimens are tested at maximum temperature for those materials which are potentially serviceable at elevated temperatures. Ten test results are obtained for the stress-strain relations at each of these conditions. Tests at intermediate temperatures are conducted to verify property changes, in which cases five specimens are tested. Ten test results are also required for the 0° - 90° shear at -65°F, 70°F - 75°F, and 160°F in the dry condition. Five tests are conducted at 70°F - 75°F to determine the stress-strain relations for Poisson's ratio. Flexure, bearing and interlaminar shear are determined in the 0° direction and dry condition at -65°F, 70°F - 75°F and 160°F. Five specimens are tested for each temperature.

A1.4 DATA PRESENTATION. Uniaxial tension, compression and shear are shown as stress-strain relations at each temperature and the properties are summarized in tabular form. Flexural, bearing and interlaminar shear properties are listed in summary tables. Poisson's ratio is shown as the response of the 0° elongation and 90° contraction to the applied tensile stress.

When ten or more results are available at a test condition, average values and the associated standard deviations are given in the tables. Stress-strain relations are plotted as an average curve and a plot of the average minus three times the standard deviation is also shown. When five to nine results are obtained from a test condition, average, maximum, and minimum values and curves are shown.

A1.4.1 Epoxy-fiberglass laminates. All data on fiberglass-epoxy systems are results obtained from the handbook test program. Properties are summarized in Tables A1.1 through A1.8. Detailed data are shown in Figures A1.1.1(a) through A1.8.5. [Four of the nine materials are known to be available.]

A1.4.2 Phenolic-fiberglass laminates. Handbook tested properties are summarized in Table A1.40 and Figures A1.40.1(a) through A1.40.5 for one fiberglass-phenolic system. [This material is available.]

A1.4.3 Silicone-fiberglass laminates. Partial handbook test results were listed in MIL-HDBK-17A for one fiberglass-silicone system. [This material is not available]

A1.4.4 Polyester-fiberglass laminates. Previous data for fiberglass-polyester laminates were listed in MIL-HDBK-17A. [None of these materials are known to be available.]

A1.4.5 Boron-epoxy laminates. Data on two boron-epoxy systems have been abstracted from the literature (Reference A1.4.5) and are presented in Tables A1.110 and A1.111 and in Figures A1.110.1(a) through A1.111.3. [One of these materials is available.]

The laminate thickness is controlled by the number of plies in the construction and the desired resin content. In general, the thickness of woven fabric laminates is maintained at eight plies, except for low resin content laminates which may require as many as ten plies. Nonwoven laminate monolayers are constructed with six plies to reduce the shear lag apparent in testing, and eight plies for the crossplied and quasi-isotropic panels.

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TABLE A1.1 *Summary of Mechanical Properties of U.S. Polymeric E-720/7781 (ECDE/0-550) Fiberglass Epoxy*

Fabrication	Lay-up: Parallel		Vacuum: None		Pressure: 55-65 PSI		Bleedout: Edge & Vertical		Cure: 2 hr/350°F		Postcure: 4 hrs/400°F		Plies: 8	
Physical Properties	Weight Percent Resin: 34.9				Avg. Specific Gravity: 1.78				Avg. Percent Voids: 2.0				Avg. Thickness: 0.082 inches	
Test Methods	Tension: ASTM-D 638 TYPE-1		Compression: MIL-HDBK-17		Shear: Rail				Flexure: ASTM-D 790		Bearing: ASTM-D 953		Interlaminar Shear: Short Beam	
Temperature	-65°F				75°F				160°F				400°F	
Condition	Dry		Wet		Dry		Wet		Dry		Wet		Dry	
	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension														
ultimate stress, ksi 0°	69.2	1.6	69.1	1.7	60.4	1.7	55.7	1.5	52.5	1.0	42.9	0.8	44.8	2.0
90°	56.0	2.0	56.5	2.0	49.0	1.8	45.9	1.4	42.3	1.2	36.9	1.1	34.9	1.6
ultimate strain, % 0°	2.93	0.08	2.70	0.11	2.43	0.14	2.12	0.08	2.05	0.08	1.61	0.06	1.80	0.20
90°	2.92	0.22	2.54	0.19	2.33	0.09	2.04	0.09	1.98	0.08	1.70	0.13	1.72	0.22
proportional limit, ksi 0°														
90°														
initial modulus, 10^6 psi 0°	3.30		3.38		3.12		3.12		2.95		2.76		2.60	
90°	2.90		3.02		2.82		2.78		2.50		2.65		2.30	
secondary modulus, 10^6 psi 0°	2.30		2.85		2.45		2.50		2.46		2.37			
90°	1.90		1.74		2.05		2.19		2.01		1.97			
Compression														
ultimate stress, ksi 0°	77.1	4.0	75.0	3.7	64.8	2.9	57.3	3.8	54.0	1.4	46.2	1.4	23.8	2.2
90°	57.2	2.7	53.9	2.7	50.2	2.9	45.2	2.4	40.8	2.9	36.2	3.1	14.7	1.6
ultimate strain, % 0°	2.48	0.16	2.44	0.15	2.14	0.11	1.99	0.09	1.86	0.08	1.62	0.06	1.12	0.22
90°	1.93	0.16	1.81	0.19	1.70	0.14	1.58	0.14	1.46	0.17	1.37	0.15	0.91	0.08
proportional limit, ksi 0°														
90°														
initial modulus, 10^6 psi 0°	3.50		3.45		3.25		3.10		3.15		3.03		2.45	
90°	3.20		3.26		3.21		3.03		2.99		2.85		1.85	
Shear														
ultimate stress, ksi 0°-90° ±45°	17.5				14.3	0.6			11.2					

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	115.6	119.4	111.5	91.7	93.4	90.3	69.4	71.1	67.2
proportional limit, ksi	0°	88.1	100.7	77.5	32.5	36.2	30.8	56.2	62.8	49.4
initial modulus, 10^6 psi	0°	2.87	2.91	2.74	3.21	3.36	3.03	2.81	2.87	2.76
Bearing										
ultimate stress, ksi	0°	74.1	78.4	70.7	60.8	64.4	58.2	50.0	53.0	47.9
stress at 4% elong., ksi	0°	32.1	34.8	29.1	23.9	34.2	20.1	18.1	21.5	15.9
Interlaminar Shear										
ultimate stress, ksi	0°	7.09	7.36	6.80	5.90	6.07	5.72	6.05	6.16	5.91

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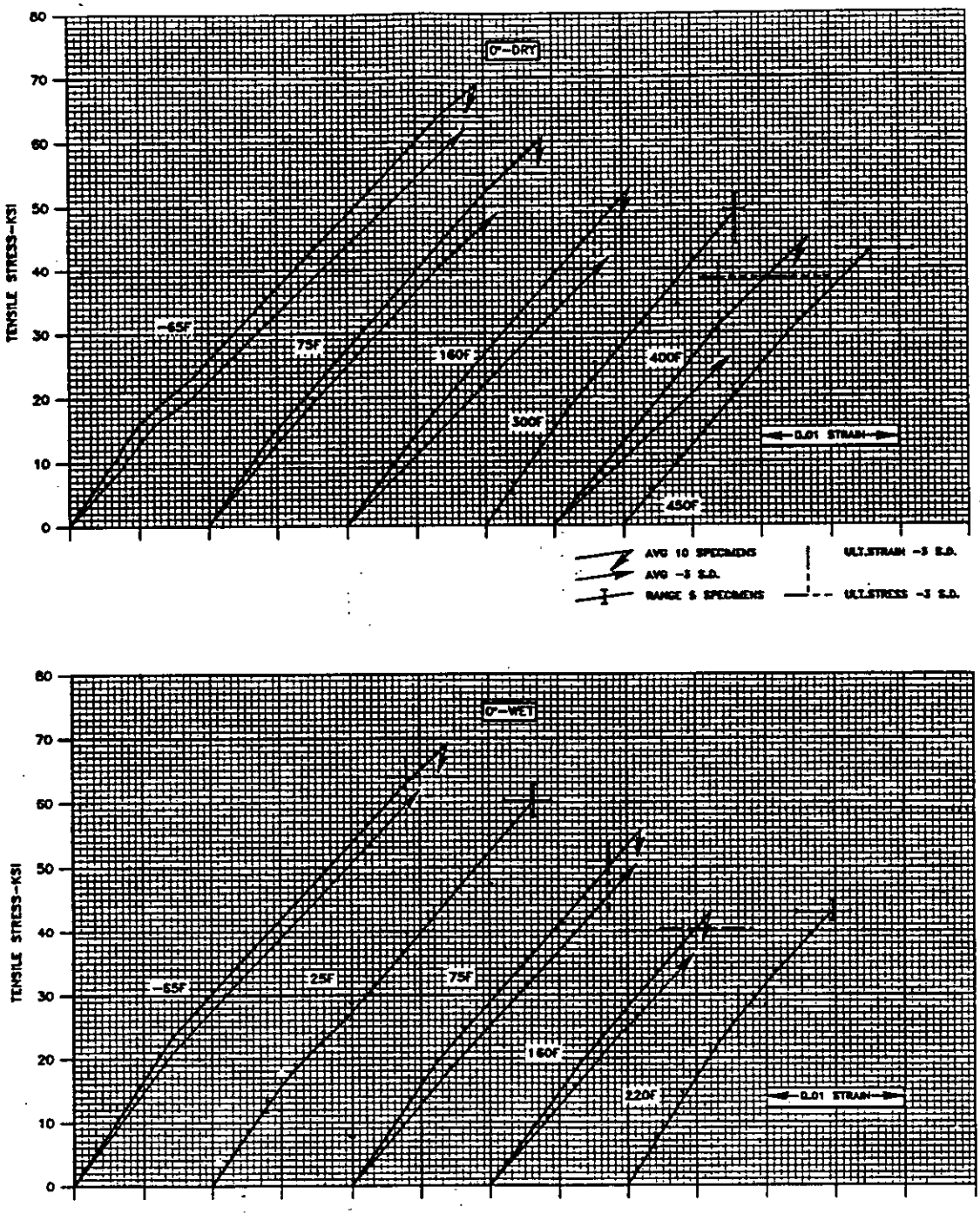


FIGURE A1.1.1(a) *Tensile stress-strain for E-720E/7781 fiberglass epoxy loaded in the 0° direction.*

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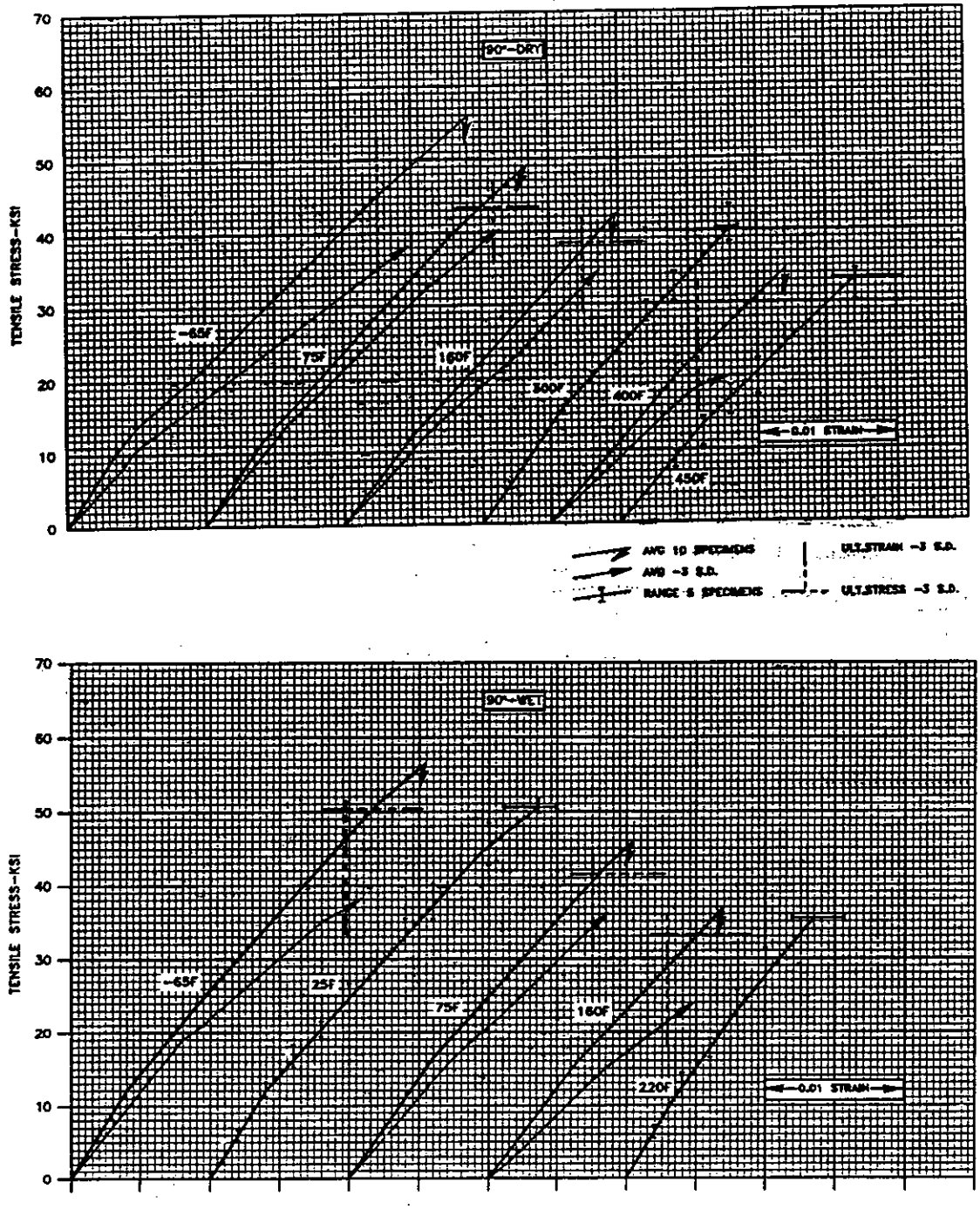
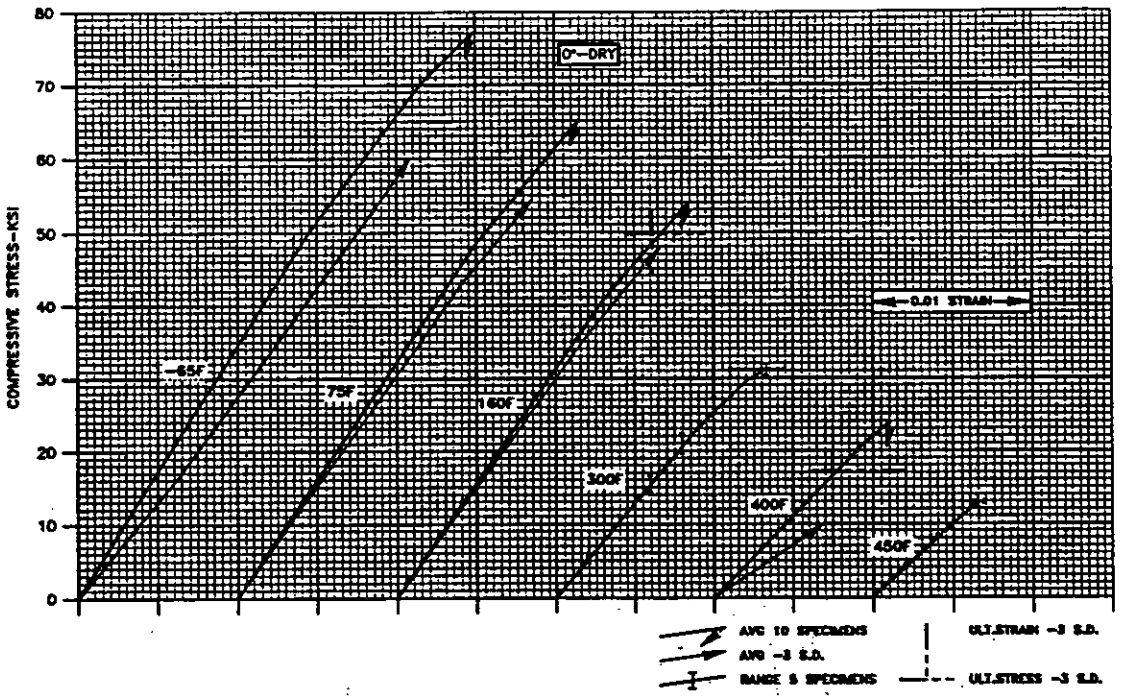


FIGURE A1.1.1(b) *Tensile stress-strain for E-720E/7781 fiberglass epoxy in the 90°direction.*

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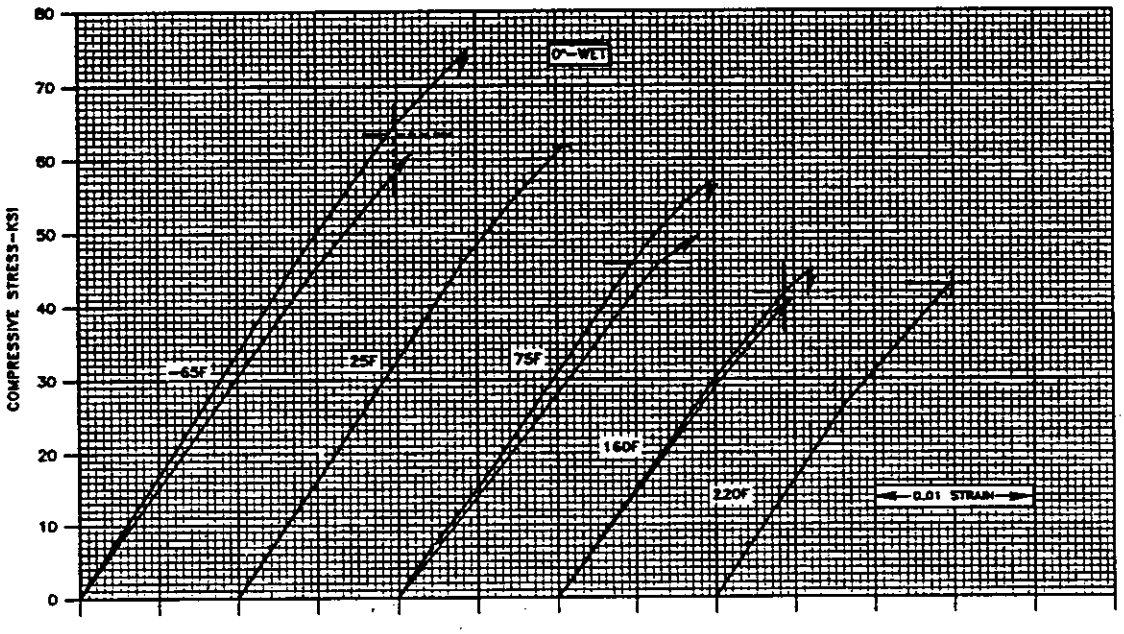


FIGURE A1.1.2(a) *Compressive stress-strain for E-720E/7781 fiberglass epoxy loaded in the 0° direction.*

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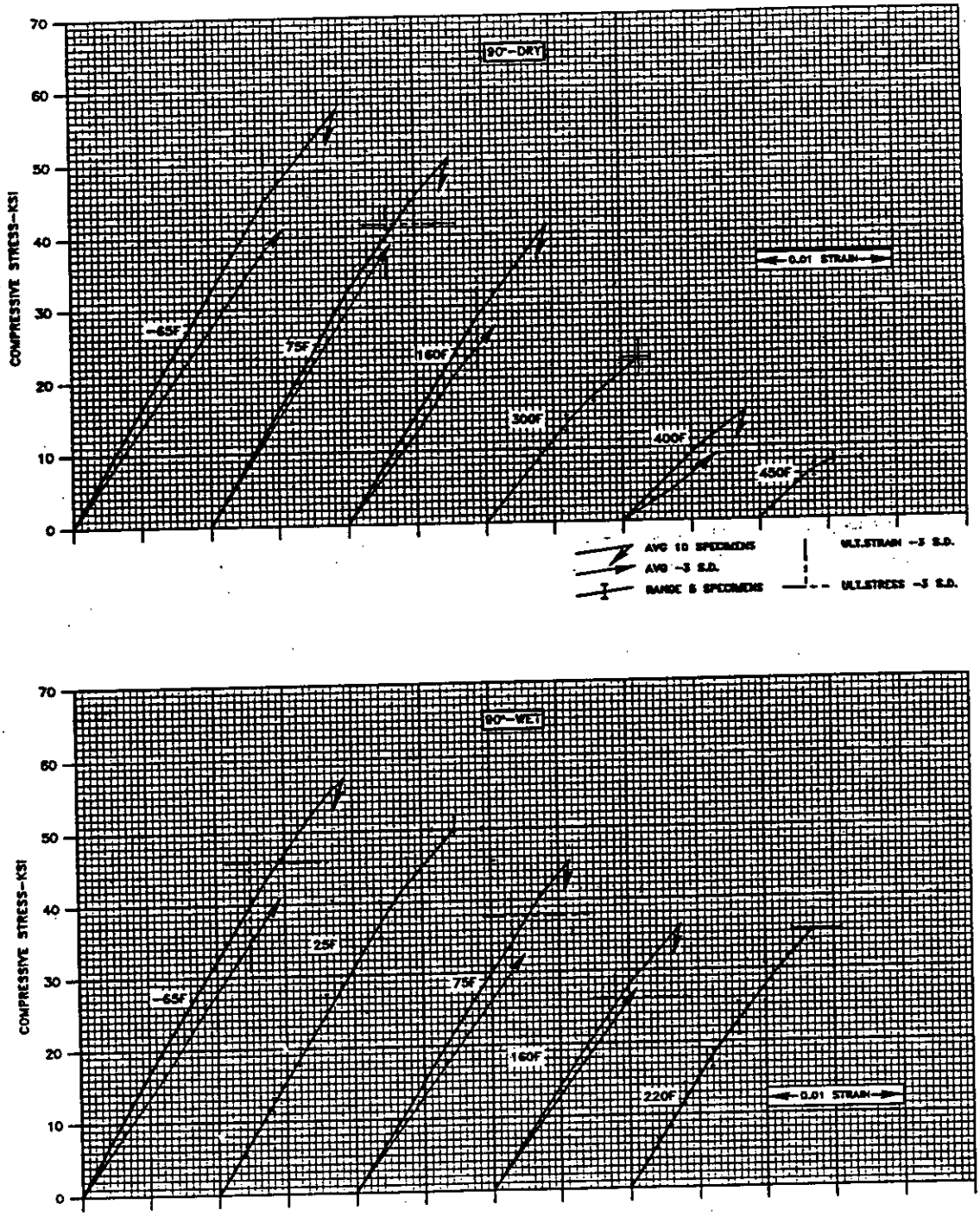


FIGURE A1.1.2(b) *Compressive stress-strain for E-720E/7781 fiberglass epoxy loaded in the 90° direction.*

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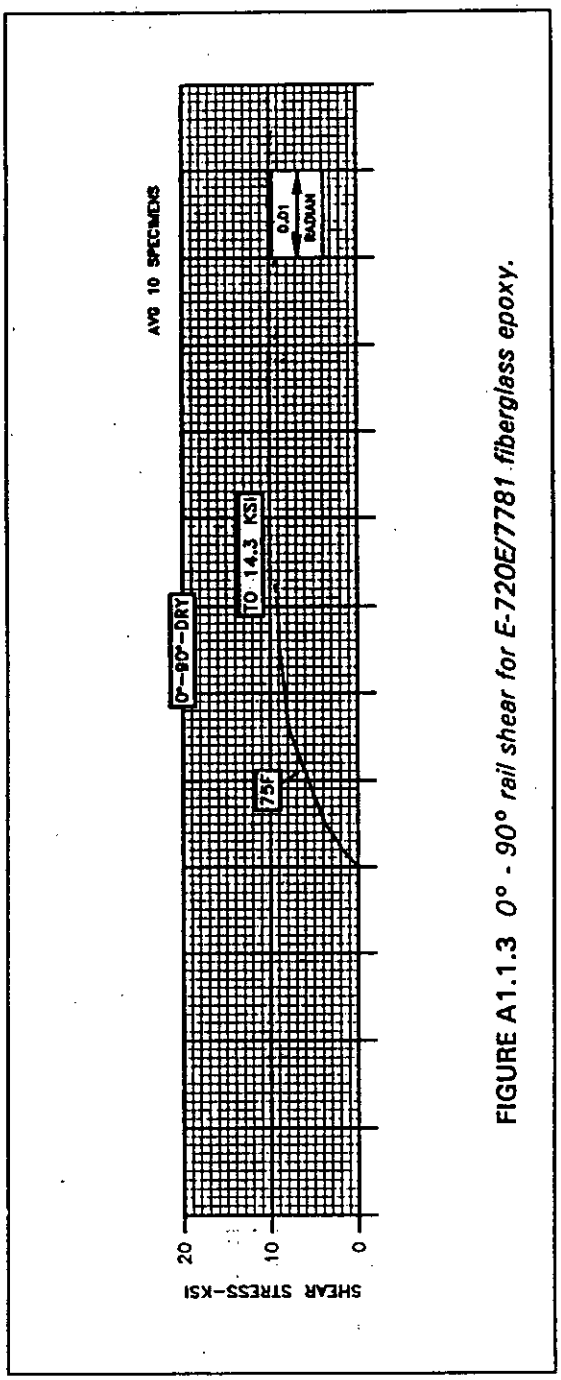


FIGURE A1.1.3 *0° - 90° rail shear for E-720E/7781 fiberglass epoxy.*

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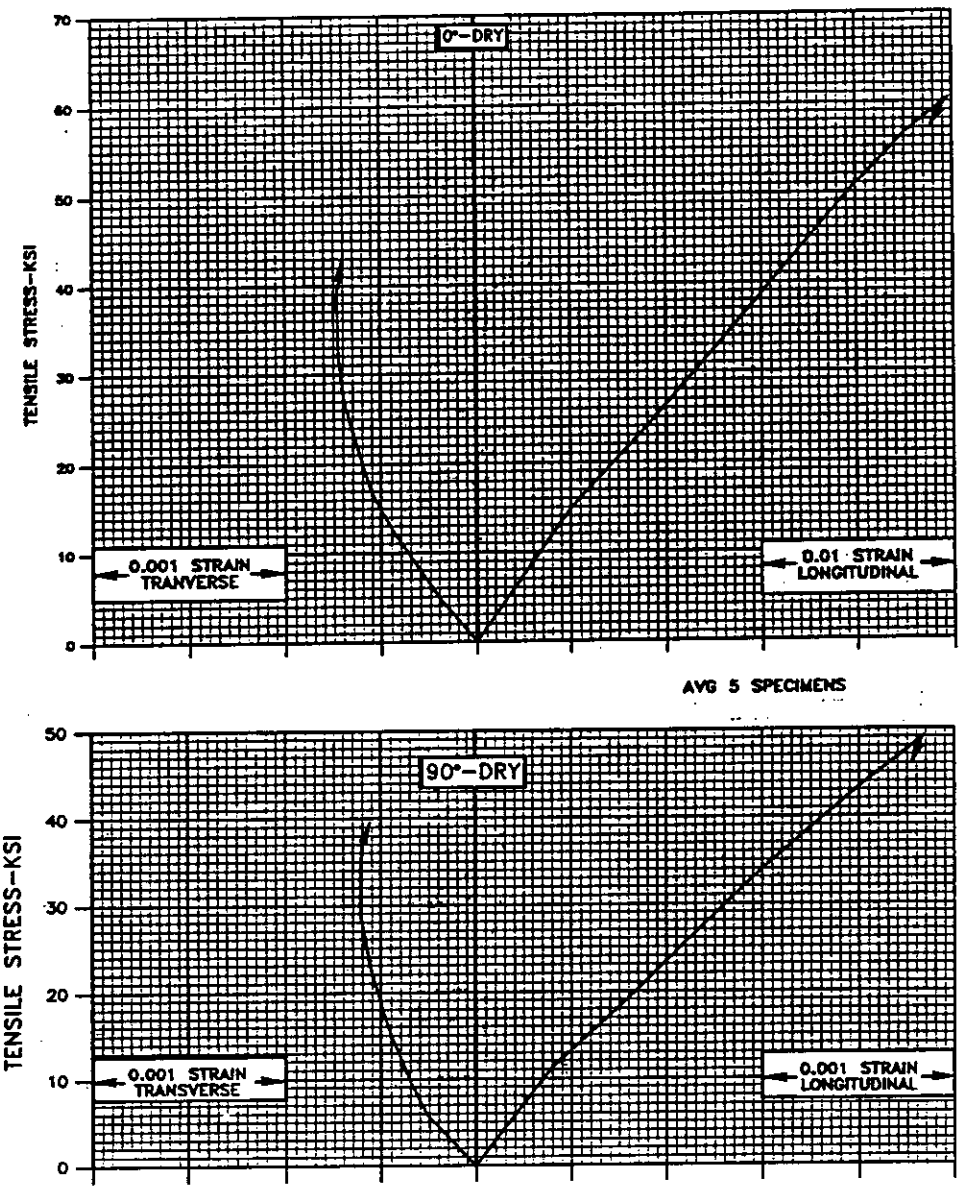


FIGURE A1.1.4 *Poisson effects for E-720E/7781 fiberglass epoxy.*

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TABLE A1.3 *Summary of mechanical properties of Hexcel F-161/7742(550) fiberglass epoxy.*

Fabrication	Lay-up: Balanced	Vacuum: 14 psi	Pressure: 35 psi	Bleedout: Pinched Edge	Cure: 2 hr/350°F	Postcure: 2 hr/350°F	Plies: 8								
Physical Properties	Weight Percent Resin: 32.4 v_f = 0.496	Avg. Specific Gravity: 1.85	Avg. Percent Voids: 3.0	Avg. Thickness: 0.086 inches											
Test Methods	Tension: ASTM-D638 TYPE 1	Compression: MIL-HDBK-17	Shear: Rail	Flexure: ASTM-D790	Bearing: ASTM-D953	Interlaminar Shear: Short Beam									
Temperature	-65°F				75°F				160°F				400°F		
Condition	Dry		Wet		Dry		Wet		Dry		Wet		Dry		
	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	
Tension															
ultimate stress, ksi	0°	111.3	1.12	107.3	3.60	95.5	7.57	87.3	5.2	80.9	4.05	71.7	2.73	74.5	5.90
	90°	9.84	0.78	9.42	0.59	8.15	0.40	7.27	0.28	6.78	0.18	6.16	0.21	6.59	0.41
ultimate strain, %	0°	2.10	0.31	2.11	0.10	1.88	0.10	1.72	0.17	1.56	0.15	1.35	0.12	1.64	0.09
	90°	2.43	0.25	2.03	0.21	1.82	0.23	1.20	0.28	1.26	0.19	0.61	0.13	1.44	0.19
proportional limit, ksi	0°	86.2		87.8		74.7		81.5		64.0		65.4		61.0	
	90°	5.6		5.0		5.2		4.8		5.0		5.0		3.0	
initial modulus, 10^6 psi	0°	5.42		5.35		5.30		5.55		5.36		5.47		4.52	
	90°	1.61		1.73		1.73		1.41		1.11		1.30		0.74	
secondary modulus, 10^6 psi	0°					5.15									
	90°					0.09									
Compression															
ultimate stress, ksi	0°	95.0	7.42	89.7	7.0	75.9	5.43	67.4	4.43	66.3	5.53	55.0	2.80	26.7	1.93
	90°	40.3	1.93	37.6	2.93	32.1	2.87	30.4	1.27	27.4	1.93	23.0	1.30	8.3	0.90
ultimate strain, %	0°	1.90	0.11	1.83	0.14	1.58	0.11	1.36	0.11	1.47	0.08	1.22	0.06	0.68	0.08
	90°	2.57	0.16	2.46	0.25	2.51	0.19	2.38	1.90	2.58	0.22	2.53	0.30	1.62	0.12
proportional limit, ksi	0°	83.0		70.0		52.2		49.8		55.6		40.8		20.0	
	90°	18.1		15.0		11.9		10.6		9.2		8.2			
initial modulus, 10^6 psi	0°	5.02		4.98		4.96		5.09		4.59		4.66		4.12	
	90°	1.91		1.88		1.65		1.77		1.46		1.37			
Shear															
ultimate stress, ksi	0°-90° ±45°	12.5				9.2	0.2			7.7					

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	203.0	210.0	196.0	160.0	163.0	155.0	138.0	142.0	135.0
proportional limit, ksi	0°	153.0	158.0	147.0	127.0	139.0	116.0	116.0	118.0	112.0
initial modulus, 10^6 psi	0°	5.71	5.80	5.63	5.18	5.27	5.10	5.43	5.46	5.32
Bearing										
ultimate stress, ksi	0°	79.4	90.2	64.8	58.8	63.2	52.7	53.7	57.5	50.6
stress at 4% elong., ksi	0°	37.9	45.6	31.5	23.0	27.1	19.5	21.9	23.6	20.5
Interlaminar Shear										
ultimate stress, ksi	0°	9.55	10.15	8.72	9.35	9.55	9.17	8.31	8.65	8.02

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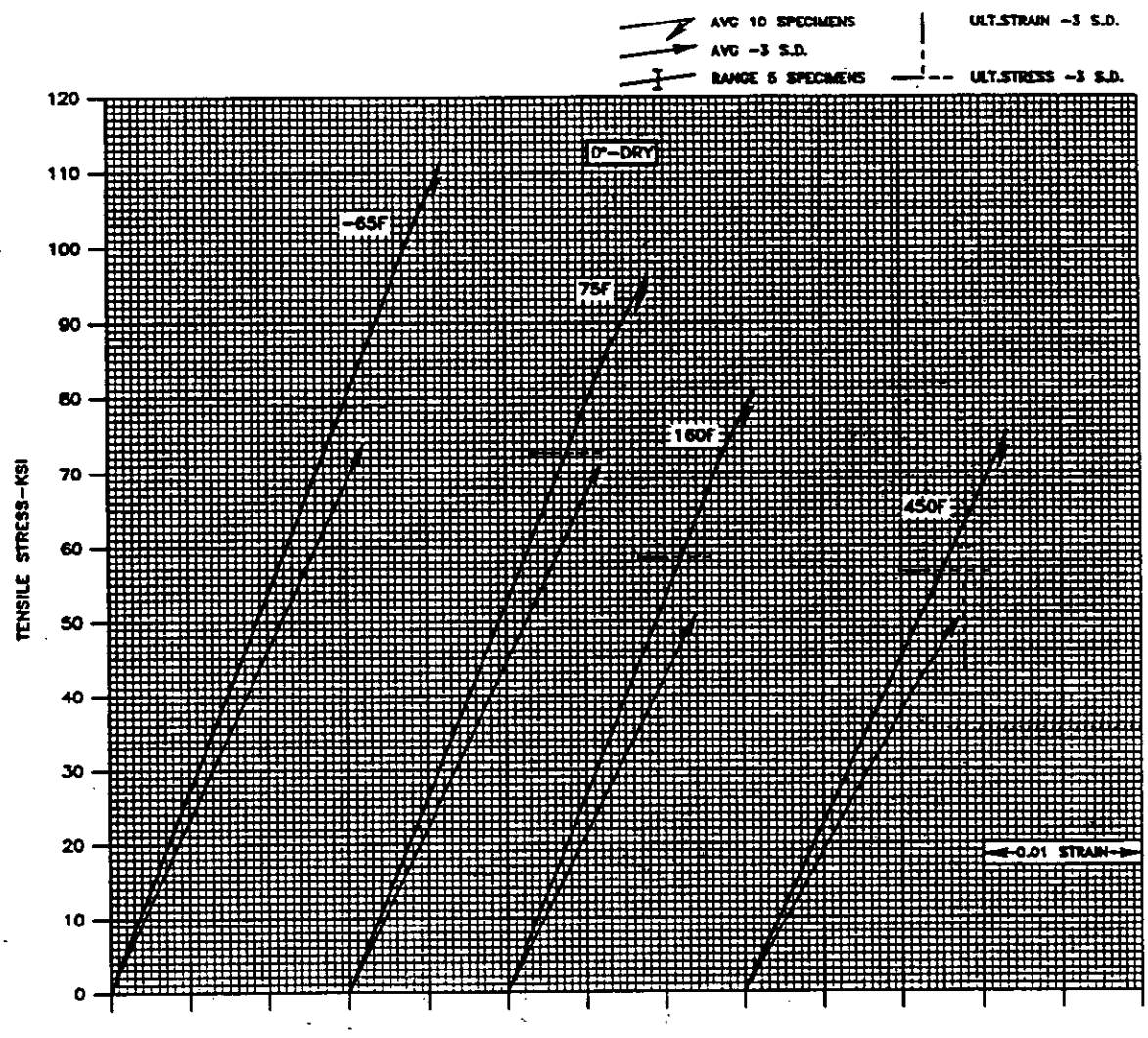


FIGURE A1.3.1(a) *Tensile stress-strain for F-161/7743 fiberglass epoxy loaded in the 0° direction*, continued on next page.

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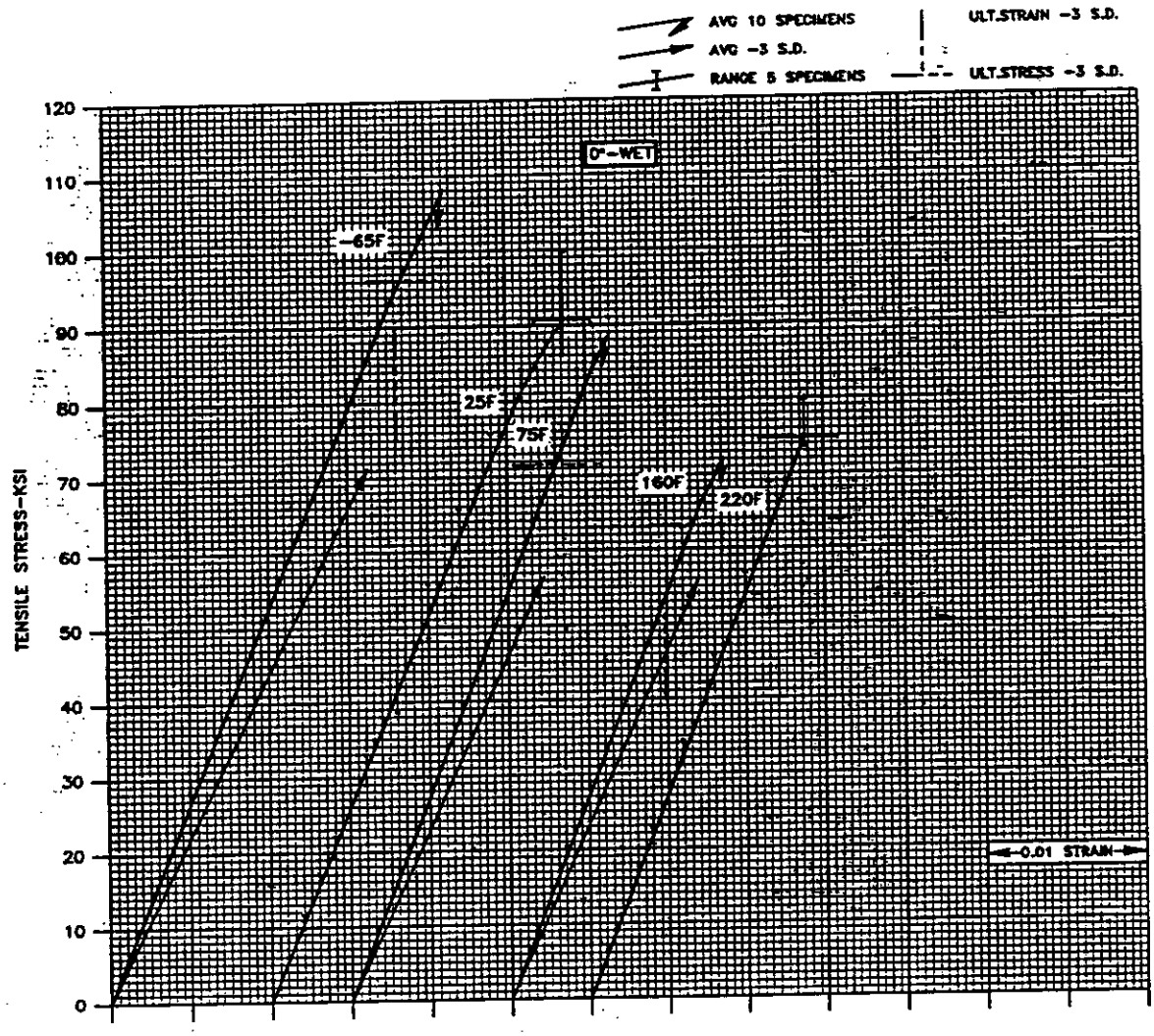


FIGURE A1.3.1(a) *Tensile stress-strain for F-161/7743 fiberglass epoxy loaded in the 0° direction,* concluded.

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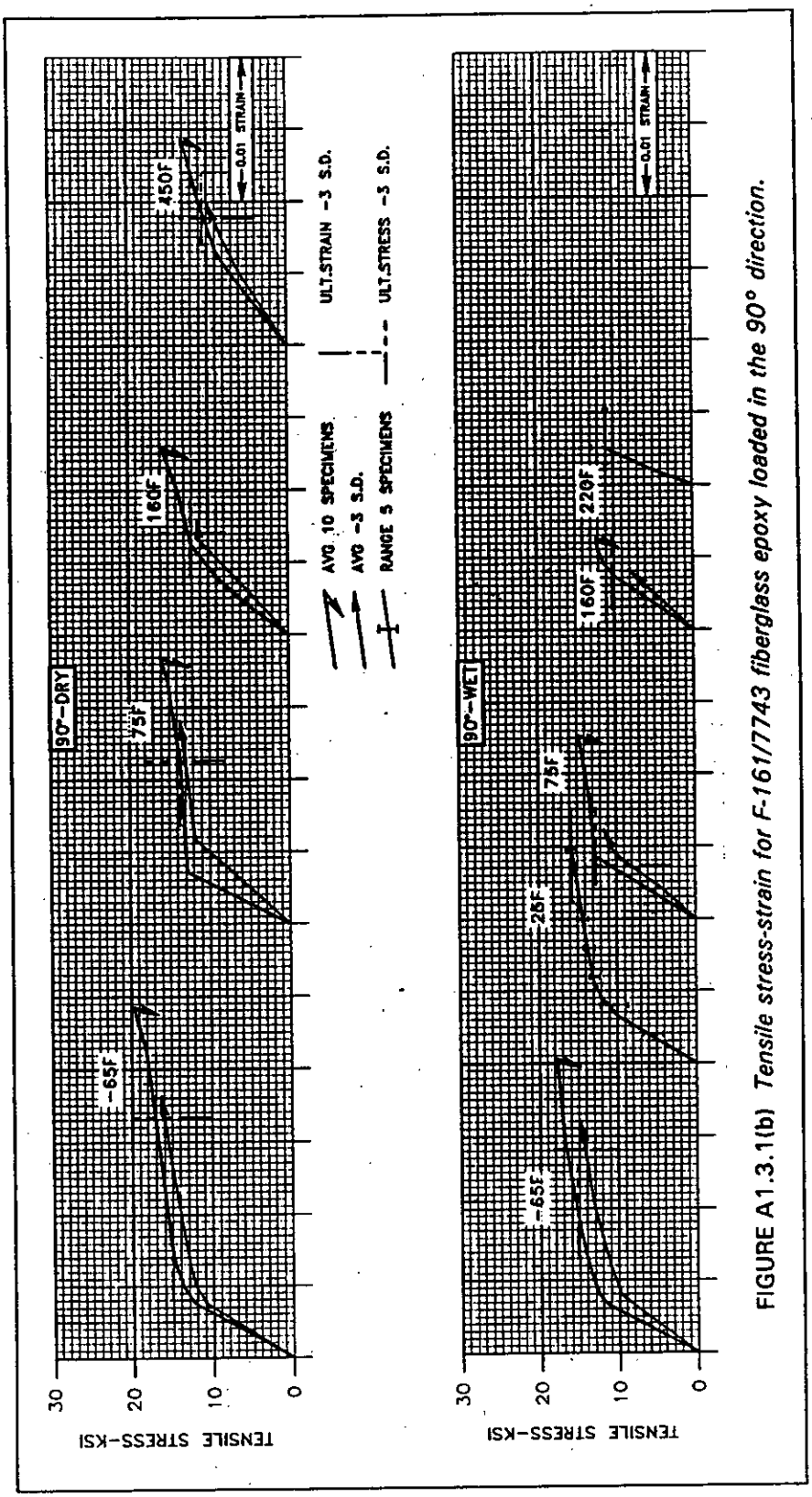


FIGURE A1.3.1(b) *Tensile stress-strain for F-161/7743 fiberglass epoxy loaded in the 90° direction.*

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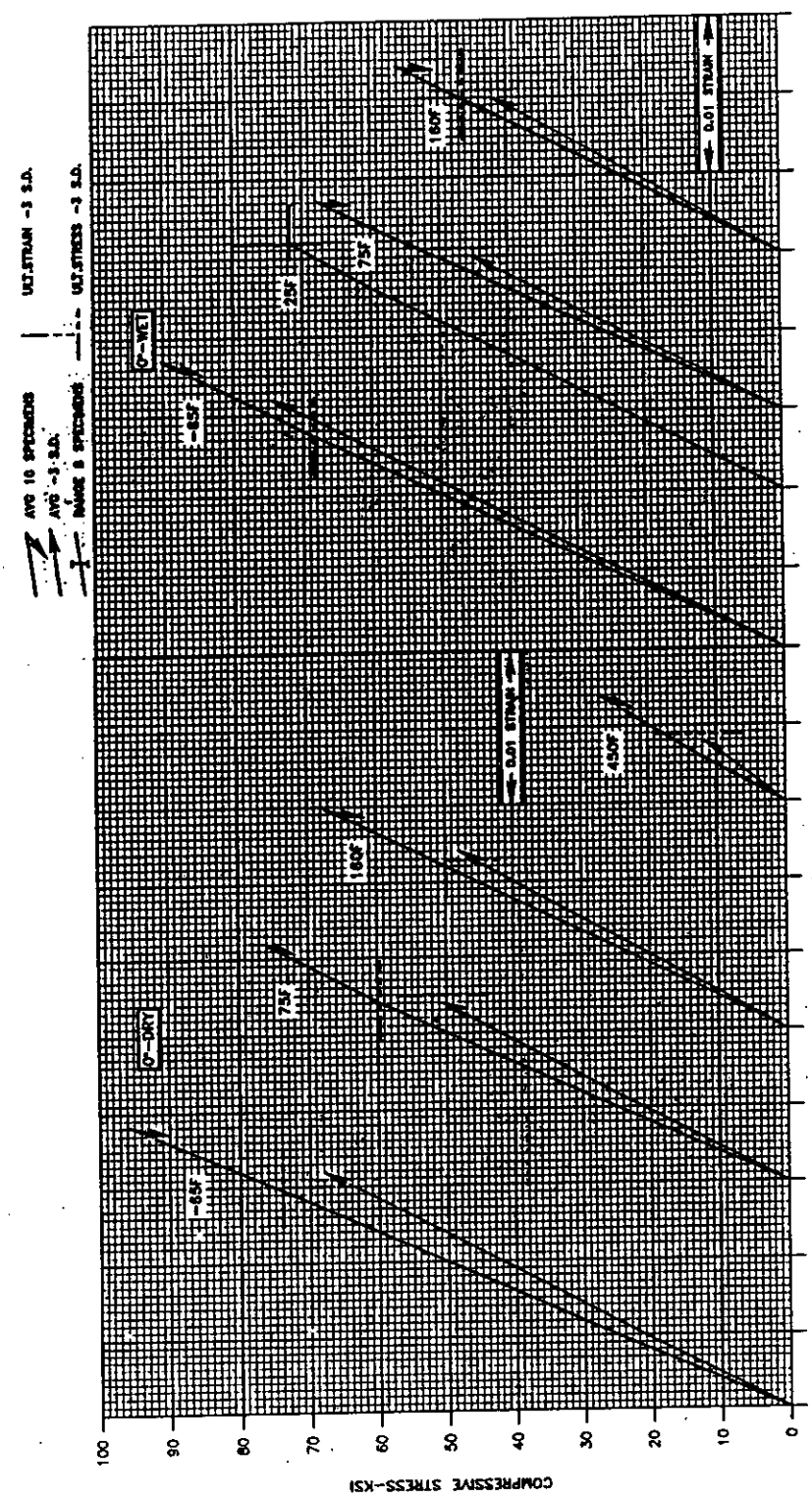


FIGURE A1.3.2(a) *Compressive stress-strain for F-161/7743 fiberglass epoxy loaded in the 0° direction.*

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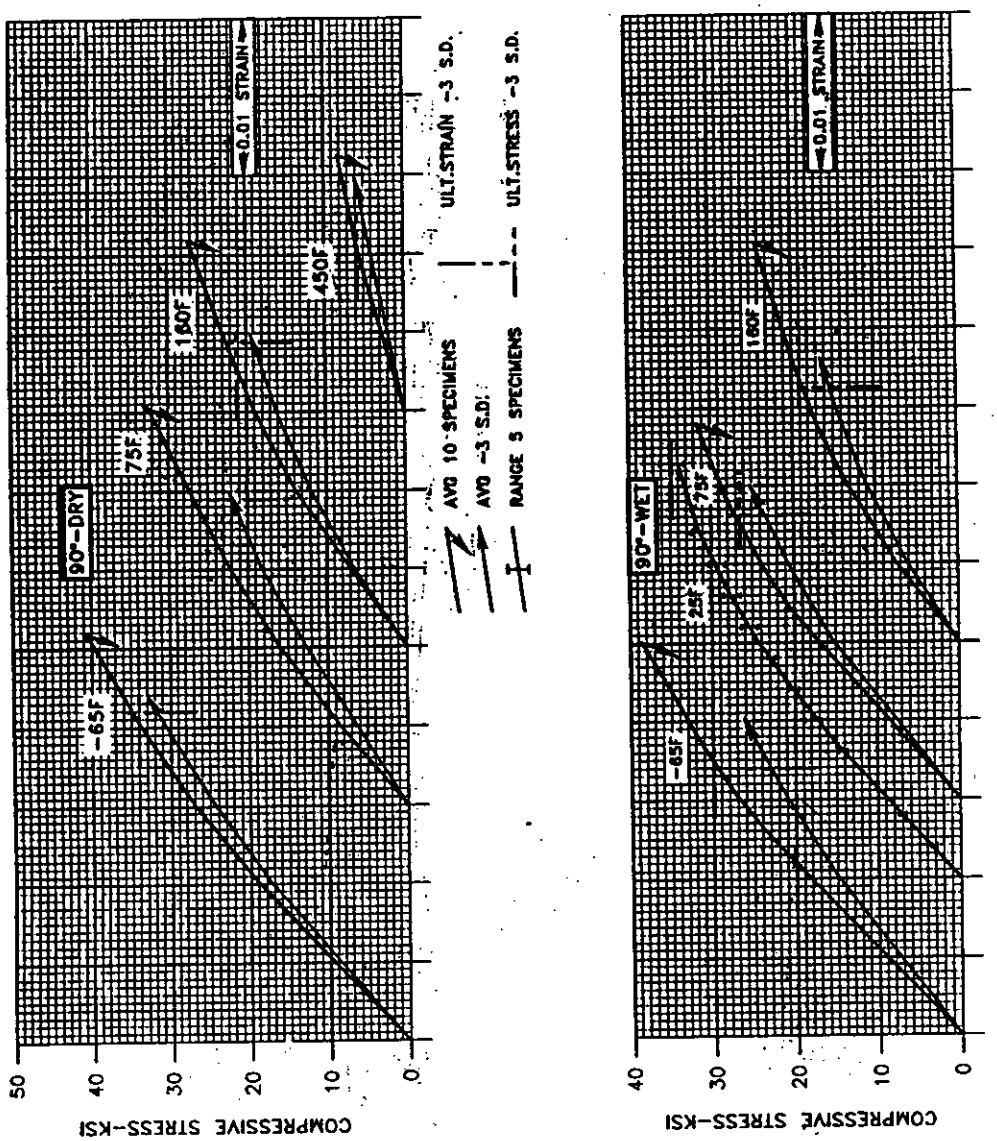


FIGURE A1.3.2(b) *Compressive stress-strain for F-161/7743 fiberglass epoxy loaded in the 90° direction.*

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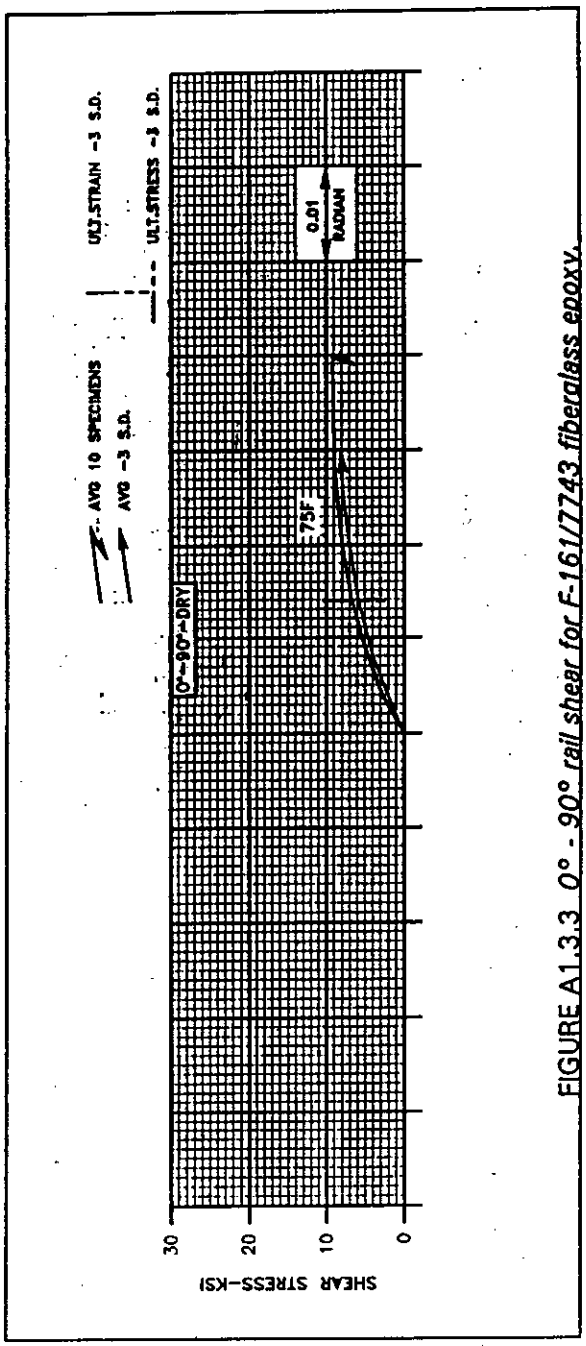


FIGURE A1.3.3 *0° - 90° rail shear for F-161/7743 fiberglass epoxy.*

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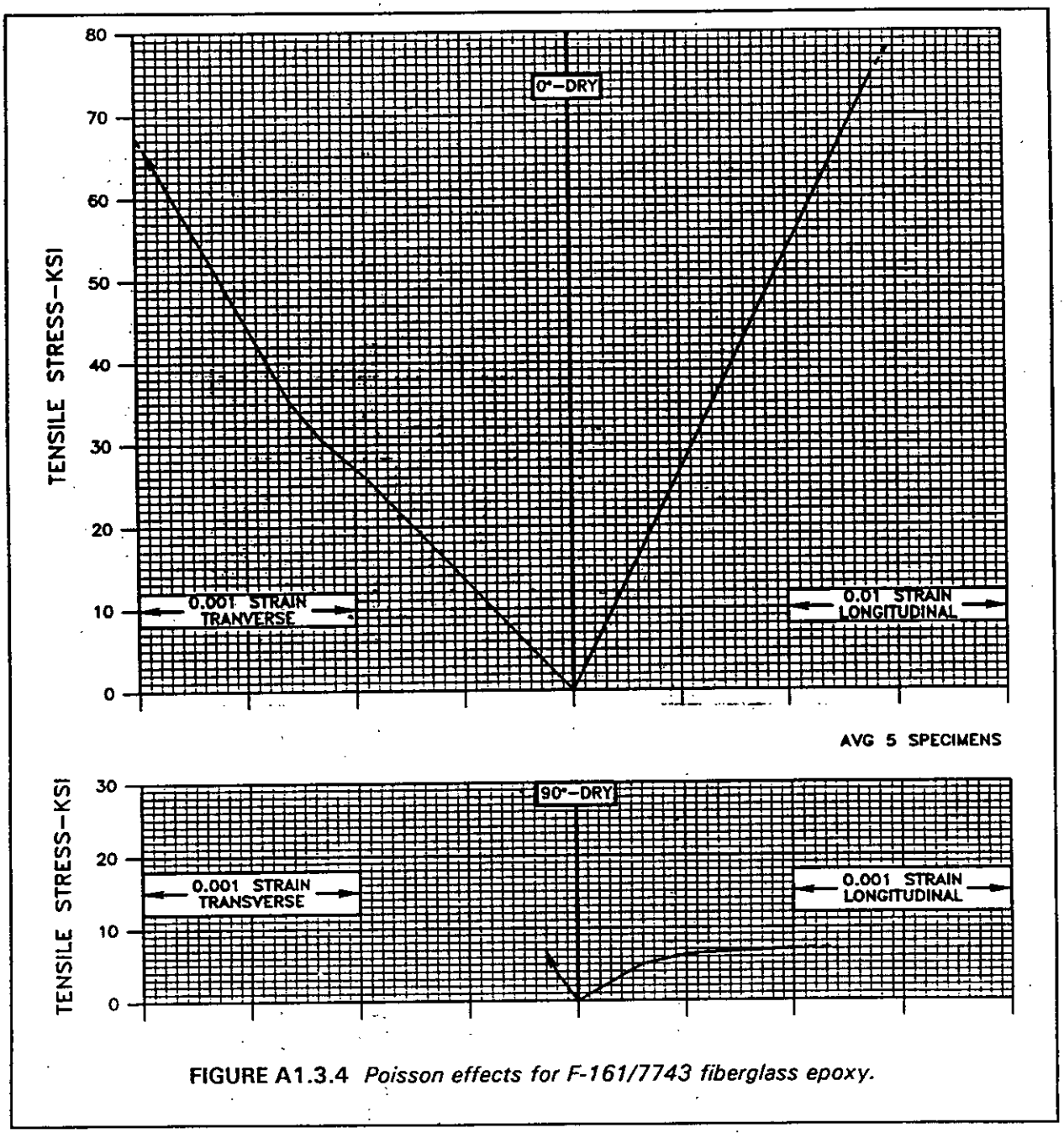


FIGURE A1.3.4 *Poisson effects for F-161/7743 fiberglass epoxy.*

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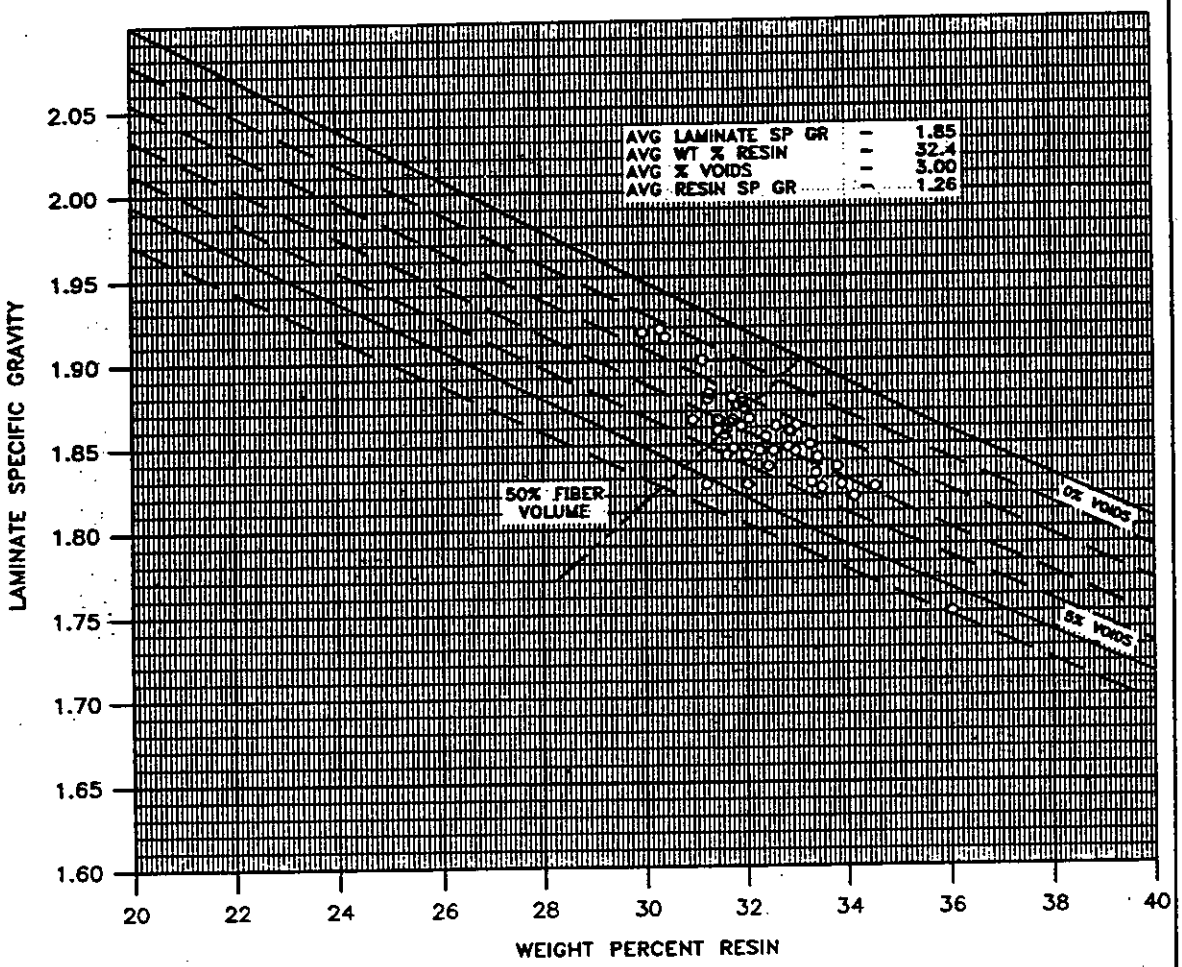


FIGURE A1.3.5 *Voids vs. resin content and specific gravity for F-161/7743 fiberglass epoxy.*

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TABLE A1.4 *Summary of mechanical properties of F-161/7781 (ECDE-1/0-550) fiberglass epoxy (26% resin)*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 55-65 psi		Bleedout: Vertical and Stepped Edge		Cure: 1 hr/350°F		Postcure: 2 hr/300°F 2.5 hr/400°F		Plies: 8 and 10	
Physical Properties		Weight Percent Resin: 26.0 v_f = 0.59				Avg. Specific Gravity: 2.01				Avg. Percent Voids: 0.5				Avg. Thickness: 0.008 inch/ply	
Test Methods		Tension: MIL-HDBK-17		Compression: MIL-HDBK-17		Shear: Picture Frame				Flexure: ASTM-D790		Bearing:		Interlaminar Shear: ASTM-D-2345	
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	92.4	5.16	80.5	10.87			61.4	3.20	65.7	3.03	50.7	5.72	59.8	3.81
	90°	67.8	10.65	62.3	5.01			50.3	2.61	53.6	5.19	46.2	2.69	35.2	5.16
ultimate strain, %	0°	2.86	2.11	2.37	0.31			1.78	0.13	1.97	0.14	1.58	0.19	1.96	0.08
	90°	2.42	3.14	1.97	0.24			1.65	0.08	1.88	0.12	1.55	0.10	1.38	0.13
proportional limit, ksi	0°														
	90°														
initial modulus, 10^6 psi	0°	4.42		4.49				4.10		3.92		3.72		3.27	
	90°	4.22		4.21				3.76		3.17		3.38		2.86	
secondary modulus, 10^6 psi	0°	3.32		3.14				3.06		3.24		3.07		2.94	
	90°	2.70		2.74				2.62		2.72		2.55		2.46	
Compression															
ultimate stress, ksi	0°	73.2	6.83	74.0	5.02			57.3	4.0	48.9	3.50	44.7	3.25	28.8	3.03
	90°	64.2	3.19	55.8	4.40			37.5	2.28	42.0	2.64	40.1	1.90	18.9	0.69
ultimate strain, %	0°	1.70	0.42	1.65	0.28			1.09	0.17	1.12	0.15	0.84	0.14	0.79	0.03
	90°	1.40	0.14	1.42	0.27			1.26	0.41	1.14	0.23	1.22	0.18	0.71	0.27
proportional limit, ksi	0°	39.0		46.0				42.0		41.0		24.0		15.0	
	90°	28.0		41.0				24.0		36.0		21.0		11.0	
initial modulus, 10^6 psi	0°	4.42		4.47				4.27		4.05		3.94		3.73	
	90°	4.02		4.19				4.12		3.68		3.40		3.07	
Shear															
ultimate stress, ksi	0°-90° ±45°	20.1	2.3					16.0	1.64	13.4	1.28				

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°				94.10	96.86	89.64			
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°				5.56	5.65	5.50			

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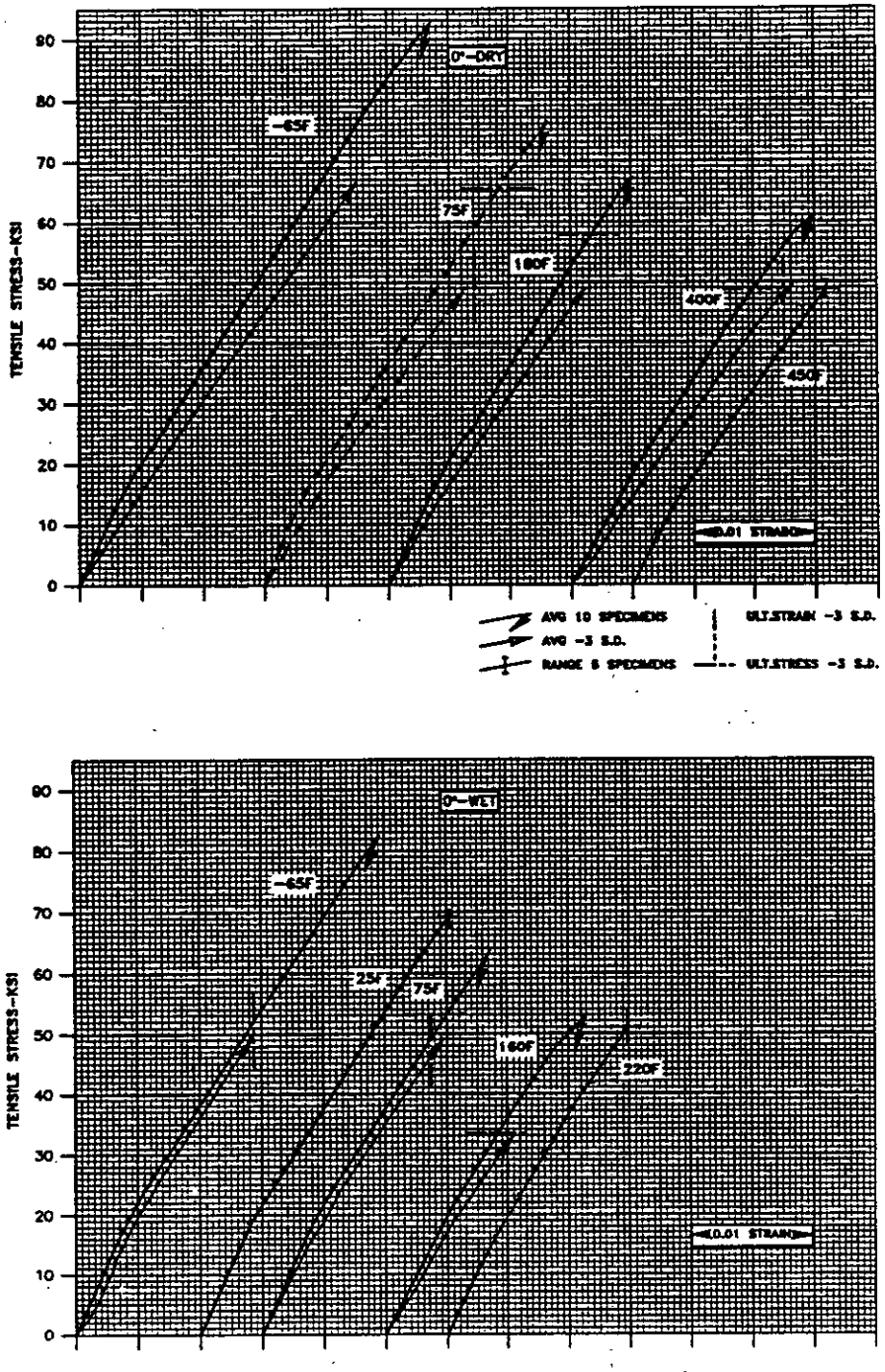


FIGURE A1.4.1(a) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (26% resin).*

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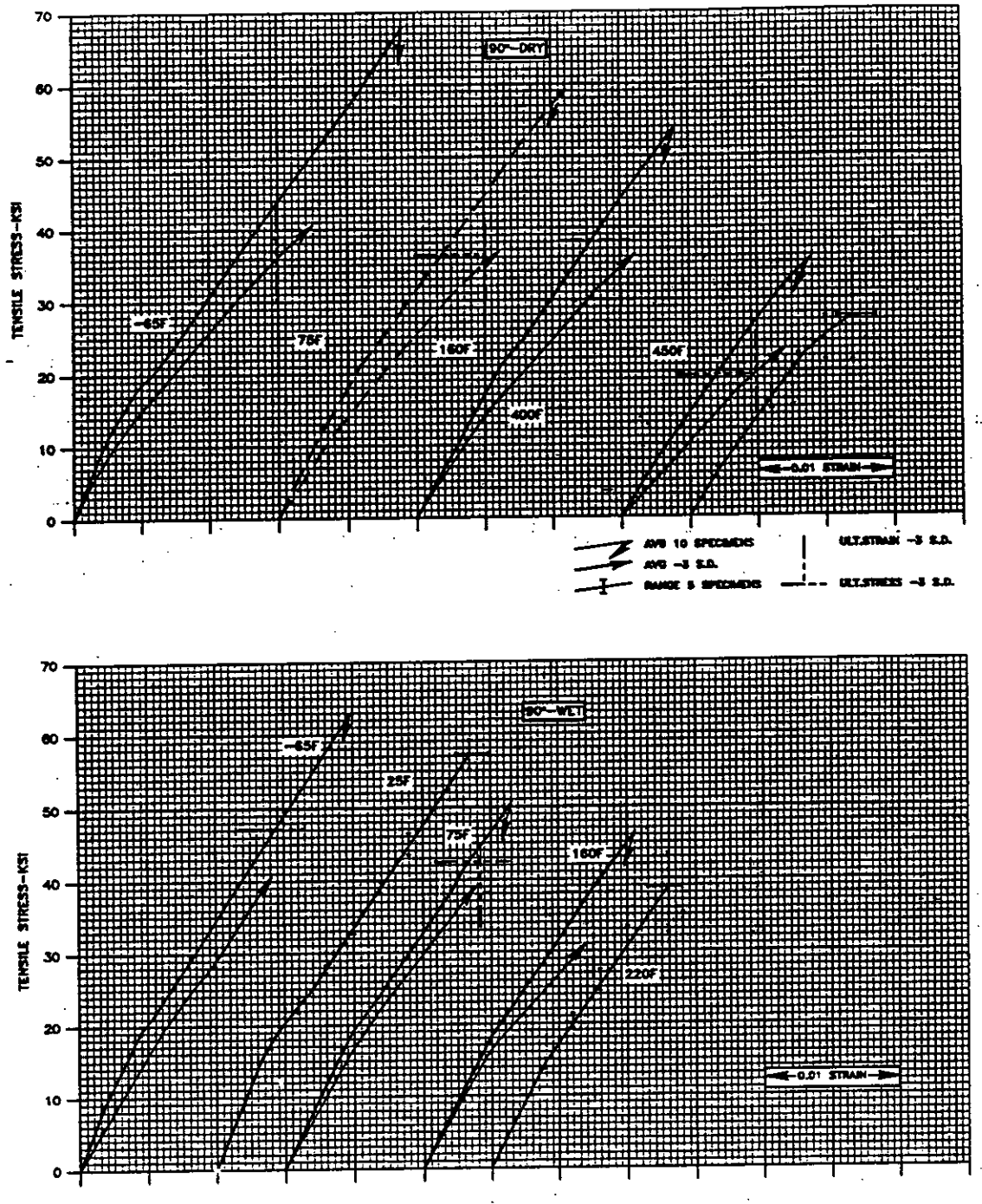


FIGURE A1.4.1(b) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (26% resin).*

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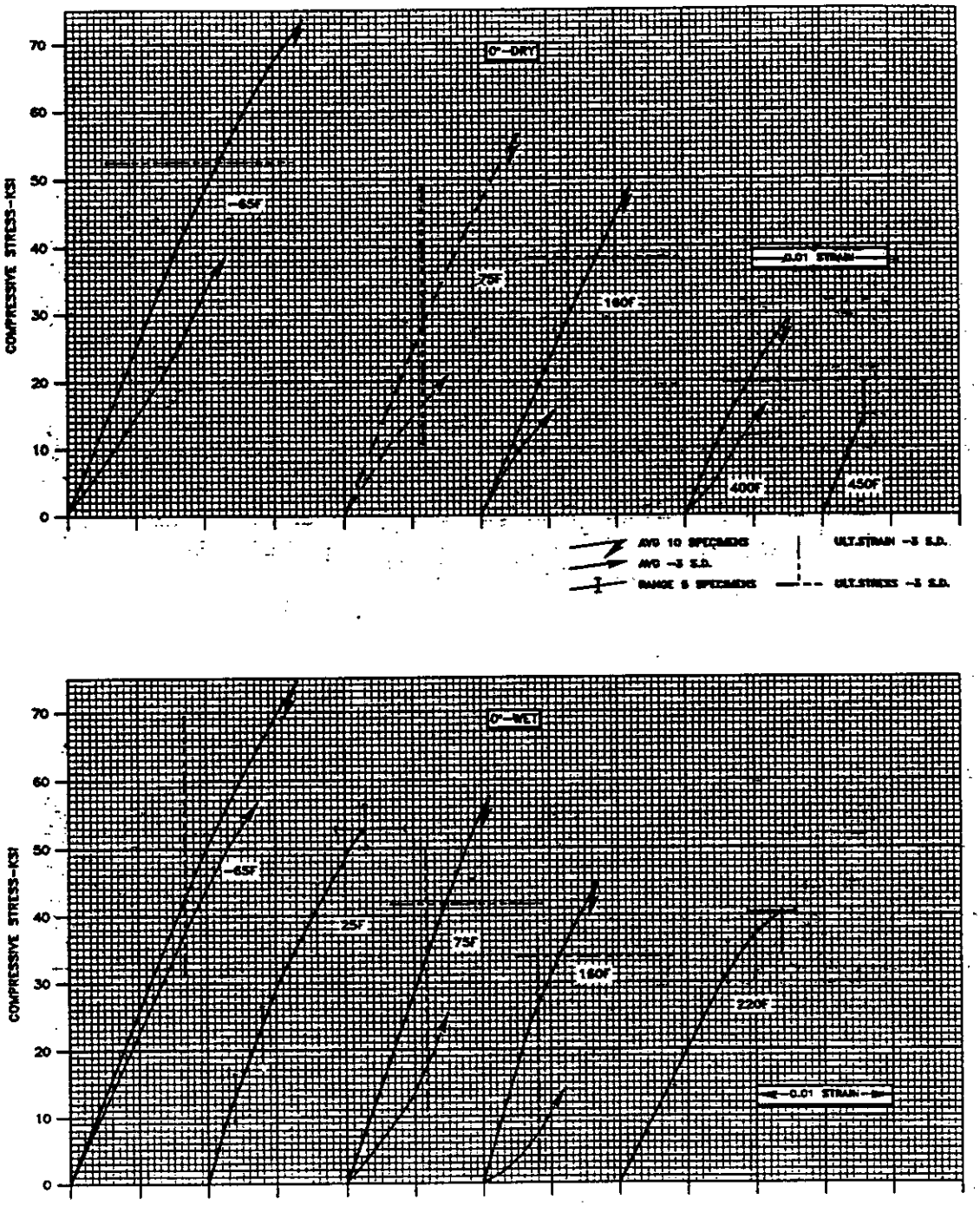


FIGURE A1.4.2(a) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (26% resin).*

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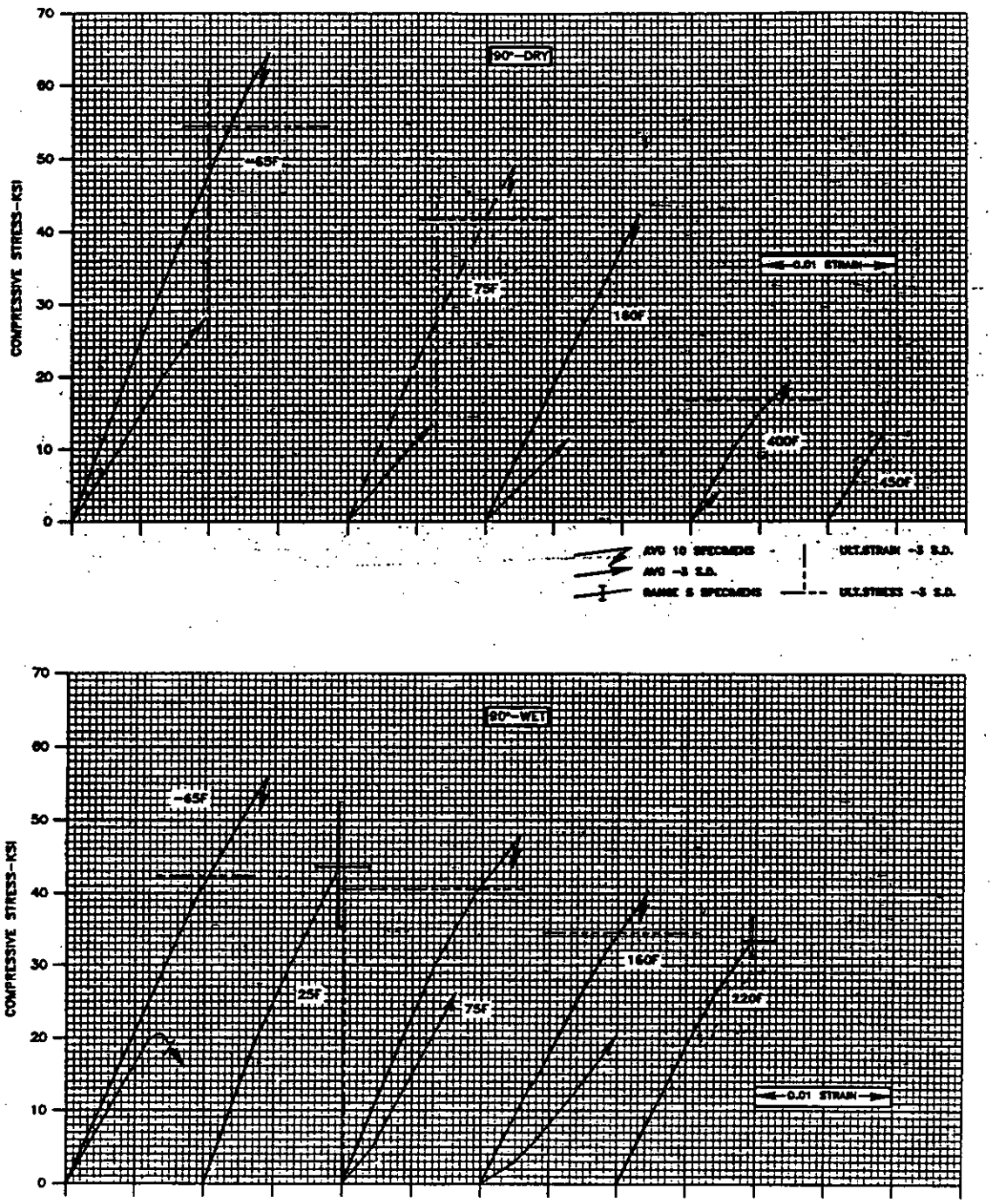


FIGURE A1.4.2(b) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (26% resin).*

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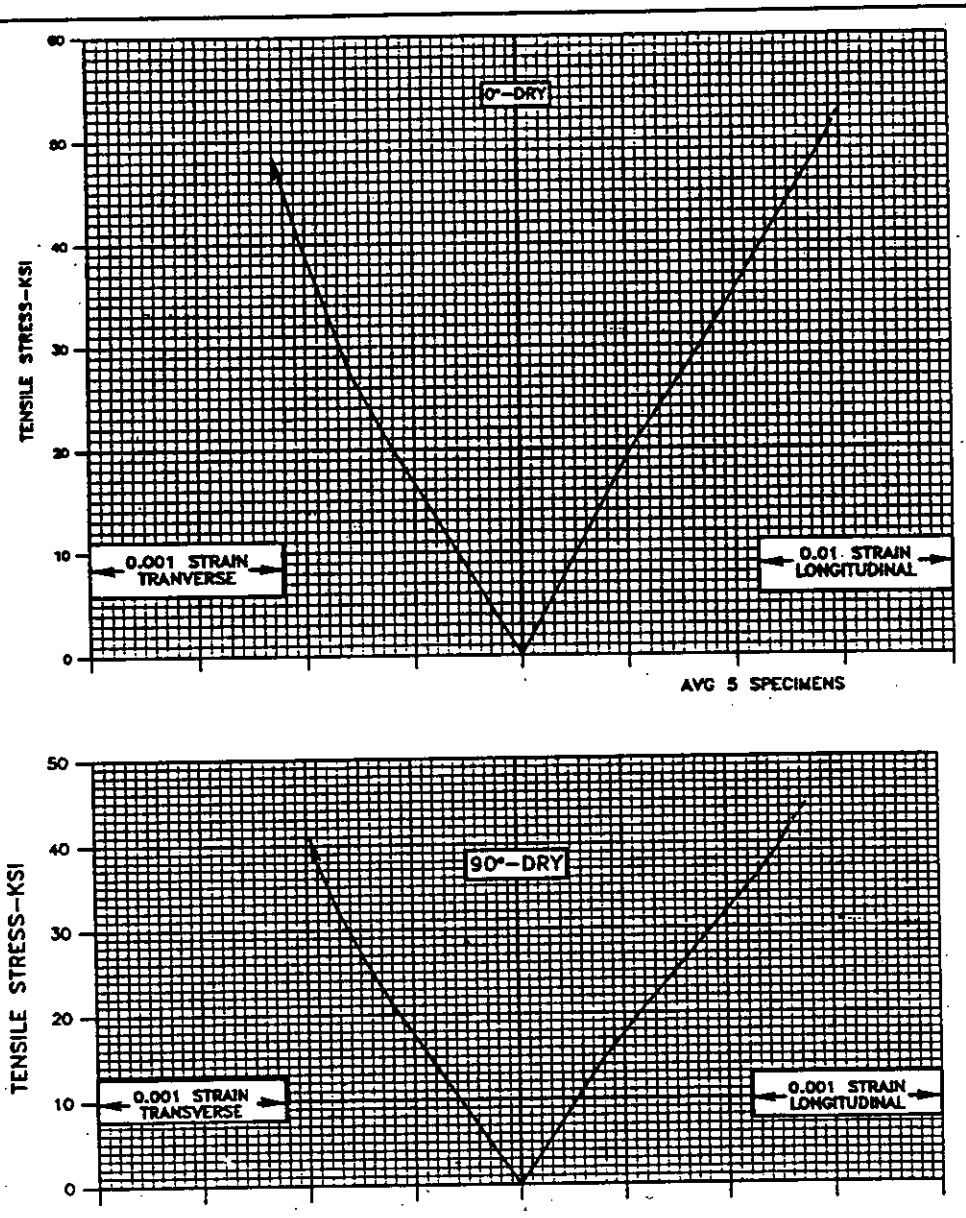


FIGURE A1.4.4 *Poisson effects for F-161/7781 fiberglass epoxy (26% resin).*

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TABLE A1.5 *Summary of mechanical properties of F-161/7781 (ECDE-1/0-550) fiberglass epoxy (31% resin)*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 55-65 psi		Bleedout: Vertical and Stepped Edge		Cure: 1 hr/350°F		Postcure: 2 hr/300°F 2.5 hr/400°F		Plies: 8 and 10	
Physical Properties		Weight Percent Resin: 31.0		Avg. Specific Gravity: 1.92				Avg. Percent Voids: 0.6				Avg. Thickness: 0.009 inch/ply			
Test Methods		Tension: MIL-HDBK-17		Compression: MIL-HDBK-17		Shear: Picture Frame		Flexure: ASTM-D790		Bearing:		Interlaminar Shear:			
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	85.2	4.68	82.3	4.97			64.0	2.04	60.1	3.75	51.4	4.23	47.3	4.87
	90°	70.0	5.24	67.9	2.98			53.5	2.91	49.3	0.95	39.8	3.50	31.0	1.95
ultimate strain, %	0°	2.93	0.14	2.53	0.18			2.10	0.06	2.02	0.10	1.66	0.17	1.66	0.18
	90°	2.50	0.21	2.41	0.22			1.90	0.11	1.86	0.06	1.47	0.09	1.25	0.09
proportional limit, ksi	0°														
	90°														
initial modulus, 10^6 psi	0°	4.22		4.30				3.84		3.69	3.72	3.65		3.09	
	90°	3.97		4.15				3.68		3.37	3.34	3.30		2.75	
secondary modulus, 10^6 psi	0°	3.13		3.01				3.03		2.97	0.04	2.88		2.94	
	90°	2.62		2.96				2.62		2.55	0.25	2.46		2.47	
Compression															
ultimate stress, ksi	0°	73.1	5.18	66.0	10.75			54.4	7.04	50.6		45.9	5.39	32.8	6.04
	90°	58.4	3.17	57.5	11.56			47.3	4.73	42.2		38.7	4.19	25.8	8.27
ultimate strain, %	0°	1.86	0.21	1.72	0.32			1.33	0.28	1.52		1.04	0.23	0.95	0.24
	90°	1.61	0.29	1.44	0.36			1.10	0.21	1.30		0.99	0.22	0.87	0.28
proportional limit, ksi	0°	44.0		38.0				33.0		32.0		25.0		16.0	
	90°	33.0		33.0				30.0		--		21.0		15.0	
initial modulus, 10^6 psi	0°	3.90		4.04				4.03		3.42		4.06		3.50	
	90°	3.56		3.84				3.96		3.23		4.01		3.07	
Shear															
ultimate stress, ksi	0°-90° ±45°	20.5	2.23					16.9	0.72	13.7	0.82				

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°				90.23	93.74	87.29			
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°				5.56	5.65	5.50			

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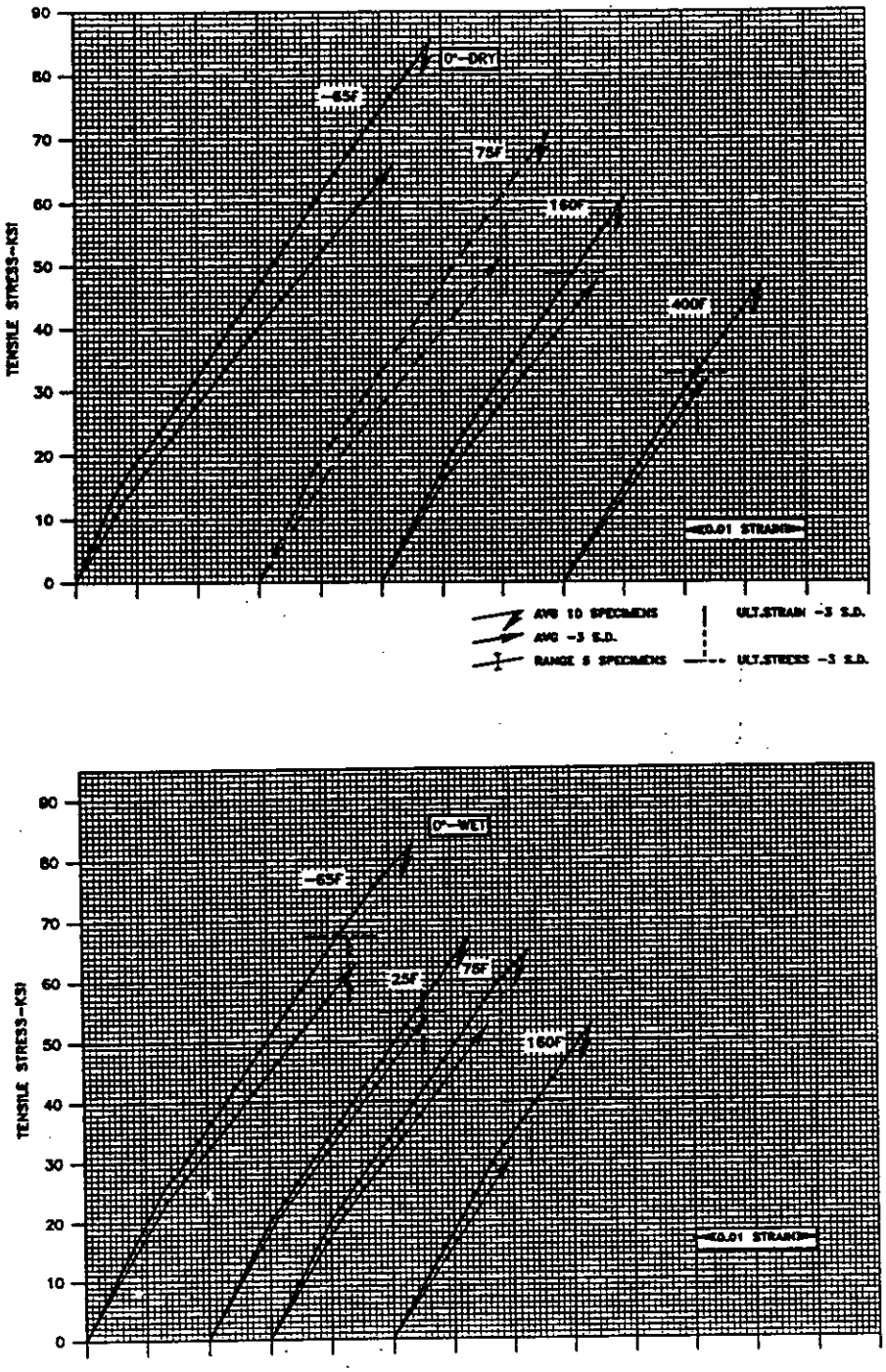


FIGURE A1.5.1(a) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (31% resin).*

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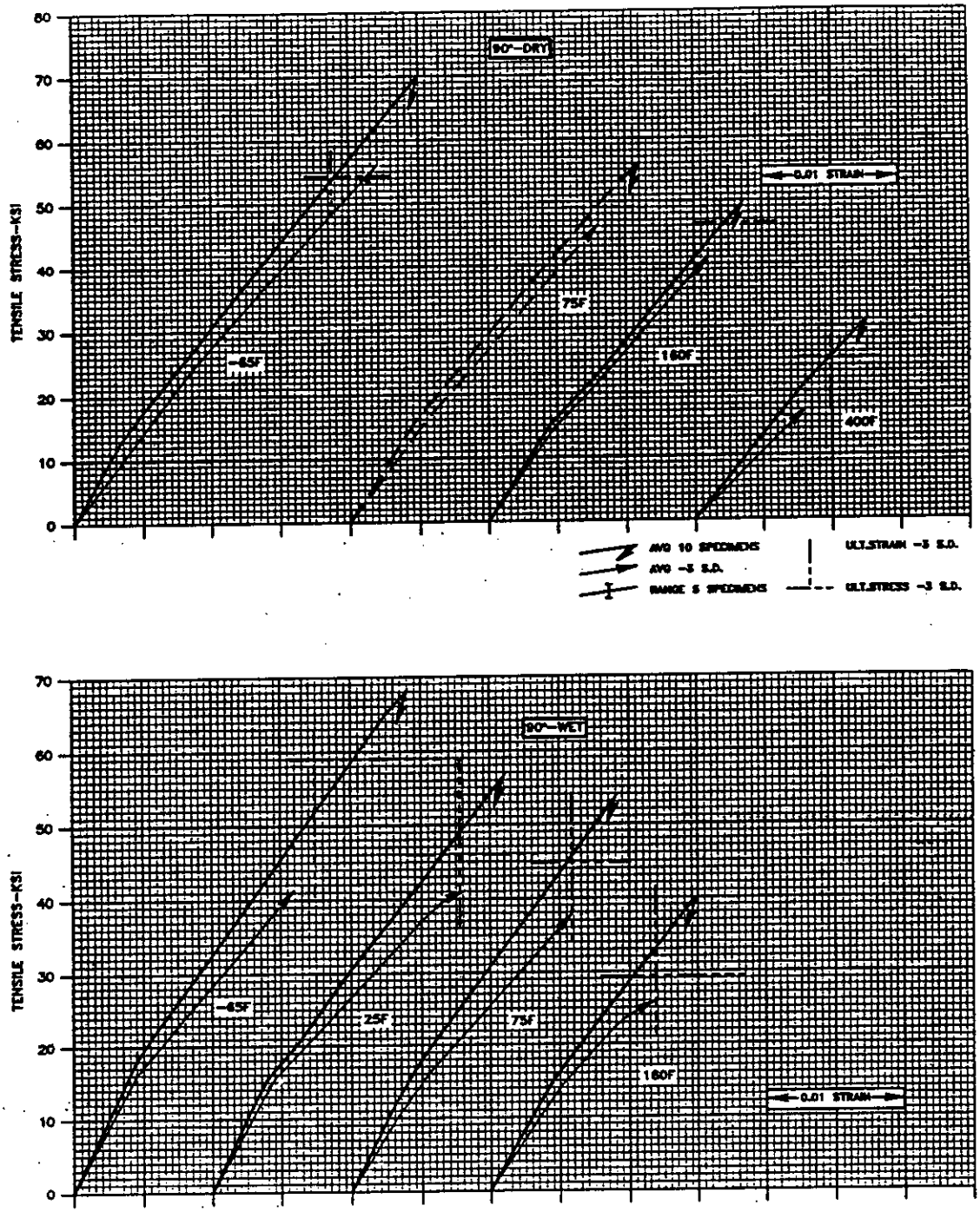


FIGURE A1.5.1(b) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (31% resin).*

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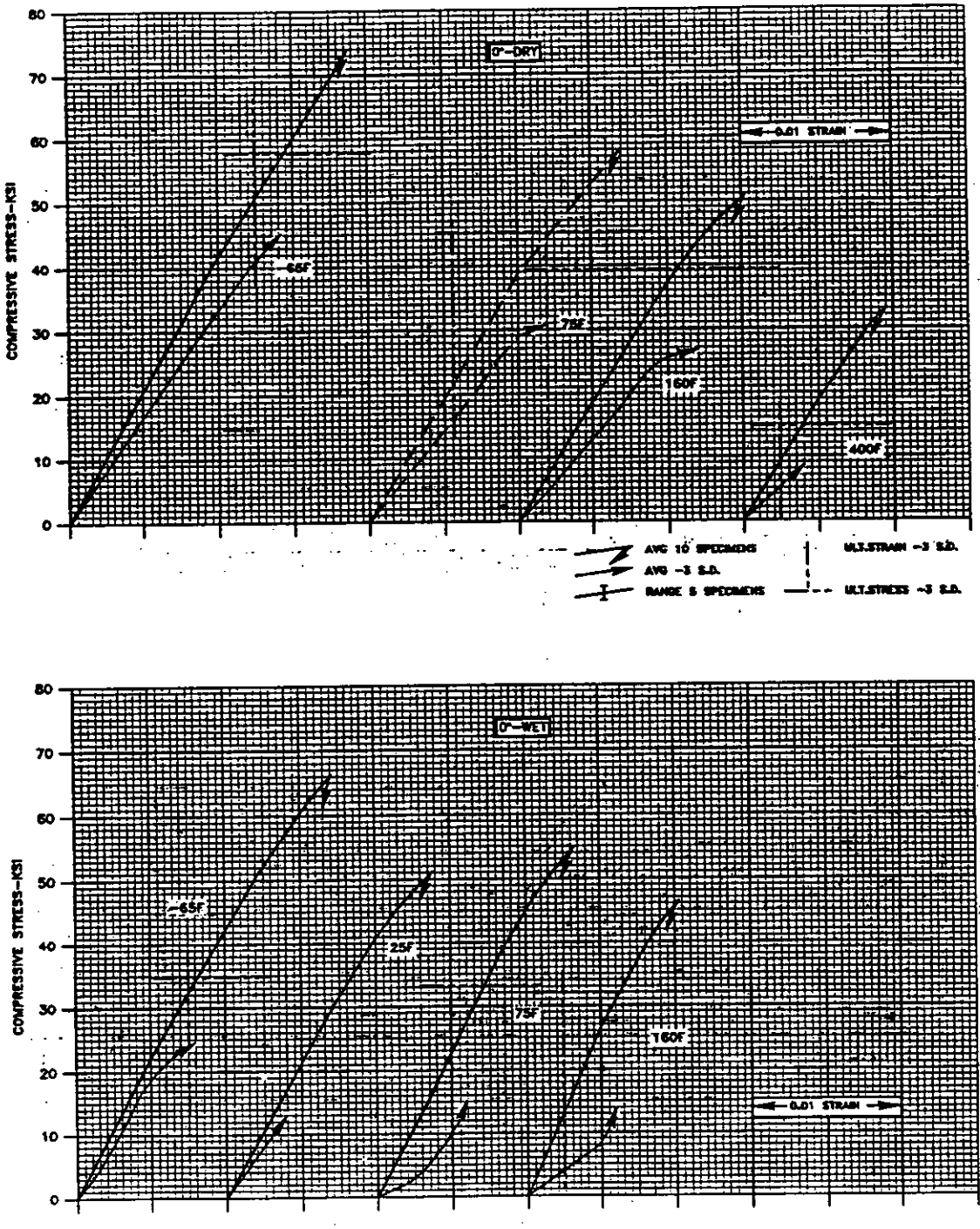


FIGURE A1.5.2(a) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (31% resin).*

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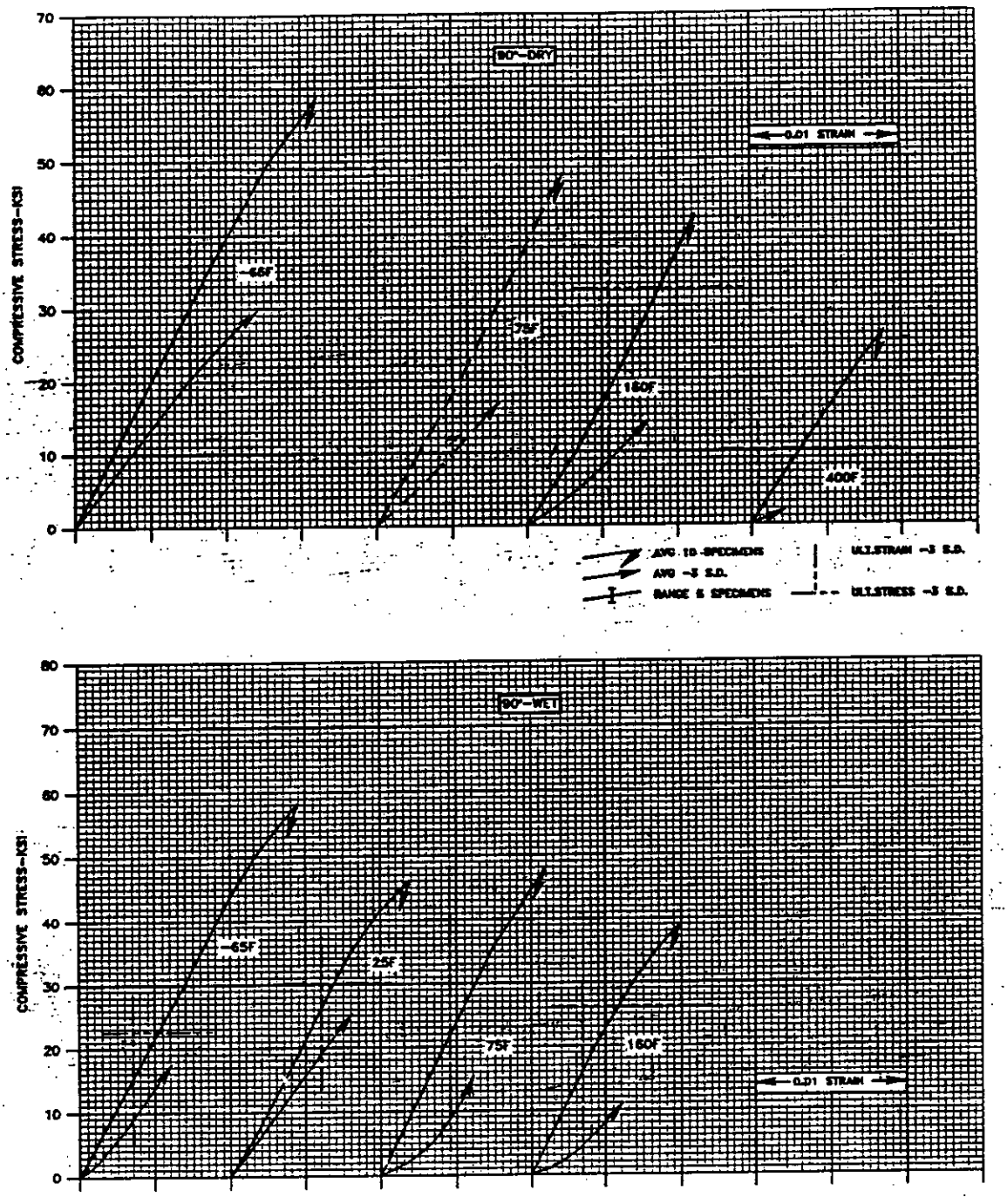


FIGURE A1.5.2(b) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (31% resin).*

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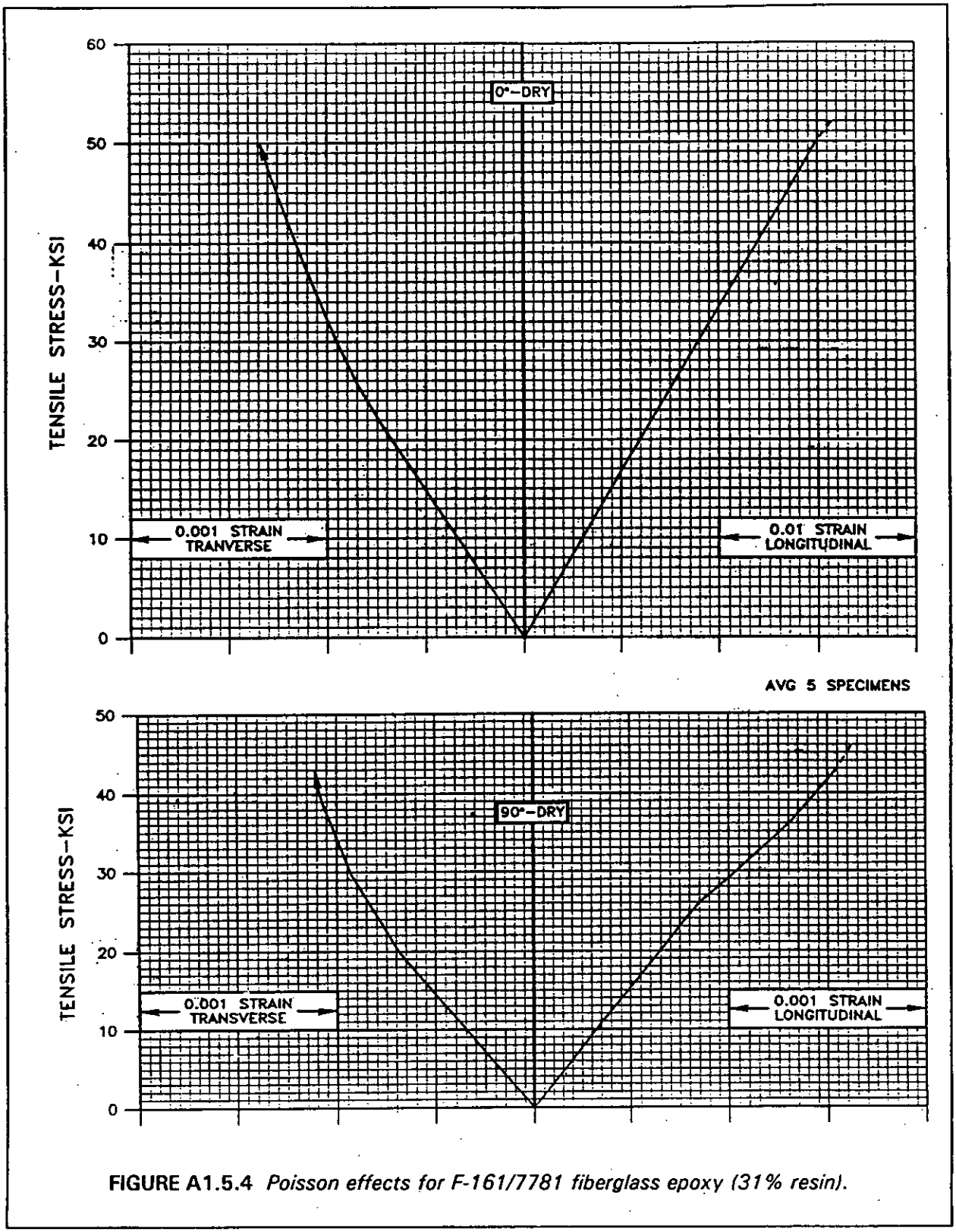


FIGURE A1.5.4 *Poisson effects for F-161/7781 fiberglass epoxy (31% resin).*

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TABLE A1.6 *Summary of mechanical properties of F-161/7781 (ECDE-1/0-550) fiberglass epoxy (36% resin)*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 55-65 psi		Bleedout: Vertical and Stepped Edge		Cure: 1 hr/350°F		Postcure: 2 hr/300°F 2.5 hr/400°F		Plies: 8	
Physical Properties		Weight Percent Resin: 35.6				Avg. Specific Gravity: 1.86				Avg. Percent Voids: 0.9				Avg. Thickness: 0.010 inch/ply	
Test Methods		Tension: MIL-HDBK-17				Compression: MIL-HDBK-17		Shear: Picture Frame		Flexure: ASTM-D790		Bearing:		Interlaminar Shear:	
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	83.9	2.85	73.0	2.89			55.5	2.57	61.9	2.24	45.0	1.85	39.2	3.40
	90°	68.7	4.19	63.9	1.61			48.9	2.67	51.9	3.25	37.6	0.99	32.0	1.44
ultimate strain, %	0°	3.30	0.18	2.79	0.02			2.12	0.14	2.61	0.08	1.59	0.07	1.45	0.13
	90°	2.80	0.18	2.41	0.05			1.95	0.09	2.18	0.19	1.50	0.05	1.35	0.08
proportional limit, ksi	0°														
	90°														
initial modulus, 10^6 psi	0°	3.84		3.81				3.58		3.25		3.35		2.96	
	90°	3.67		3.81				3.30		3.13		3.18		2.51	
secondary modulus, 10^6 psi	0°	2.81		2.75				3.04		2.49		3.04		2.74	
	90°	2.65		2.67				2.72		2.39		2.70		2.22	
Compression															
ultimate stress, ksi	0°	76.2	5.88	68.8	4.36			55.1	2.63	54.7	5.49	46.0	5.66	31.0	8.08
	90°	56.0	4.56	52.9	6.32			47.0	6.78	36.9	1.47	35.3	3.30	23.2	3.26
ultimate strain, %	0°	2.13	0.28	1.64	0.23			1.36	0.32	1.90	0.56	1.32	2.41	1.02	0.23
	90°	1.75	0.48	1.58	0.57			2.00	0.89	1.29	0.09	1.27	2.40	0.91	0.14
proportional limit, ksi	0°	28.0		24.0				24.0		32.0		22.0		17.0	
	90°	18.0		17.0				16.0		28.0		17.0		12.0	
initial modulus, 10^6 psi	0°	4.10		4.50				3.87		3.45		3.38		2.87	
	90°	4.00		4.10				3.64		2.87		2.88		2.63	
Shear															
ultimate stress, ksi	0°-90° ±45°	19.6	1.04					15.0	0.70	12.7	0.62				

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°				86.31	92.16	79.07			
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°				5.56	5.65	5.50			

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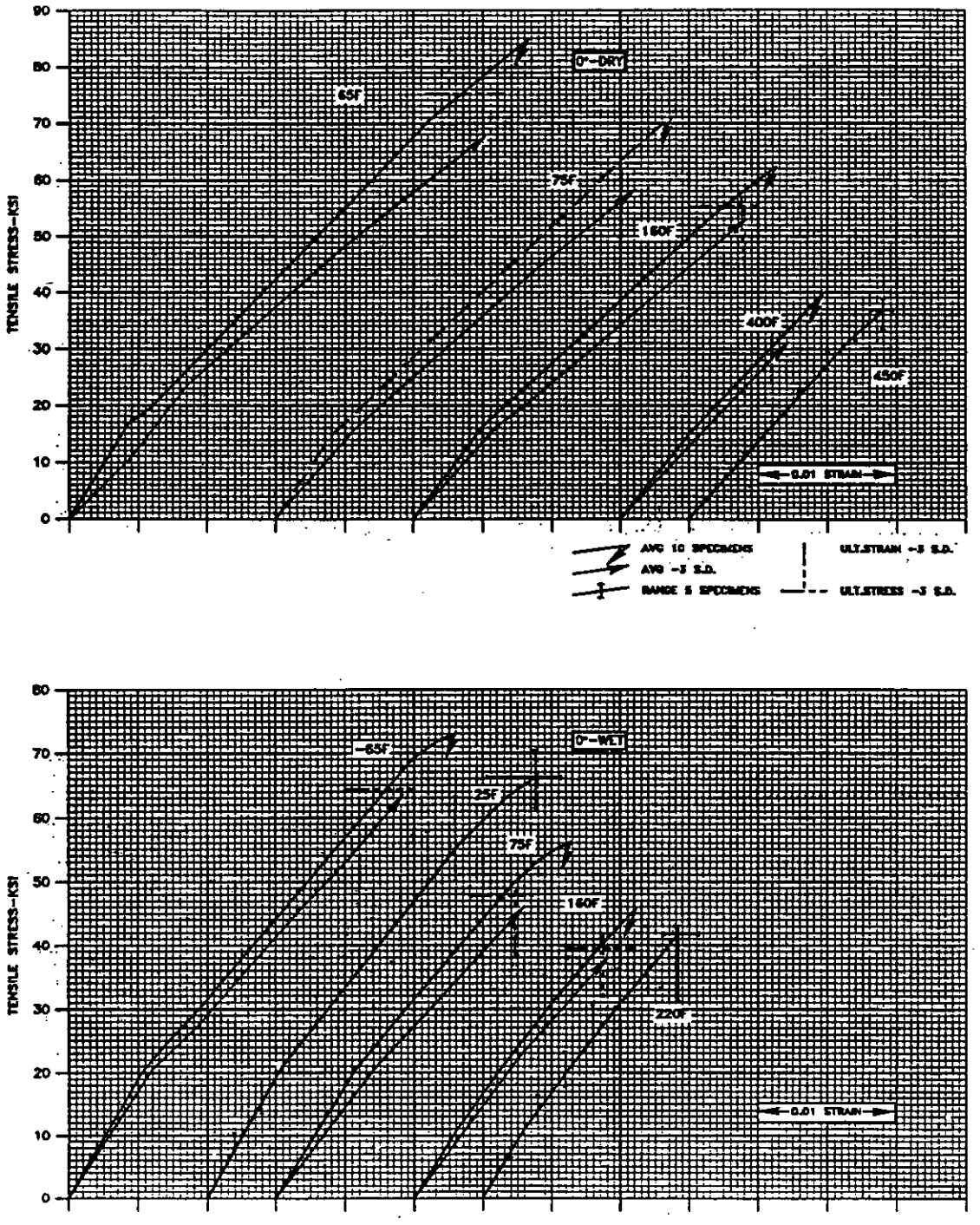


FIGURE A1.6.1(a) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (36% resin).*

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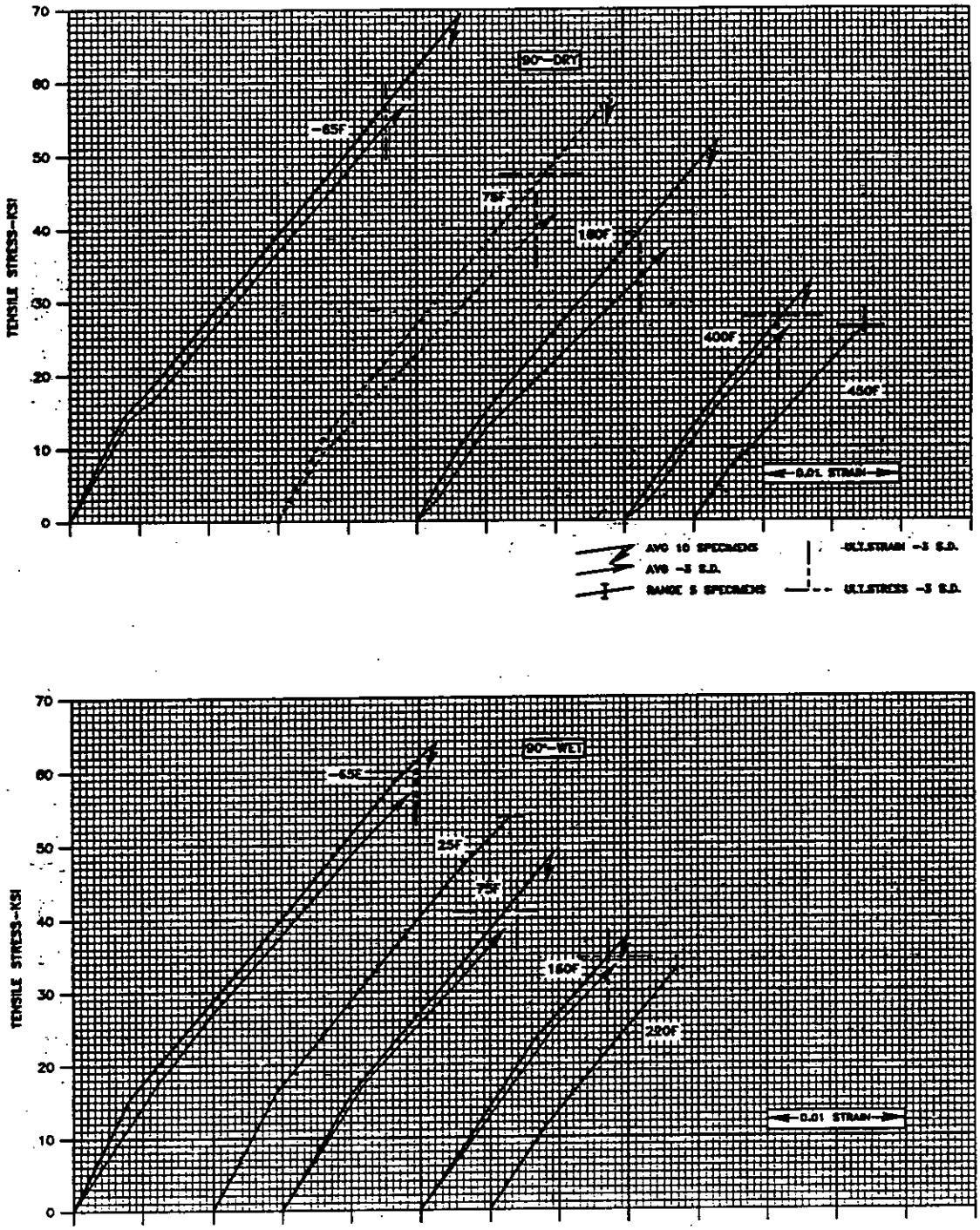


FIGURE A1.6.1(b) *Tensile stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (36% resin).*

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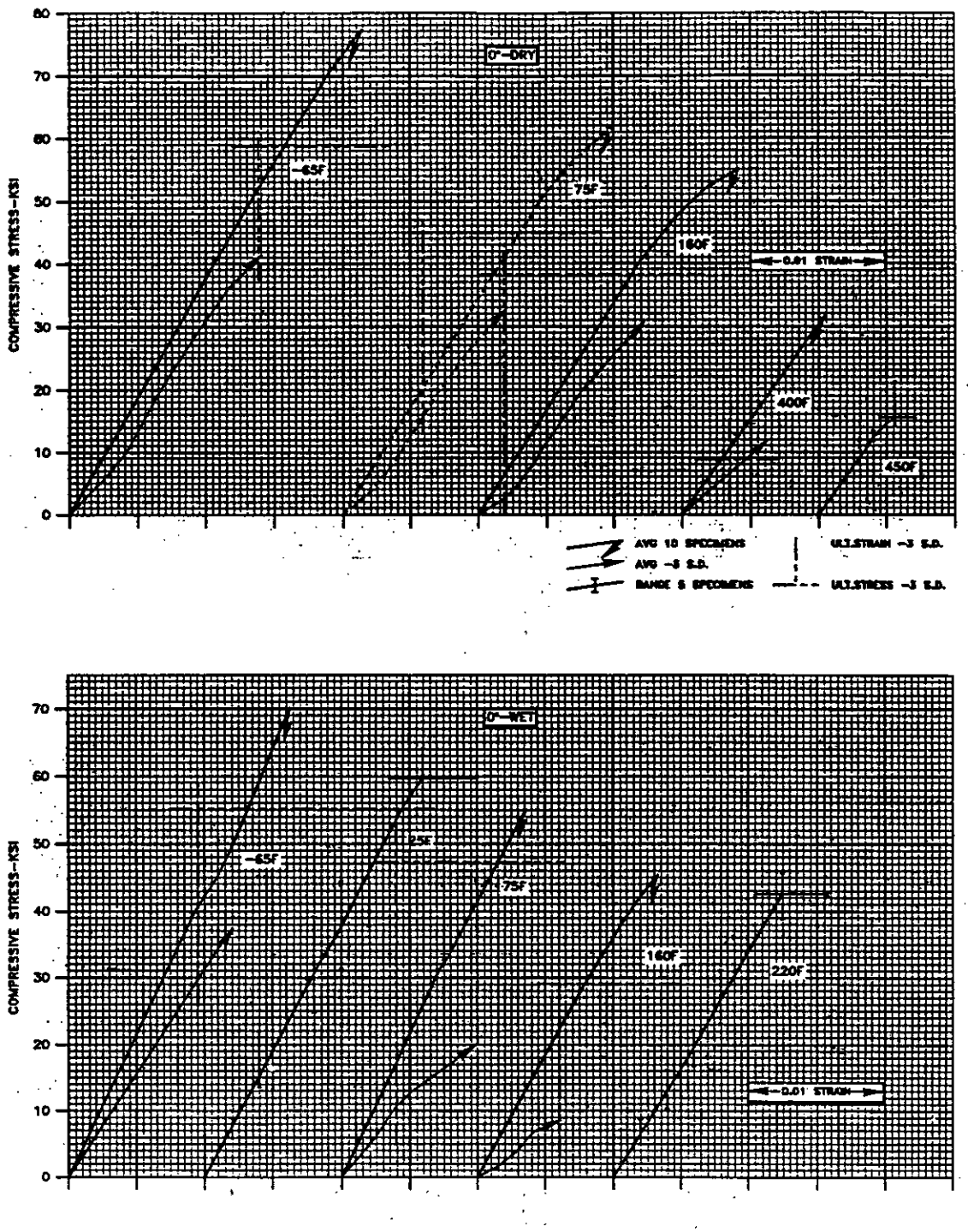


FIGURE A1.6.2(a) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 0° direction (36% resin).*

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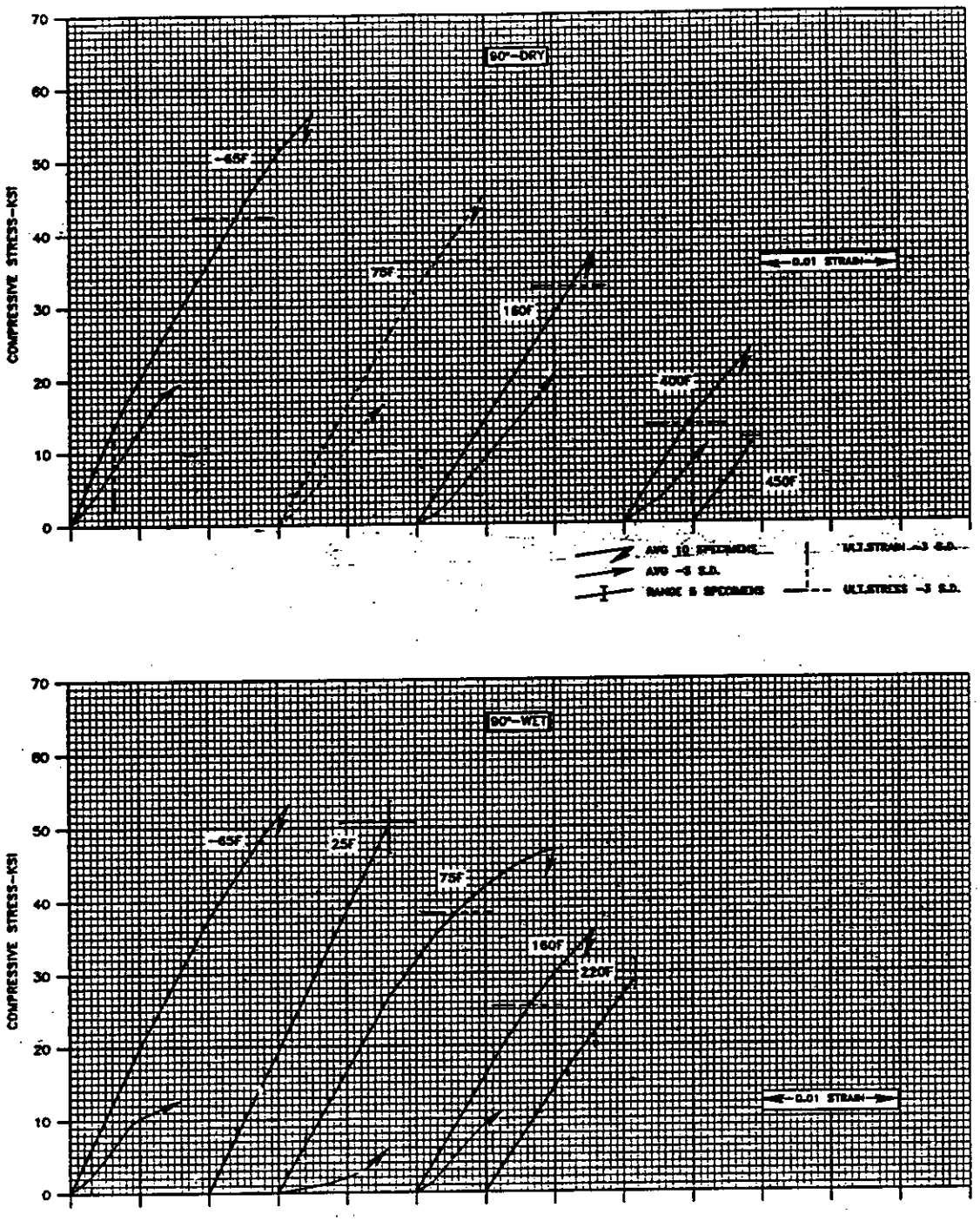


FIGURE A1.6.2(b) *Compressive stress-strain for F-161/7781 fiberglass epoxy loaded in the 90° direction (36% resin).*

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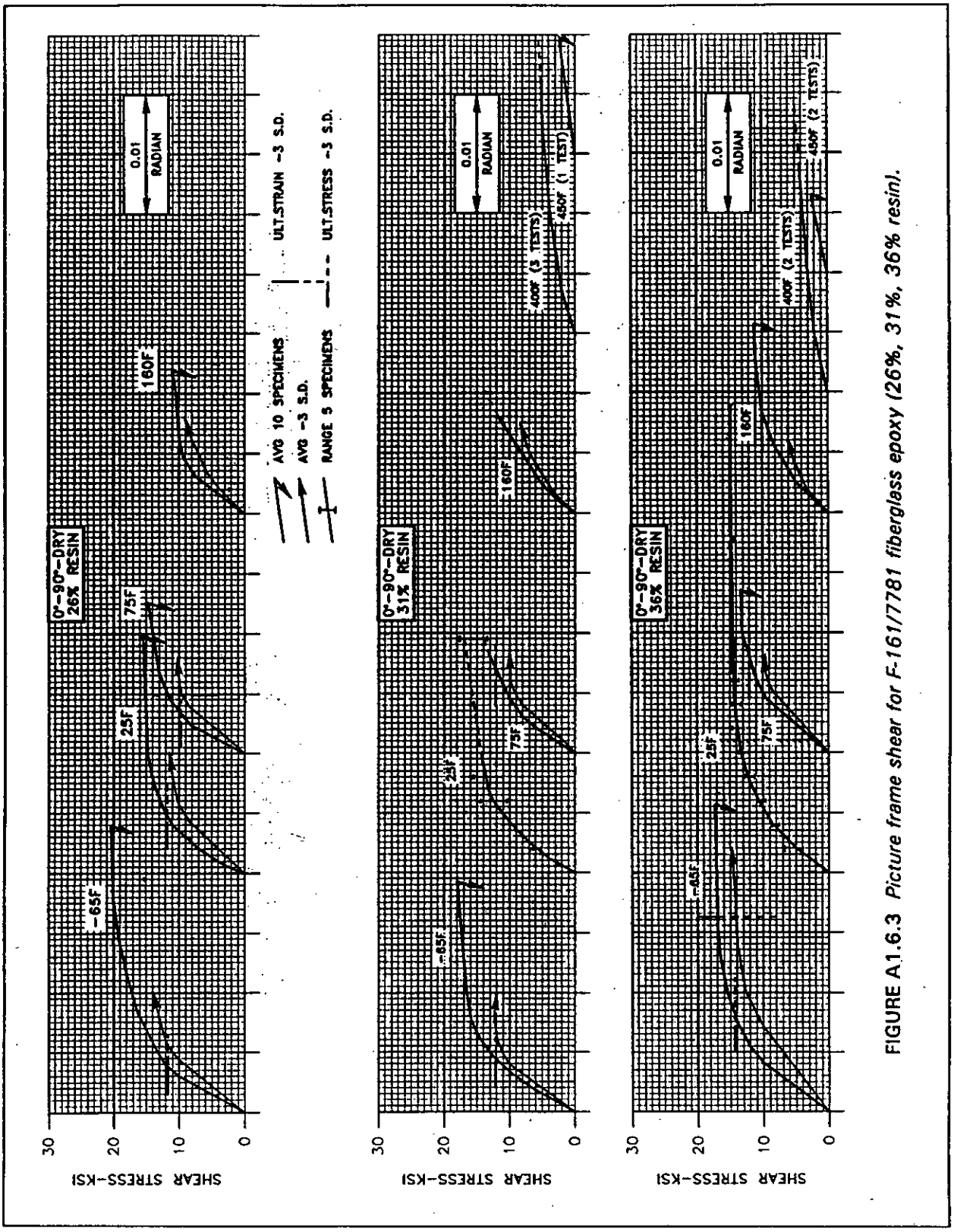


FIGURE A1.6.3 *Picture frame shear for F-161/7781 fiberglass epoxy (26%, 31%, 36% resin).*

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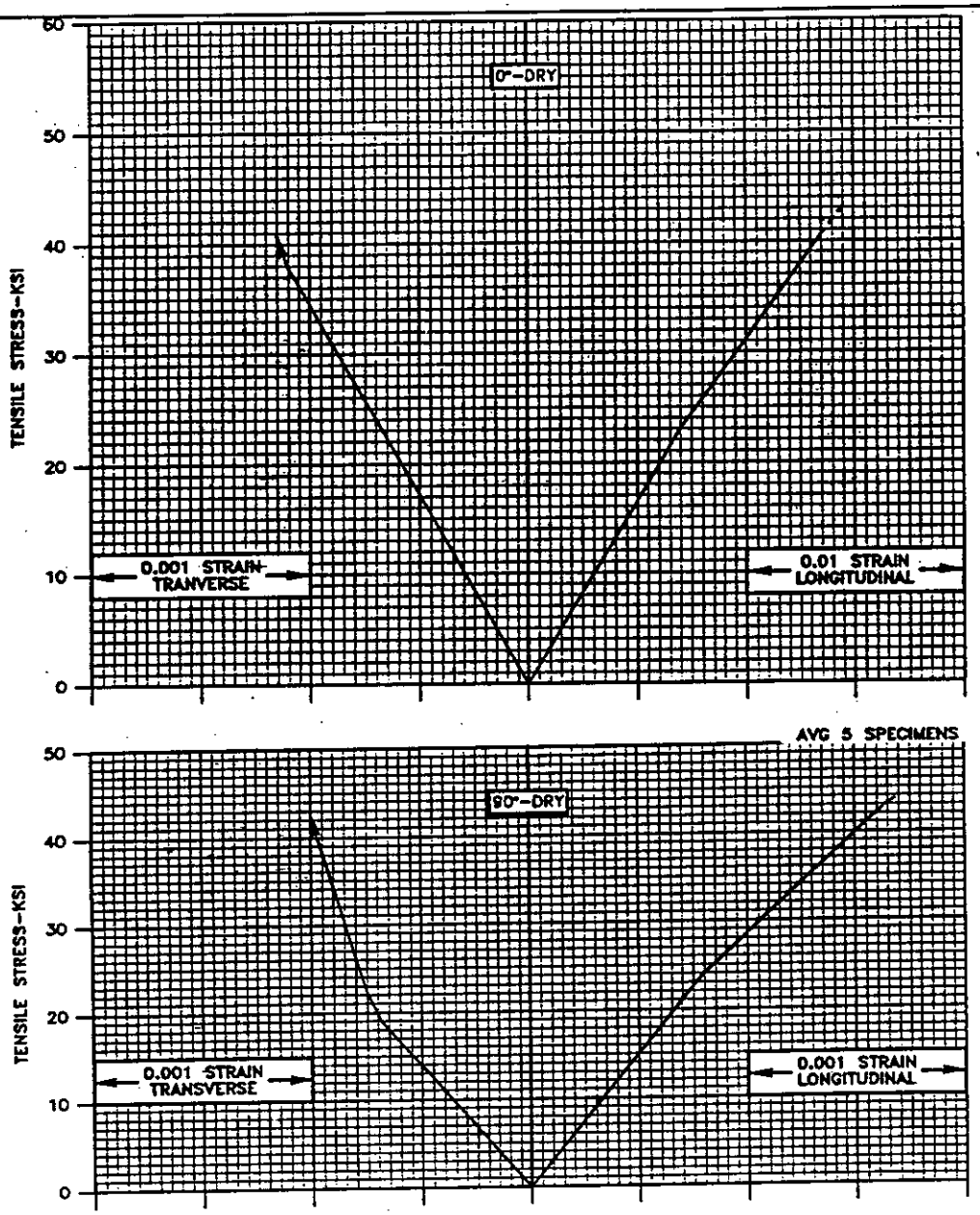


FIGURE A1.6.4 *Poisson effects for F-161/7781 fiberglass epoxy (36% resin).*

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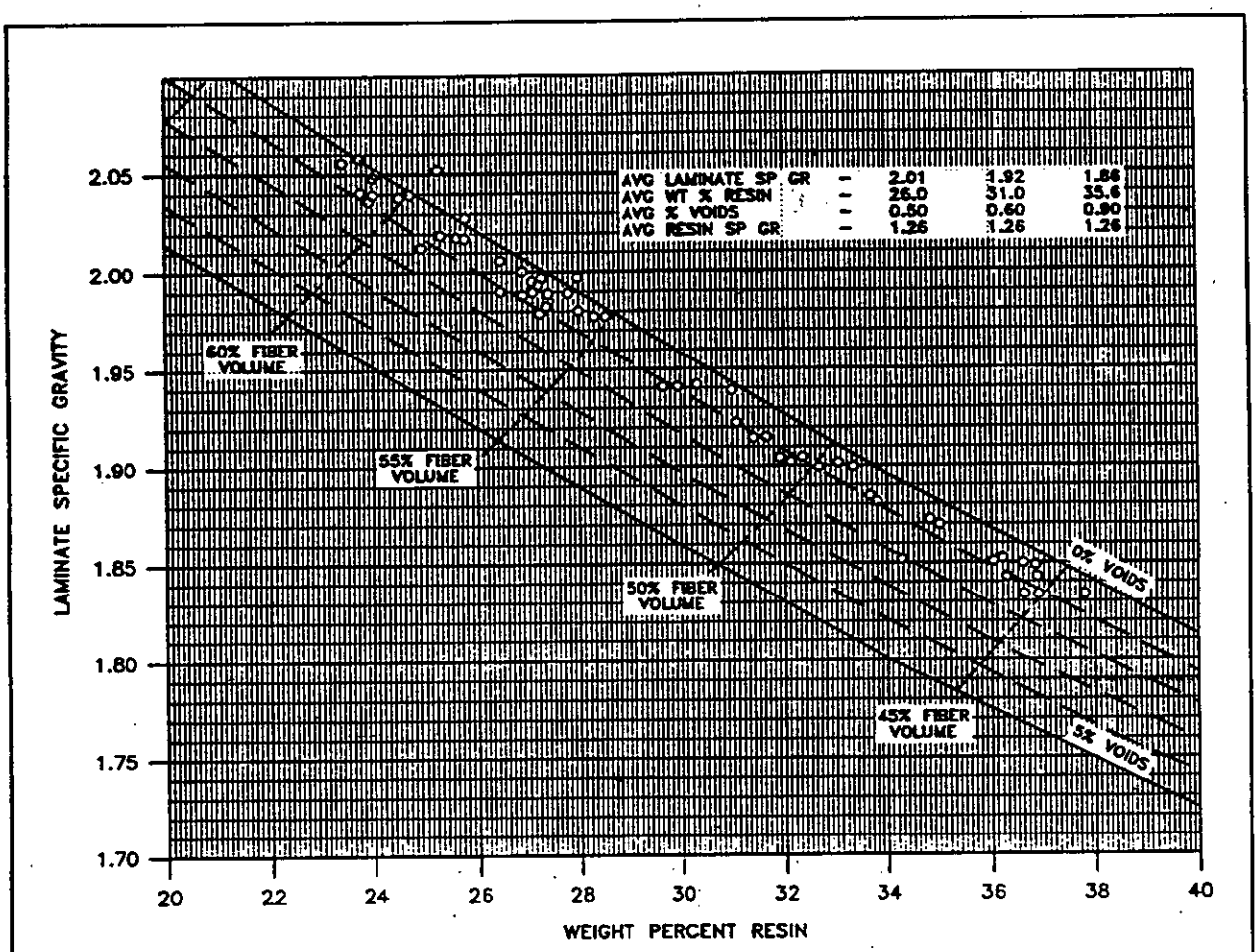


FIGURE A1.6.5 *Voids vs resin content and specific gravity for F-161/7781 fiberglass*

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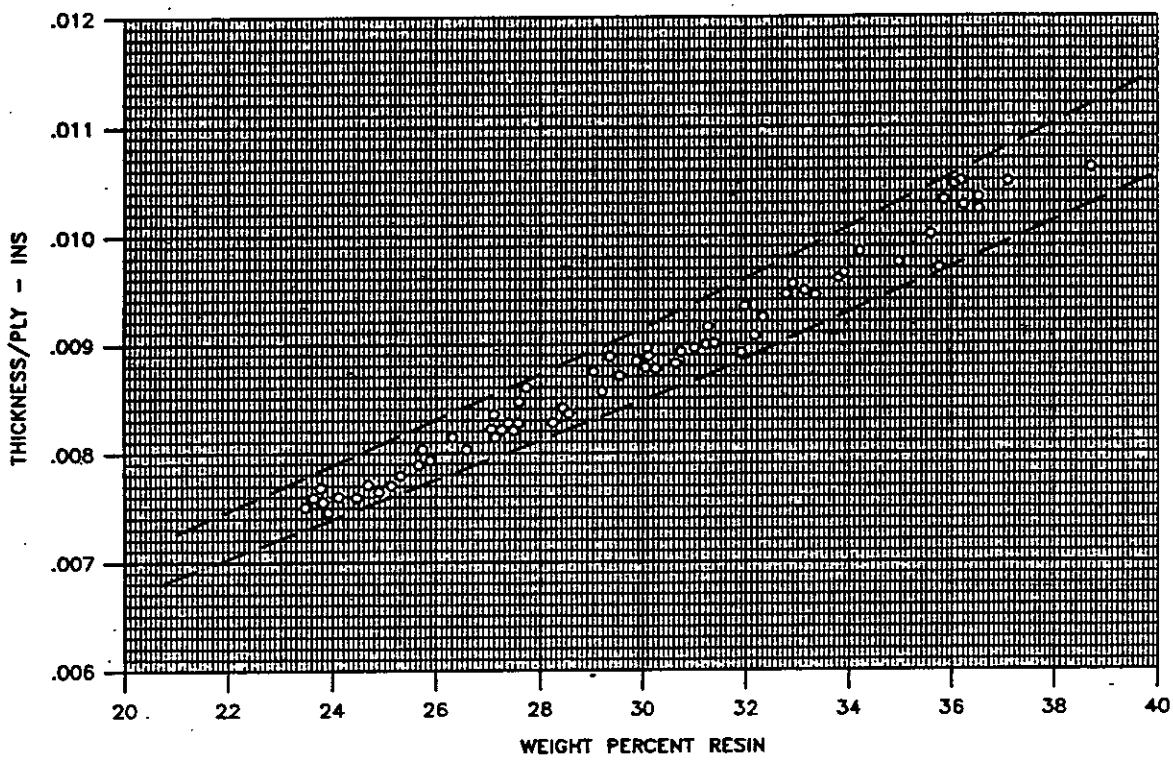


FIGURE A1.6.6 *Thickness vs resin content for F-161/7781 fiberglass epoxy.*

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TABLE A1.8 *Summary of mechanical properties of Narmco N588/7781 (ECDE-1/0-550) fiberglass epoxy.*

Fabrication		Lay-up: Balanced		Vacuum: None		Pressure: 45-55 psi		Bleedout: Vertical		Cure: Stepwise to 350°F: 1hr/350°F		Postcure: None		Plies: 8	
Physical Properties		Weight Percent Resin: 32.8 v_f = 0.51				Avg. Specific Gravity: 1.91				Avg. Percent Voids: 1.0				Avg. Thickness: 0.075 inches	
Test Methods		Tension: ASTM-D638 TYPE 1		Compression: MIL-HDBK-17		Shear: Rail		Flexure: ASTM-D790		Bearing: ASTM-D953				Interlaminar Shear: Short Beam	
Temperature		-65°F				75°F				160°F				400°F	
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry	
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD
Tension															
ultimate stress, ksi	0°	71.4	2.4	63.8	3.3	58.4	2.1	50.0	2.3	48.8	3.0	35.0	2.0	40.4	3.4
	90°	59.3	3.3	50.6	2.4	47.2	3.8	41.1	2.7	41.4	2.0	28.9	2.8	33.3	3.8
ultimate strain, %	0°	2.41	0.09	2.06	0.15	2.05	0.18	1.61	0.12	1.59	0.15	1.13	0.07	1.26	0.07
	90°	2.35	0.17	1.96	0.12	1.81	0.16	1.55	0.16	1.67	0.10	1.17	0.14	1.25	0.12
proportional limit, ksi	0°	26.6	1.7	28.7	2.5	23.3	1.1	25.4	2.8	21.0	1.7	29.9	2.0	24.3	
	90°	19.3	0.8	19.2	1.6	17.6	0.8	18.1	1.4	17.3	2.5	20.9	1.3	14.3	
initial modulus, 10^6 psi	0°	3.64		3.85		3.71		3.57		3.58		3.10		3.13	0.17
	90°	3.41		3.37		3.56		3.23		2.92		2.63		2.80	0.23
secondary modulus, 10^6 psi	0°														
	90°														
Compression															
ultimate stress, ksi	0°	99.2	5.9	87.4	5.8	74.0	3.6	63.5	3.2	59.0	2.4	49.5	1.9		
	90°	83.4	3.5	71.8	4.1	62.9	2.9	53.7	1.7	50.9	1.5	40.7	1.8		
ultimate strain, %	0°	2.52	0.26	2.30	0.25	1.89	0.15	1.65	0.19	1.60	0.12	1.38	0.06		
	90°	2.30	0.27	2.06	0.20	1.87	0.14	1.58	0.15	1.63	0.16	1.29	0.08		
proportional limit, ksi	0°	42.7	2.6	46.2	2.5	44.5	3.2	39.8	3.6	37.6	2.7	30.7	2.7		
	90°	40.8	3.8	42.4	2.7	35.3	3.7	34.4	2.3	31.2	2.4	24.4	1.6		
initial modulus, 10^6 psi	0°	4.32		4.15		4.18		4.11		3.88		3.70			
	90°	4.08		3.83		3.68		3.72		3.41		3.41			
Shear															
ultimate stress, ksi	0°-90° ±45°	22.6				16.0	1.05			13.8					

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	105.0	115.6	95.6	90.4	102.6	84.5	79.3	87.8	74.0
proportional limit, ksi	0°	69.6	75.9	59.0	68.9	72.4	64.6	64.8	72.2	57.2
initial modulus, 10^6 psi	0°	3.48	3.62	3.42	3.36	3.60	3.20	3.19	3.27	3.09
Bearing										
ultimate stress, ksi	0°	84.6	92.5	77.9	68.4	71.3	66.0	48.4	53.6	44.2
stress at 4% elong., ksi	0°	29.3	30.9	26.5	26.2	27.4	25.3	21.8	22.8	20.6
Interlaminar Shear										
ultimate stress, ksi	0°	8.84	9.16	8.56	8.35	8.56	8.05	7.39	7.72	6.47

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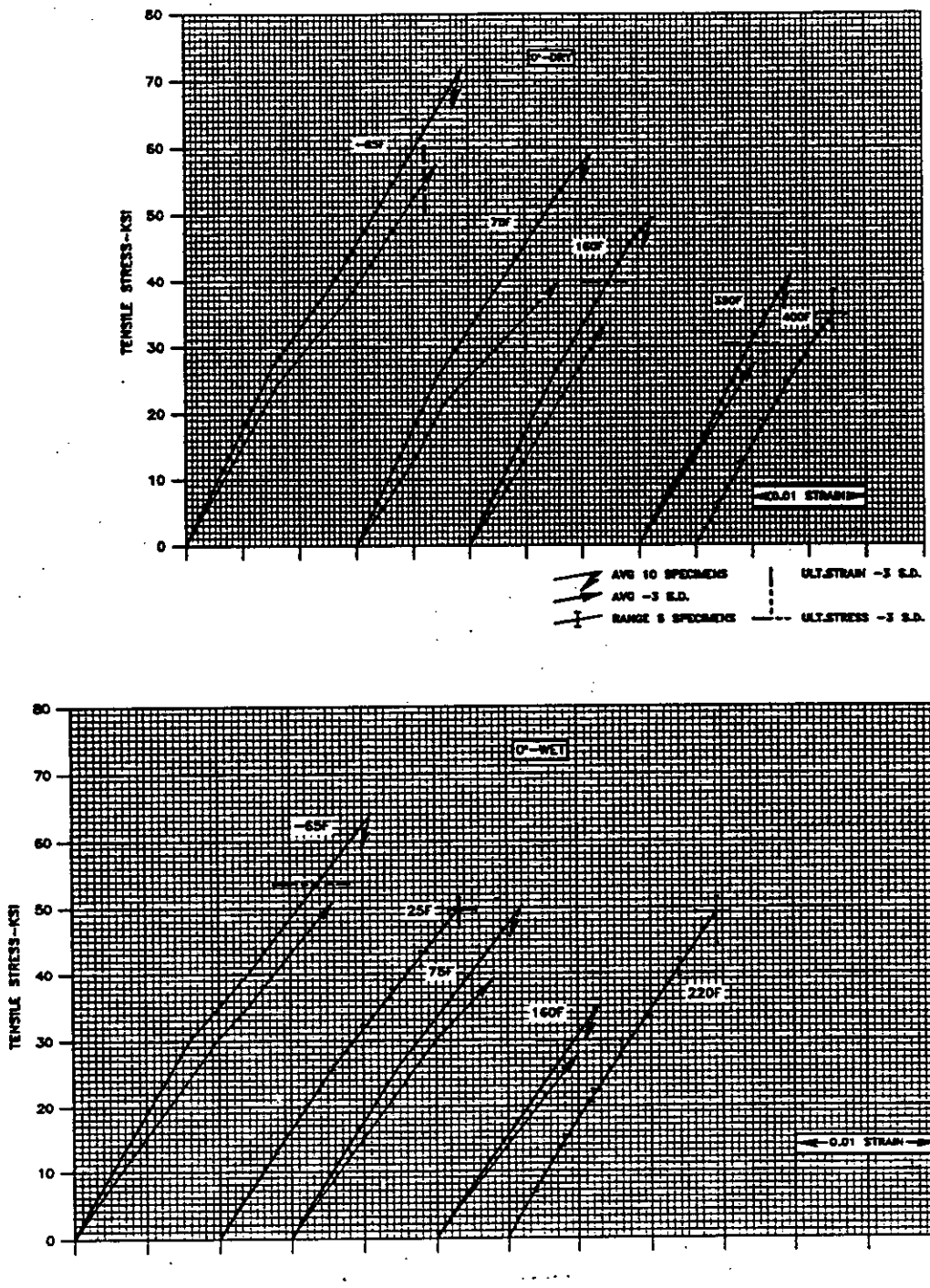


FIGURE A1.8.1(a) *Tensile stress-strain for N588/7781 fiberglass epoxy loaded in the 0° direction.*

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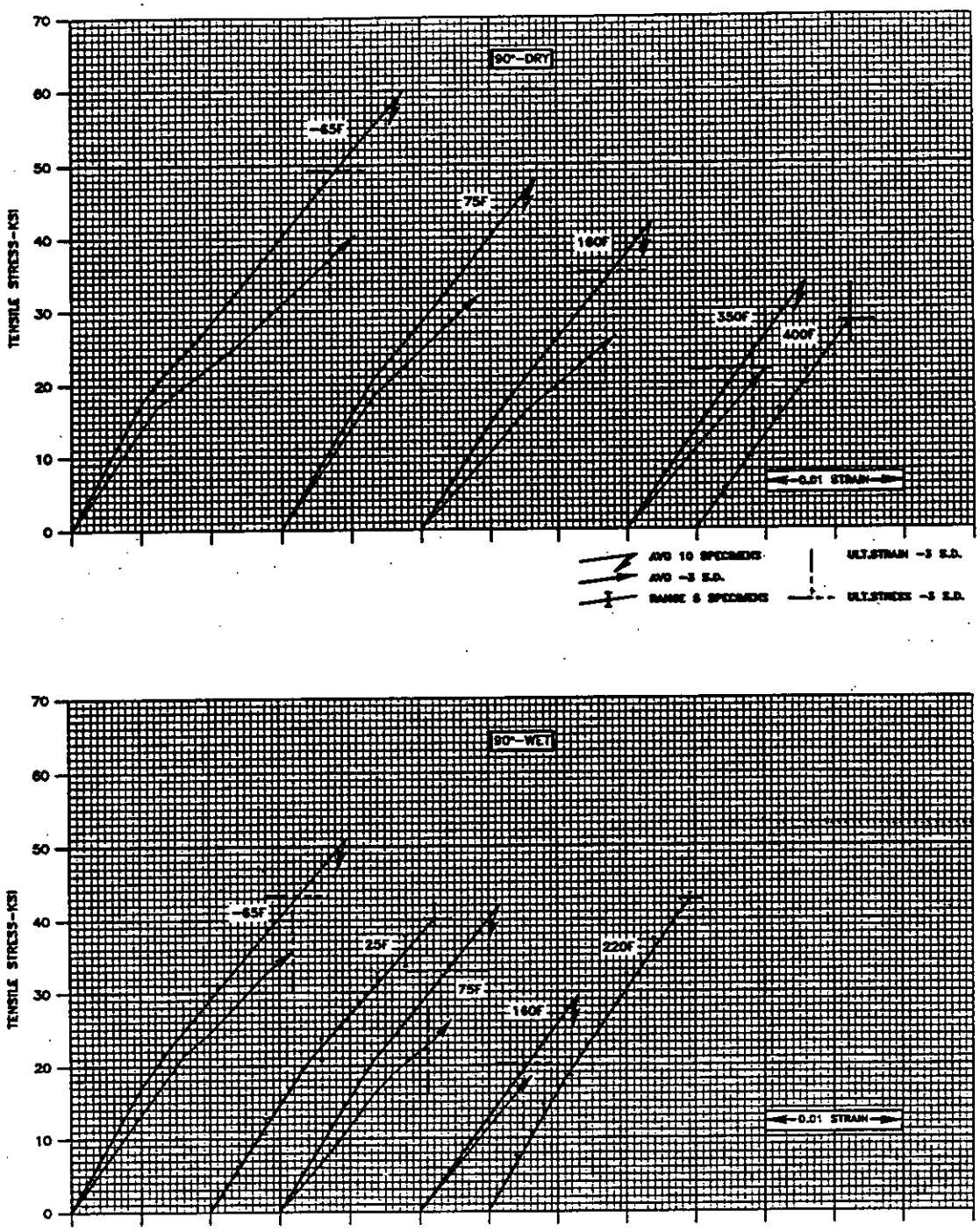


FIGURE A1.8.1(b) *Tensile stress-strain for N588/7781 fiberglass epoxy loaded in the 90° direction.*

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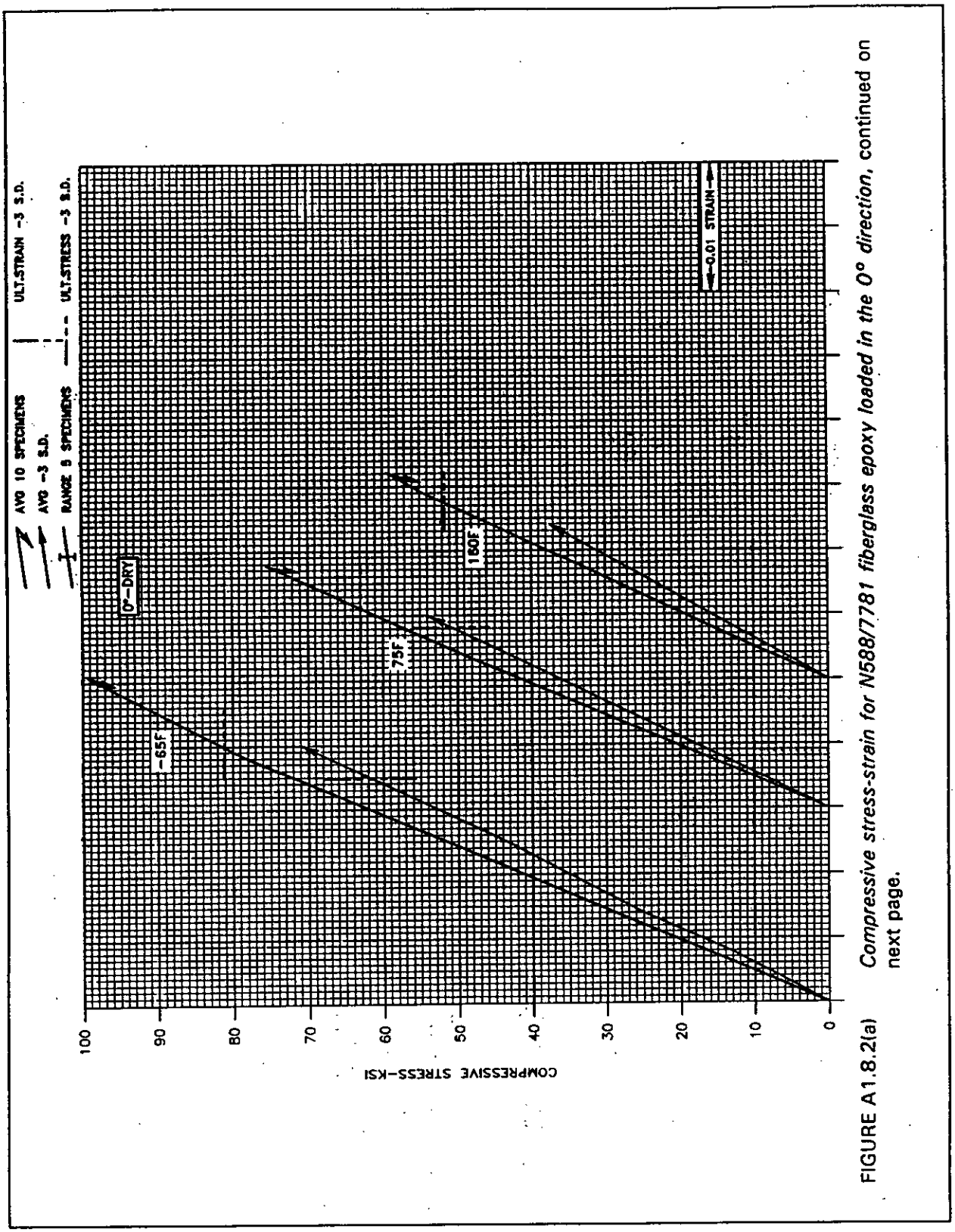


FIGURE A1.8.2(a) *Compressive stress-strain for N588/7781 fiberglass epoxy loaded in the 0° direction,* continued on next page.

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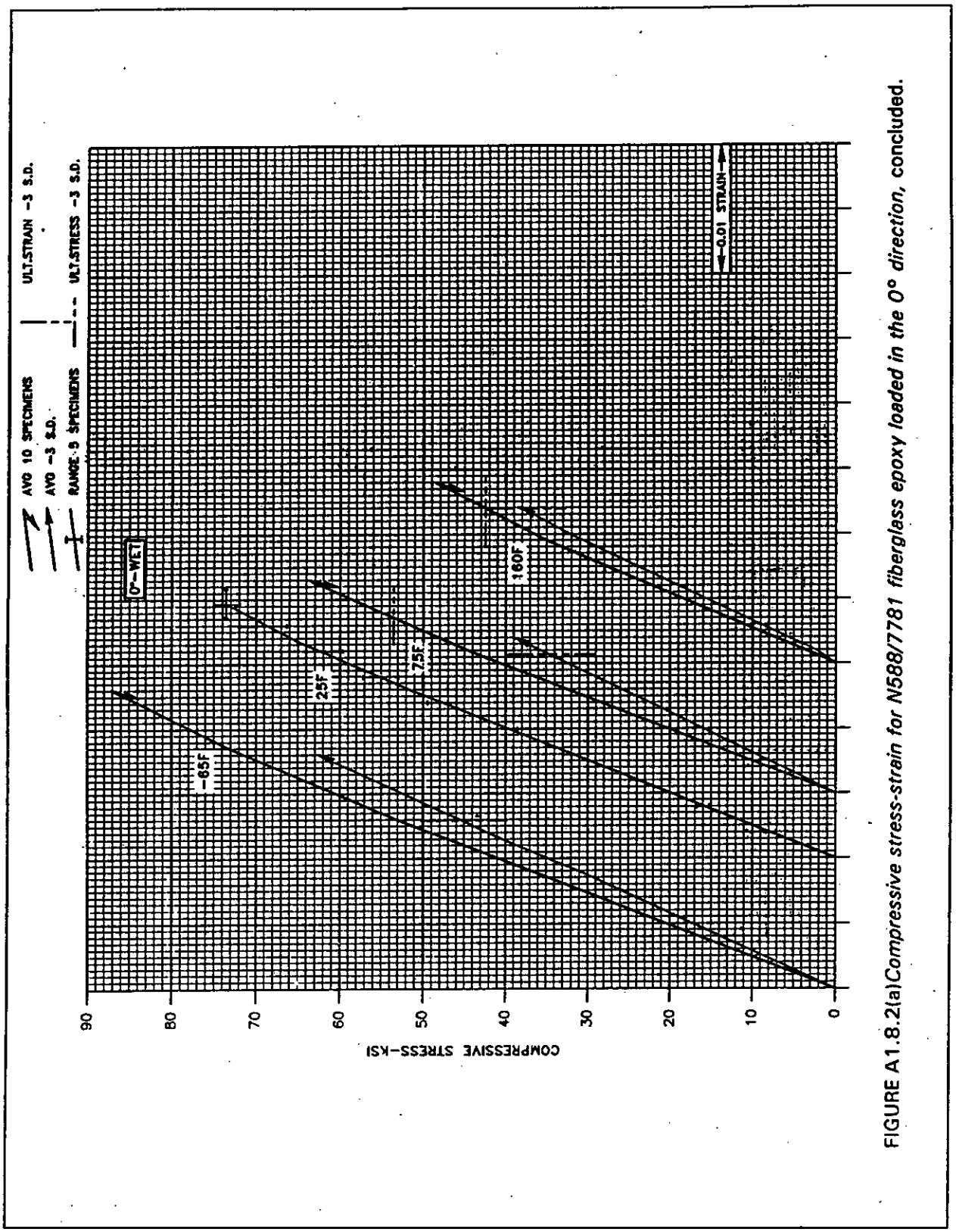


FIGURE A1.8.2(a) *Compressive stress-strain for N588/7781 fiberglass epoxy loaded in the 0° direction,* concluded.

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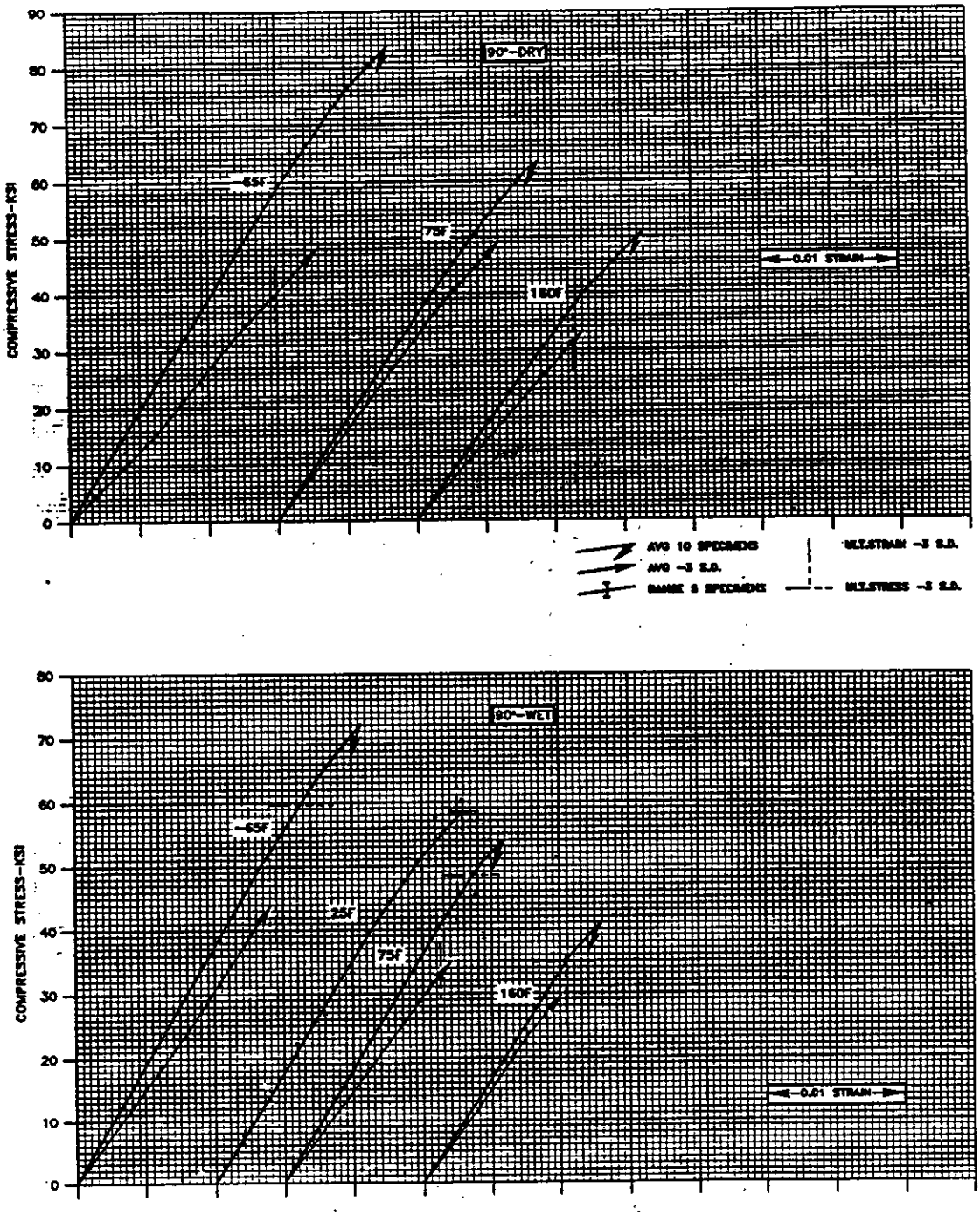


FIGURE A1.8.2(b) *Compressive stress-strain for N588/7781 fiberglass epoxy loaded in the 90° direction.*

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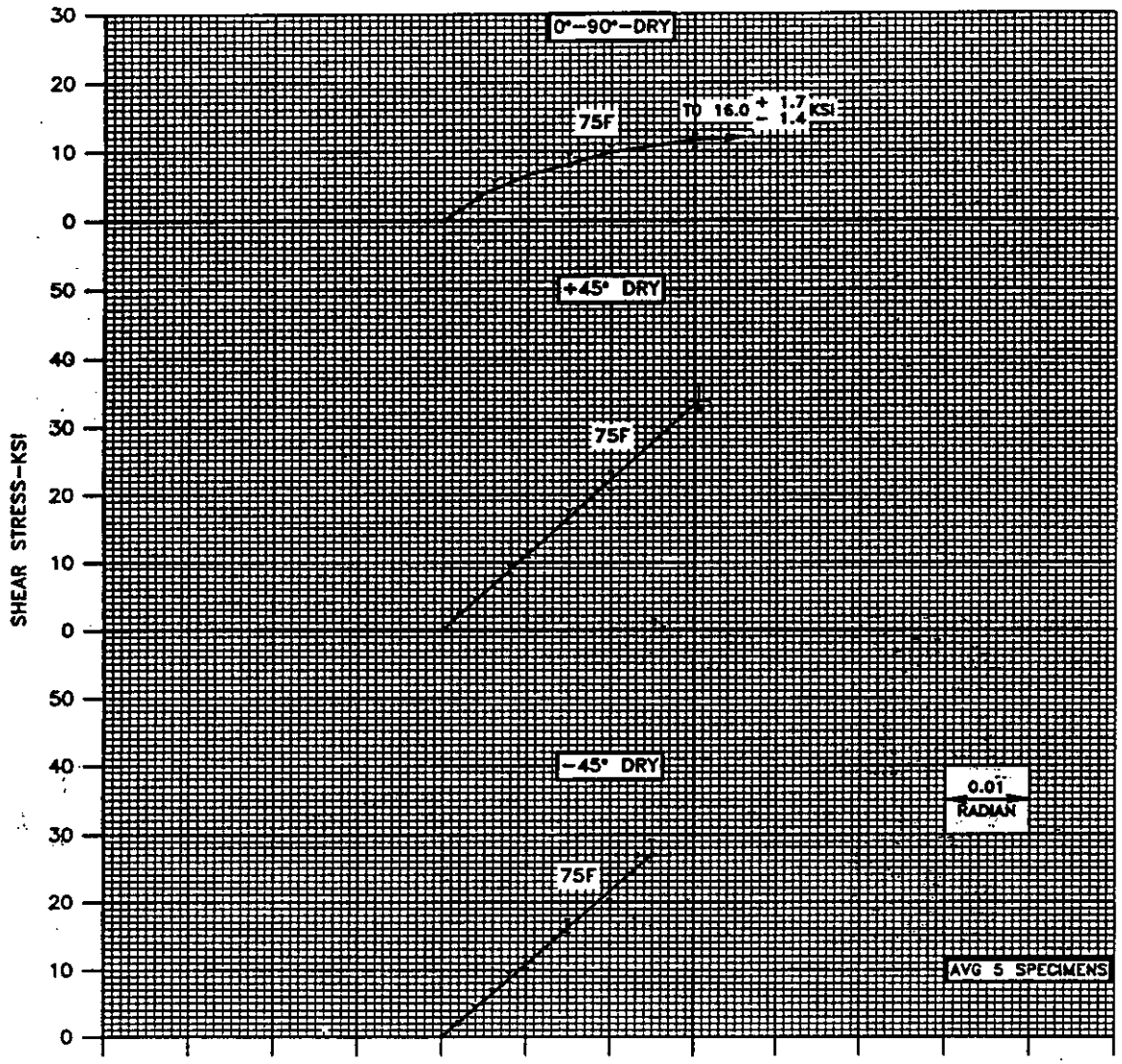


FIGURE A1.8.3 *Rail shear for N588/7781 fiberglass epoxy.*

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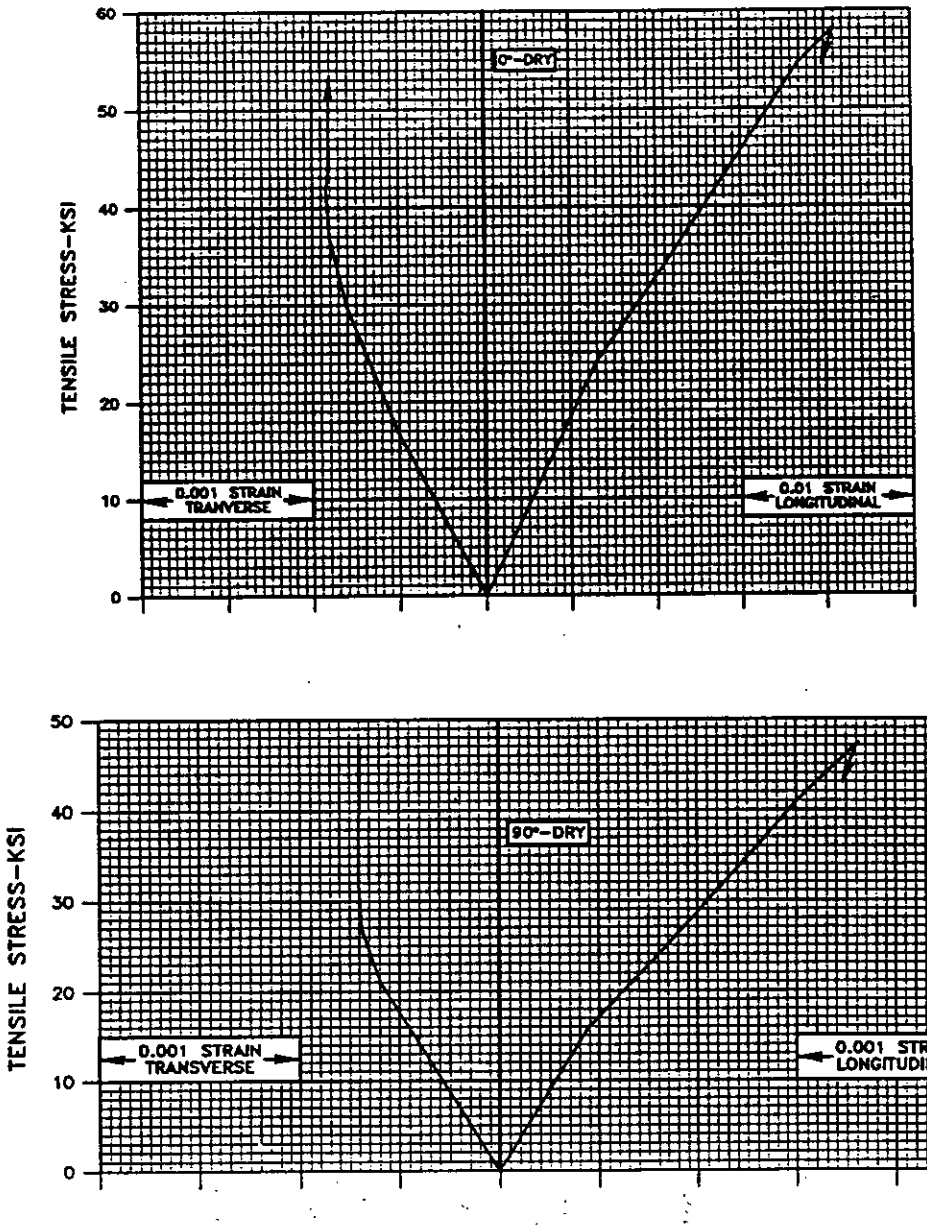


FIGURE A1.8.4 *Poisson effects for N588/7781 fiberglass epoxy.*

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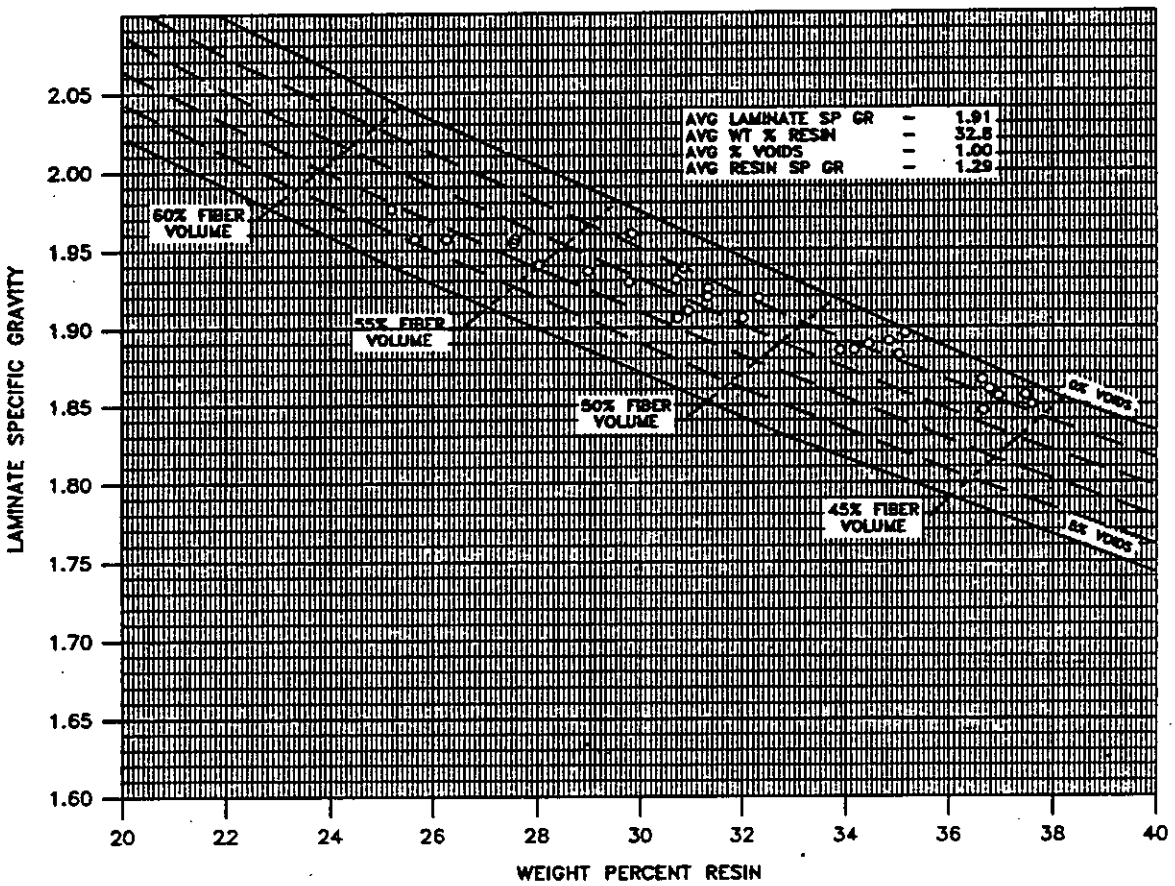


FIGURE A1.8.5 *Voids vs resin content and specific gravity for N588/7781 fiberglass epoxy.*

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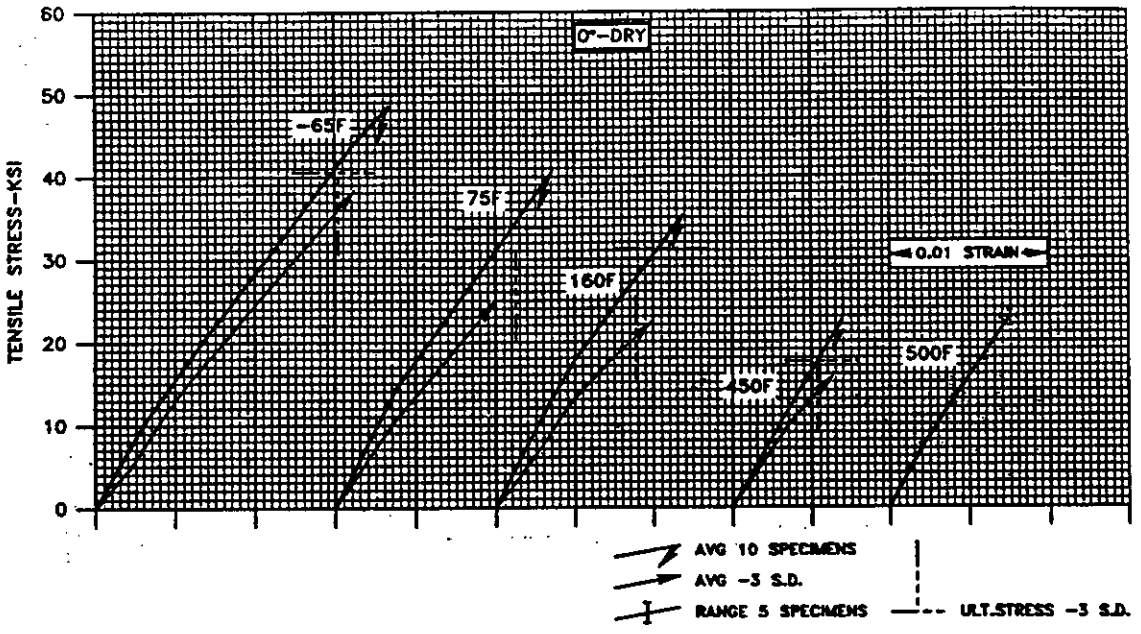
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TABLE A1.40 *Summary of mechanical properties of Narmco 506/7781 (ECDE-1/0-A1100) fiberglass phenolic.*

Fabrication		Lay-up: Balanced		Vacuum:		Pressure:		Bleedout: Vertical		Cure:		Postcure:		Plies: 8		
Physical Properties		Weight Percent Resin: 25.3 - 32.3				Avg. Specific Gravity: 1.72 - 1.85				Avg. Percent Voids: Figure 4.40.5				Avg. Thickness: 0.071 - 0.095 inches		
Test Methods		Tension: ASTM-D638 TYPE 1		Compression: MIL-HDBK-17		Shear: Rail				Flexure: ASTM-D790		Bearing: ASTM-D953		Interlaminar Shear: Short Beam		
Temperature		-65°F				75°F				160°F				400°F		
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry		
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	
Tension																
ultimate stress, ksi	0°	48.1	2.4	49.8	3.3	38.9	1.5	37.2	1.8	35.3	1.4	30.6	3.0	21.6	1.6	
	90°	37.9	1.8	40.0	2.7	31.5	1.5	32.1	1.4	27.9	1.7	26.2	2.2	21.6	1.7	
ultimate strain, %	0°	1.76	0.07	1.76	0.13	1.33	0.14	1.34	0.13	1.19	0.10	1.15	0.14	0.69	0.05	
	90°	1.63	0.08	1.65	0.13	1.26	0.15	1.32	0.07	1.11	0.07	1.11	0.14	0.78	0.06	
proportional limit, ksi	0°	13.6	0.9	18.1	1.2	13.5	0.6	17.0	1.0	13.9	1.0	14.9	0.70	9.7	1.1	
	90°	9.9	0.4	12.5	0.9	9.2	0.8	12.8	0.7	10.3	0.8	11.6	0.70	8.6	0.5	
initial modulus, 10^6 psi	0°	3.40	0.21	3.35	0.20	3.94	0.69	3.14	0.26	3.74	0.41	3.01	0.19	3.57	0.24	
	90°	3.08	0.29	3.04	0.22	3.54	0.41	2.81	0.24	3.33	0.37	2.78	0.21	3.18	0.30	
secondary modulus, 10^6 psi	0°															
	90°															
Compression																
ultimate stress, ksi	0°	66.7	6.2	65.9	5.0	59.7	4.7	54.5	7.1	50.6	2.3	49.2	4.2			
	90°	57.7	5.8	56.2	5.8	49.0	4.6	48.7	4.0	43.0	4.3	42.9	3.7			
ultimate strain, %	0°	1.85	0.09	1.69	0.18	1.58	0.14	1.49	0.12	1.45	0.06	1.40	0.12			
	90°	1.70	0.21	1.63	0.13	1.40	0.09	1.43	0.07	1.37	0.12	1.31	0.15			
proportional limit, ksi	0°	45.8	3.8	38.5	7.9	39.0	2.4	41.2	4.6	39.9	2.4	35.0	1.7			
	90°	35.2	3.8	34.4	5.0	32.6	4.4	35.5	3.0	32.4	3.1	31.1	3.3			
initial modulus, 10^6 psi	0°	3.90	0.19	4.17	0.29	3.95	0.28	3.89	0.26	3.68	0.21	3.67	0.12			
	90°	3.69	0.25	3.68	0.17	3.70	0.20	3.57	0.20	3.30	0.23	3.45	0.21			
Shear																
ultimate stress, ksi	0°-90° ±45°	13.8				12.3	0.97			11.4						

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°	68.2	72.8	65.2	58.4	64.0	52.1	52.7	56.3	47.4
proportional limit, ksi	0°	59.3	66.1	54.6	48.9	56.8	42.5	42.4	46.2	38.8
initial modulus, 10^6 psi	0°	2.97	3.04	2.88	2.89	2.99	2.78	2.97	3.06	2.82
Bearing										
ultimate stress, ksi	0°	65.7	73.2	57.0	58.9	64.0	46.8	49.5	55.8	44.5
stress at 4% elong., ksi	0°	25.1	26.0	23.7	24.5	24.9	23.8	21.6	22.6	20.7
Interlaminar Shear										
ultimate stress, ksi	0°	4.83	5.10	4.29	4.64	4.92	3.94	4.62	4.88	4.08

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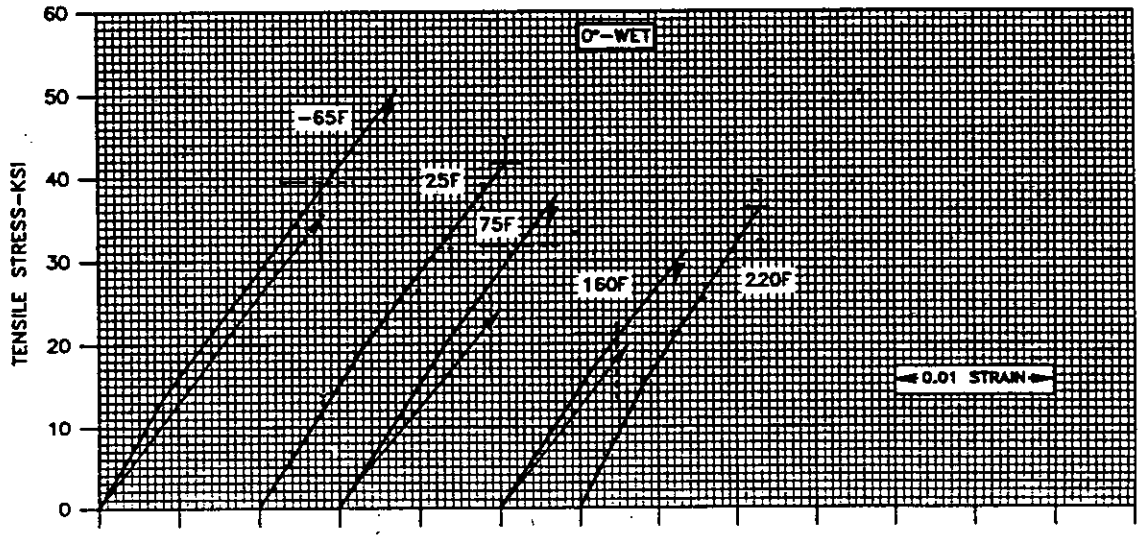


FIGURE A1.40.1(a) *Tensile stress-strain for N506/7781 fiberglass phenolic loaded in the 0° direction.*

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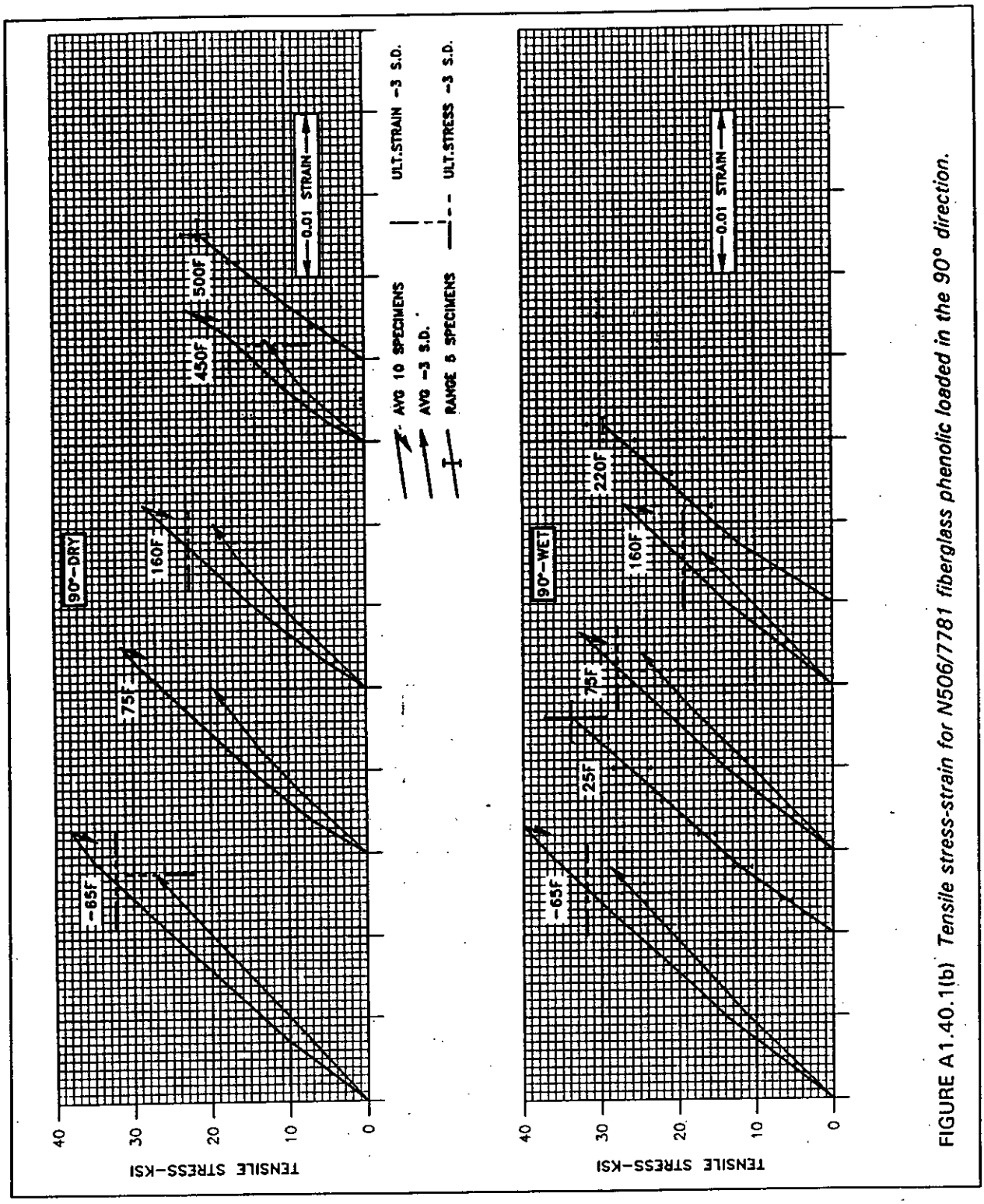


FIGURE A1.40.1(b) *Tensile stress-strain for N506/7781 fiberglass phenolic loaded in the 90° direction.*

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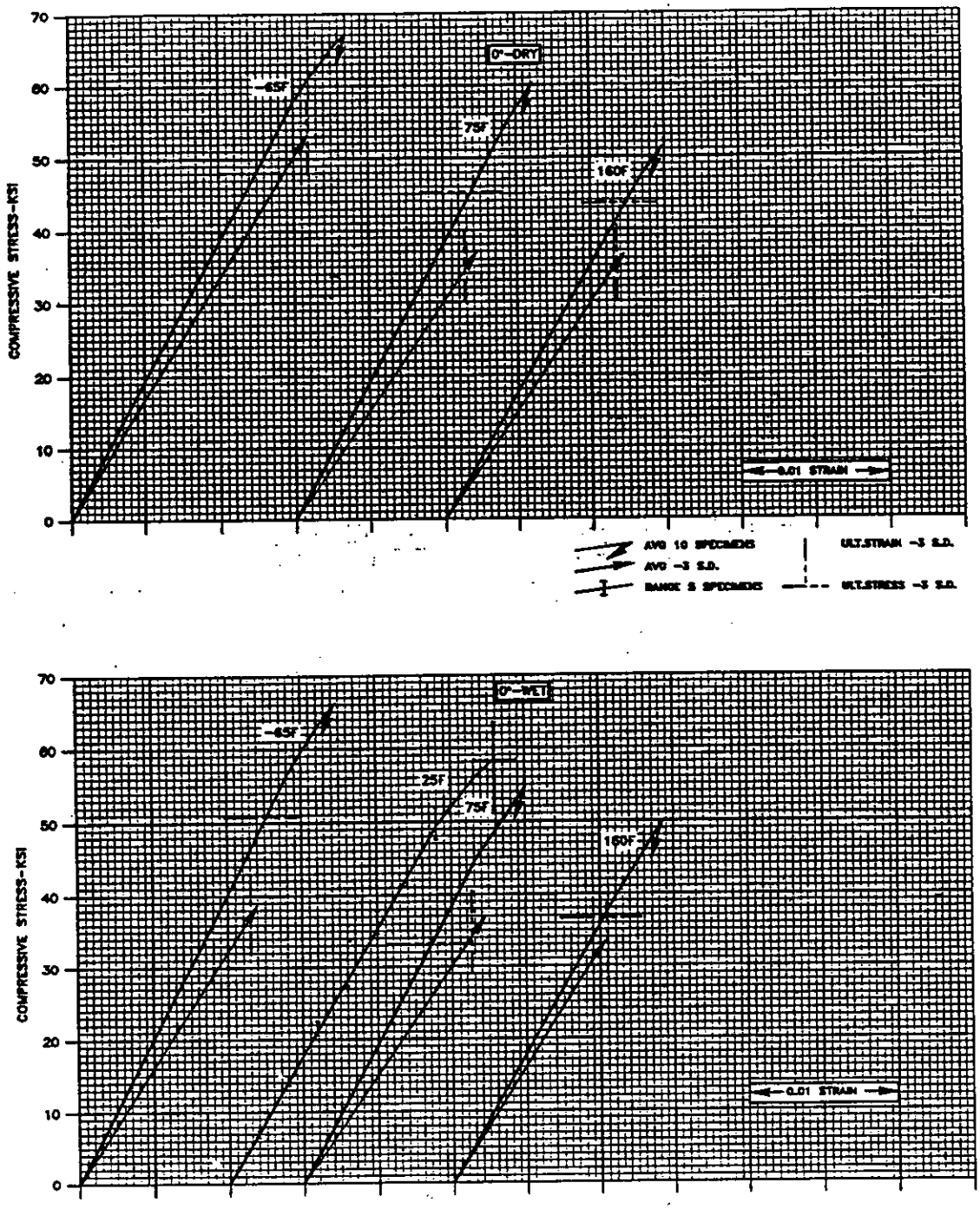


FIGURE A1.40.2(a) *Compressive stress-strain for N506/7781 fiberglass phenolic loaded in the 0° direction.*

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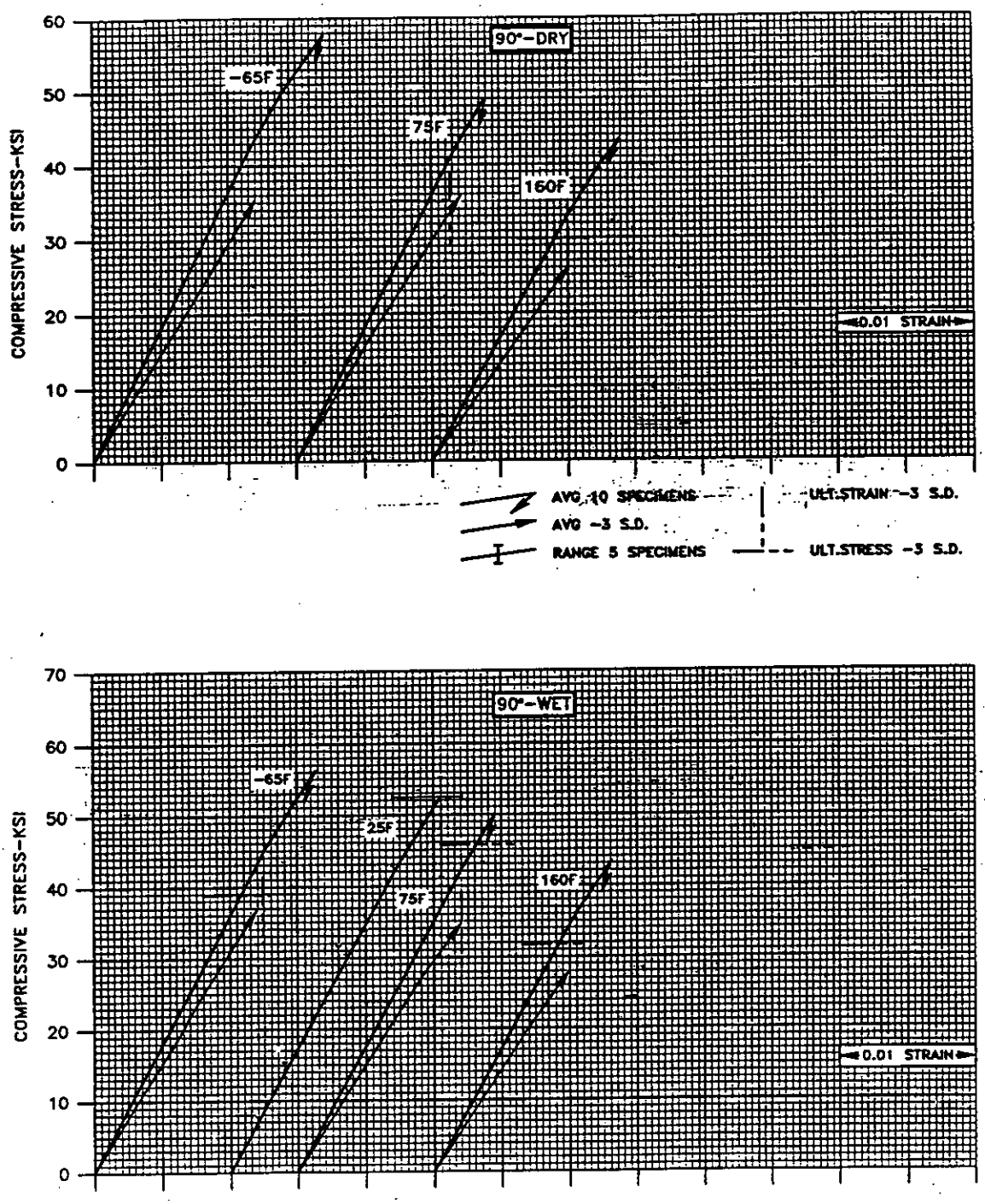


FIGURE A1.40.2(b) *Compressive stress-strain for N506/7781 fiberglass phenolic loaded in the 90° direction.*

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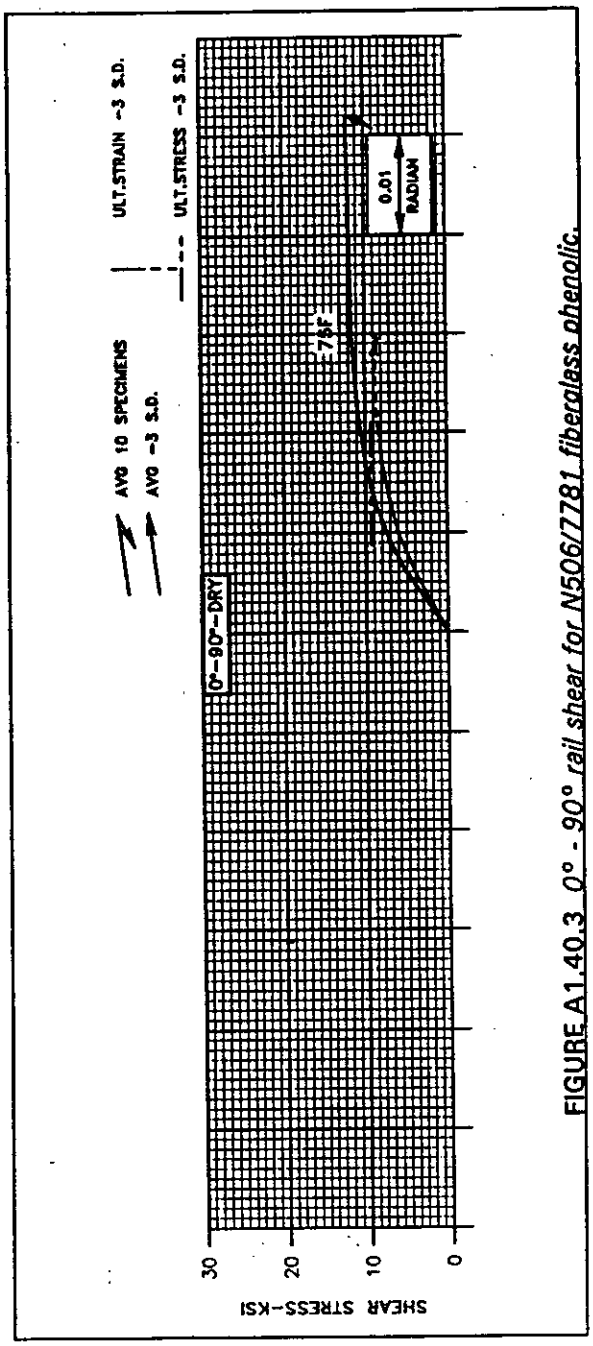


FIGURE A1.40.3 *0° - 90° rail shear for N506/7781 fiberglass phenolic.*

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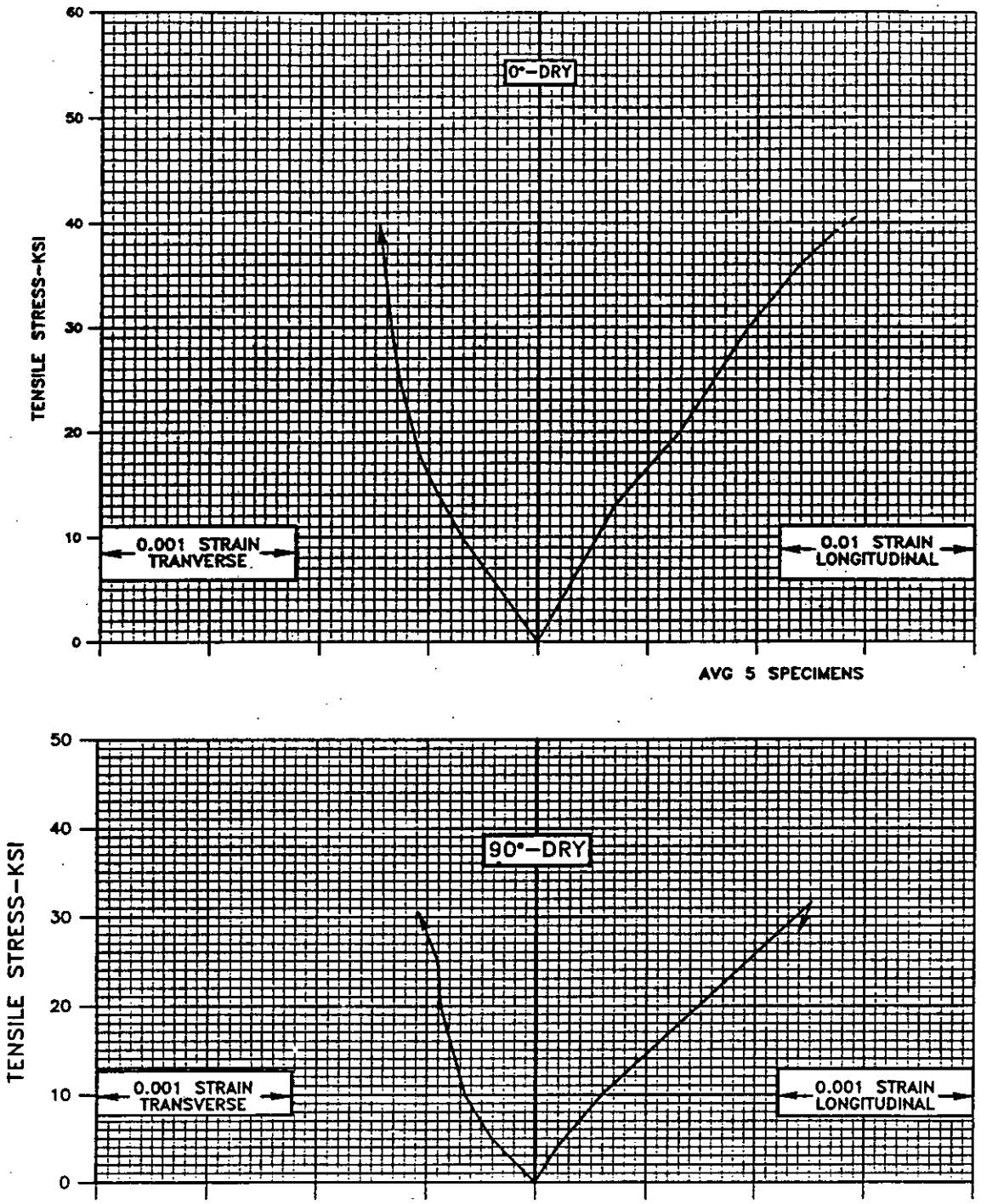


FIGURE A1.40.4 *Poisson effects for N506/7781 fiberglass phenolic.*

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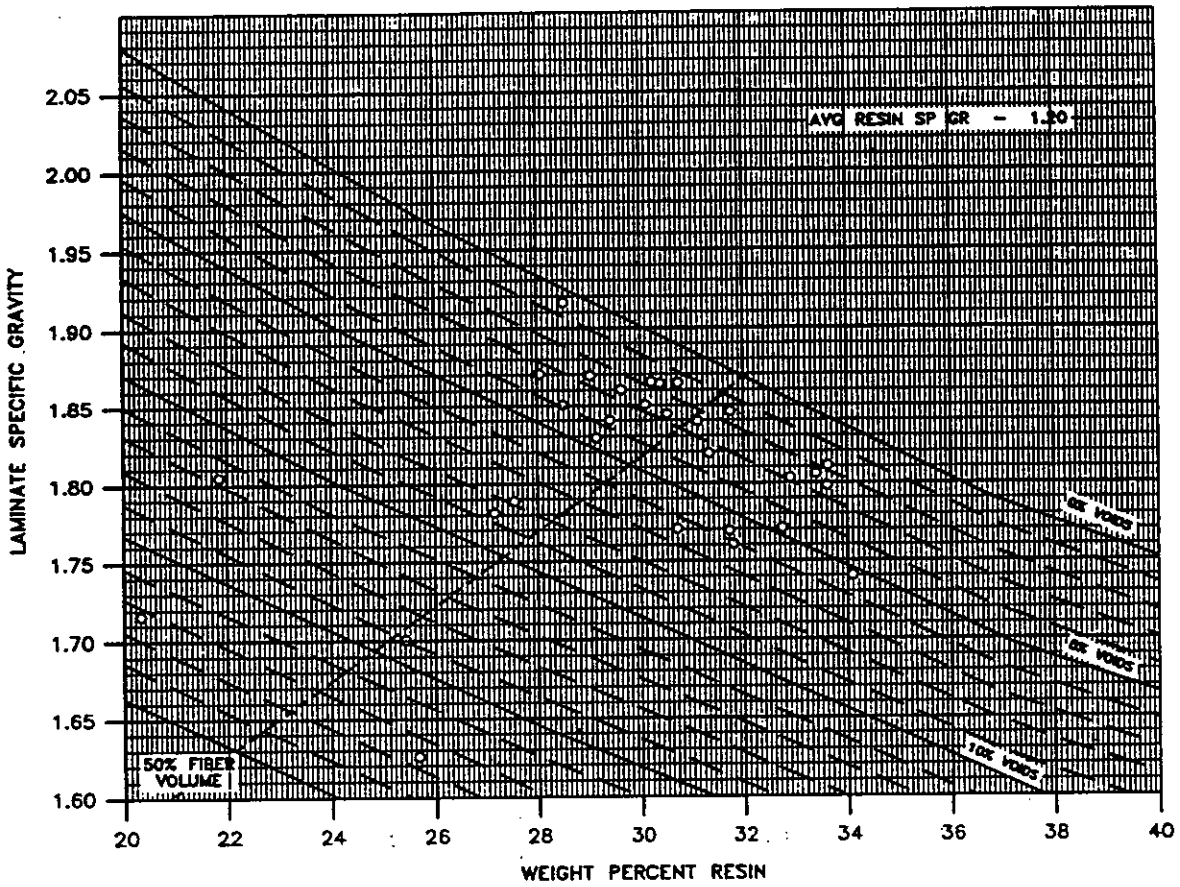


FIGURE A1.40.5 *Voids vs. resin content and specific gravity for N506/7781 fiberglass phenolic.*

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TABLE A1.110 *Summary of mechanical properties of AVCO 5505 boron-epoxy (100%-0° direction) (Tentative).*

Fabrication		Lay-up: Parallel		Vacuum: 2 ins		Pressure: 50 ± 5 psi		Bleedout:		Cure: 1.5hr/ 350°F ± 10°F		Postcure: 2hr/350°F		Plies: 6			
Physical Properties		Weight Percent Resin:		Avg. Specific Gravity:				Avg. Percent Voids:				Avg. Thickness: 0.005 in/ply					
Test Methods		Tension: Tab-ended		Compression: Sandwich Beam		Shear:		Flexure: 4 Point Loading				Bearing:		Interlaminar Shear: Short Beam			
Temperature		-67°F				75°F				260°F				375°F			
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry			
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD		
Tension																	
ultimate stress, ksi	0°	201.1				208.3				191.6				167.3			
	90°	10.5				8.7				6.5				3.3			
ultimate strain, %	0°	6390				6930				6660				6150			
	90°	3250				3710				4970				5920			
proportional limit, ksi	0°	141.8				175.5				140.0				79.5			
	90°																
initial modulus, 10^6 psi	0°	32.0				30.9				29.6				28.6			
	90°																
secondary modulus, 10^6 psi	0°																
	90°																
Compression																	
ultimate stress, ksi	0°	482.3				378.0				303.3				143.9			
	90°																
ultimate strain, %	0°	13670				10830				8920				4466			
	90°																
proportional limit, ksi	0°	333.5															
	90°																
initial modulus, 10^6 psi	0°	35.7				34.8				34.6				35.8			
	90°																
Shear																	
ultimate stress, ksi	0°-90°																
	±45°																

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°									
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°									

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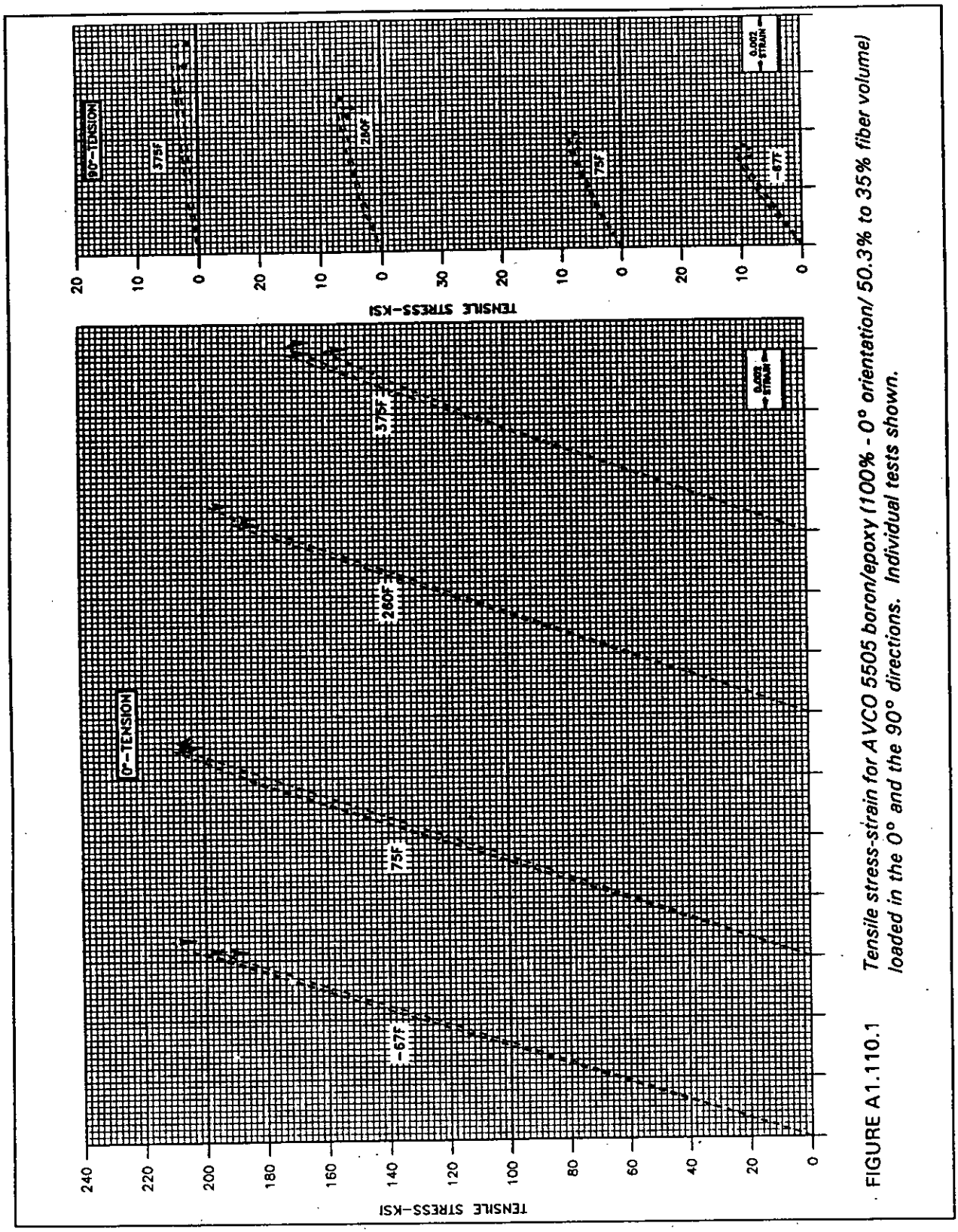


FIGURE A1.110.1 *Tensile stress-strain for AVCO 5505 boron/epoxy (100% - 0° orientation/ 50.3% to 35% fiber volume) loaded in the 0° and the 90° directions. Individual tests shown.*

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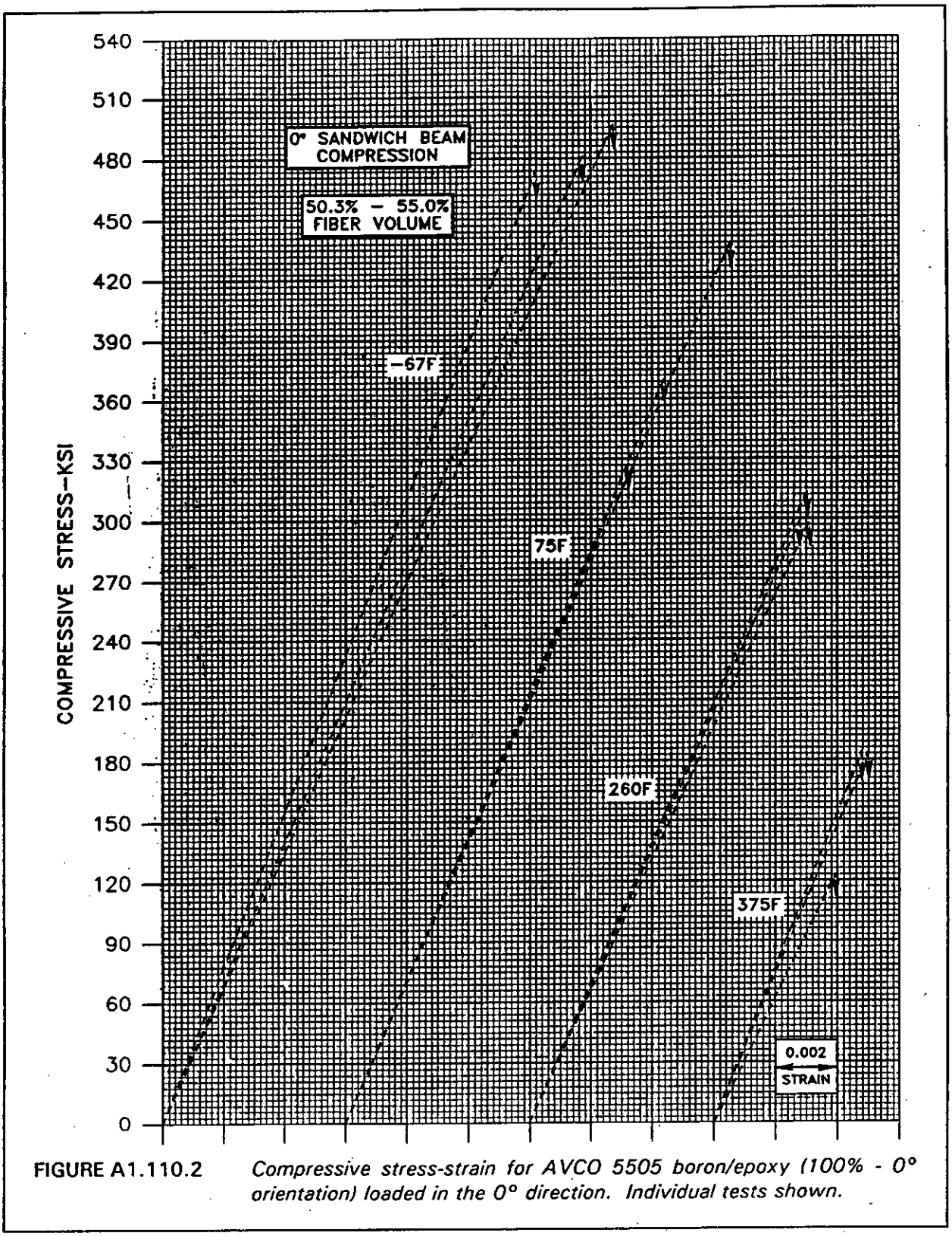


FIGURE A1.110.2 *Compressive stress-strain for AVCO 5505 boron/epoxy (100% - 0° orientation) loaded in the 0° direction. Individual tests shown.*

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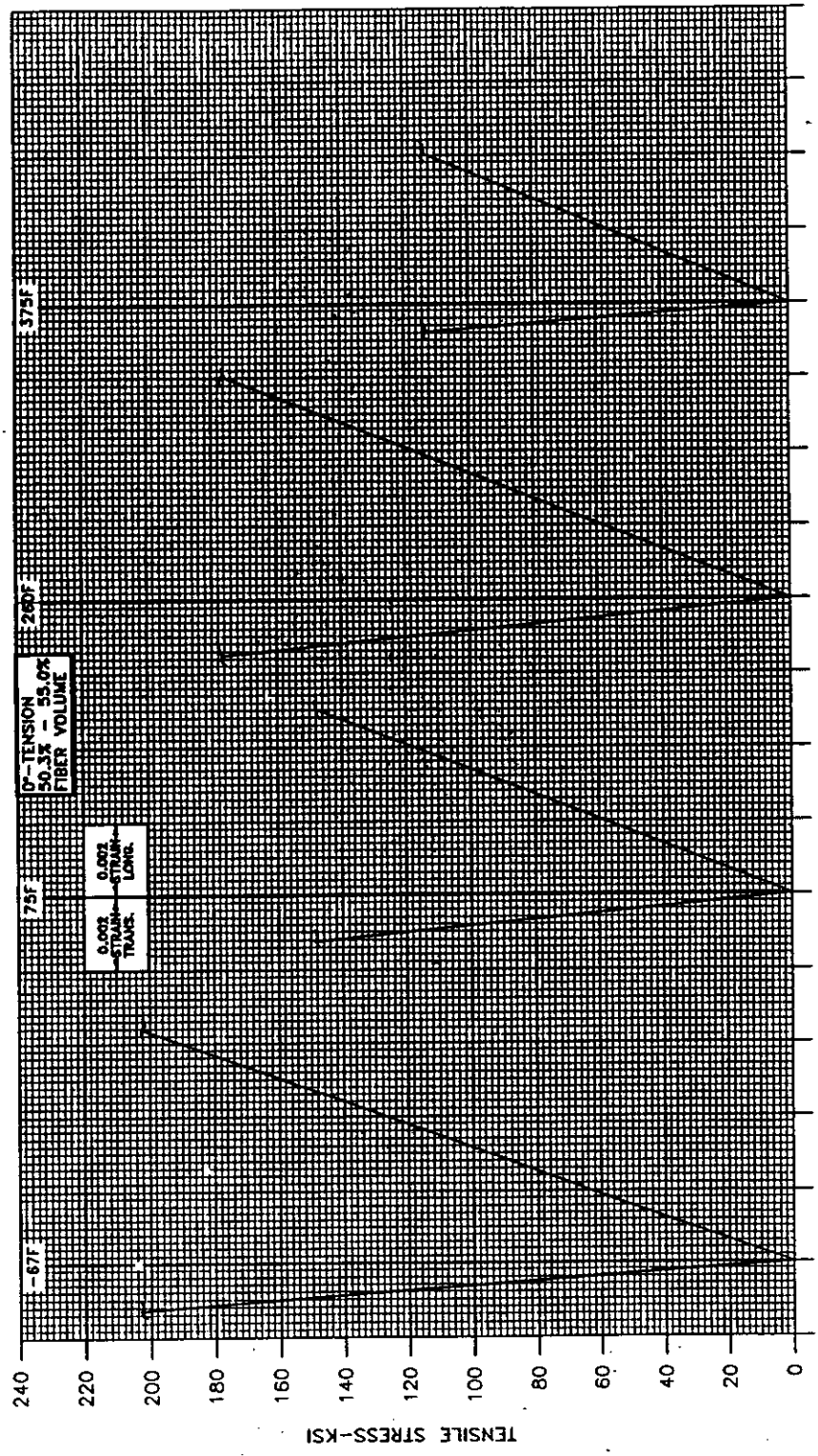


FIGURE A1.110.3 *Poisson effects for AVCO 5505 boron/epoxy (100% - 0° direction).*

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TABLE A1.111 *Summary of mechanical properties of AVCO 5505 boron-epoxy (0°-90°crossply) (Tentative).*

Fabrication		Lay-up: [2(0/90)]S		Vacuum: 2 ins		Pressure: 50 ± 5 psi		Bleedout:		Cure: 1.5hr/ 350°F ± 10°F		Postcure: 2hr/380°F		Plies: 6			
Physical Properties		Weight Percent Resin:		Avg. Specific Gravity:		Avg. Percent Voids:		Avg. Thickness: 0.005 in/ply									
Test Methods		Tension: Tab-ended		Compression:		Shear: Picture Frame		Flexure:		Bearing:		Interlaminar Shear:					
Temperature		-67°F				75°F				260°F				375°F			
Condition		Dry		Wet		Dry		Wet		Dry		Wet		Dry			
		Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD	Avg	SD		
Tension																	
ultimate stress, ksi	0°	99.9				103.9				98.5				91.9			
	90°	23.6				17.8				11.4				8.1			
ultimate strain, %	0°	5400				5710				5830				5780			
	90°	15850				24470											
proportional limit, ksi	0°	53.0				77.7				48.6				48.6			
	90°																
initial modulus, 10^6 psi	0°	18.9				18.0				17.5				16.5			
	90°																
secondary modulus, 10^6 psi	0°																
	90°																
Compression																	
ultimate stress, ksi	0°																
	90°																
ultimate strain, %	0°																
	90°																
proportional limit, ksi	0°																
	90°																
initial modulus, 10^6 psi	0°																
	90°																
Shear																	
ultimate stress, ksi	0°-90°	19.5				17.3									5.4		
	±45°	65.7				63.7									33.3		

		-65°F Dry			75°F Dry			160° Dry		
		Avg	Max	Min	Avg	Max	Min	Avg	Max	Min
Flexure										
ultimate stress, ksi	0°									
proportional limit, ksi	0°									
initial modulus, 10^6 psi	0°									
Bearing										
ultimate stress, ksi	0°									
stress at 4% elong., ksi	0°									
Interlaminar Shear										
ultimate stress, ksi	0°									

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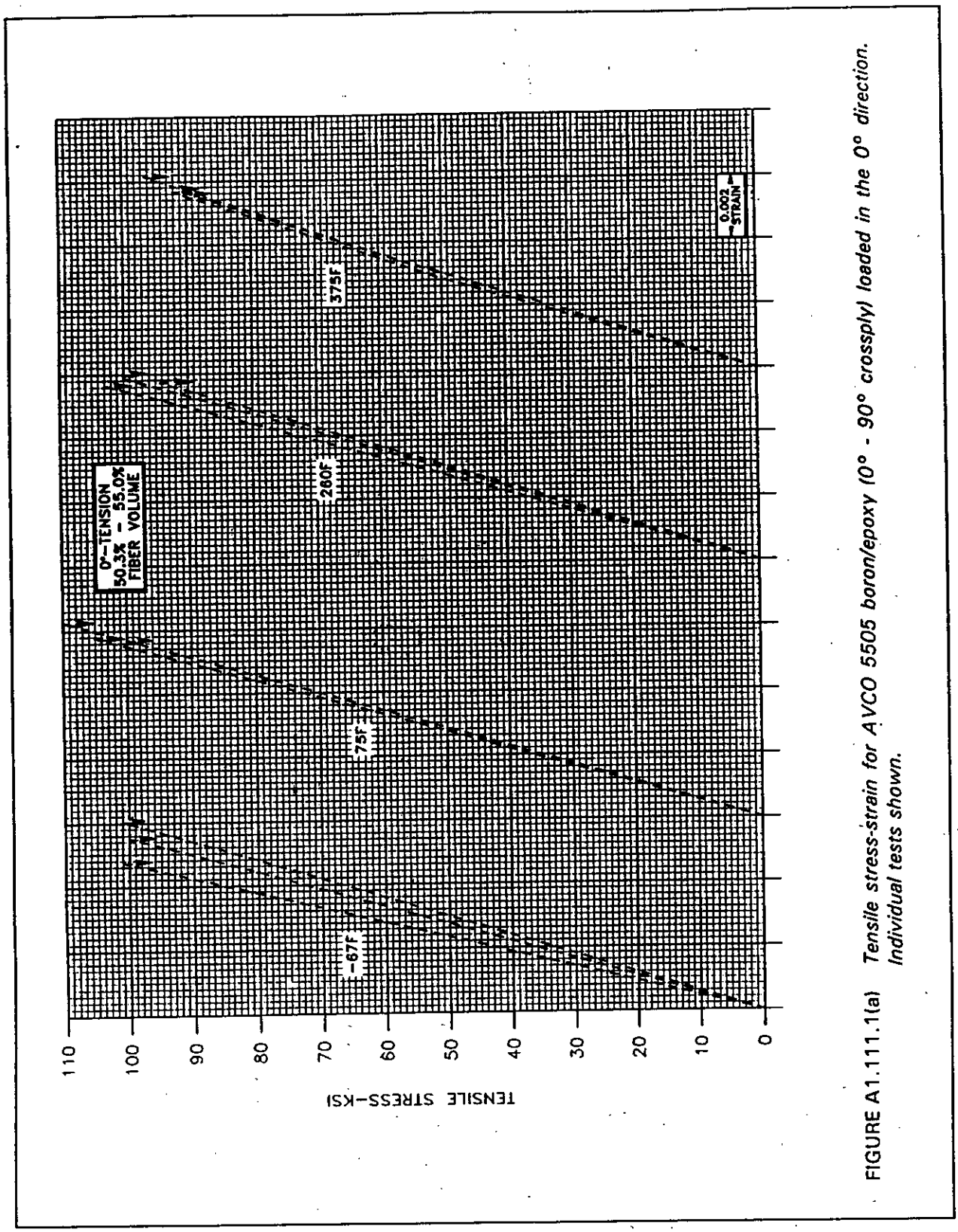


FIGURE A1.111.1(a) *Tensile stress-strain for AVCO 5505 boron/epoxy (0° - 90° crossply) loaded in the 0° direction. Individual tests shown.*

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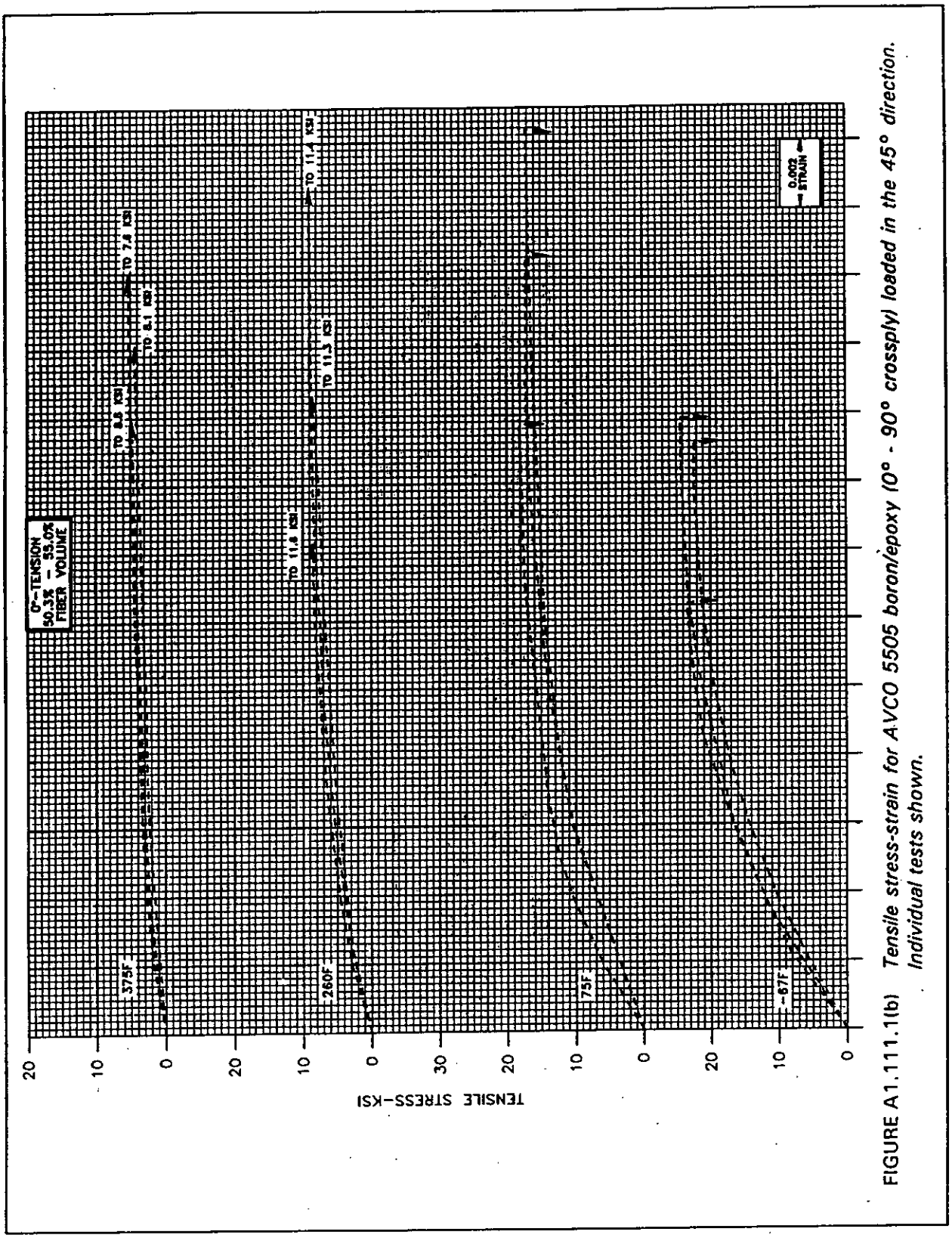


FIGURE A1.111.1(b) *Tensile stress-strain for AVCO 5505 boron/epoxy [0° - 90° crossply] loaded in the 45° direction. Individual tests shown.*

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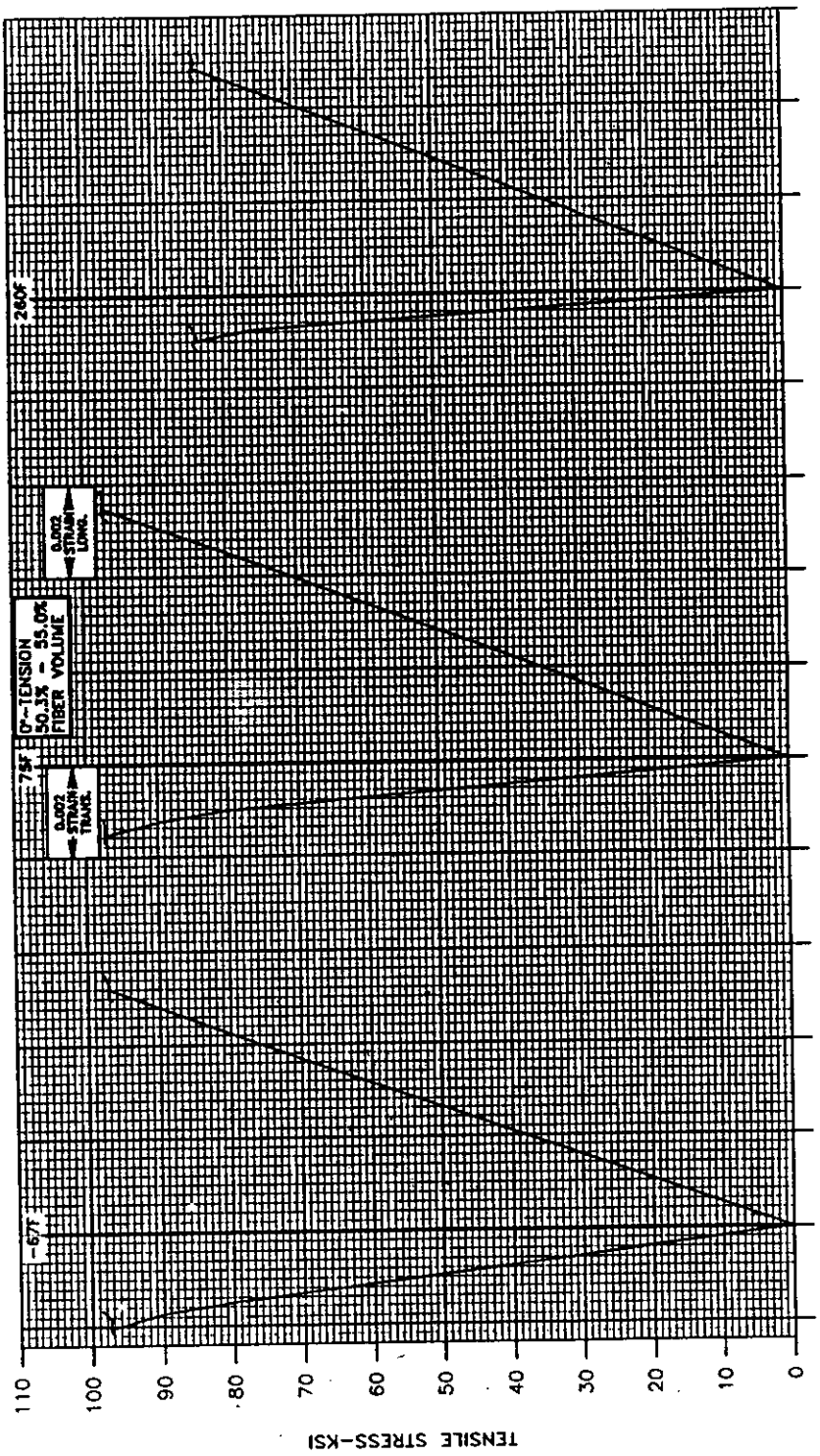


FIGURE A1.111.3 *Poisson effects for AVCO 5505 boron/epoxy (0° - 90° crossply).*

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